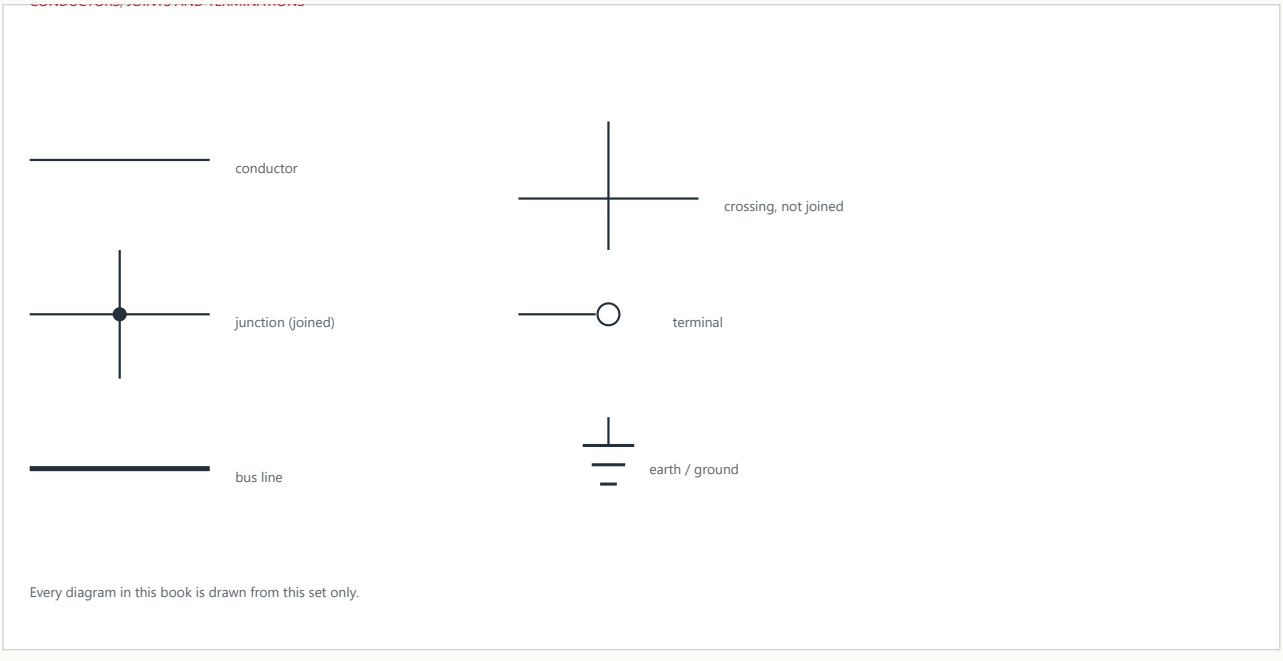
TRUEPHASE LEARNING SERIES

# Electrical Theory

A working textbook of circuits, fields, machines, electronics, power and field practice

Twenty chapters with worked examples, measurement activities, troubleshooting tables, exercises and a complete reference section. Every schematic in this book is drawn from one approved symbol set.




The conductor and termination symbols used throughout this book.

# About this book

This book develops electrical theory in the order that makes it usable. Each chapter states a model, derives the relationships that follow from it, shows the model applied to a worked problem, then puts an instrument on a real circuit to test whether the model held. Where a model breaks down, the chapter says so.

The progression runs from charge and current, through the conservation laws and network analysis, into energy storage and the transient and alternating behaviour that follows from it. Magnetism leads to transformers and rotating machines. Semiconductors lead to rectification, amplification, logic and regulated supplies. The last chapters address three-phase distribution, safety and fault clearance, control and sensing, and the diagnostic method that ties the whole subject together.

### How each chapter is built

- An opening statement of why the material matters and what you will be able to do.
- Explanatory sections with equations set out and their conditions of validity stated.
- Worked examples showing the full reasoning, not only the arithmetic.
- A practical activity with equipment list, procedure, results table and discussion.
- A troubleshooting table pairing symptoms with the measurement that identifies the cause.
- Exercises with selected answers, a summary, an equation list and a vocabulary check.

### A note on codes and standards

Numerical requirements in electrical work are set by codes that are revised on a cycle. Where this book gives a limit, a disconnection time or a permitted method, treat it as illustrating the method rather than as the authority. The authority is the edition of the code in force where the work is carried out. The physics in this book does not change between editions; the numbers sometimes do.

> **SAFETY**
> The practical activities in this book are specified at low voltage wherever the principle allows it, because a principle demonstrated at 12 V is the same principle at 400 V with none of the risk. Where mains voltage is unavoidable, the activity says so and states the precautions. Do not substitute a higher voltage for convenience.
> Isolate, lock off and prove dead before working on any circuit. Prove the test instrument on a known source before and after use.

# Contents

Chapter starts are linked in the PDF bookmarks panel. Each chapter is self-contained but assumes the chapters before it.


01 Charge, voltage, current and energy

02 Kirchhoff's laws and circuit analysis

03 Series, parallel and divider networks

04 Capacitance and transient response

05 Inductance and magnetic energy storage

06 Alternating quantities and their measurement

07 Impedance, phase and power factor

08 Resonance and filters

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10 Transformers

11 Rotating machines

12 Diodes and rectification

13 Transistors as switches and amplifiers

14 Operational amplifiers

15 Digital logic and signals

16 Power supplies and regulation

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18 Safety, fault paths and protection

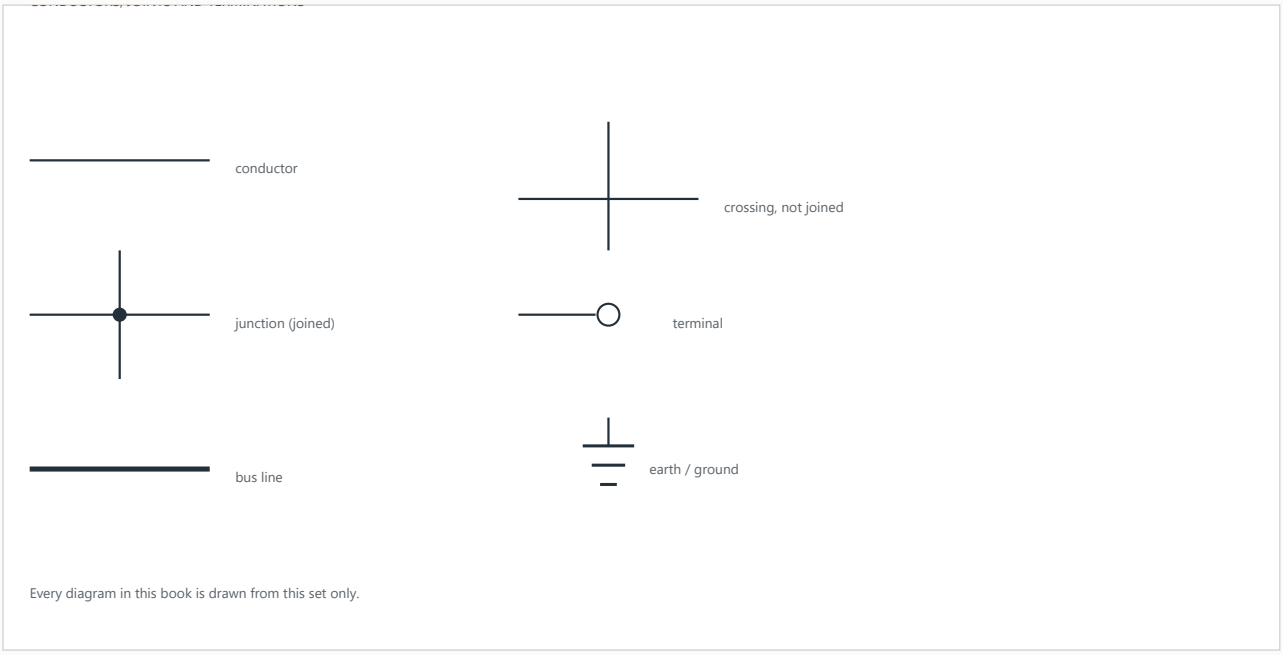
19 Control, sensing and signal conversion

20 Diagnosis, design review and lasting habits

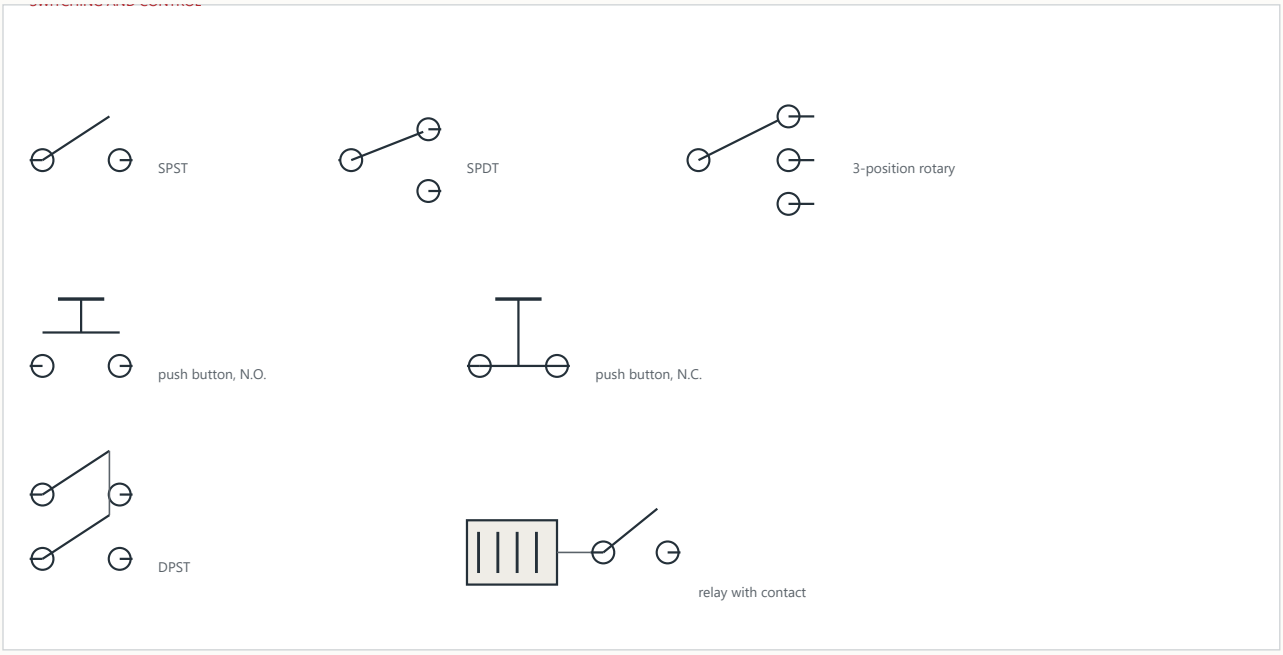
R Reference section: equations, units, symbols, checklists, glossary

# The symbol set used in this book

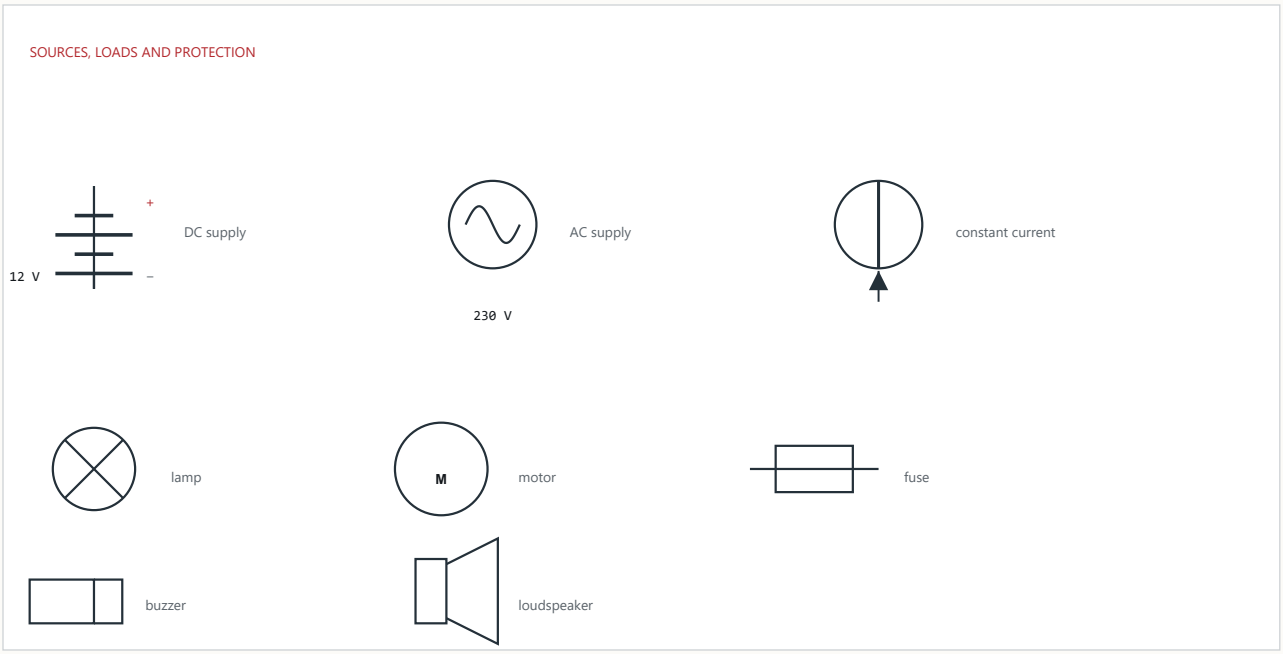
Every schematic in this book is drawn from the set below and from nothing else. The symbols follow the conventions of the referenced symbol standards, redrawn as vector line work so that they reproduce at any size. Learn this page and every diagram in the book becomes readable.



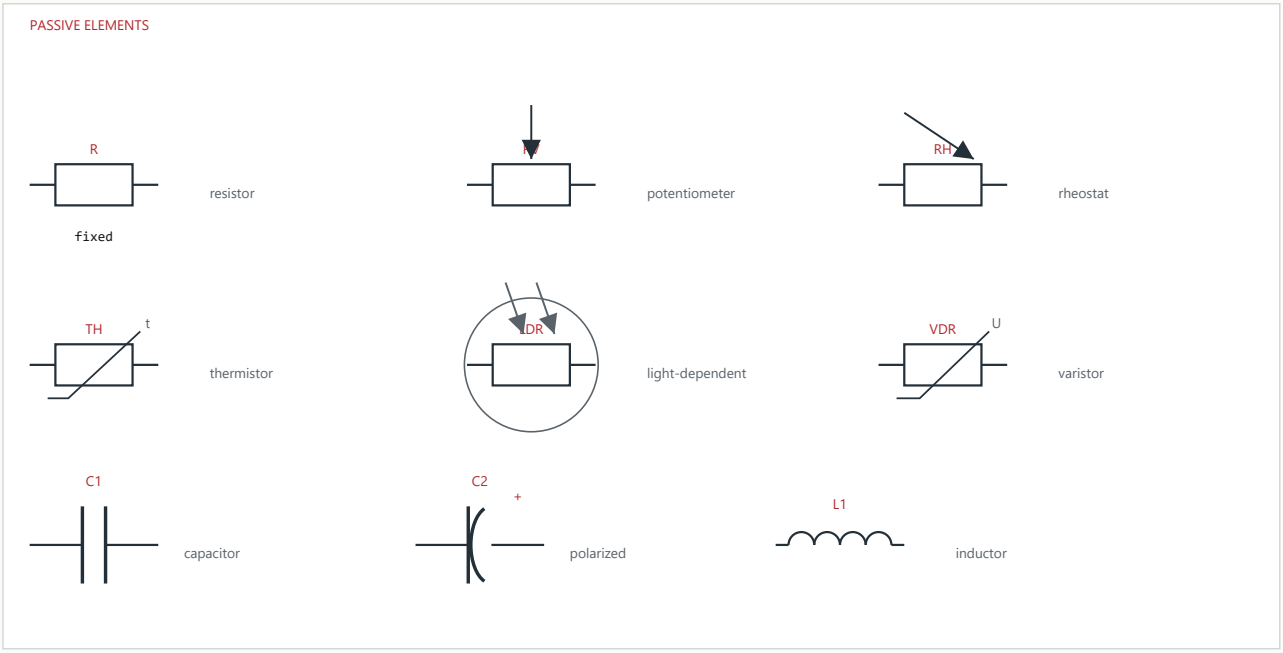

Conductors, joints and terminations.



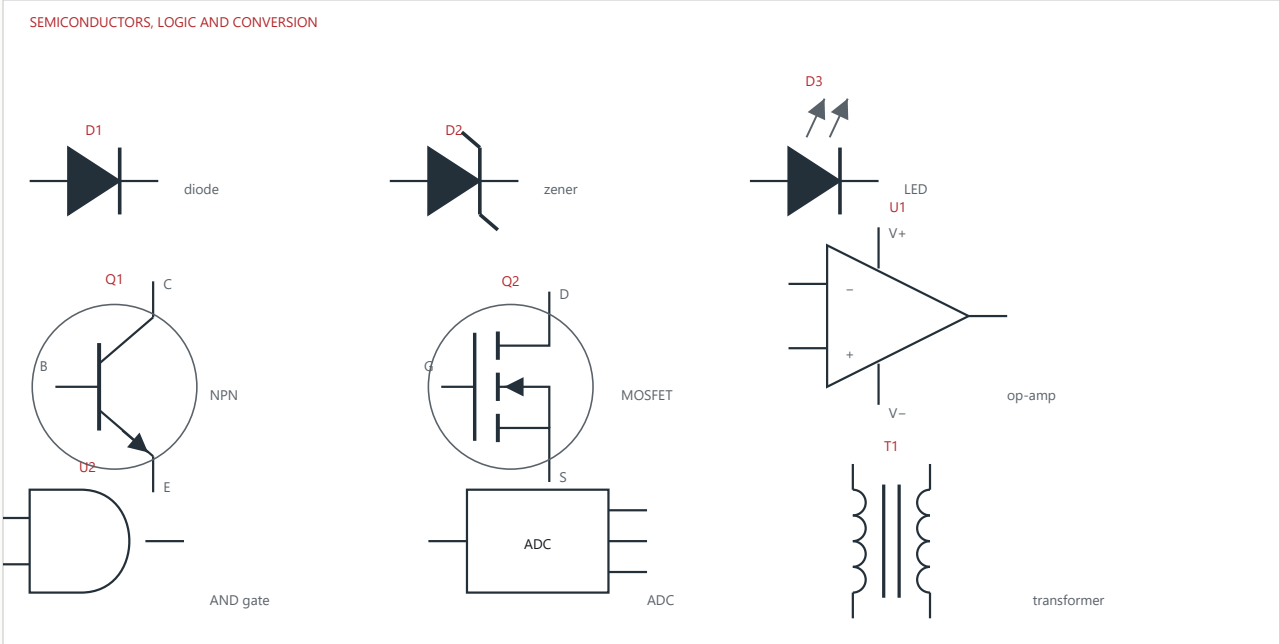

Switching and control devices.




Sources, loads and protection.




Passive components and semiconductors.




Logic, amplifiers, machines and instruments.

CHAPTER 01

# Charge, voltage, current and energy

The four quantities that every later chapter depends on

WHY THIS CHAPTER MATTERS

Every circuit calculation you will ever make reduces to four quantities and the relationships between them. Get these definitions precise now and the rest of the book becomes bookkeeping; leave them vague and even Ohm's law feels like guesswork.

## What you will be able to do

- State what charge, current, voltage, power and energy measure, and in which unit.
- Convert between coulombs, amperes, joules, volts and watts without a calculator.
- Choose a reference node and assign consistent polarity and current directions.
- Connect an ammeter and a voltmeter correctly and predict how each loads the circuit.
- Estimate the energy a load consumes over time and the cost of running it.

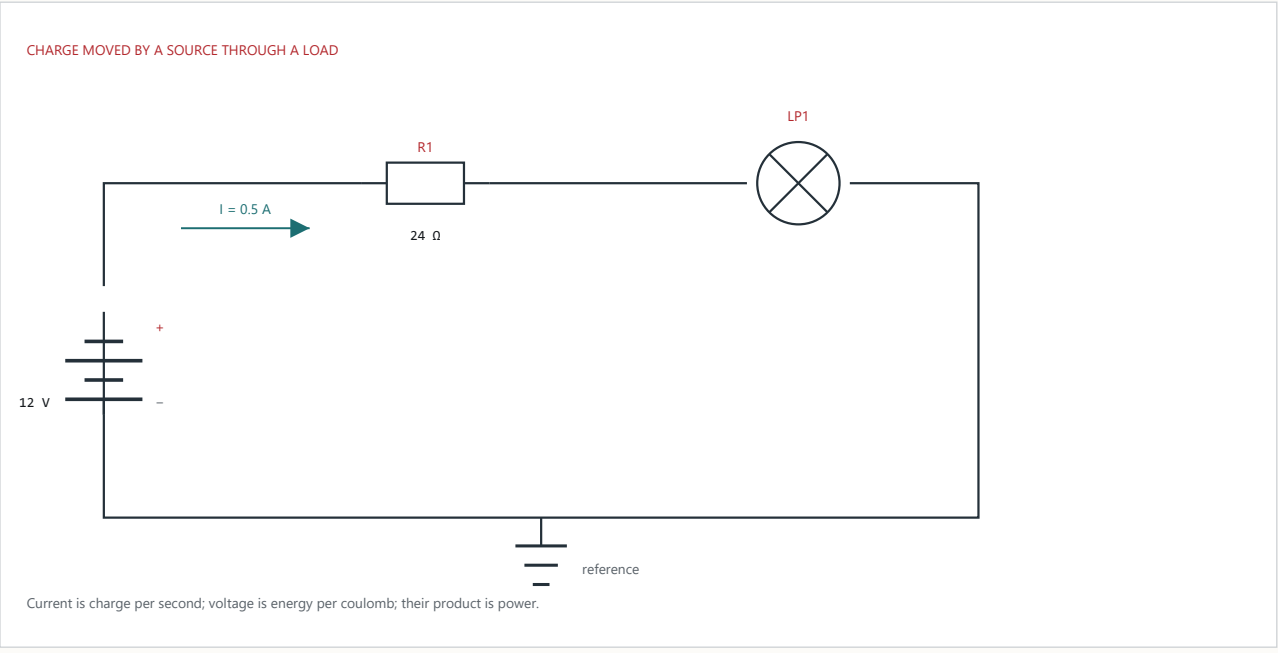



Fig. 1.1 The quantities of a simple circuit. The source maintains a potential difference across the load; charge circulates; energy is transferred at the load.

## Charge is the quantity being moved

Electric charge, symbol $Q$, is measured in coulombs (C). One coulomb is the charge carried by about $6.24 \times 10^{18}$ electrons. The electron charge itself is $1.602 \times 10^{-19}$ C, negative by convention. Charge is conserved: it is never created or destroyed in a circuit, only relocated. That single fact is what later becomes Kirchhoff's current law.

A battery does not store charge in any useful quantity; it stores chemical energy and uses it to push charge that is already present in the conductor. A copper conductor is already crowded with mobile electrons before the supply is connected. Switching on does not inject new carriers, it organises the drift of the carriers already there, which is why a lamp lights apparently instantly even though individual electrons drift at a fraction of a millimetre per second.

> WHY DRIFT IS SLOW BUT SWITCHING IS FAST
> Drift velocity in a 2.5 $mm^2$ conductor carrying 10 A is roughly 0.3 mm/s.
> The electric field that sets the drift propagates at a large fraction of the speed of light, so the far end of the circuit responds in nanoseconds.
> Practical consequence: conductor length changes resistance and voltage drop, not response time.

## Current is charge flow per unit time

Current, symbol $I$, is the rate at which charge passes a chosen cross-section. One ampere is one coulomb per second. Because current is defined through a surface, it is meaningless to speak of the current *at* a component without saying through which terminal; it is equally meaningless to speak of current flowing *to* a load, because the same current must return.

Conventional current is drawn from the positive terminal, through the external circuit, to the negative terminal. Electrons in a metal move the opposite way. Both descriptions give identical answers, and every schematic in this book uses conventional current. The only rule that matters is that you declare a reference direction on the drawing and keep it; a negative answer then simply means the real flow opposes your arrow.

Direction conventions worth writing on your drawing

- Mark one node as the reference (0 V) before anything else.
- Draw a current arrow on every branch you intend to solve for.
- Place the voltmeter's black lead at the reference node and leave it there.
- A negative reading is information, not a mistake: it means the arrow is reversed.

```
I = Q / t          Q = I t

1 A = 1 C/s        1 mA = 10^-3 A
```

## Voltage is energy per unit charge

Voltage, symbol $V$ or $U$, is the work done per unit charge in moving charge between two points. One volt is one joule per coulomb. Voltage is therefore always a difference between two points, which is why a meter has two leads and why the phrase “the voltage at the motor” is incomplete until you name the second point.

Treating voltage as a difference removes a large family of beginner errors. A conductor floating at 480 V relative to earth but at 0 V relative to its neighbour will register nothing on a meter placed across the pair, and will still injure anyone who bridges it to earth. Potential is relative; hazard is not.

```
V = W / Q          W = V Q

1 V = 1 J/C
```

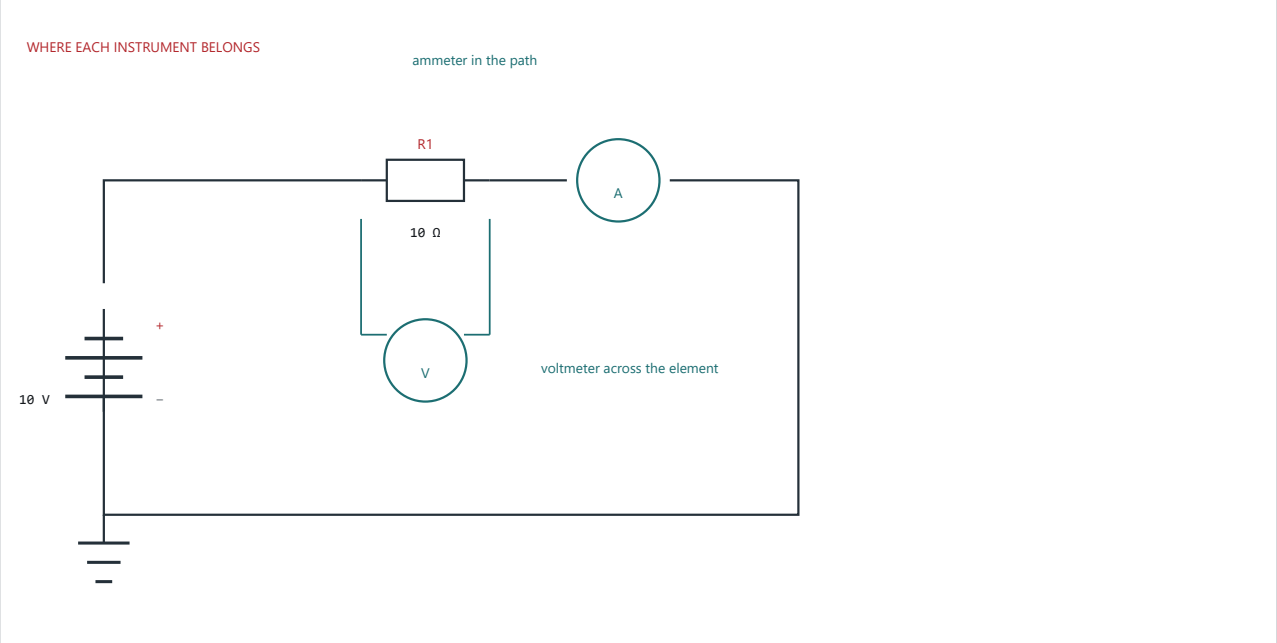



Fig. 1.2 Correct instrument connections. The voltmeter sits across the component, the ammeter is inserted in the branch so that the branch current passes through it.

## Power and energy: the quantities on the bill

Power, symbol *P*, is the rate of energy transfer, measured in watts. One watt is one joule per second. In a circuit the product of the voltage across an element and the current through it gives the power that element absorbs. Energy is power accumulated over time, and utilities sell it in kilowatt-hours: one kilowatt sustained for one hour, equal to 3.6 megajoules.

Power is where theory becomes thermal reality. A resistor dissipating 2 W in free air will reach a surface temperature high enough to discolour a circuit board; the same 2 W spread over a heatsink is unremarkable. Always calculate dissipation, then ask whether the part can shed it.

```
P = V I            W = P t

1 kWh = 3.6 MJ = 3.6 × 10^6 J
```

| Quantity | Symbol | Unit | Defining relation |
|---|---|---|---|
| Charge | Q | coulomb, C | Q = I t |
| Current | I | ampere, A | I = Q / t |

| Quantity | Symbol | Unit | Defining relation |
|---|---|---|---|
| Voltage | V | volt, V | V = W / Q |
| Power | P | watt, W | P = V I |
| Energy | W | joule, J | W = P t |
| Resistance | R | ohm, Ω | R = V / I |

These six rows are the vocabulary of the whole book. Every equation that follows is either one of them rearranged or a statement about how several of them combine.

## Prefixes, and why they cause more errors than algebra

Most field mistakes are decimal mistakes. A result expressed as 0.000047 F invites error; the same value as 47 µF does not. Learn to carry prefixes through the calculation rather than converting everything to base units and back.

A useful habit is to pair prefixes so that they cancel. Volts divided by kilohms gives milliamperes directly. Milliamperes multiplied by kilohms gives volts. Microfarads multiplied by kilohms gives milliseconds. These three pairings cover most of the arithmetic in electronics work.

| Prefix | Symbol | Multiplier | Typical use |
|---|---|---|---|
| mega | M | $10^{6}$ | MΩ insulation resistance, MW generation |
| kilo | k | $10^{3}$ | kΩ resistors, kW loads, kV distribution |
| milli | m | $10^{-3}$ | mA signal currents, mH chokes, mV offsets |
| micro | µ | $10^{-6}$ | µF capacitors, µA leakage |
| nano | n | $10^{-9}$ | nF coupling capacitors, ns switching |
| pico | p | $10^{-12}$ | pF stray capacitance, crystal loading |

THE THREE CANCELLING PAIRS

V ÷ kΩ = mA, so 12 V across 4.7 kΩ is 2.55 mA with no powers of ten.
mA × kΩ = V, so 2.55 mA through 1.8 kΩ drops 4.59 V.
µF × kΩ = ms, so 100 µF with 10 kΩ gives τ = 1000 ms.

## Energy accounting around a loop

Because voltage is energy per unit charge, a complete trip around any loop must return a charge to its starting potential. Energy supplied by sources equals energy absorbed by everything else, including the resistance of the conductors themselves. This is the conservation statement that becomes Kirchhoff's voltage law in the next chapter.

Treating conductor resistance as part of the accounting, rather than as an annoyance, explains a great deal of field behaviour: why a long run dims a lamp, why a motor starting surge makes lights flicker, and why voltage measured at the panel differs from voltage measured at the load.

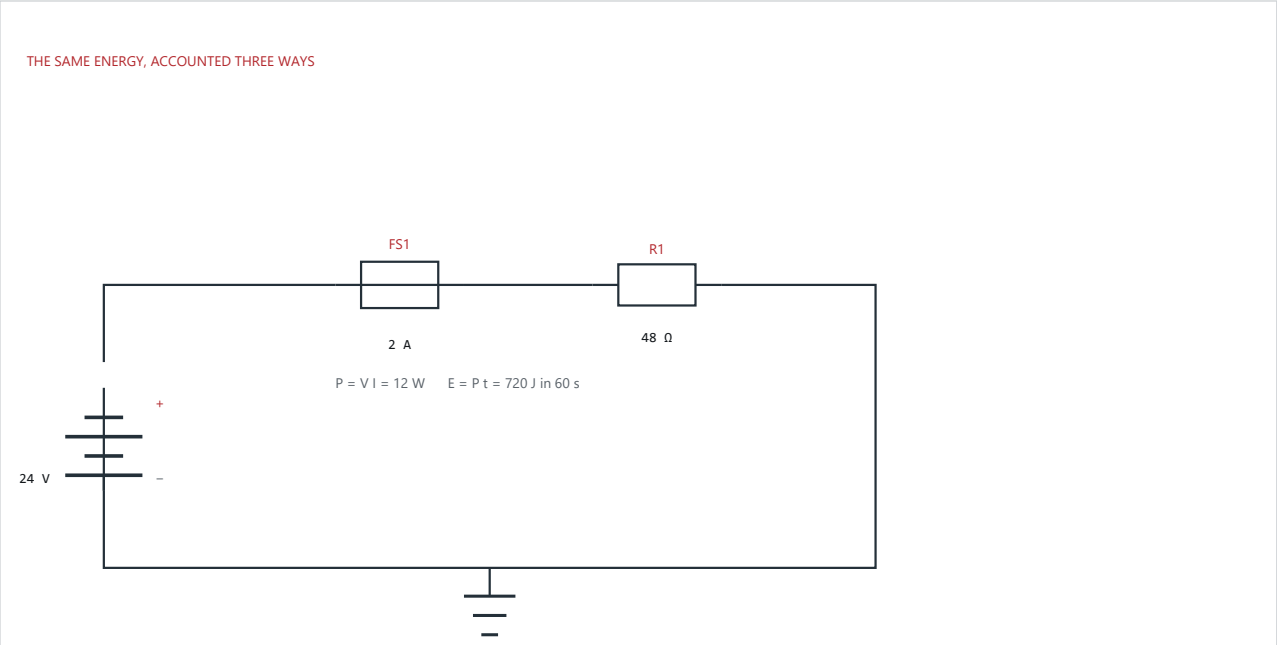



Fig. 1.3 Energy flow in a supply, conductor and load. The conductors are not ideal: part of the supplied energy is dissipated before it reaches the intended load.

## Worked examples

Work each example with a pencil before reading the solution. Sketch the circuit, mark the reference node, and write the governing law before substituting numbers.

EXAMPLE 1.1 Charge delivered by a lamp circuit

**Given.** A 24 V lamp draws a steady 0.75 A and is left on for 20 minutes.

**Find.** The charge circulated, the power and the energy transferred.

Convert the time to seconds

$$t = 20 \times 60 = 1200 \text{ s}$$

Charge from current and time

$$Q = I\,t = 0.75 \times 1200 = 900 \text{ C}$$

Power from voltage and current

$$P = V\,I = 24 \times 0.75 = 18 \text{ W}$$

Energy from power and time

$$W = P\,t = 18 \times 1200 = 21\,600 \text{ J} = 21.6 \text{ kJ}$$

Energy as a billed quantity

$$W = 21\,600 \,/\, 3.6\times10^{6} = 0.006 \text{ kWh}$$

**Check.** Cross-check through charge: W = V Q = 24 × 900 = 21 600 J. The two routes agree, which confirms the time conversion.

EXAMPLE 1.2 Voltage drop along a supply cable

**Given.** A load draws 16 A through a two-core run whose total loop resistance is 0.21 Ω. The panel measures 232 V.

**Find.** The voltage at the load, the power lost in the cable and the percentage drop.

Drop across the conductors

$$V_{cable} = I\,R = 16 \times 0.21 = 3.36\ \text{V}$$

Voltage reaching the load

$$V_{load} = 232 - 3.36 = 228.64\ \text{V}$$

Power dissipated in the cable

$$P_{cable} = I^2\,R = 16^2 \times 0.21 = 53.8\ \text{W}$$

Drop as a percentage

$$3.36 / 232 \times 100 = 1.45\ \%$$

**Check.** 1.45 % is inside the 3 % branch-circuit guidance of NFPA 70 (2023) 210.19(A) Informational Note 4, so the run is acceptable on voltage-drop grounds. Note that the 53.8 W lost in the cable is real heat inside the wiring method.

EXAMPLE 1.3 Running cost of a continuous load

**Given.** A 1.8 kW immersion heater runs 2.5 hours per day. Energy costs 0.19 per kWh.

**Find.** Daily energy, monthly cost over 30 days, and the current at 240 V.

Daily energy

$$W = 1.8 \times 2.5 = 4.5\ \text{kWh}$$

Monthly energy

$$4.5 \times 30 = 135\ \text{kWh}$$

Monthly cost

$$135 \times 0.19 = 25.65$$

Current drawn

$$I = P / V = 1800 / 240 = 7.5\ \text{A}$$

**Check.** 7.5 A on a 240 V circuit is consistent with a 15 A branch circuit at 50 % loading, and the heater is a continuous load for sizing purposes.

## Practical activity: measuring current, voltage and power in a lamp circuit

AIM

Establish by measurement that the current entering a load equals the current leaving it, that voltage is a difference between two points, and that the product of the two predicts the power the lamp actually dissipates.

### Equipment

- Regulated DC supply, 0–30 V, current limit set to 1 A.
- Two digital multimeters, or one meter used twice with the supply switched off between moves.
- Filament lamp, 24 V 10 W, in a holder with screw terminals.
- Series resistor, 10 Ω 5 W, to make the supply current limit meaningful.
- Insulated test leads, a screwdriver, and a notebook.

### Procedure

1. Draw the circuit before touching the equipment. Mark the reference node and the intended current direction.
2. With the supply off and set to zero, connect the lamp in series with the 10 Ω resistor.
3. Insert the ammeter in the branch between the resistor and the lamp. The branch current must pass through the meter, so the circuit has to be opened to admit it.
4. Connect the voltmeter across the lamp only. Leave its black lead on the reference node.
5. Raise the supply in 4 V steps to 24 V, recording both readings at each step.
6. Move the ammeter to the return conductor and repeat two of the steps. The reading must not change; this is conservation of charge, measured.
7. Switch off, then place the voltmeter across the resistor and confirm that the two component voltages add to the supply voltage.
8. Return the supply to zero and disconnect before dismantling.

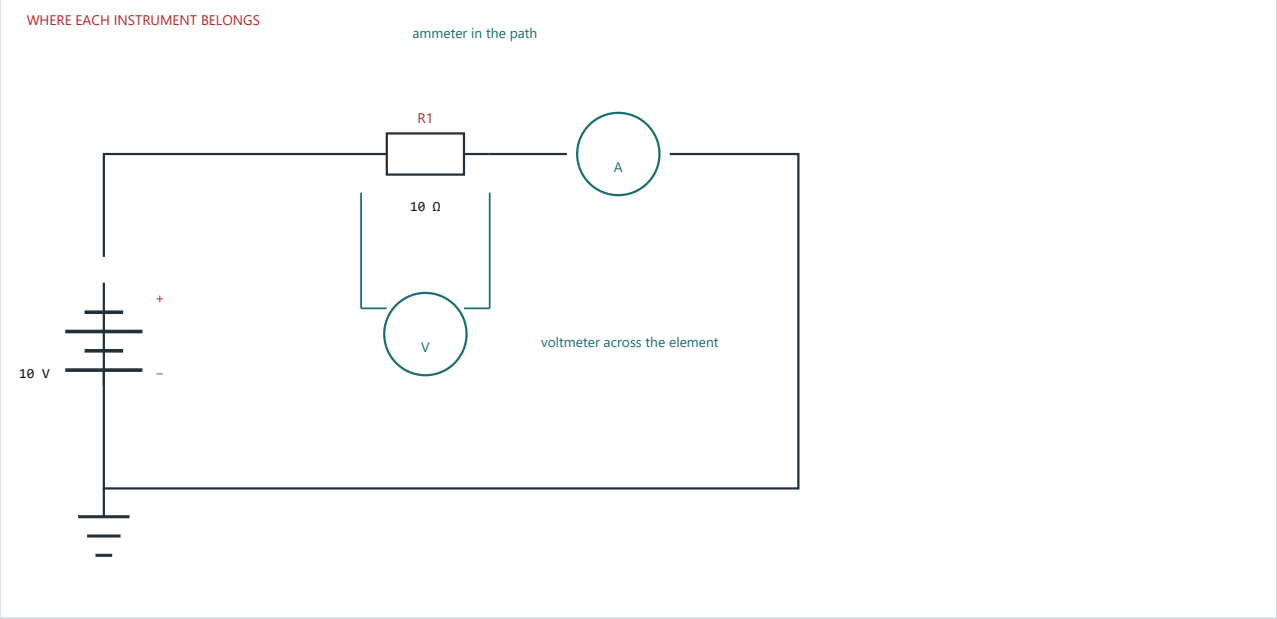



Fig. 1.L Lab connections for the measurement sequence. Opening the branch to insert the ammeter is the step most often done wrongly.

### Results table

| Supply (V) | Lamp voltage (V) | Current (A) | P = V I (W) | Lamp resistance (Ω) |
|---|---|---|---|---|
| 4.0 | | | | |
| 8.0 | | | | |
| 12.0 | | | | |
| 16.0 | | | | |
| 20.0 | | | | |
| 24.0 | | | | |

### Discussion

1. Plot lamp current against lamp voltage. The line is not straight. Explain why, in terms of filament temperature.
2. Compute lamp resistance at 4 V and at 24 V. State the ratio and account for it.
3. Did the ammeter reading change when you moved it to the return conductor? What would a change have told you about your connections?
4. At 24 V, compare the measured power with the lamp's 10 W rating and comment on the series resistor's share.
5. The ammeter has its own resistance. Estimate the error it introduces at 24 V and say whether it matters here.

> SAFE-WORK REMINDERS
> Keep the supply below 30 V and the current limit at 1 A throughout this activity.
> Switch off and verify zero before changing any connection; moving an ammeter in a live circuit is how meters and fingers are damaged.
> The series resistor becomes hot. Allow it to cool before handling.
> Never place an ammeter across a source. The meter is a near short circuit on its current ranges.

## Troubleshooting guide

Faults are found by narrowing the circuit, not by replacing parts. Each row below pairs an observed symptom with the measurement that separates the likely causes.

| Symptom | Likely cause | Decisive test | Correct action |
|---|---|---|---|
| Lamp dark, meter shows full supply voltage across the lamp | Open filament or open holder contact | Measure resistance of the cold lamp out of circuit | Replace the lamp; inspect the holder for burnt contacts |
| Lamp dim, voltage at the lamp well below the panel voltage | Excessive drop in the supply run or a high-resistance joint | Measure voltage at the panel and at the load under load | Re-terminate the joint or increase conductor size |
| Ammeter reads zero but the lamp is lit | Meter on the wrong jack or a blown current fuse | Short the meter leads on the current range | Move the lead to the correct jack; replace the meter fuse |
| Meter fuse blows the moment the supply is energised | Ammeter connected across the source rather than in series | Trace the meter leads against the drawing | Open the branch and insert the ammeter in it |
| Readings drift upward over a minute | Filament heating, so resistance is rising | Record the reading at a fixed time after switch-on | Report readings with the timing stated |
| Voltage reading unstable, current steady | Poor voltmeter lead contact | Wiggle the probes while watching the display | Clean and retighten the measuring points |

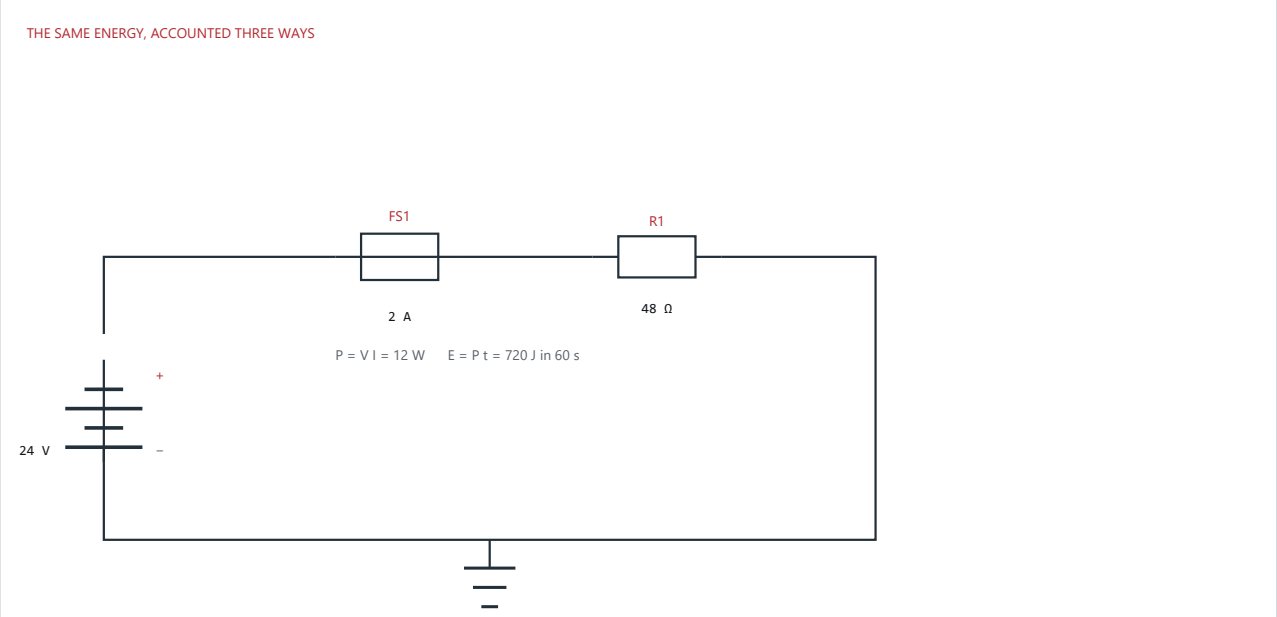



Fig. 1.T Where the supplied energy ends up. A fault that adds unwanted resistance moves energy out of the load and into the wiring.

FIELD HABITS THAT PREVENT CALLBACKS

State the reference point in every reading you write down. “228 V” means nothing; “228 V line to neutral at the load terminals” is a measurement.

Predict before you probe. A meter confirms or refutes a prediction; used without one it only produces numbers.

Record the supply voltage at the same instant as the branch current, because both move together when a fault is present.

Treat conductor resistance as a component. It appears in the loop whether or not you drew it.

## Exercises

Answer from first principles. Selected answers follow; the reasoning matters more than the number.

1. A conductor passes 3.2 A for 4 minutes. How much charge has moved?
2. A charge of 540 C is moved through a potential difference of 12 V. Find the energy transferred.
3. Convert: 0.047 F to microfarads; 2 200 000 Ω to megohms; 0.0056 A to milliamperes.
4. A 9 V battery delivers 150 mA. Find the power and the energy supplied in 5 minutes.
5. A 2.4 kW load runs on 240 V. Find the current and the resistance.
6. A cable run has a loop resistance of 0.34 Ω and carries 22 A. Find the voltage drop and the power lost in the conductors.
7. A load is measured at 11.6 V while the supply terminals read 12.0 V. What is dropped in the wiring, and what fraction of the supply is that?
8. Explain, using the definition of voltage, why a bird on a single energised conductor is not injured.
9. A 60 W lamp runs 6 hours a day. Find the monthly energy at 30 days in kWh.
10. An ammeter has 0.1 Ω of internal resistance and is inserted in a branch of total resistance 4.9 Ω fed from 12 V. Find the current with and without the meter, and the percentage error introduced.

## Selected answers

| # | Answer | Reason |
|---|---|---|
| 1 | 768 C | Q = I t with t = 240 s |
| 2 | 6.48 kJ | W = V Q = 12 × 540 |
| 3 | 47 000 µF; 2.2 MΩ; 5.6 mA | Shift the decimal point by the prefix exponent |
| 4 | 1.35 W; 405 J | P = V I, then W = P t with t = 300 s |
| 5 | 10 A; 24 Ω | I = P / V, then R = V / I |
| 6 | 7.48 V; 165 W | V = I R, then P = $I^2$ R |
| 7 | 0.4 V; 3.3 % | Difference divided by the supply voltage |
| 8 | No potential difference across the bird | Both feet are at essentially the same potential, so no appreciable current flows through it |
| 9 | 10.8 kWh | 0.06 kW × 6 h × 30 d |
| 10 | 2.449 A with the meter, 2.4 A without, 2.0 % error | The meter adds series resistance; compare 12/4.9 with 12/5.0 |

## Chapter summary

- Charge is the quantity moved, current its rate of flow, voltage the energy carried per unit charge, and power the rate at which that energy is delivered.
- Voltage exists only between two points; every reading needs a stated reference.
- Current is defined through a surface, so it must be measured in series and is the same entering and leaving a two-terminal element.
- Conductors are components. Their resistance takes a share of the supplied energy and converts it to heat inside the wiring method.
- Prefix arithmetic causes more field errors than algebra does; use cancelling pairs such as V ÷ kΩ = mA.
- Energy billed in kilowatt-hours is the same quantity as energy in joules: 1 kWh = 3.6 MJ.

### Equations introduced in this chapter

| Relationship | Expression | When it applies |
|---|---|---|
| Charge and current | $Q = I\,t$ | Steady current over an interval t |
| Voltage and energy | $W = V\,Q$ | Work to move charge Q through a difference V |
| Power | $P = V\,I$ | Instantaneous, or steady DC values |
| Energy | $W = P\,t$ | Constant power over an interval t |
| Resistance | $R = V / I$ | Defines resistance from a measured pair |
| Conductor loss | $P = I^2\,R$ | Heat generated in a run of known resistance |

### Vocabulary check

| Term | Meaning in this chapter |
|---|---|
| Reference node | The point chosen as 0 V, against which all other potentials are quoted |
| Conventional current | Flow direction taken from positive to negative in the external circuit |
| Drift velocity | Average speed of charge carriers along a conductor, typically under 1 mm/s |
| Potential difference | Energy per unit charge between two named points |
| Meter loading | The change a measuring instrument causes in the circuit it is measuring |

LOOKING AHEAD

You now have the quantities and the habit of declaring references and directions. The next chapter turns the two conservation statements behind them — charge is conserved at a junction, energy is conserved around a loop — into Kirchhoff's two laws, the tools that solve circuits Ohm's law alone cannot.

CHAPTER 02

# Kirchhoff's laws

Conservation of charge at a node, conservation of energy around a loop

WHY THIS CHAPTER MATTERS

Ohm's law describes one component. Kirchhoff's two laws describe how components constrain each other, and they are the only tools that will solve a network no amount of series-parallel reduction can simplify.

## What you will be able to do

- State both laws and identify the conservation principle behind each.
- Assign and keep consistent reference directions and loop senses.
- Write and solve node equations for a two-node network.
- Write and solve mesh equations for a two-loop network.
- Use a measured set of readings to detect an inconsistent or faulty circuit.

THE CURRENT LAW AT ONE NODE

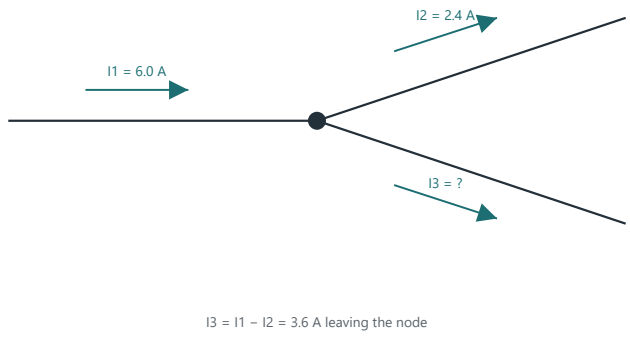



Fig. 2.1 Kirchhoff's current law at a single node. The algebraic sum of the currents entering the node is zero; charge does not accumulate there.

## The current law: charge cannot pile up

Kirchhoff's current law, KCL, states that the algebraic sum of currents entering any node is zero. Equivalently, the sum of currents arriving equals the sum leaving. The law follows directly from conservation of charge: a node is a point with no capacity to store charge, so whatever arrives must depart in the same instant.

To apply KCL you must first commit to arrow directions. Any choice works provided you keep it, because the algebra will correct a wrong guess with a negative sign. The common failure is not a bad guess but an inconsistent one: changing a direction halfway through a solution guarantees a wrong answer.

```
Σ I_in = Σ I_out

I_1 + I_2 − I_3 − I_4 = 0
```

WHAT COUNTS AS A NODE

A node is any stretch of conductor at one potential, however it is drawn on the page.
Two junctions joined only by plain wire are the same node, so they share one equation.
Counting nodes correctly is how you know in advance how many equations you need: one fewer than the node count.

## The voltage law: energy cannot be created in a loop

Kirchhoff's voltage law, KVL, states that the algebraic sum of potential differences around any closed loop is zero. Because voltage is energy per unit charge, a charge returned to its starting point must have gained and lost equal energy. Sources supply, resistances absorb, and the books must balance.

In practice KVL means you can walk a loop on the drawing, adding a voltage rise when you pass from minus to plus through a source and subtracting a drop when you pass through a resistance in the direction of its current arrow. Return to the start and the total is zero.

```
Σ V_rises = Σ V_drops

V_s − I R_1 − I R_2 = 0
```

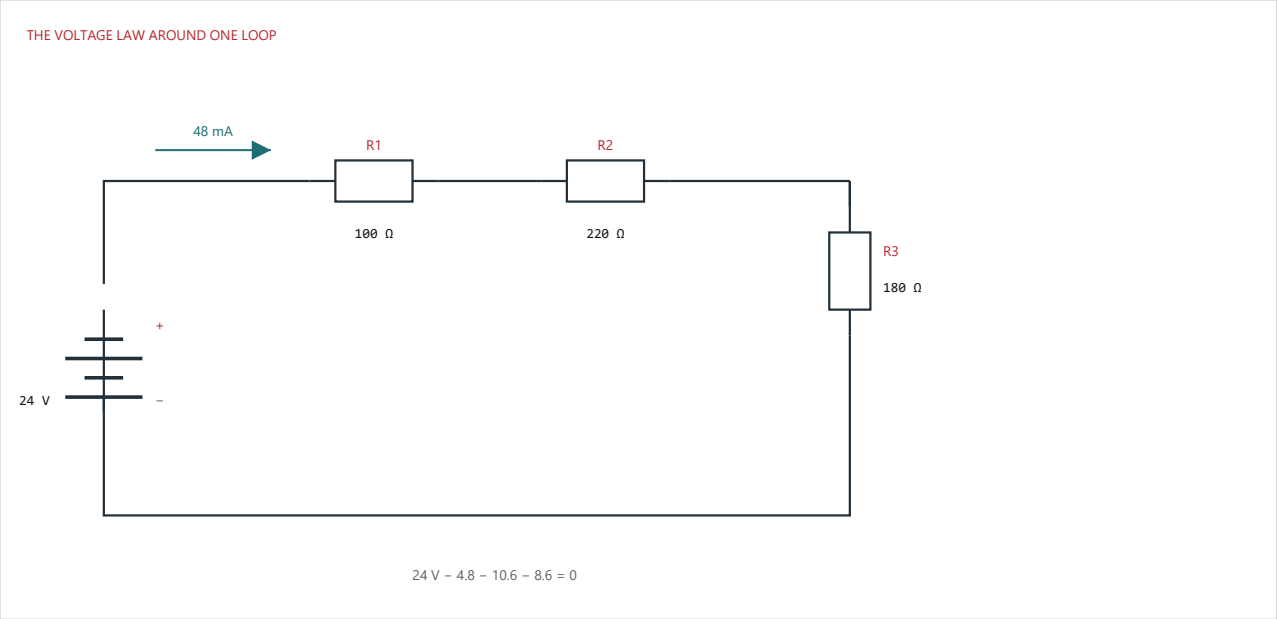



Fig. 2.2 Walking a loop with KVL. The supply voltage is distributed among the series elements, and the signed sum around the closed path is zero.

## Node-voltage analysis

Node analysis takes the node potentials as the unknowns. Choose a reference node, usually the one with the most connections, call it zero, then write KCL at each remaining node with every branch current expressed by Ohm's law as a potential difference divided by a resistance. The result is one equation per unknown node.

Node analysis suits circuits with many parallel branches and few loops, which describes most distribution and sensor wiring. It is also the method that mirrors measurement, because a voltmeter referenced to one point reads node potentials directly.

1. Identify and merge all nodes; count them.
2. Choose the reference node and mark it 0 V.
3. Label the remaining node potentials V_a, V_b and so on.
4. At each labelled node write KCL with every current as (V_this − V_other) / R.
5. Solve the simultaneous equations, then back-substitute for branch currents.
6. Check by confirming KVL around any loop you did not use.

## Mesh-current analysis

Mesh analysis takes circulating loop currents as the unknowns. Assign a current to each independent loop, all in the same rotational sense, then write KVL around each loop. A resistance shared between two meshes carries the difference of the two mesh currents, which is where most sign errors occur.

Mesh analysis suits circuits with few loops and many series elements. For a planar network the number of meshes equals the number of independent equations, so a two-loop circuit gives two equations in two unknowns regardless of how many components it contains.

$$\text{Loop 1:}\quad V_s = I_1(R_1+R_3) - I_2 R_3$$

$$\text{Loop 2:}\quad 0 = -I_1 R_3 + I_2(R_2+R_3)$$

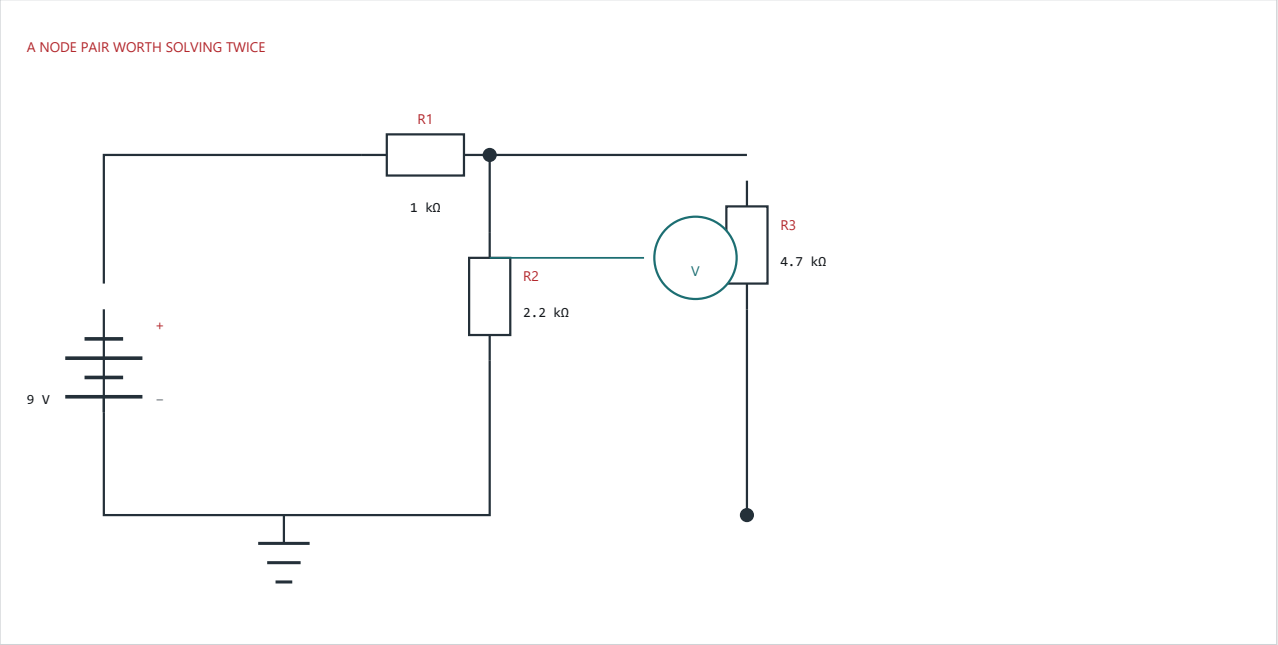



Fig. 2.3 A bridge network. No amount of series or parallel reduction simplifies it, so node or mesh analysis is required.

## Choosing a method, and checking the answer

Count before you commit. If a circuit has fewer nodes than meshes, use node analysis; if fewer meshes than nodes, use mesh analysis. Either method gives the same answer, so the choice is only about arithmetic effort.

Every solution should be checked against the law you did not use. If you solved by node equations, verify KVL around a loop. If you solved by mesh equations, verify KCL at a node. A solution that satisfies both laws and gives positive power absorbed by every resistance is almost certainly right.

| Situation | Preferred method | Reason |
|---|---|---|
| Many parallel branches | Node analysis | Fewer unknowns, matches voltmeter readings |
| Few loops, long series strings | Mesh analysis | Fewer unknowns, avoids many node merges |
| Bridge or ladder | Either, by count | Neither reduces by series-parallel alone |
| One source, simple series-parallel | Reduction then Ohm's law | No simultaneous equations needed |

THREE CHECKS THAT CATCH ALMOST EVERY ERROR

Does every resistance absorb positive power? A negative value means a sign slip.
Does the total source power equal the total dissipated power?
Does the unused law hold? Node solutions must satisfy KVL and vice versa.

## Worked examples

Work each example with a pencil before reading the solution. Sketch the circuit, mark the reference node, and write the governing law before substituting numbers.

EXAMPLE 2.1 Current law at a supply node

**Given.** Three branches leave a distribution node. Two carry 6.2 A and 3.4 A away from the node; the feed carries current toward it.

**Find.** The feed current, and the third branch current if it measures 1.1 A away.

Write KCL with inbound positive

```
I_feed − 6.2 − 3.4 − 1.1 = 0
```

Solve for the feed

```
I_feed = 10.7 A
```

Confirm the balance

```
10.7 = 6.2 + 3.4 + 1.1
```

**Check.** A clamp meter on the feed should read 10.7 A. A reading materially below that, with branch readings unchanged, indicates an unrecorded branch or a leakage path.

EXAMPLE 2.2 Voltage law around a two-resistor loop

**Given.** A 24 V source feeds $R_1$ = 180 Ω in series with $R_2$ = 120 Ω.

**Find.** The loop current and the voltage across each resistor.

Apply KVL around the loop

```
24 − 180 I − 120 I = 0
```

Collect terms

```
300 I = 24
```

Loop current

```
I = 0.08 A = 80 mA
```

Drops from Ohm's law

$V_1 = 0.08 \times 180 = 14.4$ V ; $V_2 = 0.08 \times 120 = 9.6$ V

**Check.** 14.4 + 9.6 = 24.0 V, equal to the source, so KVL is satisfied around the loop.

EXAMPLE 2.3 Two-node network by node analysis

**Given.** A 12 V source feeds node A through 100 Ω. Node A connects to the reference through 200 Ω and to node B through 50 Ω. Node B connects to the reference through 150 Ω.

**Find.** The node potentials and the current in the 50 Ω link.

KCL at A

```
(12 − V_a)/100 = V_a/200 + (V_a − V_b)/50
```

KCL at B

```
(V_a − V_b)/50 = V_b/150
```

From B, express V_a

```
V_a = V_b (1 + 50/150) = 1.3333 V_b
```

Substitute into A and solve

```
V_b ≈ 3.13 V, V_a ≈ 4.17 V
```

Link current

```
I = (4.17 − 3.13)/50 ≈ 20.8 mA
```

**Check.** Check KVL on the source branch: 12 − 0.0783×100 ≈ 4.17 V, matching V_a, so the solution is consistent.

# Practical activity: verifying both of Kirchhoff's laws on a two-loop network

AIM

Build a network that cannot be reduced by inspection, predict every branch current and node voltage from the two laws, then confirm the predictions by measurement within the tolerance of the components.

## Equipment

- DC supply, 0–20 V, current limited to 0.5 A.
- Resistors: 100 Ω, 150 Ω, 220 Ω, 330 Ω, 470 Ω, all 0.5 W.
- Two multimeters, breadboard or terminal strip, and marked test leads.
- Squared paper for the circuit drawing and the prediction table.

## Procedure

1. Draw the two-loop network and label every node and branch before building it.
2. Assign a reference node, a current arrow to every branch and a sense to each loop.
3. Predict all branch currents and node voltages using node or mesh analysis. Record the predictions before any measurement.
4. Build the circuit and set the supply to 12.0 V measured at its terminals.
5. Measure every node voltage with the black lead held on the reference node throughout.
6. Measure each branch current by opening that branch only, then restoring it before moving on.
7. At each node, sum your measured currents and record the residual.
8. Around each loop, sum your measured voltages and record the residual.

A NODE PAIR WORTH SOLVING TWICE

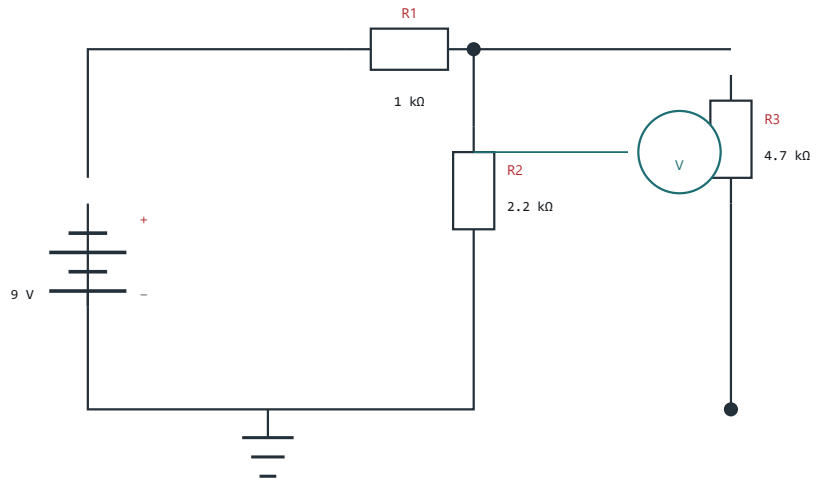



Fig. 2.L The network for this activity. Node and branch labels must be written on your own drawing before the circuit is energised.

## Results table

| Branch | Predicted I (mA) | Measured I (mA) | Node | Predicted V | Measured V |
|---|---|---|---|---|---|
| 1 | | | A | | |
| 2 | | | B | | |
| 3 | | | C | | |
| 4 | | | D | | |
| 5 | | | — | | |

### Discussion

1. How close is each node residual to zero? Express the worst one as a percentage of the largest current at that node.
2. How close is each loop residual to zero? Account for the difference using the resistor tolerance bands.
3. Measure the resistors individually and recompute your predictions. Does the agreement improve?
4. Which measurement disturbed the circuit most, the voltage readings or the current readings? Explain in terms of meter resistance.
5. If one node residual were 15 % while all others were near zero, what single fault would you suspect and what would you measure next?

> **SAFE-WORK REMINDERS**
> Keep the supply at or below 20 V and the current limit at 0.5 A.
> De-energise before opening a branch to insert the ammeter.
> Check resistor power ratings against your predicted currents before energising.
> Never leave an ammeter connected across the supply.

## Troubleshooting guide

Faults are found by narrowing the circuit, not by replacing parts. Each row below pairs an observed symptom with the measurement that separates the likely causes.

| Symptom | Likely cause | Decisive test | Correct action |
|---|---|---|---|
| Node currents do not sum to zero | An unrecorded branch or a leakage path to earth | Clamp or series-measure every conductor at that node | Trace the extra path; test insulation resistance de-energised |
| Loop voltages do not sum to zero | A high-resistance joint acting as an unseen resistor | Measure voltage across each joint under load | Re-terminate the joint and re-measure |
| A branch current is negative in reality but positive in prediction | Reference arrow drawn opposite to the true flow | Compare the drawing with the physical wiring | Correct the drawing, not the measurement |
| Node voltages all read low | Supply sagging under load, or source internal resistance | Measure supply terminal voltage loaded and unloaded | Use a stiffer supply or account for internal resistance |
| Readings change when the meter is moved | Meter loading on a high-resistance node | Compare a high-impedance meter against the original | Use a higher input impedance meter and state it with the reading |
| Two nodes read identical voltages unexpectedly | The two are actually one node, or a component is short circuited | Measure resistance between them de-energised | Correct the node count or replace the shorted part |

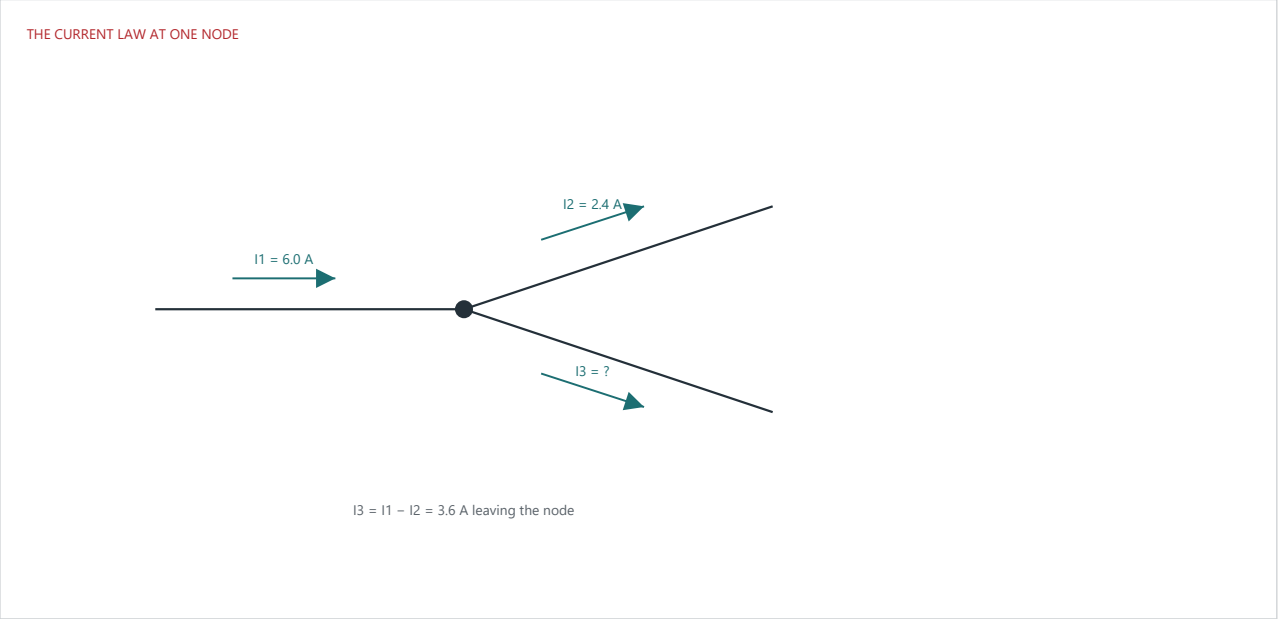



Fig. 2.T A node under test. Every conductor at the node must be accounted for before the current law can be used as a fault indicator.

FIELD HABITS THAT PREVENT CALLBACKS

Write the node and branch labels on the drawing before energising; labels invented during fault-finding are rarely consistent.

Hold the voltmeter reference lead in one place for a whole set of readings.

Record the supply terminal voltage alongside every set, so sag is visible later.

Treat a non-zero node residual as a real finding and chase it; it almost always means a path you have not drawn.

## Exercises

Answer from first principles. Selected answers follow; the reasoning matters more than the number.

1. Four conductors meet at a node. Three carry 2.1 A, 3.6 A and 0.8 A away. Find the fourth current and state its direction.
2. A 36 V source drives a series loop of 220 Ω and 380 Ω. Find the current and both drops.
3. In a loop containing a 12 V and a 5 V source in opposition with a single 68 Ω resistor, find the current.
4. State how many independent node equations a circuit with six nodes requires.
5. State how many independent mesh equations a planar circuit with four meshes requires.
6. Two resistors, 150 Ω and 450 Ω, are in parallel across 18 V. Verify KCL at the top node.
7. A loop sums to −0.6 V instead of zero with 24 V applied. Express the discrepancy as a percentage and suggest a cause.
8. Write, but do not solve, the node equations for a circuit with nodes A and B, a 20 V source feeding A through 120 Ω, a 240 Ω from A to reference, a 60 Ω from A to B and a 180 Ω from B to reference.
9. Explain why a resistance shared by two meshes carries the difference of the mesh currents.
10. A bridge has 1 kΩ in three arms and 1.2 kΩ in the fourth, fed from 10 V. Explain, without full calculation, why the detector branch carries current.

## Selected answers

| # | Answer | Reason |
|---|---|---|
| 1 | 6.5 A toward the node | Inbound must equal the 6.5 A total outbound |
| 2 | 60 mA; 13.2 V and 22.8 V | I = 36/600, then V = I R for each |
| 3 | 103 mA | Net driving voltage is 12 − 5 = 7 V across 68 Ω |
| 4 | Five | One fewer than the number of nodes |
| 5 | Four | One per independent mesh in a planar network |
| 6 | 120 mA + 40 mA = 160 mA total | Branch currents 18/150 and 18/450 sum to the supply current |
| 7 | 2.5 %; a high-resistance joint | 0.6/24; an unmodelled series resistance absorbs the balance |
| 8 | (20−V_a)/120 = V_a/240 + (V_a−V_b)/60 and (V_a−V_b)/60 = V_b/180 | KCL at each labelled node with Ohm's law substituted |
| 9 | Both mesh currents traverse it, in opposite senses | Their superposition is the net branch current, hence the difference |
| 10 | The arm ratios are unequal, so the bridge is unbalanced | 1/1 ≠ 1/1.2, so the two divider midpoints sit at different potentials |

## Chapter summary

- KCL is conservation of charge at a node; KVL is conservation of energy around a loop.
- Both laws require committed reference directions, and a negative answer means the arrow was drawn the other way.
- Node analysis needs one equation fewer than the node count; mesh analysis needs one per mesh.
- Choose the method with fewer unknowns, then check the solution against the law you did not use.
- Bridge and ladder networks cannot be reduced by series-parallel alone, which is the practical reason these laws exist.
- A non-zero node or loop residual in measurement is a fault indicator, not experimental noise.

### Equations introduced in this chapter

| Relationship | Expression | When it applies |
|---|---|---|
| Current law | $\Sigma I_{in} = \Sigma I_{out}$ | At any node, for any instant |
| Voltage law | $\Sigma V = 0$ around a loop | Any closed path, signs by traverse direction |
| Node equation | $\Sigma (V_n - V_k)/R_{nk} = 0$ | KCL written with Ohm's law substituted |
| Mesh equation | $\Sigma V_s = \Sigma I R$ around loop | KVL with shared resistors carrying mesh differences |
| Independent nodes | $n - 1$ | n is the merged node count |
| Independent meshes | $b - n + 1$ | b branches, n nodes, planar network |

### Vocabulary check

| Term | Meaning in this chapter |
|---|---|
| Node | A stretch of conductor at a single potential, however drawn |
| Mesh | A loop that encloses no other loop in a planar drawing |
| Reference direction | The arrow you commit to before solving, which fixes the sign convention |
| Residual | The amount by which a measured sum fails to reach zero |
| Bridge | A four-arm network whose detector branch current depends on the arm ratios |

LOOKING AHEAD

With both conservation laws available you can solve any resistive network. The next chapter applies them to the two arrangements that occur most often, series and parallel, and derives the divider rules that let you skip the simultaneous equations entirely.

CHAPTER 03

# Series, parallel and divider rules

Reducing networks by inspection, and the two shortcuts worth memorising

WHY THIS CHAPTER MATTERS

Most real circuits are series-parallel combinations. Learning to reduce them by inspection, and to use the divider rules, removes simultaneous equations from the majority of your work and makes mental estimation possible on site.

## What you will be able to do

- Compute equivalent resistance for series, parallel and mixed networks.
- Apply the voltage divider and current divider rules correctly, including their conditions.
- Predict how adding a load changes a divider's output.
- Measure resistance correctly, including why the circuit must be de-energised.
- Use resistivity to find the resistance of a conductor run from its dimensions.

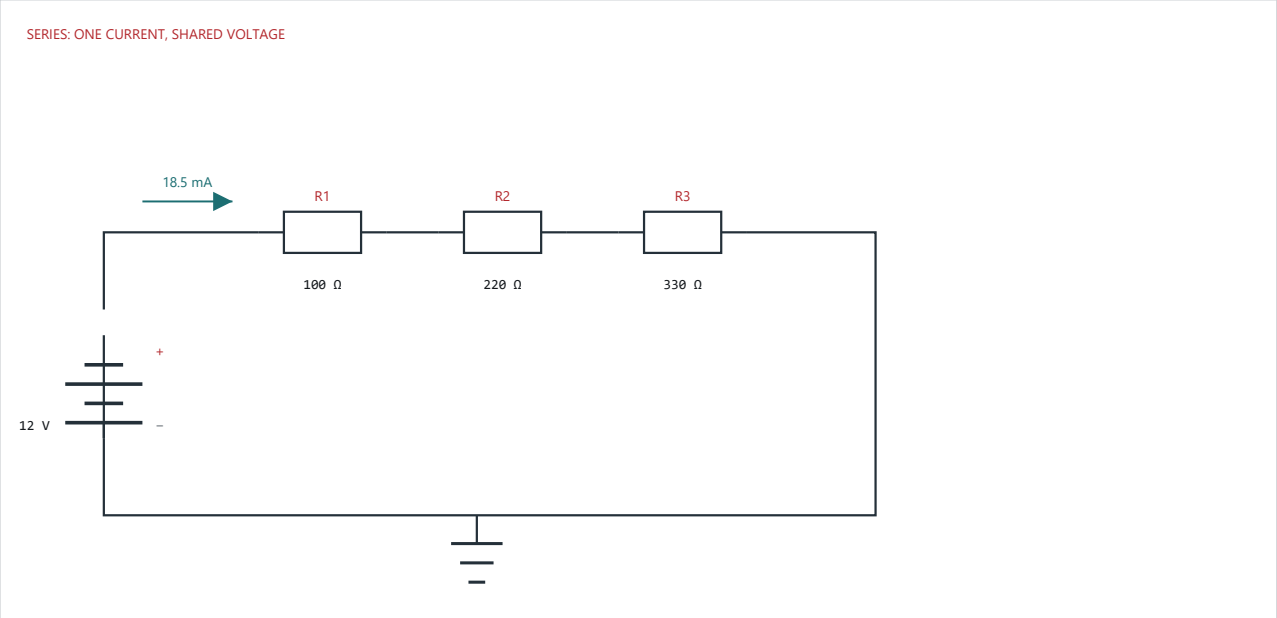



Fig. 3.1 A series string. The same current passes through every element, and the applied voltage divides in proportion to resistance.

## Series: one current, shared voltage

Elements are in series when they carry the identical current because there is only one path. The equivalent resistance is the sum, and the applied voltage divides in direct proportion to each resistance. The largest resistance therefore takes the largest share of the voltage, which is the basis of the divider rule.

Series behaviour explains why a single open circuit stops everything, and why a high-resistance joint anywhere in a string starves the entire load. It also explains why measuring current anywhere in a series string gives the same answer.

```
R_eq = R_1 + R_2 + ... + R_n

I is identical in every element
```

## Parallel: one voltage, shared current

Elements are in parallel when they share both end nodes and therefore the identical voltage. Conductances add, so the reciprocal of the equivalent resistance is the sum of the reciprocals. The equivalent is always smaller than the smallest branch, which is a useful sanity check on any parallel calculation.

For two resistors the product-over-sum form is quicker. For n identical resistors the equivalent is simply R divided by n. Both shortcuts are worth memorising because they let you check a calculator result instantly.

```
1/R_eq = 1/R_1 + 1/R_2 + ... + 1/R_n

Two only:  R_eq = R_1 R_2 / (R_1 + R_2)

n equal:   R_eq = R / n
```

CHECKS THAT CATCH PARALLEL ERRORS

The equivalent must be smaller than the smallest branch.
Adding a branch always lowers the equivalent resistance and raises the total current.
If two equal resistors give anything other than half their value, the arithmetic is wrong.

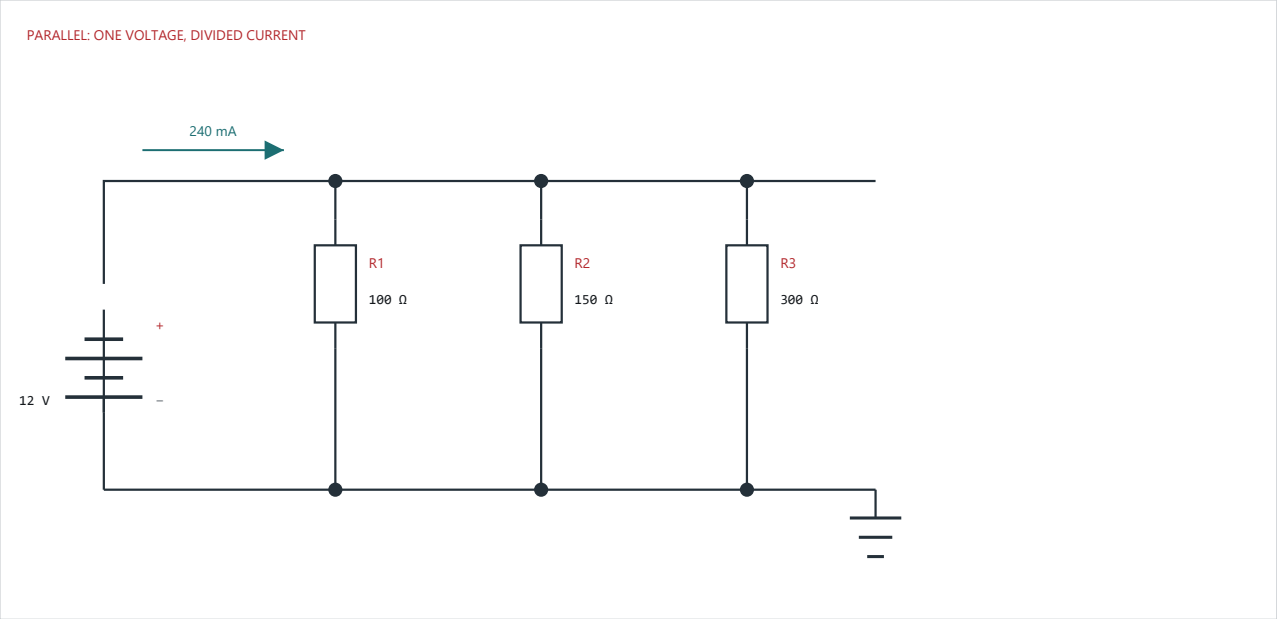



Fig. 3.2 A parallel group. Every branch sees the same voltage and the branch currents sum to the supply current.

## The voltage divider rule

In a series pair the voltage across one element is the supply voltage multiplied by that element's resistance over the total. This holds only while no current is drawn from the junction, which is the condition beginners most often forget.

A divider loaded by a finite resistance no longer follows the simple ratio. The loaded output is found by first combining the lower resistor with the load in parallel, then reapplying the rule. If the load resistance is at least ten times the lower resistor the error stays under ten percent, which is the usual design guideline.

```
V_out = V_in × R_2 / (R_1 + R_2)

Loaded:  R_2' = R_2 R_L / (R_2 + R_L)
```

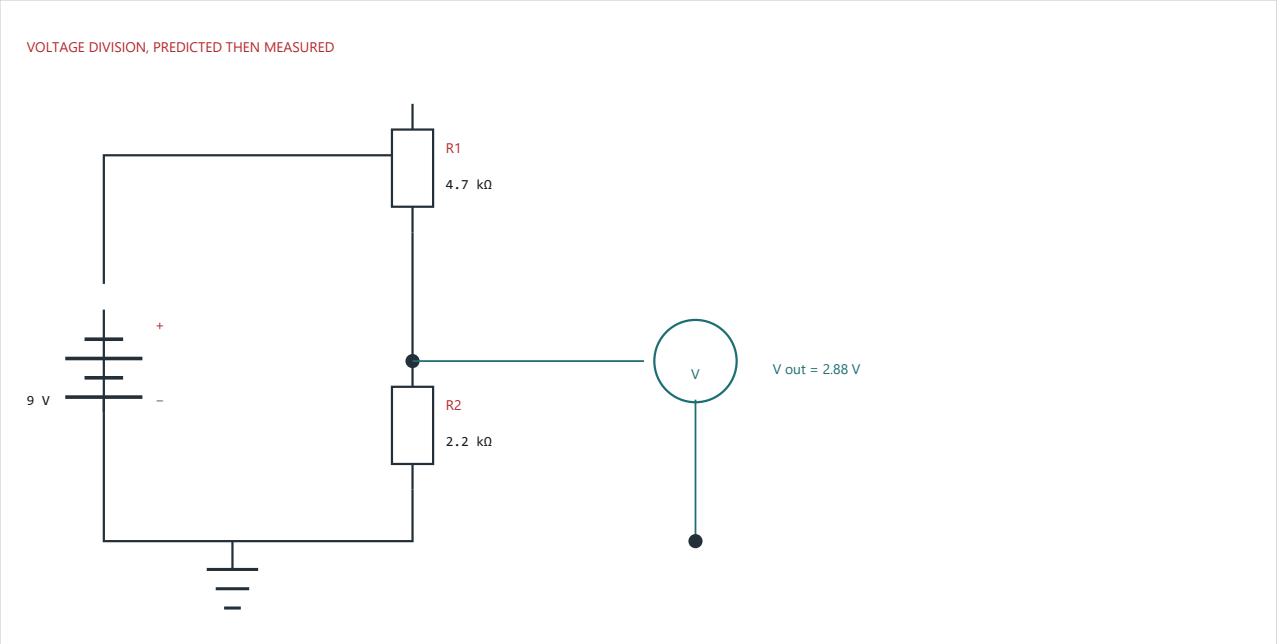



Fig. 3.3 A voltage divider with and without a load. Drawing current from the junction lowers the output below the unloaded ratio.

## The current divider rule

In a parallel pair the current in one branch is the total current multiplied by the opposite resistance over the sum. The inversion is deliberate: the smaller resistance takes the larger current, which is the opposite of the voltage divider's behaviour and a frequent source of confusion.

The current divider is the rule that explains parallel fault paths. When an unintended low-resistance path appears alongside the intended load, it takes most of the current, which is both why fuses operate and why earth faults are dangerous.

```
I_1 = I_total × R_2 / (R_1 + R_2)

The smaller resistance carries the larger current
```

## Resistivity: resistance from conductor dimensions

The resistance of a uniform conductor is proportional to its length and inversely proportional to its cross-sectional area, with resistivity as the material constant. Doubling the run doubles the resistance; doubling the area halves it.

Resistivity rises with temperature in metals, which is why a motor winding measures higher resistance hot than cold, and why insulation resistance measurements must be recorded with the temperature noted. A measured winding resistance can in fact be used as a thermometer.

```
R = ρ L / A

R_hot = R_cold [1 + α (T_hot − T_cold)]
```

| Material | Resistivity at 20 °C (Ω·m) | Temperature coefficient (per °C) | Typical use |
|---|---|---|---|
| Silver | $1.59 \times 10^{-8}$ | 0.0038 | Contacts, plating |
| Copper | $1.68 \times 10^{-8}$ | 0.00393 | Conductors, windings |
| Aluminium | $2.65 \times 10^{-8}$ | 0.00403 | Feeders, busbar |
| Nichrome | $1.10 \times 10^{-6}$ | 0.0004 | Heating elements |
| Carbon | $3.5 \times 10^{-5}$ | −0.0005 | Brushes, resistors |

## Worked examples

Work each example with a pencil before reading the solution. Sketch the circuit, mark the reference node, and write the governing law before substituting numbers.

EXAMPLE 3.1 Mixed series-parallel reduction

**Given.** A 24 V source feeds 100 Ω in series with a parallel pair of 300 Ω and 600 Ω.

**Find.** The equivalent resistance, the supply current and the current in each parallel branch.

Combine the parallel pair

```
R_p = 300×600/900 = 200 Ω
```

Add the series element

```
R_eq = 100 + 200 = 300 Ω
```

Supply current

```
I = 24/300 = 0.08 A = 80 mA
```

Voltage across the pair

```
V_p = 0.08 × 200 = 16 V
```

Branch currents

$I_{300} = 16/300 = 53.3$ mA ; $I_{600} = 16/600 = 26.7$ mA

**Check.** 53.3 + 26.7 = 80 mA, equal to the supply current, so KCL is satisfied at the junction.

EXAMPLE 3.2 A divider that sags under load

**Given.** A 12 V supply drives a divider of $R_1$ = 10 kΩ and $R_2$ = 10 kΩ. A 10 kΩ load is then connected across the output.

**Find.** The unloaded and loaded output voltages, and the percentage error.

Unloaded ratio

$$V = 12 \times 10/20 = 6.00\ V$$

Combine $R_2$ with the load

$$R_2' = 10 \times 10/20 = 5\ k\Omega$$

Reapply the rule

$$V = 12 \times 5/15 = 4.00\ V$$

Error introduced

$$(6.00 - 4.00)/6.00 = 33\ \%$$

**Check.** The load equals $R_2$, far below the ten-times guideline, so a third of the output is lost. Raising the load to 100 kΩ would give 5.71 V, an error of under 5 %.

EXAMPLE 3.3 Resistance of a conductor run

**Given.** A 60 m run of copper with a 2.5 $mm^2$ cross-section carries current to a load, and the return conductor is the same length.

**Find.** The loop resistance at 20 °C and the drop at 16 A.

Total conductor length

$$L = 2 \times 60 = 120\ m$$

Convert the area

$$A = 2.5\ mm^2 = 2.5 \times 10^{-6}\ m^2$$

Apply the resistivity relation

$$R = 1.68 \times 10^{-8} \times 120 / 2.5 \times 10^{-6} = 0.806\ \Omega$$

Voltage drop at 16 A

$$V = 16 \times 0.806 = 12.9\ V$$

Drop as a percentage of 230 V

$$12.9/230 \times 100 = 5.6\ \%$$

**Check.** 5.6 % exceeds the 3 % branch-circuit guidance in the NFPA 70 (2023) 210.19(A) informational note, so this run needs a larger conductor.

## Practical activity: series and parallel resistance, and a loaded divider

> AIM
> Confirm the series and parallel equivalence formulas against measurement, then demonstrate experimentally that drawing current from a divider lowers its output below the simple ratio.

### Equipment

- DC supply, 0–20 V, current limited to 0.25 A.
- Resistors: two 10 kΩ, one 1 kΩ, one 100 kΩ, one 330 Ω, one 470 Ω.
- Two multimeters, one used as an ohmmeter and one as a voltmeter.
- Breadboard or terminal strip and marked leads.

### Procedure

1. With the supply disconnected, measure and record every resistor individually. Use these values, not the printed ones, in your predictions.
2. Connect the 330 Ω and 470 Ω in series. Predict, then measure, the equivalent resistance with the circuit de-energised.
3. Connect the same pair in parallel. Predict, then measure, the equivalent resistance.
4. Build the divider from two 10 kΩ resistors and set the supply to 12.0 V.
5. Measure the unloaded output voltage and compare it with the predicted 6 V.
6. Connect the 1 kΩ as a load across the output, measure, and record the sag.
7. Replace the load with 10 kΩ, then 100 kΩ, measuring each time.
8. Plot output voltage against load resistance and mark where the error falls below 5 %.

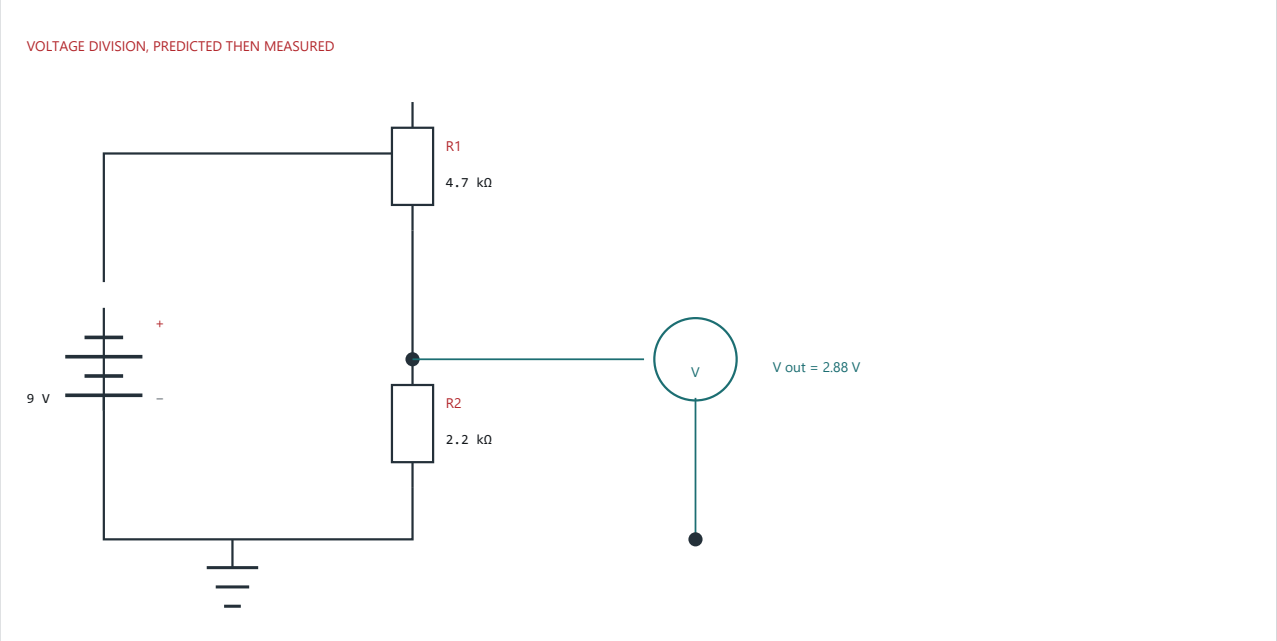



Fig. 3.L The divider under test. The load is connected across the output, between the junction and the reference node.

### Results table

| Load (Ω) | Predicted V_out | Measured V_out | Error (%) | Comment |
|---|---|---|---|---|
| none (open) | 6.00 | | | |
| 100 k | | | | |
| 10 k | | | | |
| 1 k | | | | |
| 470 | | | | |
| 330 | | | | |

### Discussion

1. At which load resistance did the error first exceed 5 %? Compare that with the ten-times guideline.
2. Why must the ohmmeter measurements be made with the supply disconnected?
3. Your voltmeter has a finite input resistance. Treat it as a load and estimate the error it introduces on the unloaded reading.
4. Measured series equivalents are usually very close to prediction while parallel equivalents deviate more. Suggest why.
5. Design a divider that delivers 5 V from 12 V into a 10 kΩ load with under 2 % error, and state your resistor values.

> **SAFE-WORK REMINDERS**
> Keep the supply at or below 20 V with the current limit at 0.25 A.
> Never apply an ohmmeter to an energised circuit; the reading is meaningless and the meter may be damaged.
> Check the dissipation in the lowest-value resistor before energising.
> Discharge and de-energise before changing any connection.

## Troubleshooting guide

Faults are found by narrowing the circuit, not by replacing parts. Each row below pairs an observed symptom with the measurement that separates the likely causes.

| Symptom | Likely cause | Decisive test | Correct action |
|---|---|---|---|
| Load runs at reduced output, divider output sags | The divider is loaded far below the ten-times guideline | Measure the output open and loaded | Lower the divider resistances or add a buffer stage |
| Parallel group draws far more current than predicted | A branch is short circuited or an extra branch is present | Isolate and measure each branch resistance | Repair the shorted branch; account for the extra load |
| Series string dead, full supply voltage across one point | An open element or joint in the string | Measure voltage across each element in turn | Replace the open element or re-terminate the joint |
| Measured resistance much higher than the marked value | Measuring in circuit, with parallel paths included | Disconnect one end of the component and re-measure | Measure components out of circuit, de-energised |
| Long run delivers low voltage under load only | Conductor resistance, correctly predicted by resistivity | Measure at the panel and the load, both under load | Increase conductor size or shorten the run |
| Winding resistance reads high on a hot machine | Normal temperature coefficient of copper | Record the winding temperature with the reading | Correct the value to 20 °C before comparing |

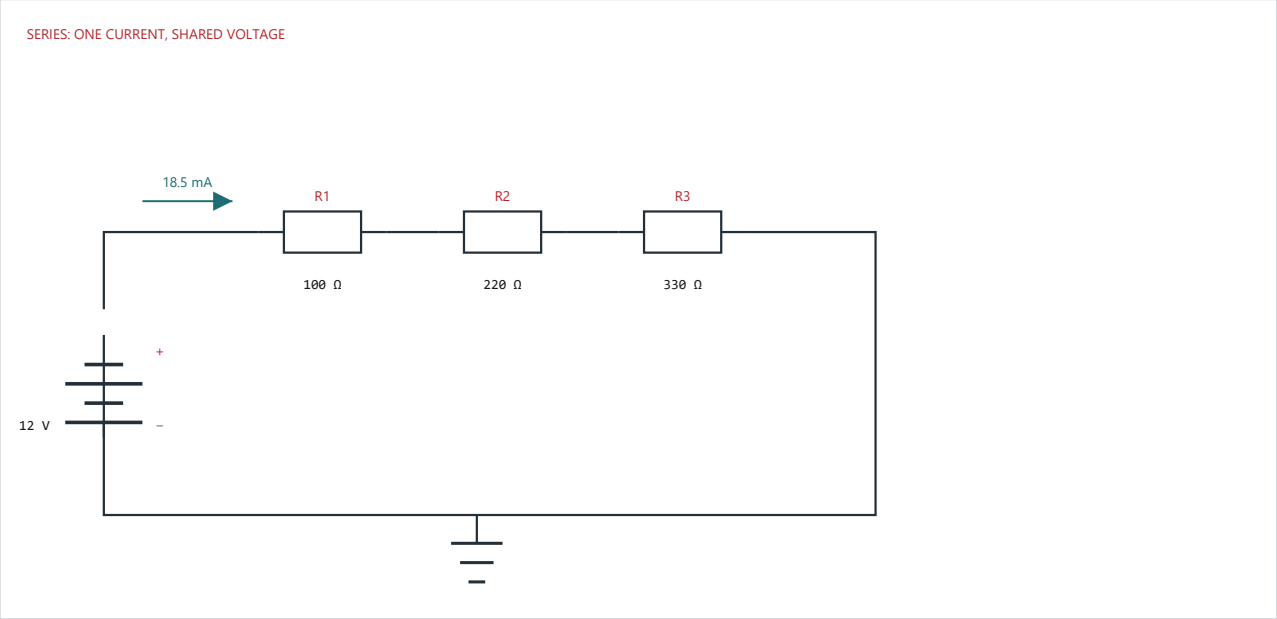



Fig. 3.T Fault location in a series string. The open element is the one with the supply voltage across it while the others read nothing.

FIELD HABITS THAT PREVENT CALLBACKS

Measure resistance out of circuit and de-energised, or you are measuring the whole network.
Use measured resistor values in predictions; tolerance bands are wide enough to matter.
Treat the voltmeter as a load whenever the circuit resistance is above about 100 kΩ.
Record conductor length and size when you log a voltage drop, because the two together explain the reading.

## Exercises

Answer from first principles. Selected answers follow; the reasoning matters more than the number.

1. Find the equivalent of 220 Ω, 330 Ω and 470 Ω in series.
2. Find the equivalent of 220 Ω and 330 Ω in parallel.
3. Find the equivalent of four 1 kΩ resistors in parallel.
4. A 20 V supply feeds 1.2 kΩ in series with 1.8 kΩ. Find the voltage across each.
5. A 2 A total current divides between 50 Ω and 150 Ω in parallel. Find each branch current.
6. A 9 V supply drives a divider of two 4.7 kΩ resistors loaded by 4.7 kΩ. Find the output.
7. A 100 m length of copper of 1.5 $mm^2$ cross-section is used as a single conductor. Find its resistance at 20 °C.
8. A copper winding measures 4.0 Ω at 20 °C. Find its resistance at 75 °C.
9. Explain why a parallel equivalent must always be less than the smallest branch.
10. Design a divider giving 3.3 V from 12 V with a total current near 1 mA, and state your values.

Selected answers

| # | Answer | Reason |
|---|---|---|
| 1 | 1020 Ω | Series resistances add |
| 2 | 132 Ω | Product over sum: 220×330/550 |
| 3 | 250 Ω | R/n for n equal resistors |
| 4 | 8 V and 12 V | Divider ratios 1.2/3.0 and 1.8/3.0 of 20 V |
| 5 | 1.5 A and 0.5 A | Current divider inverts: the smaller resistance takes more |
| 6 | 3.0 V | $R_2$ in parallel with the load gives 2.35 kΩ, then 9×2.35/7.05 |
| 7 | 1.12 Ω | $R = \rho L/A$ with L = 100 m, $A = 1.5\times10^{-6}$ $m^2$ |
| 8 | 4.86 Ω | R = 4.0[1 + 0.00393×55] |
| 9 | Adding any branch adds conductance | Total conductance exceeds that of the smallest branch alone, so the resistance is lower |
| 10 | About 8.7 kΩ and 3.3 kΩ | Ratio 3.3/12 with a total near 12 kΩ for 1 mA |

## Chapter summary

- Series elements share one current and divide the voltage; parallel elements share one voltage and divide the current.
- A parallel equivalent is always lower than the smallest branch, which is a one-second check.
- The voltage divider rule holds only for an unloaded output; a load must be combined with the lower resistor first.
- The current divider inverts the resistance ratio, so the smaller resistance carries the larger current.
- Conductor resistance follows $R = \rho L/A$ and rises with temperature, so runs and hot windings both read higher.
- A load of at least ten times the lower divider resistor keeps the output error under about ten percent.

### Equations introduced in this chapter

| Relationship | Expression | When it applies |
|---|---|---|
| Series equivalent | $R_{eq} = \Sigma R$ | One current path only |
| Parallel equivalent | $1/R_{eq} = \Sigma 1/R$ | Shared end nodes, one voltage |
| Two in parallel | $R_1R_2/(R_1+R_2)$ | Quick form for a pair |
| Voltage divider | $V_{out} = V_{in} R_2/(R_1+R_2)$ | Unloaded output only |
| Current divider | $I_1 = I R_2/(R_1+R_2)$ | Two parallel branches |
| Resistivity | $R = \rho L/A$ | Uniform conductor at a stated temperature |
| Temperature correction | $R_T = R_0[1+\alpha\Delta T]$ | Metals over moderate ranges |

### Vocabulary check

| Term | Meaning in this chapter |
|---|---|
| Equivalent resistance | The single resistance that would draw the same current from the same supply |
| Loaded divider | A divider whose output supplies current, so the simple ratio no longer applies |
| Conductance | The reciprocal of resistance, in siemens; it adds for parallel branches |
| Resistivity | The material property linking resistance to length and cross-section |
| Temperature coefficient | The fractional change in resistance per degree of temperature rise |

LOOKING AHEAD

Series and parallel reduction handles resistive networks. The next chapter introduces the capacitor, an element whose current depends on the rate of change of voltage, and with it the idea of a circuit whose behaviour unfolds over time.

CHAPTER 04

# Capacitors and RC circuits

Stored charge, stored energy, and circuits whose behaviour unfolds over time

WHY THIS CHAPTER MATTERS

A capacitor is the first element whose current depends not on the voltage across it but on how fast that voltage is changing. That single difference introduces time into circuit analysis and underlies timing, filtering, smoothing and every transient you will meet.

## What you will be able to do

- State what capacitance measures and compute stored charge and energy.
- Combine capacitors in series and parallel, noting that the rules invert the resistor rules.
- Compute the time constant of an RC circuit and sketch charge and discharge curves.
- Predict the voltage or current at a stated number of time constants.
- Discharge and test a capacitor safely before handling it.

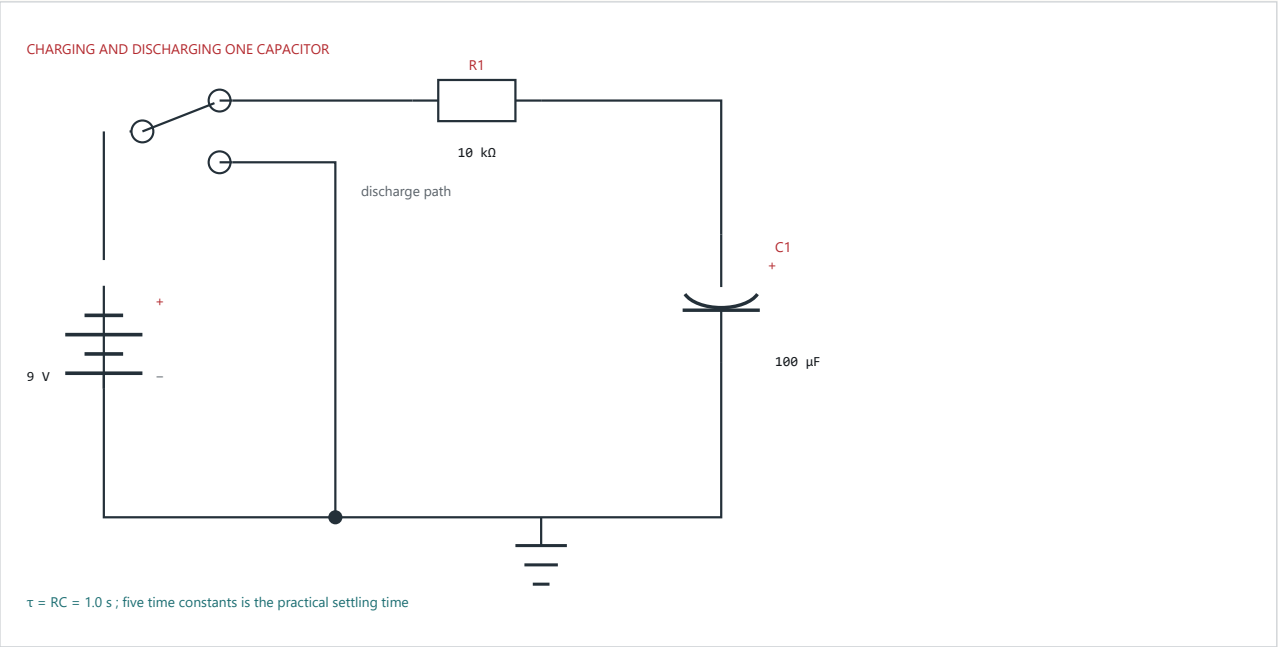



Fig. 4.1 An RC charging circuit. The resistor limits the rate at which charge accumulates, so the capacitor voltage rises exponentially toward the supply.

## Capacitance, charge and stored energy

Capacitance, symbol *C*, is the charge stored per volt applied, measured in farads. One farad is one coulomb per volt, an enormous quantity, so practical values run from picofarads to millifarads. A capacitor stores energy in the electric field between its plates, and that energy is proportional to the square of the voltage.

Because the energy depends on voltage squared, a capacitor charged to twice the voltage holds four times the energy. This is why a large capacitor in a drive or power supply remains a genuine hazard after the supply is removed, and why discharge is a required step rather than a precaution.

$$C = Q / V \qquad Q = C\,V$$

$$W = \tfrac{1}{2}\,C\,V^2$$

CAPACITOR DISCHARGE IS A SAFETY STEP, NOT AN OPTIONAL ONE

A 1000 µF capacitor at 400 V stores 80 J, enough to cause serious injury.
Discharge through a resistor sized for the energy, never by shorting with a screwdriver.
Verify zero volts with a meter after discharging, then verify the meter on a known source.

## Combining capacitors

Capacitors in parallel add, because the plate areas effectively add. Capacitors in series combine reciprocally, because the same charge must pass through each and the voltages add. The rules are therefore exactly the inverse of the resistor rules, which is worth stating explicitly because it is a common examination trap.

A practical consequence is that series capacitors divide the applied voltage in inverse proportion to their capacitance, so the smallest capacitor sees the largest voltage. Equalising resistors are fitted across series electrolytics for exactly this reason.

$$\text{Parallel:}\quad C_{eq} = C_1 + C_2 + \ldots + C_n$$

$$\text{Series:}\quad \frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \ldots + \frac{1}{C_n}$$

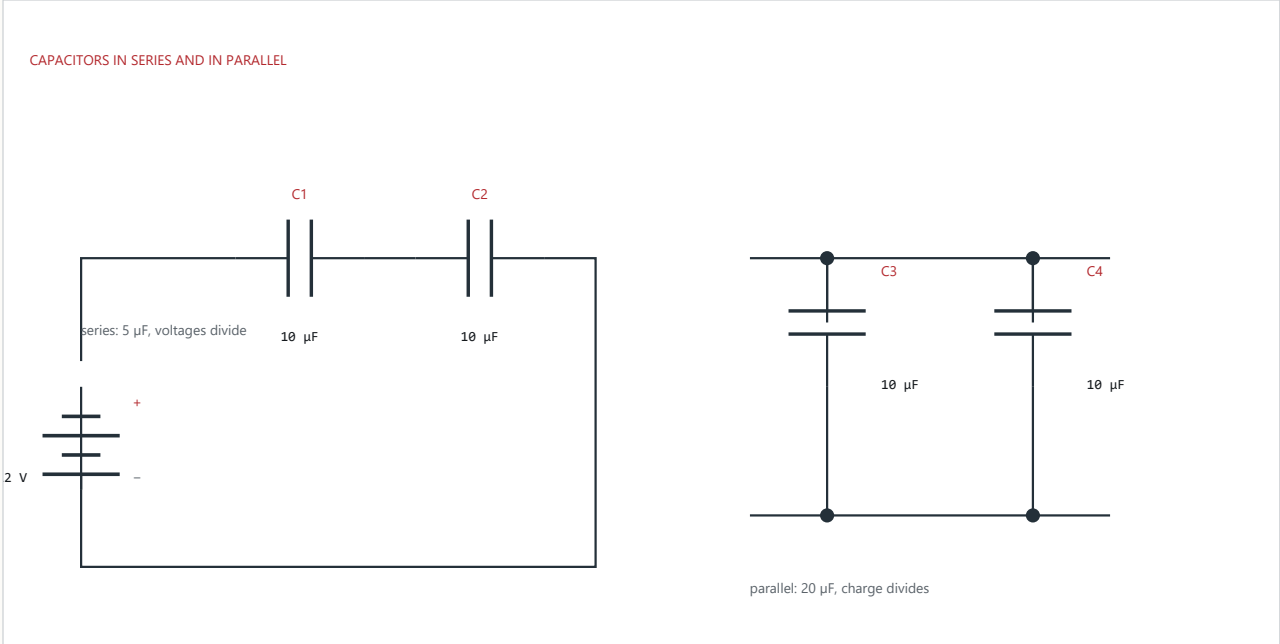



Fig. 4.2 Series and parallel capacitor groups. In series the same charge passes through each, so the smallest capacitance takes the largest share of the voltage.

## The time constant

When a capacitor charges through a resistance the product of the two has units of time and is called the time constant, written τ. It is the time to reach about 63.2 per cent of the final change, and the curve is effectively complete after five time constants, by which point 99.3 per cent of the change has occurred.

The same time constant governs charging and discharging. On discharge the voltage falls to 36.8 per cent of its initial value in one time constant. Memorising the five-step table below lets you read any RC transient without a calculator.

```
τ = R C

Charging:    v(t) = V(1 − e^(−t/τ))

Discharging: v(t) = V₀ e^(−t/τ)
```

| Elapsed time | Charged to | Remaining on discharge | Practical reading |
|---|---|---|---|
| 1 τ | 63.2 % | 36.8 % | Clearly rising, far from final |
| 2 τ | 86.5 % | 13.5 % | Most of the change has happened |
| 3 τ | 95.0 % | 5.0 % | Within 5 % of final |
| 4 τ | 98.2 % | 1.8 % | Indistinguishable on most meters |
| 5 τ | 99.3 % | 0.7 % | Treated as complete |

## Current during a transient

At the instant of switching, an uncharged capacitor behaves as a short circuit: the full supply appears across the resistor and the current is at its maximum. As the capacitor charges the resistor voltage falls and so does the current, decaying with the same time constant. Once fully charged the capacitor passes no steady current at all, behaving as an open circuit to DC.

These two limiting behaviours, short circuit at the first instant and open circuit at steady state, let you find the initial and final conditions of any RC circuit without solving a differential equation. The exponential simply connects the two.

```
i(0) = V / R          i(t) = (V/R) e^(-t/τ)

Steady state: i = 0, v_C = V
```

TWO LIMITS THAT SOLVE MOST PROBLEMS BY INSPECTION

At t = 0 an uncharged capacitor is a short circuit, so all the supply is across the resistance.
At steady state a capacitor is an open circuit, so no current flows in its branch.
Find both limits first, then join them with the exponential.

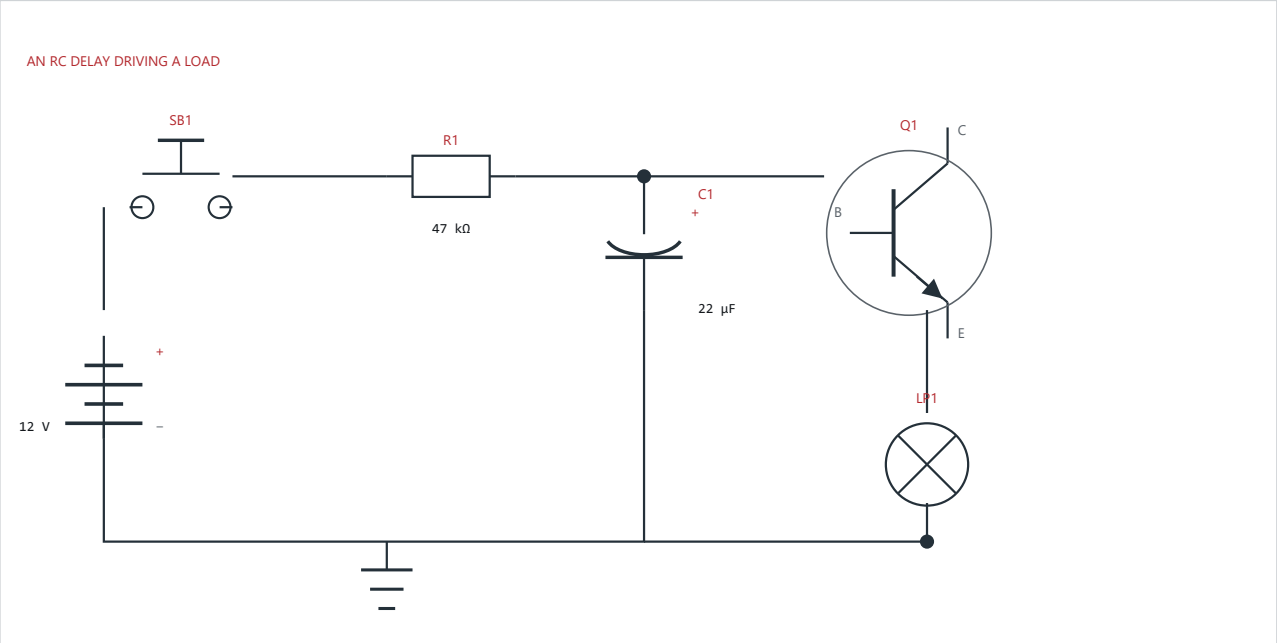



Fig. 4.3 Voltage and current during charging. The current starts at its maximum and decays as the capacitor voltage rises.

## Practical capacitors and their imperfections

A real capacitor is not purely capacitive. It has series resistance, which causes heating under ripple current, leakage resistance, which slowly discharges it, and series inductance, which limits its high-frequency usefulness. Electrolytic types are polarised and will fail, sometimes violently, if reverse connected.

Capacitors also age. An electrolytic that has lost electrolyte shows reduced capacitance and raised series resistance, which in a power supply appears as excessive output ripple long before the part fails outright.

Measuring series resistance is therefore more diagnostic than measuring capacitance alone.

| Type | Typical range | Polarised | Common application |
|---|---|---|---|
| Ceramic | 1 pF – 1 µF | No | Decoupling, high frequency |
| Film | 1 nF – 10 µF | No | Timing, audio, motor run |
| Aluminium electrolytic | 1 µF – 10 mF | Yes | Smoothing, bulk storage |
| Tantalum | 0.1 µF – 100 µF | Yes | Compact decoupling |
| Motor start | 20 µF – 400 µF | No | Single-phase motor starting |

## Worked examples

Work each example with a pencil before reading the solution. Sketch the circuit, mark the reference node, and write the governing law before substituting numbers.

EXAMPLE 4.1 Charge and energy stored

**Given.** A 470 µF capacitor is charged to 25 V.

**Find.** The stored charge and the stored energy.

Charge from capacitance and voltage

$$Q = C\,V = 470\times10^{-6} \times 25 = 0.01175\ \text{C}$$

Energy from the square law

$$W = \tfrac{1}{2}\,C\,V^{2} = 0.5 \times 470\times10^{-6} \times 625$$

Evaluate

$$W = 0.147\ \text{J}$$

**Check.** Doubling the voltage to 50 V would give 0.587 J, four times the energy, confirming the square law. At 400 V the same capacitor would hold 37.6 J, firmly in hazardous territory.

EXAMPLE 4.2 Time constant and charging time

**Given.** A 100 µF capacitor charges through 10 kΩ from a 12 V supply.

**Find.** The time constant, the voltage after 1 s, and the time to reach 95 % of final.

Time constant

$$\tau = R\,C = 10\,000 \times 100\times10^{-6} = 1.0\ \text{s}$$

Voltage after one time constant

$$v = 12(1 - e^{-1}) = 12 \times 0.632 = 7.58\ \text{V}$$

Time to 95 %

$$\text{three time constants} = 3.0\ \text{s}$$

Initial current

$$i(0) = 12/10\,000 = 1.2\ \text{mA}$$

**Check.** Using the cancelling pair µF × kΩ = ms: 100 × 10 = 1000 ms = 1 s, which confirms the time constant without powers of ten.

EXAMPLE 4.3 Series capacitors sharing voltage

**Given.** A 10 µF and a 22 µF capacitor are connected in series across 200 V.

**Find.** The equivalent capacitance and the voltage across each.

Series combination

$1/C = 1/10 + 1/22 = 0.1455 \text{ per } \mu F$

Equivalent

$C_{eq} = 6.875\ \mu F$

Charge is common

$Q = C_{eq} V = 6.875 \times 10^{-6} \times 200 = 1.375\ mC$

Voltage on each

$V_{10} = 1.375\ m/10\ \mu = 137.5\ V\ ;\ V_{22} = 1.375\ m/22\ \mu = 62.5\ V$

**Check.** 137.5 + 62.5 = 200 V, satisfying KVL. The smaller capacitor carries the larger voltage, so its voltage rating governs the design.

EXAMPLE 4.4 Discharge to a safe level

**Given.** A 1000 µF capacitor at 320 V is discharged through a 2.2 kΩ bleed resistor.

**Find.** The time constant, the time to fall below 50 V, and the peak power in the resistor.

Time constant

$\tau = 2200 \times 1000 \times 10^{-6} = 2.2\ s$

Required ratio

$50/320 = 0.156,\ \text{and}\ \ln(1/0.156) = 1.86$

Time to 50 V

$t = 1.86 \times 2.2 = 4.1\ s$

Peak resistor power

$P = V^2/R = 320^2/2200 = 46.5\ W \text{ momentarily}$

**Check.** The resistor must tolerate a brief 46.5 W pulse even though the average is low. Verify zero with a meter after discharging; never assume the bleed resistor is intact.

## Practical activity: charging and discharging a capacitor through a known resistance

> AIM
> Measure a charge and a discharge curve, extract the time constant from the data, and compare it with the product of the measured resistance and capacitance.

### Equipment

- DC supply, 12 V regulated.
- Capacitor, 1000 µF electrolytic, voltage rating at least 25 V.
- Resistors, 10 kΩ and 22 kΩ, 0.5 W.
- Multimeter with at least 1 MΩ input resistance, and a stopwatch.
- SPDT switch, or a flying lead used carefully, and a 1 kΩ bleed resistor.

### Procedure

1. Measure and record the resistor and capacitor values. Compute the expected time constant before building anything.
2. Observe polarity: the electrolytic's negative terminal goes to the reference node.
3. Wire the capacitor in series with the 10 kΩ resistor, with the switch selecting either the supply or a short to the reference through the resistor.
4. Connect the voltmeter directly across the capacitor and leave it there for the whole run.
5. With the capacitor discharged, switch to the supply and start the stopwatch together.
6. Record the capacitor voltage every 2 s for 60 s.
7. Switch to the discharge position and record every 2 s for a further 60 s.
8. Repeat the charge run with the 22 kΩ resistor and compare the two time constants.
9. Discharge fully through the bleed resistor and verify zero volts before dismantling.

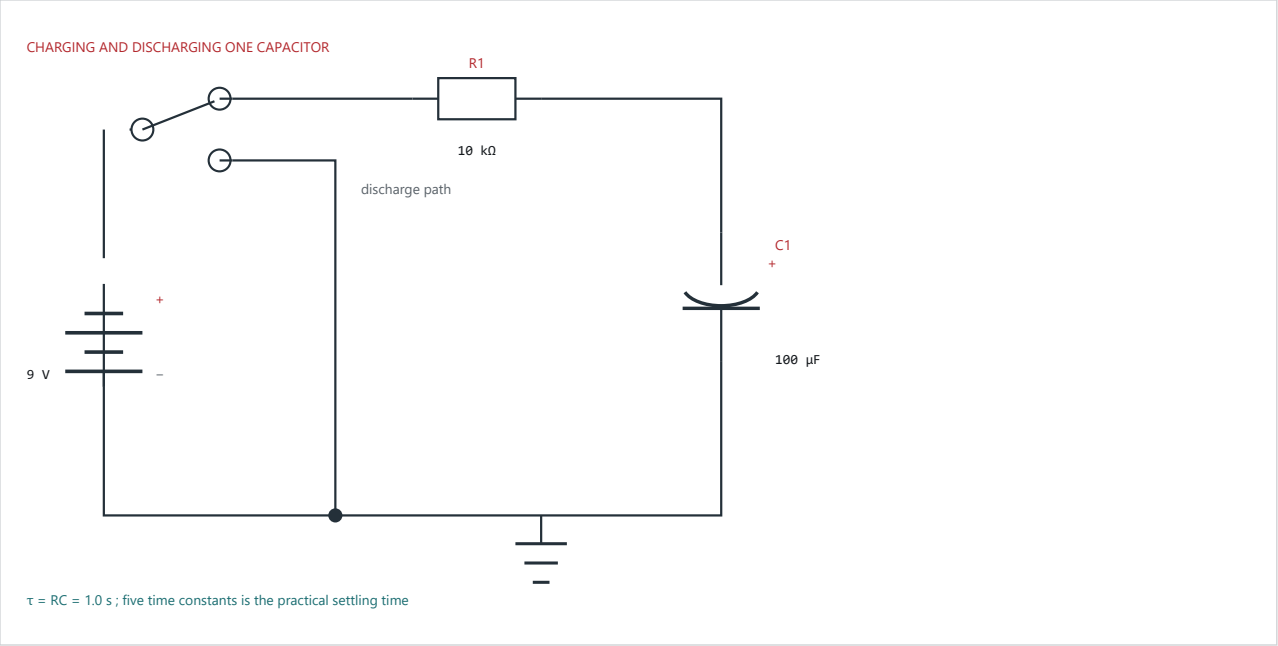



Fig. 4.L The charge and discharge arrangement. The voltmeter stays across the capacitor throughout, and its input resistance appears in parallel with the capacitor.

## Results table

| Time (s) | Charge v (V) | Predicted v (V) | Discharge v (V) | Predicted v (V) |
|---|---|---|---|---|
| 0 | | 0.00 | | 12.00 |
| 5 | | | | |
| 10 | | 7.58 | | |
| 20 | | | | |
| 30 | | 11.40 | | |
| 60 | | 11.97 | | |

## Discussion

1. From your charge data, read the time to reach 7.58 V. How does it compare with the computed time constant?
2. From your discharge data, read the time to fall to 4.42 V. Does it match the same time constant?
3. Your voltmeter's input resistance is in parallel with the capacitor during discharge. Estimate the error it introduces.
4. Compare the two time constants from the 10 kΩ and 22 kΩ runs. Is the ratio what you expected?
5. Electrolytic capacitors are often 20 % tolerance. Compute the capacitance implied by your measured time constant and compare it with the marked value.

SAFE-WORK REMINDERS

The electrolytic is polarised. Reverse connection can cause it to vent or burst.
Keep the supply at 12 V; this activity does not require or permit higher voltages.
Discharge through the bleed resistor and confirm zero volts before touching the terminals.
Do not short the capacitor with a tool; the current is high enough to weld and to spray metal.

## Troubleshooting guide

Faults are found by narrowing the circuit, not by replacing parts. Each row below pairs an observed symptom with the measurement that separates the likely causes.

| Symptom | Likely cause | Decisive test | Correct action |
|---|---|---|---|
| Capacitor voltage rises far more slowly than predicted | Capacitance higher than marked, or resistance higher than measured | Measure both components and recompute the time constant | Use measured values; replace the component if it is out of tolerance |
| Capacitor will not hold charge after the supply is removed | Internal leakage, or the voltmeter is loading it | Disconnect the meter and retest after a delay | Replace a leaky capacitor; use a higher impedance meter |
| Power supply output ripple has risen over time | Smoothing electrolytic has lost capacitance and gained series resistance | Measure ripple on the rail and the capacitor's series resistance | Replace the electrolytic with an equal or better ripple rating |
| Electrolytic is swollen or has vented | Reverse polarity, overvoltage or excessive ripple heating | Inspect, then confirm the applied polarity and voltage | Replace and correct the cause before re-energising |
| Series electrolytics fail one at a time | Unequal voltage sharing, so one exceeded its rating | Measure the voltage across each while energised | Fit equalising resistors across each capacitor |
| Motor will not start, run capacitor suspected | Open or low-capacitance run capacitor | Discharge, then measure capacitance out of circuit | Replace with the same microfarad and voltage rating |

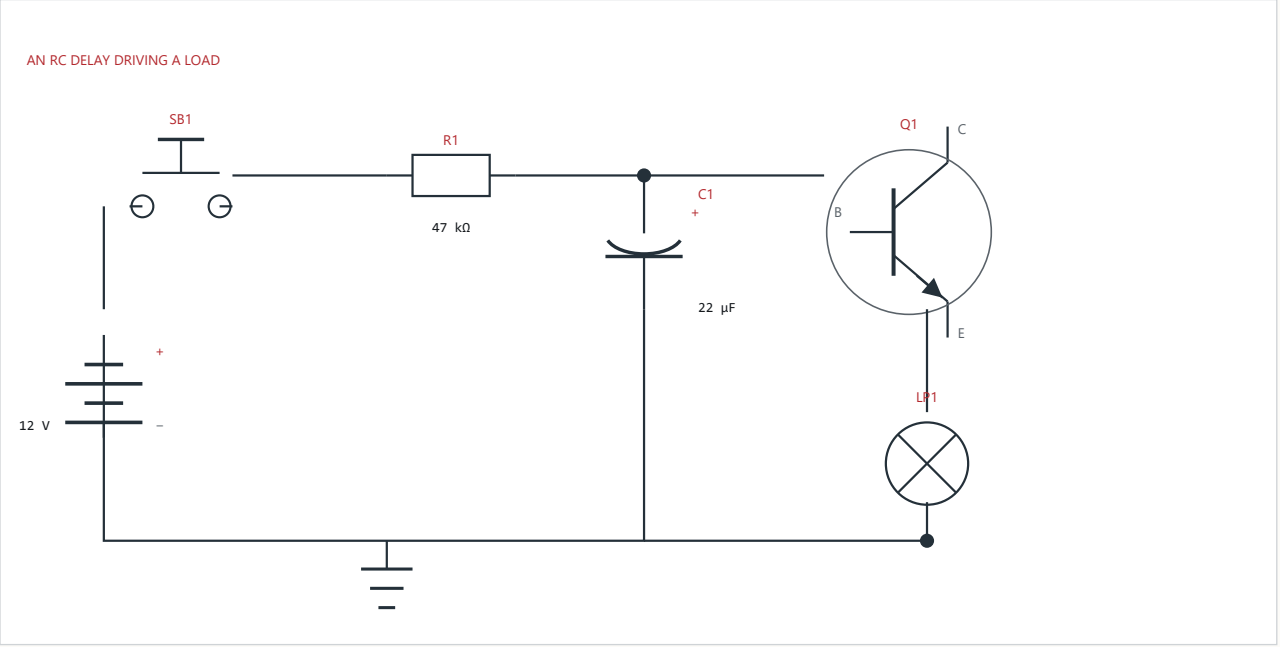



Fig. 4.T Reading a transient on an instrument. The time constant is found from the 63 % point on charge or the 37 % point on discharge.

FIELD HABITS THAT PREVENT CALLBACKS

Treat every large capacitor as charged until you have discharged it and verified zero.
Record the resistance and capacitance you actually measured, not the printed values.
When a transient looks wrong, check the meter's input resistance before suspecting the circuit.
On a power supply, measure ripple rather than capacitance; it fails first and is easier to see.

## Exercises

Answer from first principles. Selected answers follow; the reasoning matters more than the number.

1 A 220 µF capacitor is charged to 40 V. Find the stored charge and energy.
2 Find the equivalent of 10 µF and 15 µF in parallel, and in series.
3 Find the time constant of 47 kΩ with 10 µF.
4 A 22 µF capacitor charges through 100 kΩ from 15 V. Find the voltage after 2.2 s.
5 How long does a transient take to be 99 % complete, in time constants?
6 A capacitor discharges from 100 V to 36.8 V in 0.5 s. Find the time constant.
7 Two capacitors, 4.7 µF and 1 µF, are in series across 150 V. Find the voltage across each.
8 Find the initial current when an uncharged 100 µF capacitor is connected through 470 Ω to 24 V.
9 Explain why a capacitor passes no steady DC current but does pass alternating current.
10 A 2200 µF capacitor at 400 V must reach under 60 V within 5 s. Find the largest bleed resistor.

Selected answers

| # | Answer | Reason |
|---|---|---|
| 1 | 8.8 mC; 0.176 J | $Q = CV$, then $W = \frac{1}{2}CV^2$ |
| 2 | 25 µF parallel; 6 µF series | Parallel adds; series is reciprocal |
| 3 | 0.47 s | $\tau = RC$, using 47 × 10 = 470 ms |
| 4 | 9.48 V | $\tau = 2.2$ s, so $t = 1\tau$ and $v = 15 \times 0.632$ |
| 5 | About 4.6 time constants | $\ln(100) \approx 4.6$ |
| 6 | 0.5 s | Falling to 36.8 % defines one time constant |
| 7 | 26.3 V and 123.7 V | Inverse to capacitance; the 1 µF takes the larger share |
| 8 | 51.1 mA | $i(0) = V/R$, the capacitor is momentarily a short circuit |
| 9 | DC gives no rate of change once charged | Current requires dv/dt, which alternating voltage supplies continuously |
| 10 | About 1.2 kΩ | Needs $\tau \le 2.6$ s, so $R \le 2.6/2200$ µF |

## Chapter summary

- Capacitance is charge stored per volt; stored energy follows the square of the voltage, so high-voltage capacitors are hazardous long after isolation.
- Capacitors add in parallel and combine reciprocally in series, the inverse of the resistor rules.
- Series capacitors divide voltage inversely to capacitance, so the smallest sees the most volts.
- The time constant RC sets the pace of every RC transient: 63 % in one, 99 % in five.
- An uncharged capacitor is momentarily a short circuit; a fully charged one is an open circuit to DC.
- Real capacitors have series resistance and leakage, and in power supplies these degrade before the part fails outright.

### Equations introduced in this chapter

| Relationship | Expression | When it applies |
|---|---|---|
| Capacitance | $C = Q/V$ | Defines capacitance from charge and voltage |
| Stored energy | $W = \frac{1}{2}CV^2$ | Any capacitor at a steady voltage |
| Parallel capacitance | $C_{eq} = \Sigma C$ | Shared end nodes |
| Series capacitance | $1/C_{eq} = \Sigma 1/C$ | Common charge path |
| Time constant | $\tau = RC$ | Single resistance charging a single capacitance |
| Charging voltage | $v = V(1 - e^{(-t/\tau)})$ | From an uncharged start |
| Discharging voltage | $v = V_0 e^{(-t/\tau)}$ | From an initial charge |

### Vocabulary check

| Term | Meaning in this chapter |
|---|---|
| Time constant | The product RC, the time to complete 63.2 % of a change |
| Transient | The time-varying response between two steady states |
| Steady state | The condition reached once all rates of change have ceased |
| Series resistance | The unwanted internal resistance that heats a capacitor under ripple |
| Bleed resistor | A resistor fitted to discharge a capacitor safely after isolation |

LOOKING AHEAD

The capacitor opposes change in voltage. The next chapter introduces the inductor, which opposes change in current, and shows why switching an inductive circuit produces voltages far higher than the supply.

CHAPTER 05

# Inductors and RL circuits

Stored magnetic energy, induced voltage, and why switching inductive loads is violent

WHY THIS CHAPTER MATTERS

An inductor opposes change in current. That opposition stores energy in a magnetic field and, when a switch interrupts the current, releases it as a voltage spike many times the supply. Understanding this explains contact erosion, relay suppression diodes and most switching damage.

## What you will be able to do

- State what inductance measures and compute induced voltage from a rate of current change.
- Compute stored magnetic energy and the time constant of an RL circuit.
- Explain why interrupting inductive current produces a large reverse voltage.
- Select and connect a suppression diode across a DC coil correctly.
- Recognise inductive behaviour in chokes, coils, contactors and motor windings.

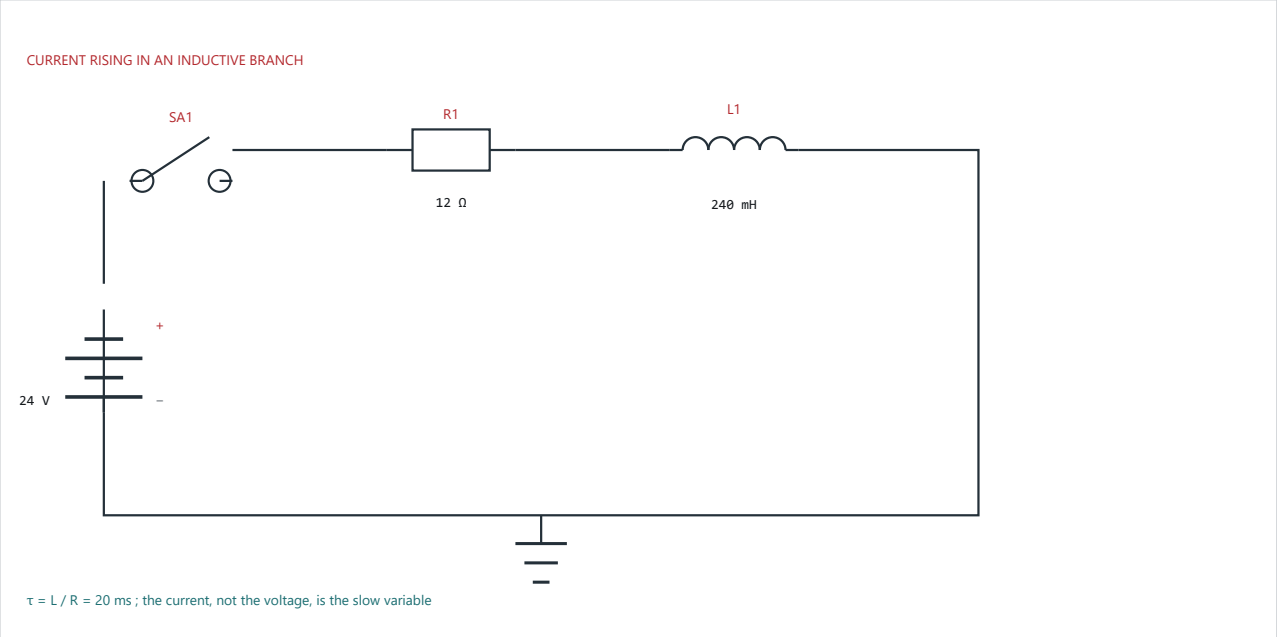



Fig. 5.1 An RL circuit. The inductor opposes the change in current, so the current rises exponentially rather than instantly.

## Inductance and induced voltage

Inductance, symbol *L*, measured in henries, relates the voltage induced across a coil to the rate of change of current through it. A one-henry inductor develops one volt when the current changes at one ampere per second. The induced voltage opposes the change that produced it, which is Lenz's law and the reason inductors are said to resist change.

The practical consequence is immediate: the faster you try to change the current, the larger the voltage that appears. A switch opening a coil circuit attempts to bring the current to zero in microseconds, so the induced voltage can reach hundreds or thousands of volts even from a 24 V supply.

```
v = L (di/dt)

W = ½ L I^2
```

WHY RELAY CONTACTS ERODE

Opening a coil circuit forces a very large di/dt, so v = L di/dt becomes large.
The resulting voltage strikes an arc across the opening contacts.
The arc carries the stored energy ½LI$^2$ into the contact metal, eroding it.

## Combining inductors, and the RL time constant

Inductors in series add and in parallel combine reciprocally, following the same pattern as resistors. The arithmetic is only valid when the coils share no magnetic coupling; coupled coils must be treated as a transformer instead.

In an RL circuit the time constant is the inductance divided by the resistance. The current rises toward its final value with the same exponential shape as capacitor voltage, reaching 63.2 per cent in one time constant and effectively settling after five. Note that raising the resistance shortens an RL transient, the opposite of the RC case.

```
Series:  L_eq = Σ L      Parallel: 1/L_eq = Σ 1/L

τ = L / R

i(t) = (V/R)(1 - e^(-t/τ))
```

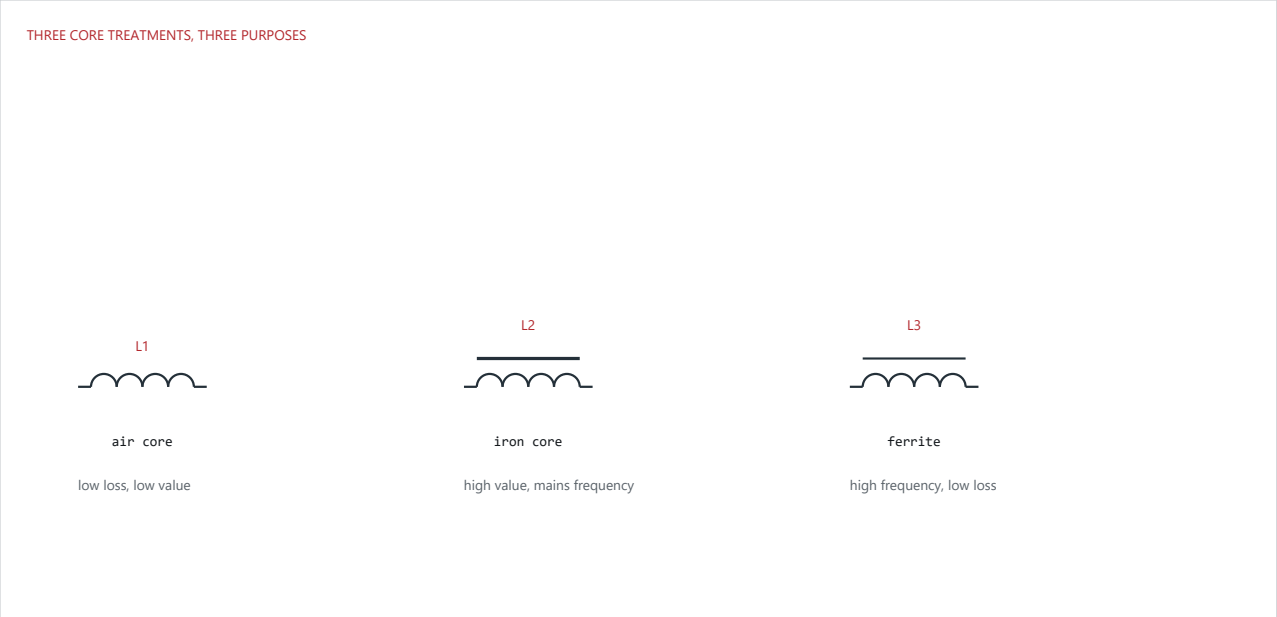



Fig. 5.2 A choke in a supply line. It passes direct current but opposes rapid changes, which is the basis of inductive filtering.

## Limiting behaviour at the two instants

At the first instant after switching, an inductor carrying no current behaves as an open circuit, because any sudden current would demand an infinite voltage. At steady state, with current no longer changing, it behaves as a plain resistance equal to its winding resistance. These are the mirror image of the capacitor's limits.

Knowing both limits lets you solve any RL problem by inspection: find the initial condition, find the final condition, and join them with the exponential governed by L over R.

| Element | At the first instant | At steady state | Energy stored in |
|---|---|---|---|
| Capacitor, uncharged | Short circuit | Open circuit | Electric field |
| Capacitor, charged | Source at its stored voltage | Open circuit | Electric field |
| Inductor, no current | Open circuit | Winding resistance only | Magnetic field |
| Inductor, carrying current | Source holding that current | Winding resistance only | Magnetic field |

## Suppressing the switching transient

A diode connected across a DC coil, reverse biased in normal operation, provides a path for the inductor current the instant the switch opens. The current circulates through the diode and decays harmlessly, clamping the coil voltage to roughly one diode drop above the supply. This is the flyback or freewheel diode, and fitting it correctly is one of the highest-value habits in control wiring.

Polarity is critical. Connected the wrong way the diode short circuits the supply through the coil and will fail immediately, usually taking the driving transistor with it. On AC coils a diode cannot be used; a resistor-capacitor snubber or a varistor is fitted instead.

Suppression choices by supply type

- DC coil: a diode across the coil, cathode to the positive side.
- AC coil: a resistor-capacitor snubber across the contacts or the coil.
- Either: a varistor across the coil, clamping both polarities at a chosen voltage.
- Semiconductor drivers: many include an internal clamp, but check before relying on it.

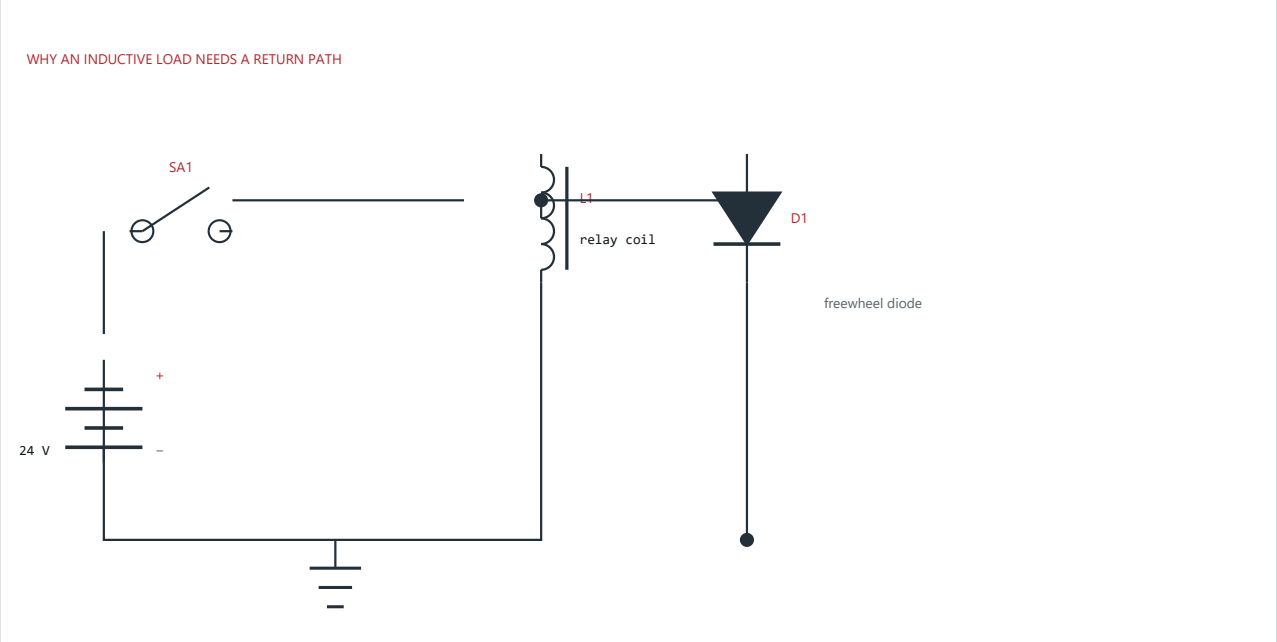



Fig. 5.3 A flyback diode across a DC coil. The diode is reverse biased while the coil is energised and conducts only during the switching transient.

## Where inductance appears in practice

Deliberate inductance appears in chokes, filter reactors, ballasts, relay and contactor coils, solenoids and transformer windings. Incidental inductance appears in every conductor loop, which is why high-current cable runs are kept close together and why long leads cause ringing on fast switching edges.

Motor windings are strongly inductive, which is why a motor draws a large starting current that lags the voltage, and why opening a motor circuit under load stresses the switching device far more than opening a resistive one of the same rating.

READING A COMPONENT AS ITS MODEL

A coil is an inductance in series with its own winding resistance, so an ohmmeter reads the resistance only. An open coil reads infinite resistance; a shorted turn reads lower than the nameplate value. Winding resistance alone cannot confirm inductance, but it does find most coil faults.

## Worked examples

Work each example with a pencil before reading the solution. Sketch the circuit, mark the reference node, and write the governing law before substituting numbers.

EXAMPLE 5.1 Induced voltage from a switching event

**Given.** A 0.4 H coil carries 2 A. A switch reduces the current to zero in 1 ms.

**Find.** The average induced voltage and the energy that must be absorbed.

Rate of current change

$$di/dt = 2 / 0.001 = 2000 \text{ A/s}$$

Induced voltage

$$v = L\, di/dt = 0.4 \times 2000 = 800 \text{ V}$$

Stored energy

$$W = \tfrac{1}{2} L I^2 = 0.5 \times 0.4 \times 4 = 0.8 \text{ J}$$

**Check.** 800 V from a low-voltage supply is why an unsuppressed coil destroys contacts and semiconductors. A flyback diode diverts the 0.8 J into the coil's own resistance instead.

EXAMPLE 5.2 RL time constant and current rise

**Given.** A 250 mH coil of 50 Ω winding resistance is connected to 24 V.

**Find.** The time constant, the final current, and the current after 5 ms.

Time constant

$$\tau = L/R = 0.25/50 = 5 \text{ ms}$$

Final current

$$I = V/R = 24/50 = 0.48 \text{ A}$$

Current at one time constant

$$i = 0.48 \times 0.632 = 0.303 \text{ A}$$

Time to settle

$$5\tau = 25 \text{ ms}$$

**Check.** Note that raising the resistance would shorten the transient and lower the final current, the opposite of the RC relationship where resistance lengthens the transient.

EXAMPLE 5.3 Sizing a flyback diode

**Given.** A 24 V contactor coil draws 150 mA and has an inductance of 1.2 H.

**Find.** The diode's required reverse voltage and forward current, and the decay time constant.

Reverse voltage seen in normal operation

`24 V, so choose at least 100 V rating`

Forward current at the instant of switching

`Equal to the coil current, 150 mA`

Coil resistance

`R = 24/0.15 = 160 Ω`

Decay time constant

`τ = 1.2/160 = 7.5 ms`

Energy dissipated per operation

$W = \tfrac{1}{2} \times 1.2 \times 0.15^2 = 13.5\ \text{mJ}$

**Check.** A 1 A 100 V diode is ample. The decay takes about 37 ms to complete, which slightly delays contactor drop-out, a known and usually acceptable side effect of freewheel suppression.

# Practical activity: current rise in an inductive circuit and the effect of a flyback diode

> AIM
> Observe that current in an inductive branch rises gradually rather than instantly, measure the time constant, and demonstrate that a correctly connected diode removes the switching spike.

### Equipment

- DC supply, 24 V, current limited to 0.5 A.
- Inductor or relay coil of known inductance, roughly 0.1–1 H, with its resistance measured.
- Series resistor, 100 Ω 1 W, used as a current-sensing shunt.
- Oscilloscope, or a multimeter with a peak-hold function.
- Diode, 1N4007 or similar, and a toggle switch.

### Procedure

1. Measure and record the coil resistance and the series resistor value.
2. Compute the expected time constant from L over the total circuit resistance.
3. Wire the coil in series with the 100 Ω resistor and the switch across the supply.
4. Connect the oscilloscope across the 100 Ω resistor, so the trace shows current directly.
5. Close the switch and capture the rising current. Measure the time to 63 % of final.
6. Move the oscilloscope across the switch contacts and capture the voltage when the switch opens.
7. Record the peak voltage observed. This is the unsuppressed transient.
8. Fit the diode across the coil, cathode to the positive side, and repeat the opening capture.
9. Reverse the diode briefly only as a thought exercise on paper; do not actually energise it reversed.

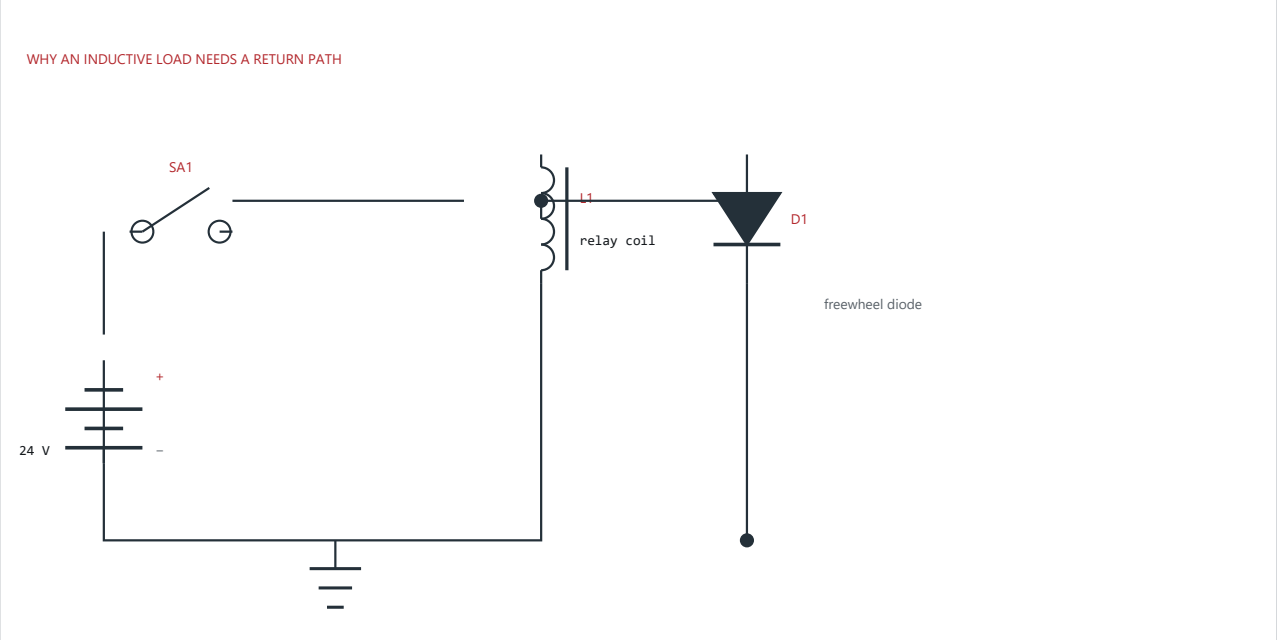



Fig. 5.L The test circuit. The sensing resistor converts current to a voltage the instrument can display, and the diode is fitted across the coil only.

### Results table

| Condition | Peak switch voltage (V) | Time to 63 % (ms) | Predicted τ (ms) | Comment |
|---|---|---|---|---|
| No diode, closing | — | | | |
| No diode, opening | | — | | |
| Diode fitted, opening | | — | | |
| Diode fitted, closing | — | | | |
| Repeat, no diode | | | | |
| Repeat, diode | | | | |

### Discussion

1. By what factor did the diode reduce the peak switching voltage?
2. Did fitting the diode change the current rise on closing? Explain why or why not.
3. Compute the coil inductance implied by your measured time constant and compare it with the marked value.
4. Explain, using $v = L\,di/dt$, why the opening transient is far larger than the closing one.
5. What would happen electrically if the diode were connected the other way round, and which component would fail first?

> **SAFE-WORK REMINDERS**
> Keep the supply at 24 V with the current limit at 0.5 A.
> The unsuppressed opening transient can exceed several hundred volts at the switch; do not touch the contacts while switching.
> Observe diode polarity. A reversed diode becomes a short circuit across the supply.
> The coil stores energy; allow it to decay before disconnecting.

## Troubleshooting guide

Faults are found by narrowing the circuit, not by replacing parts. Each row below pairs an observed symptom with the measurement that separates the likely causes.

| Symptom | Likely cause | Decisive test | Correct action |
|---|---|---|---|
| Relay contacts pitted and welding after short service | Unsuppressed inductive switching transient and arcing | Observe the switch voltage on an oscilloscope while opening | Fit a flyback diode on DC, or a snubber or varistor on AC |
| Driving transistor fails repeatedly on a coil output | Missing or reversed suppression diode | Check the diode's presence and polarity against the drawing | Fit the diode with its cathode to the positive supply |
| Coil reads open on an ohmmeter | Broken winding or an open internal joint | Measure resistance at the coil terminals de-energised | Replace the coil or the whole device |
| Coil resistance well below the nameplate value | Shorted turns from insulation failure | Compare with a known good coil of the same part number | Replace; do not attempt repair |
| Contactor drops out sluggishly after a diode was added | Freewheel current sustains the coil flux briefly | Measure the decay time against the mechanical requirement | Accept it, or use a varistor instead for faster clamping |
| Electronic equipment misbehaves when a nearby coil switches | Radiated or conducted switching transient | Correlate the fault with the switching event | Suppress at the source and separate the wiring routes |

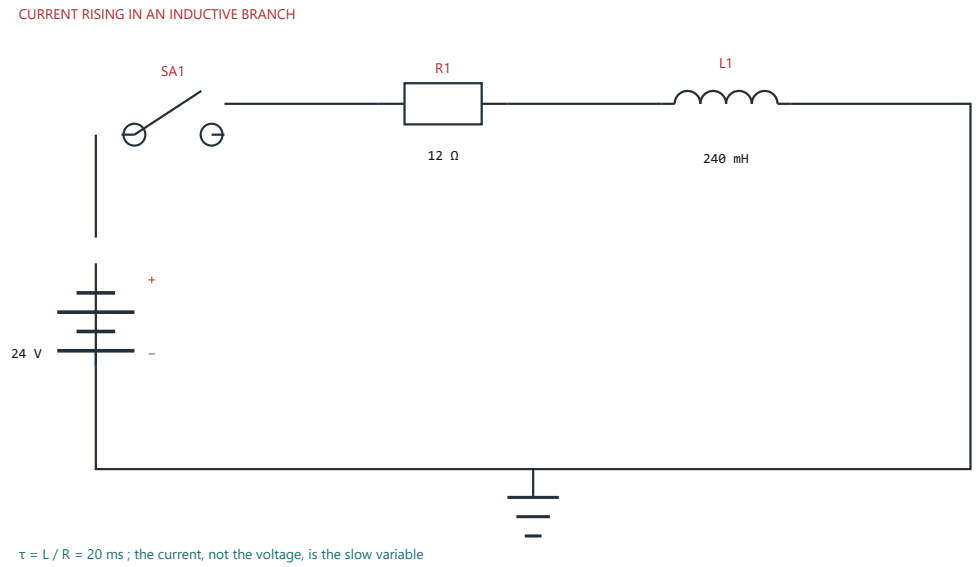



Fig. 5.T Measuring an inductive transient. A sensing resistor converts the branch current into a voltage the instrument can record.

FIELD HABITS THAT PREVENT CALLBACKS

Fit suppression at the coil, not at the panel; the transient must be clamped where it is generated.
Check diode polarity twice before energising a coil output; a reversed diode is an immediate short circuit.
Measure coil resistance when a device misbehaves; it finds open and shorted windings quickly.
Expect inductive loads to stress switchgear far more than resistive loads of the same current rating.

## Exercises

Answer from first principles. Selected answers follow; the reasoning matters more than the number.

1. A 0.8 H coil's current changes at 50 A/s. Find the induced voltage.
2. Find the energy stored in a 0.5 H inductor carrying 3 A.
3. Find the time constant of a 400 mH coil with 80 Ω total circuit resistance.
4. A 12 V supply feeds a 100 mH coil of 24 Ω. Find the final current and the time constant.
5. Find the equivalent of 100 mH and 150 mH in series, and in parallel.
6. A coil carrying 1.5 A has its current interrupted in 0.5 ms. With L = 0.3 H, find the induced voltage.
7. State the correct polarity for a flyback diode across a DC coil and explain why.
8. Explain why a diode cannot be used to suppress an AC-operated coil.
9. An RL circuit reaches 95 % of final current in 60 ms. Find its time constant.
10. A motor winding measures 3.2 Ω cold. State two separate faults that would make it read much lower, and how you would distinguish them.

Selected answers

| # | Answer | Reason |
|---|---|---|
| 1 | 40 V | $v = L\,di/dt$ |
| 2 | 2.25 J | $W = \frac{1}{2}LI^2$ |
| 3 | 5 ms | $\tau = L/R$ |
| 4 | 0.5 A; 4.17 ms | $I = V/R$, then $\tau = L/R$ |
| 5 | 250 mH series; 60 mH parallel | Series adds; parallel is reciprocal |
| 6 | 900 V | $di/dt = 3000$ A/s, then $v = L\,di/dt$ |
| 7 | Cathode to the positive side, reverse biased when energised | It then conducts only when the coil voltage reverses at switch-off |
| 8 | The diode would conduct on alternate half cycles | It would short the supply for half of every cycle, so a snubber or varistor is used instead |
| 9 | 20 ms | 95 % corresponds to three time constants |
| 10 | Shorted turns, or a measurement across the wrong terminals | Compare against a known good machine and verify the terminal identification first |

## Chapter summary

- Inductance relates induced voltage to the rate of change of current, so fast changes produce large voltages.
- Stored magnetic energy is $\frac{1}{2}LI^2$ and must go somewhere when the circuit is interrupted.
- The RL time constant is L over R, so increasing resistance shortens the transient.
- An inductor is momentarily an open circuit and eventually just its winding resistance, the mirror of a capacitor.
- A correctly polarised flyback diode clamps DC coil transients; AC coils need a snubber or varistor.
- Inductive loads stress contacts and semiconductors far more than resistive loads of equal current.

### Equations introduced in this chapter

| Relationship | Expression | When it applies |
|---|---|---|
| Induced voltage | $v = L\,di/dt$ | Any inductance with changing current |
| Stored energy | $W = \frac{1}{2}LI^2$ | Inductor carrying a steady current |
| Time constant | $\tau = L/R$ | Single inductance with series resistance |
| Current rise | $i = (V/R)(1 - e^{(-t/\tau)})$ | From zero initial current |
| Series inductance | $L_{eq} = \Sigma L$ | Uncoupled coils |
| Parallel inductance | $1/L_{eq} = \Sigma\, 1/L$ | Uncoupled coils |

### Vocabulary check

| Term | Meaning in this chapter |
|---|---|
| Inductance | The property relating induced voltage to the rate of change of current |
| Lenz's law | The induced voltage opposes the change in current that produced it |
| Flyback diode | A diode across a DC coil that carries the current during switch-off |
| Snubber | A resistor-capacitor network that absorbs a switching transient |
| Varistor | A voltage-dependent resistor that clamps transients of either polarity |

LOOKING AHEAD

Capacitors and inductors both respond to change rather than to steady values. The next chapter introduces alternating current, where the voltage changes continuously, and develops the language of frequency, phase and root-mean-square values.

CHAPTER 06

# Alternating current fundamentals

Waveforms, frequency, phase and the meaning of a root-mean-square value

WHY THIS CHAPTER MATTERS

Supply systems alternate, so every quantity you measure on them is an average of a changing value. Knowing exactly which average your meter reports, and what phase means, is the difference between a correct reading and a plausible wrong one.

## What you will be able to do

- Describe a sinusoid by amplitude, frequency, period and phase.
- Convert between peak, peak-to-peak, average and root-mean-square values.
- Explain why the root-mean-square value is the one that determines heating.
- Read period and amplitude from an oscilloscope display and compute frequency.
- Recognise when a meter's averaging method will give a misleading result.

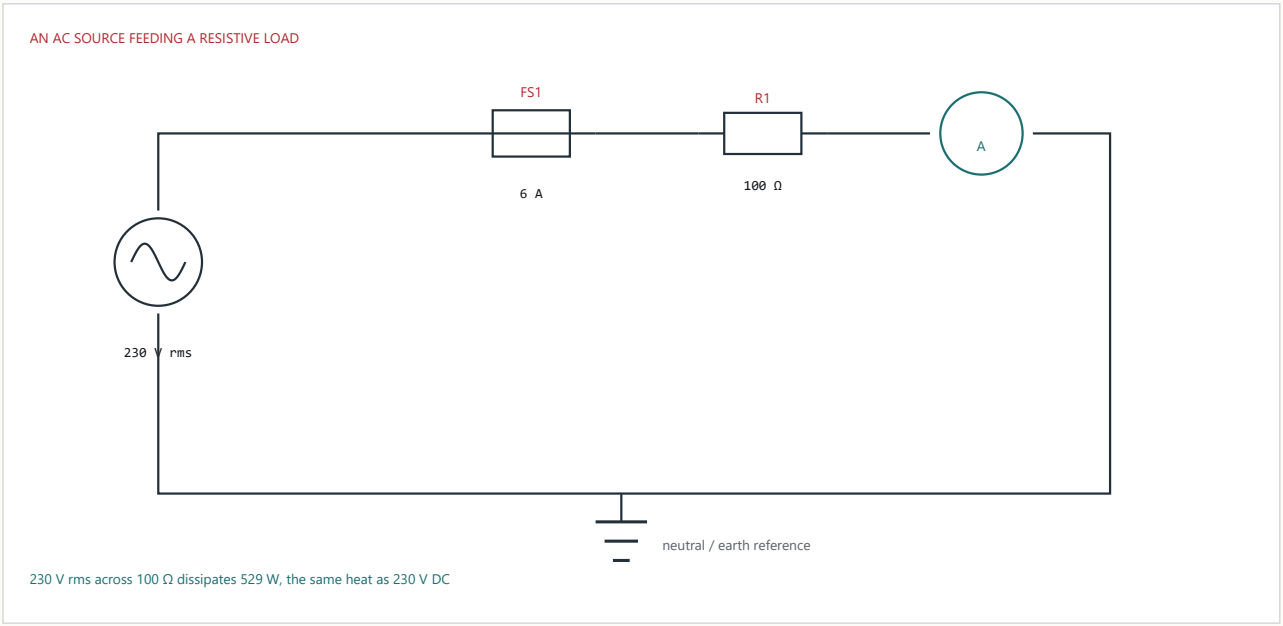



Fig. 6.1 A sinusoidal supply driving a resistive load. The instantaneous values change continuously, so a single figure must be defined before it can be quoted.

## Describing a sinusoid

A sinusoidal voltage is fully described by three numbers: its amplitude, its frequency and its phase relative to some reference. The period is the time for one complete cycle and is the reciprocal of the frequency. At 60 Hz the period is 16.67 ms; at 50 Hz it is 20 ms.

Phase expresses where in its cycle a waveform sits compared with another of the same frequency. It is meaningless to speak of the phase of a single isolated waveform, just as it is meaningless to speak of a voltage at a single point: both need a reference.

```
v(t) = V_peak sin(2π f t + φ)

T = 1 / f          ω = 2π f
```

## Peak, average and root-mean-square

The peak value is the largest instantaneous value. The peak-to-peak value is twice that for a symmetrical wave. The average over a full cycle of a symmetrical alternating wave is zero, which is why the rectified average is quoted instead. None of these is the value that determines heating.

The root-mean-square value is defined as the equivalent direct voltage that would produce the same heating in a resistance. For a sinusoid it is the peak divided by the square root of two, about 0.707 of peak. Because heating depends on the square of the instantaneous value, this is the only average that predicts power correctly.

```
V_rms = V_peak / √2 = 0.707 V_peak

V_peak = 1.414 V_rms

V_avg(rectified) = 0.637 V_peak
```

| Quantity | For a sinusoid | At 120 V rms | At 230 V rms |
|---|---|---|---|
| Root-mean-square | V_rms | 120 V | 230 V |
| Peak | 1.414 V_rms | 170 V | 325 V |
| Peak to peak | 2.828 V_rms | 339 V | 651 V |
| Rectified average | 0.900 V_rms | 108 V | 207 V |
| Form factor | 1.11 | — | — |
| Crest factor | 1.414 | — | — |

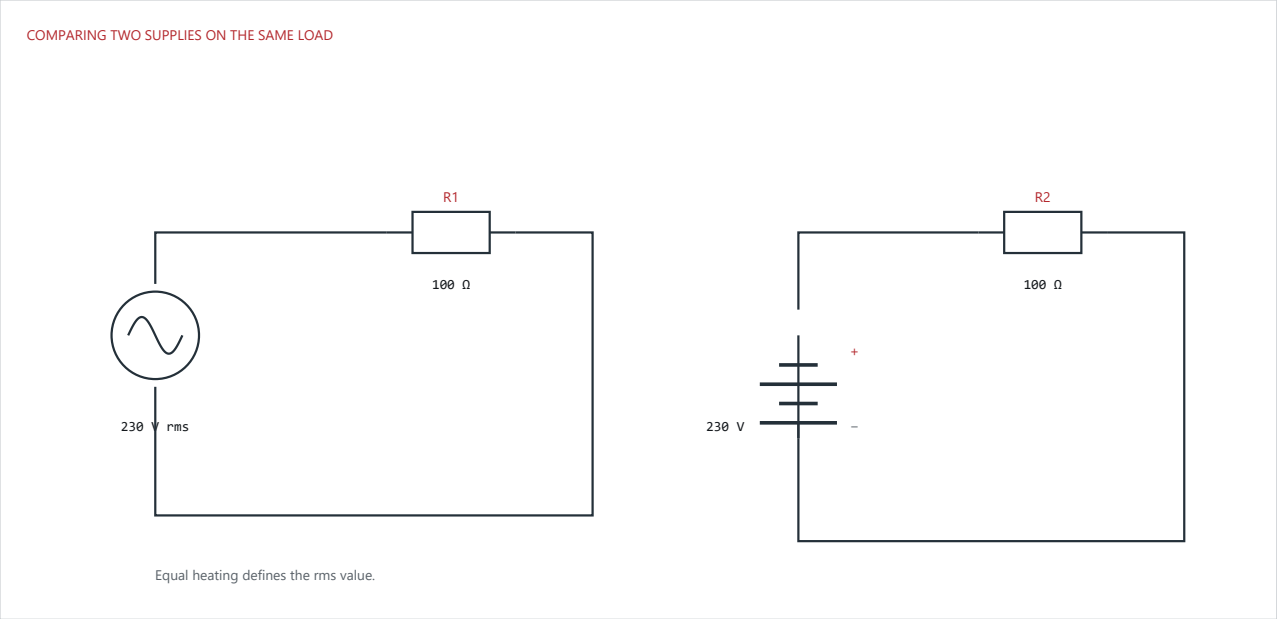



Fig. 6.2 Peak, peak-to-peak and root-mean-square values of a sinusoid. Only the root-mean-square value predicts heating in a resistance.

## Why the root-mean-square value is the useful one

Instantaneous power in a resistance is the square of the instantaneous voltage divided by the resistance. Averaging that over a cycle requires the mean of the squared voltage, and taking the square root of that mean gives a voltage that can be used in the familiar DC power formulas. That is precisely the definition of root-mean-square.

The practical consequence is that all the DC power relationships work unchanged on alternating quantities provided every value used is a root-mean-square value. Mixing a peak voltage with a root-mean-square current is one of the most common errors in AC calculation.

$$P = V_{rms}\, I_{rms} \quad \text{(resistive load)}$$

$$P = V_{rms}^2 / R = I_{rms}^2 R$$

THE RULE THAT PREVENTS MOST AC ARITHMETIC ERRORS

Convert everything to root-mean-square before any power calculation.
Use peak values only for insulation stress and semiconductor voltage ratings.
Never multiply a peak voltage by a root-mean-square current.

## What your meter actually measures

An averaging meter rectifies the waveform, measures the average and multiplies by 1.11 to display a root-mean-square value. This is correct only for an undistorted sinusoid. On a distorted waveform, such as the current drawn by electronic loads, the same meter can read substantially low.

A true root-mean-square meter squares, averages and takes the root of the actual waveform, so it is correct regardless of shape, within its crest factor limit. On modern installations with electronic loads a true root-mean-square instrument is not a refinement but a requirement.

| Meter type | Reads correctly on | Error on distorted current | When to use |
|---|---|---|---|
| Averaging, rms calibrated | Clean sinusoids | Typically 10–40 % low | Simple linear loads |
| True rms | Any waveform within crest factor | Small | Electronic loads, drives, lighting |
| Peak reading | Peak value directly | Not applicable | Insulation and device ratings |
| Clamp, averaging | Clean sinusoidal current | Can be large | Motor circuits only |

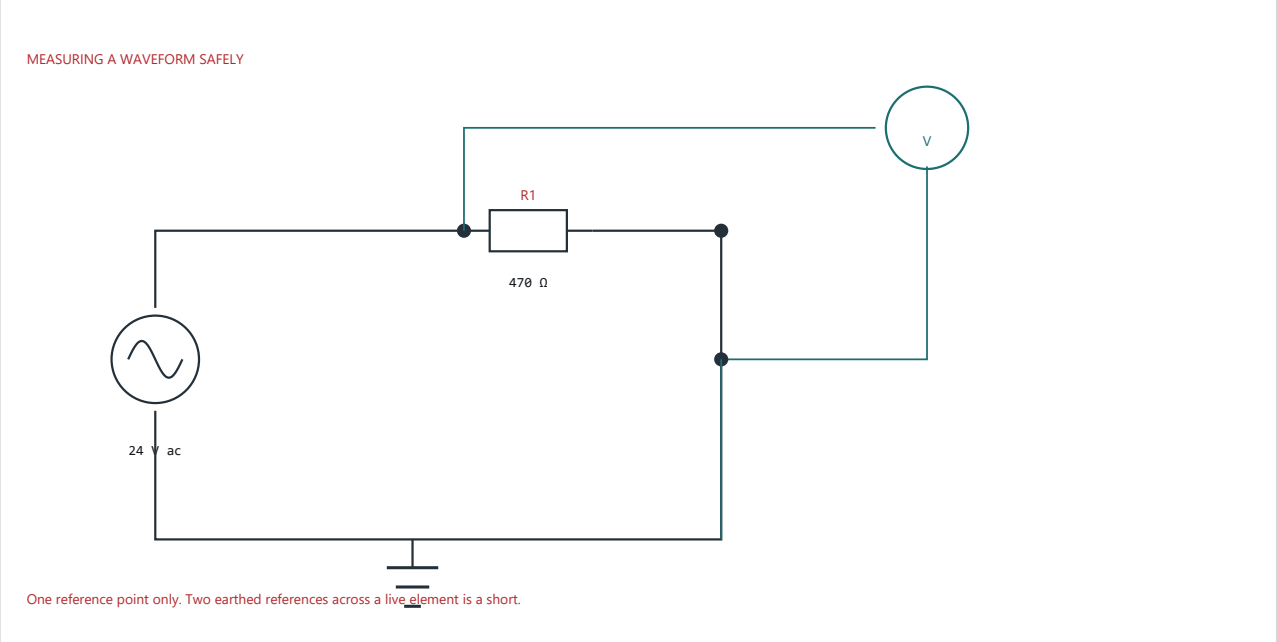



Fig. 6.3 Reading a waveform on an oscilloscope. Period is read horizontally and amplitude vertically, and the frequency follows from the period.

## Reading an oscilloscope

An oscilloscope displays instantaneous voltage against time, so it shows the shape that a meter reduces to a single number. Read the period by counting horizontal divisions for one complete cycle and multiplying by the time base setting, then invert for frequency. Read amplitude by counting vertical divisions and multiplying by the volts-per-division setting.

The oscilloscope is the only instrument that reveals waveform distortion, clipping, notching and transients. When a meter reading is inconsistent with expectation, looking at the waveform usually explains it immediately.

1. Set the coupling to DC unless you specifically want to remove the offset.
2. Adjust the time base so two or three cycles fill the screen.
3. Count divisions for one full cycle and multiply by the time base to get the period.
4. Invert the period to obtain the frequency.
5. Count peak-to-peak divisions and multiply by the volts per division.
6. Divide the peak-to-peak value by 2.828 to obtain the root-mean-square value of a sinusoid.

## Worked examples

Work each example with a pencil before reading the solution. Sketch the circuit, mark the reference node, and write the governing law before substituting numbers.

EXAMPLE 6.1 Peak and root-mean-square conversion

**Given.** An oscilloscope shows a sinusoid of 340 V peak to peak at a period of 16.67 ms.

**Find.** The peak value, the root-mean-square value and the frequency.

Peak from peak to peak

```
V_peak = 340/2 = 170 V
```

Root-mean-square value

```
V_rms = 170/1.414 = 120.2 V
```

Frequency from period

```
f = 1/0.01667 = 60 Hz
```

**Check.** 120 V at 60 Hz is a standard North American supply, which confirms the reading. Note that insulation must withstand 170 V peak, not 120 V.

EXAMPLE 6.2 Power in a resistive AC load

**Given.** A 2.4 kΩ heating element is supplied at 230 V rms.

**Find.** The current, the power and the peak voltage the element insulation sees.

Current from Ohm's law in rms terms

```
I = 230/2400 = 95.8 mA
```

Power

```
P = V I = 230 × 0.0958 = 22.0 W
```

Check by the squared form

```
P = 230^2/2400 = 22.0 W
```

Peak voltage

```
V_peak = 230 × 1.414 = 325 V
```

**Check.** Both power routes agree because every value used was root-mean-square. The 325 V peak is the figure that matters for insulation rating.

EXAMPLE 6.3 An averaging meter on a distorted current

**Given.** An electronic load draws a distorted current. A true rms clamp reads 6.4 A; an averaging clamp reads 4.9 A on the same conductor.

**Find.** The error of the averaging meter and the consequence for a 230 V supply calculation.

Absolute difference

$$6.4 - 4.9 = 1.5 \text{ A}$$

Error relative to the true value

$$1.5/6.4 \times 100 = 23\ \%\ \text{low}$$

Apparent power from the true reading

$$S = 230 \times 6.4 = 1472 \text{ VA}$$

Apparent power from the averaging reading

$$S = 230 \times 4.9 = 1127 \text{ VA}$$

**Check.** The averaging meter understates the loading by about 345 VA, which could lead to an undersized conductor or an overloaded circuit being declared acceptable.

# Practical activity: measuring peak and root-mean-square values of an alternating supply

> AIM
> Compare oscilloscope-derived peak values with meter-displayed root-mean-square values on both a clean sinusoid and a deliberately distorted waveform, and quantify the difference between averaging and true rms instruments.

## Equipment

- Signal generator capable of sine, square and triangle output at 50–1000 Hz.
- Oscilloscope with calibrated probes.
- One averaging multimeter and one true rms multimeter.
- Load resistor, 1 kΩ 1 W, and a breadboard.
- Optionally a low-voltage transformer output for a mains-frequency sinusoid.

## Procedure

1. Set the generator to a 1 V rms sinusoid at 100 Hz into the 1 kΩ load.
2. Measure the peak-to-peak voltage on the oscilloscope and compute the root-mean-square value.
3. Record both meter readings and compare all three figures.
4. Measure the period on the oscilloscope and compute the frequency; compare with the generator setting.
5. Switch the generator to a square wave of the same peak amplitude and repeat all readings.
6. Switch to a triangle wave and repeat again.
7. Tabulate the ratio of each meter reading to the oscilloscope-derived true value for each shape.
8. Return the generator to a low amplitude before disconnecting.

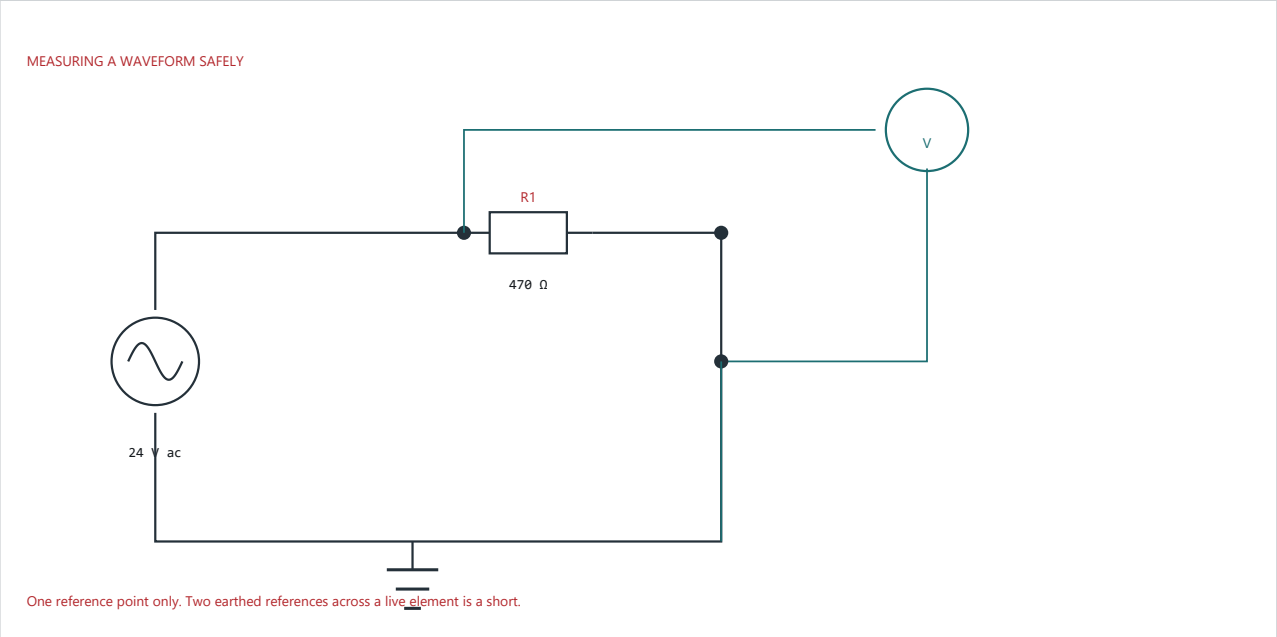



Fig. 6.L The measurement setup. The oscilloscope and both meters observe the same waveform across the same load.

## Results table

| Waveform | Scope V p-p | True rms computed | Averaging meter | True rms meter |
|---|---|---|---|---|
| Sine 100 Hz | | | | |
| Square 100 Hz | | | | |
| Triangle 100 Hz | | | | |
| Sine 1 kHz | | | | |
| Square 1 kHz | | | | |
| Triangle 1 kHz | | | | |

### Discussion

1. For which waveform did the averaging meter agree best with the computed value? Explain why.
2. For a square wave the true root-mean-square value equals the peak. Did your true rms meter confirm this?
3. Compute the form factor of each waveform from your data and compare with the theoretical values of 1.11 for sine and 1.00 for square.
4. At 1 kHz, did either meter's accuracy change? Check both meters' stated bandwidth.
5. A distorted load current reads 23 % low on an averaging clamp. Describe one installation decision this error could invalidate.

> **SAFE-WORK REMINDERS**
> Keep all signal levels below 10 V; this activity requires no hazardous voltages.
> If a mains-derived transformer output is used, treat the primary side as live and do not probe it.
> Check the oscilloscope probe attenuation setting before trusting an amplitude reading.
> Confirm both meters are set to AC volts, not DC, before recording.

## Troubleshooting guide

Faults are found by narrowing the circuit, not by replacing parts. Each row below pairs an observed symptom with the measurement that separates the likely causes.

| Symptom | Likely cause | Decisive test | Correct action |
|---|---|---|---|
| Meter reads low on an electronic load | Averaging meter used on a distorted current | Compare against a true rms instrument on the same conductor | Use a true rms meter and re-record the loading |
| Oscilloscope amplitude is ten times wrong | Probe attenuation setting does not match the instrument setting | Check the probe switch against the channel menu | Set both to the same attenuation and re-measure |
| Waveform shows flattened peaks | Supply clipping, or the instrument input is overdriven | Reduce the vertical sensitivity and re-observe | Investigate the supply; reduce the probe loading |
| Frequency reads half or double the expected value | Triggering on the wrong edge or counting half a cycle | Count a full cycle between identical points on the trace | Re-measure the period between two matching zero crossings |
| AC voltage reading drifts while the load is steady | Meter on the wrong range, or supply voltage genuinely fluctuating | Log readings over a minute and compare with another meter | Identify whether the instrument or the supply is varying |
| Calculated power far exceeds the measured value | Peak and root-mean-square values mixed in the calculation | Recheck every value used for which average it represents | Convert all quantities to root-mean-square and recalculate |

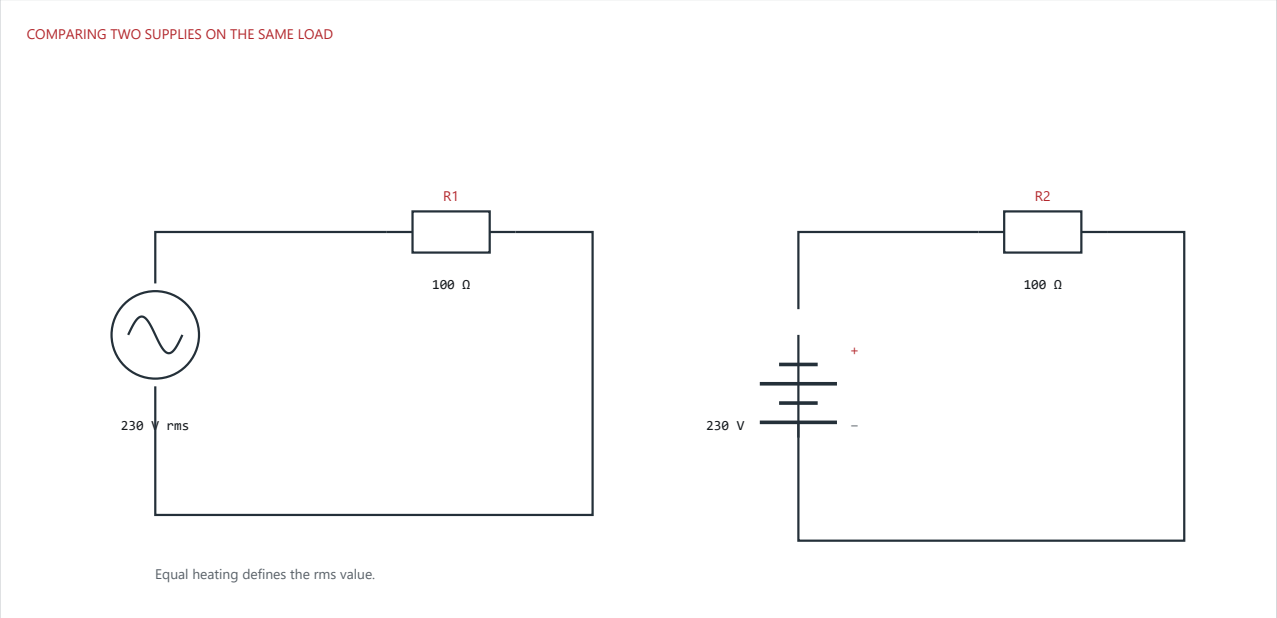



Fig. 6.T Distinguishing the averages. Peak matters for insulation, root-mean-square matters for heating, and the rectified average matters only to the meter's internal circuit.

FIELD HABITS THAT PREVENT CALLBACKS

Write which average a number is whenever you record it; “170 V” is ambiguous and “170 V peak” is a measurement.

Use a true rms instrument on any circuit containing electronic loads, drives or modern lighting.

Check probe attenuation before believing an oscilloscope amplitude.

Look at the waveform whenever a meter reading contradicts your prediction; the shape usually explains it.

## Exercises

Answer from first principles. Selected answers follow; the reasoning matters more than the number.

1. A sinusoid has a peak of 170 V. Find its root-mean-square and peak-to-peak values.
2. A 230 V rms supply feeds a 46 Ω element. Find the current and the power.
3. Find the period of a 50 Hz supply and of a 400 Hz aircraft supply.
4. An oscilloscope shows 8 divisions peak to peak at 50 V per division. Find the root-mean-square value for a sinusoid.
5. A waveform occupies 5 divisions per cycle at 2 ms per division. Find the frequency.
6. Explain why the average of a symmetrical sinusoid over a full cycle is zero.
7. A square wave has a peak of 10 V. State its root-mean-square value and explain the difference from a sinusoid.
8. An averaging meter reads 4.2 A where the true value is 5.5 A. Find the percentage error.
9. Find the peak voltage that a 480 V rms system's insulation must withstand.
10. Explain why mixing a peak voltage with a root-mean-square current gives a power figure that is too high, and by what factor for a sinusoid.

Selected answers

| # | Answer | Reason |
|---|---|---|
| 1 | 120.2 V rms; 340 V peak to peak | Divide by 1.414; double the peak |
| 2 | 5.0 A; 1150 W | I = V/R, then P = V I, all in rms terms |
| 3 | 20 ms; 2.5 ms | T = 1/f |
| 4 | 141.4 V rms | 400 V peak to peak, so 200 V peak, divided by 1.414 |
| 5 | 100 Hz | Period is 10 ms, so f = 1/0.01 |
| 6 | Equal positive and negative halves | They cancel exactly over a complete cycle |
| 7 | 10 V rms | The magnitude is constant, so the mean of the square equals the square of the peak |
| 8 | 23.6 % low | 1.3/5.5 expressed as a percentage |
| 9 | 679 V peak | 480 × 1.414 |
| 10 | Too high by a factor of 1.414 | The peak exceeds the root-mean-square value by √2 |

## Chapter summary

- A sinusoid needs amplitude, frequency and phase to be fully described, and phase is always relative to a reference.
- The root-mean-square value is the equivalent direct value for heating, and equals the peak divided by $\sqrt{2}$ for a sinusoid.
- All the DC power relationships hold on AC provided every quantity used is root-mean-square.
- Peak values govern insulation stress and semiconductor ratings; root-mean-square values govern heating and conductor sizing.
- Averaging meters are calibrated for clean sinusoids and read low on distorted waveforms.
- An oscilloscope is the only instrument that shows waveform shape, which is often the explanation for a surprising meter reading.

### Equations introduced in this chapter

| Relationship | Expression | When it applies |
|---|---|---|
| Instantaneous value | $v = V_p \sin(\omega t + \varphi)$ | Sinusoidal quantity at time t |
| Period and frequency | $T = 1/f$ | Any periodic waveform |
| Angular frequency | $\omega = 2\pi f$ | Radians per second |
| Root-mean-square | $V_{rms} = V_p/\sqrt{2}$ | Sinusoids only |
| Rectified average | $V_{avg} = 0.637\, V_p$ | Sinusoids only |
| Resistive AC power | $P = V_{rms} I_{rms}$ | Purely resistive load |

### Vocabulary check

| Term | Meaning in this chapter |
|---|---|
| Root-mean-square | The equivalent direct value that produces the same heating |
| Crest factor | The ratio of peak to root-mean-square value, 1.414 for a sinusoid |
| Form factor | The ratio of root-mean-square to rectified average, 1.11 for a sinusoid |
| True rms meter | An instrument that computes the root-mean-square of the actual waveform |
| Phase | The displacement of one waveform relative to another of the same frequency |

LOOKING AHEAD

You can now describe and measure an alternating quantity. The next chapter adds reactive elements to alternating circuits, where current no longer peaks with voltage and the concept of impedance replaces simple resistance.

CHAPTER 07

# Impedance, phase and power factor

Reactance, the impedance triangle, and why apparent power exceeds real power

WHY THIS CHAPTER MATTERS

Add a capacitor or an inductor to an alternating circuit and the current no longer peaks with the voltage. The opposition is then impedance rather than resistance, and the product of voltage and current is no longer the power delivered. Power factor is the bookkeeping that reconciles them.

## What you will be able to do

- Compute inductive and capacitive reactance at a stated frequency.
- Combine resistance and reactance into an impedance magnitude and phase angle.
- Distinguish real, reactive and apparent power and state the unit of each.
- Compute power factor and explain its effect on conductor and supply loading.
- Select a correction capacitor to raise the power factor of an inductive load.

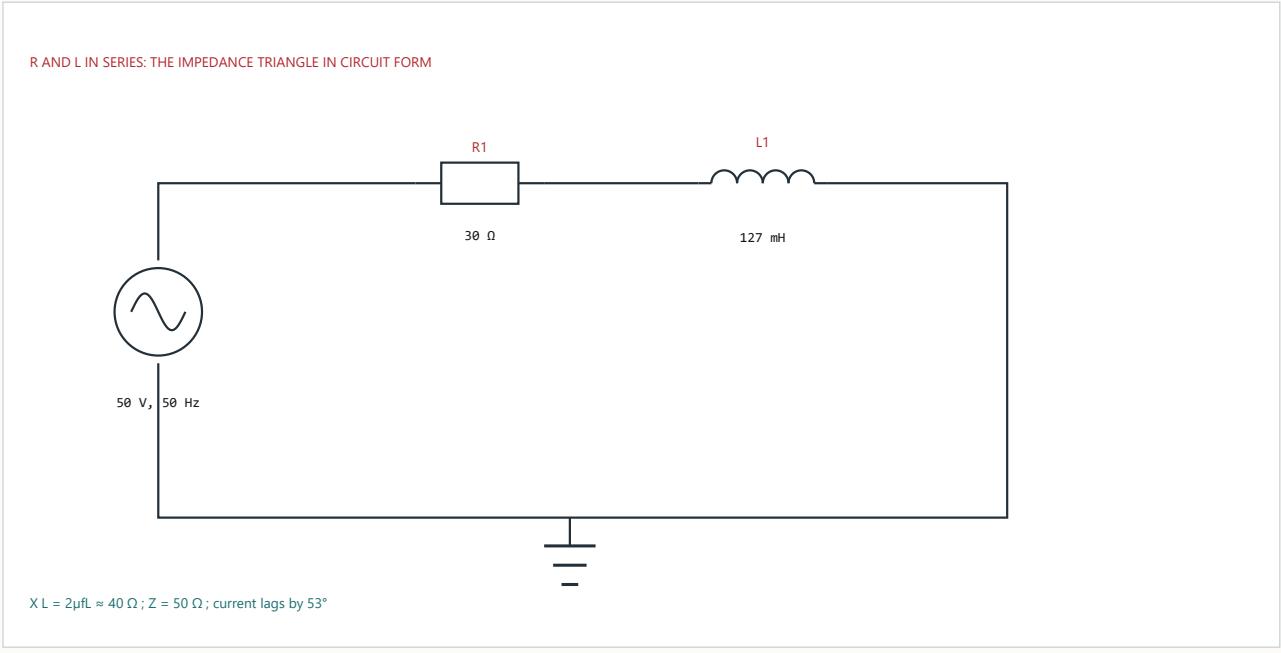



Fig. 7.1 A resistive-inductive series circuit. The current is common to both elements, but the voltages across them are not in phase with each other.

## Reactance: frequency-dependent opposition

An inductor's opposition rises with frequency, because a faster rate of change produces a larger induced voltage. A capacitor's opposition falls with frequency, because a faster change moves more charge per second. Both are measured in ohms and called reactance, but neither dissipates energy: reactance stores and returns it each cycle.

Because reactance depends on frequency, the same component behaves differently on a 50 Hz supply, a 60 Hz supply and the harmonic frequencies that electronic loads inject. A capacitor sized for correction at the fundamental presents a much lower reactance to the fifth harmonic, which is why correction equipment is sometimes detuned.

```
X_L = 2π f L          X_C = 1 / (2π f C)

Both in ohms; neither dissipates power
```

| Element | Reactance | As frequency rises | Current relative to voltage |
|---|---|---|---|
| Resistance | R, constant | Unchanged | In phase |
| Inductance | $X_L = 2\pi fL$ | Increases | Lags by 90 ° |
| Capacitance | $X_C = 1/(2\pi fC)$ | Decreases | Leads by 90 ° |

## Impedance as a magnitude and an angle

Resistance and reactance do not add arithmetically because they act 90 degrees apart. They combine as the sides of a right triangle whose hypotenuse is the impedance magnitude and whose angle is the phase difference between voltage and current. This is the impedance triangle, and it is the single most useful picture in alternating current work.

The same triangle, scaled by the current, gives the voltage triangle, and scaled by the current squared, gives the power triangle. Learning one shape therefore gives you three sets of relationships at no extra cost.

```
Z = √(R² + X²)

ϕ = arctan(X / R)

I = V / Z
```

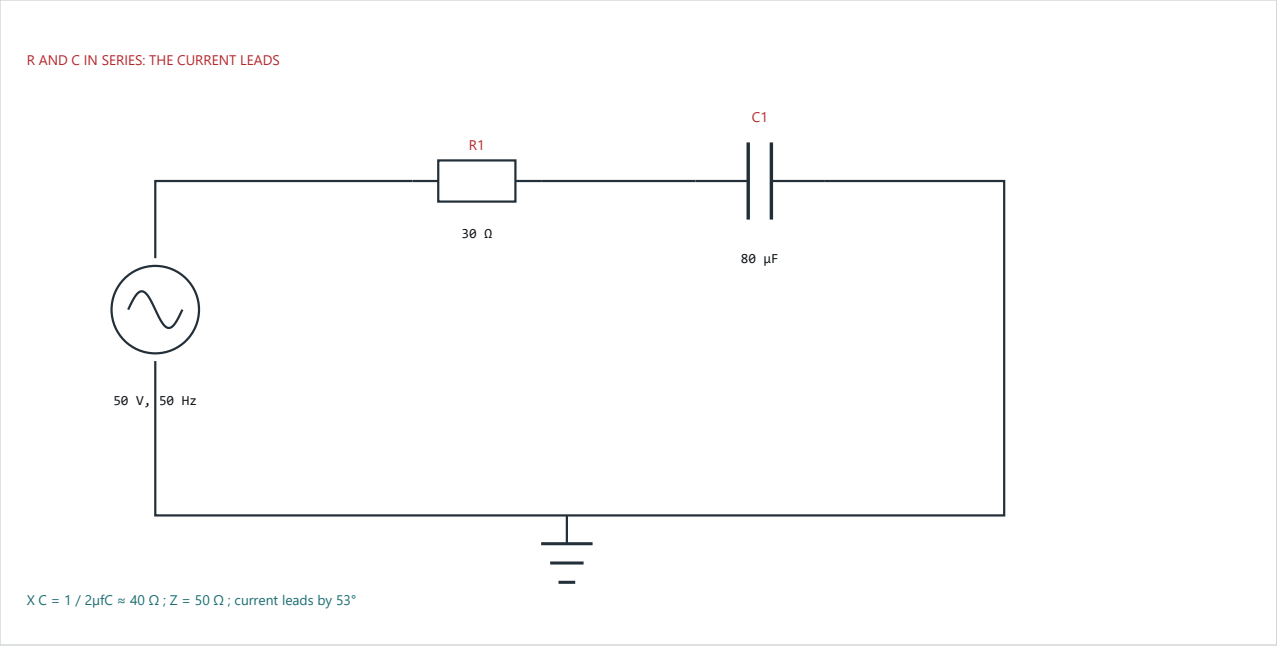



Fig. 7.2 A resistive-capacitive series circuit. The capacitor voltage lags the resistor voltage by 90 degrees, so the two add vectorially, not arithmetically.

## Real, reactive and apparent power

Real power, in watts, is the energy actually converted to heat, light or mechanical work. Reactive power, in reactive volt-amperes, is energy borrowed and returned each cycle by reactive elements; it does no useful work but it does occupy conductor capacity. Apparent power, in volt-amperes, is the simple product of voltage and current and is what the supply must actually deliver.

The three form a right triangle in the same proportions as the impedance triangle. Power factor is the ratio of real to apparent power, equal to the cosine of the phase angle for a sinusoidal supply. A load at 0.7 power factor draws about 43 per cent more current than a unity-power-factor load doing the same work.

```
P = V I cosϕ   (watts, W)

Q = V I sinϕ   (reactive volt-amperes, var)

S = V I         (volt-amperes, VA)

S = √(P² + Q²)      pf = P / S = cosϕ
```

WHY A POOR POWER FACTOR COSTS MONEY EVEN THOUGH IT DOES NO WORK

Conductors and transformers are sized by current, which is set by apparent power. At 0.7 power factor, delivering 7 kW needs 10 kVA of supply capacity. The extra current produces real $I^2R$ losses in the distribution system.

## Correcting the power factor

An inductive load draws lagging reactive power. A capacitor draws leading reactive power, so a correctly sized capacitor connected in parallel supplies the reactive component locally and the supply current falls.

The real power is unchanged, because correction does not reduce the work being done.

Correction is sized from the reactive power to be cancelled, not from the kilowatts. Overcorrection should be avoided: it produces a leading power factor, can raise the voltage at light load and can resonate with system inductance at harmonic frequencies.

1. Determine the real power P and the existing power factor, giving $\varphi_1$.
2. Compute the existing reactive power $Q_1 = P \tan\varphi_1$.
3. Choose the target power factor, giving $\varphi_2$, and compute $Q_2 = P \tan\varphi_2$.
4. The capacitor must supply $Q_c = Q_1 - Q_2$.
5. Convert to capacitance using $C = Q_c / (2\pi f V^2)$.
6. Verify the resulting current and confirm the correction is not excessive at light load.

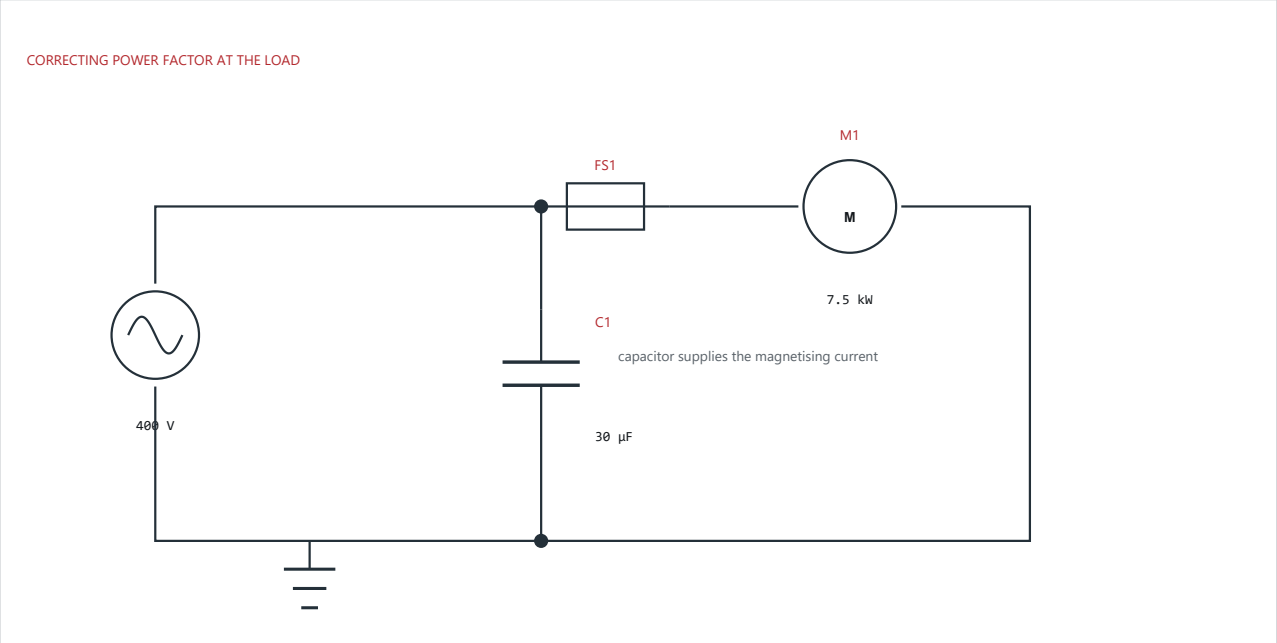



Fig. 7.3 Parallel capacitive correction of an inductive load. The capacitor supplies the reactive current locally, so the supply conductor carries less.

## Series resistance-inductance-capacitance behaviour

When all three elements appear in series the two reactances oppose each other, so the net reactance is their difference. Below a particular frequency the capacitive reactance dominates and the circuit appears capacitive; above it the inductive reactance dominates. At the frequency where they are equal the reactances cancel entirely.

At that frequency the impedance reduces to the resistance alone, the current reaches its maximum and the voltage and current are in phase. This is series resonance, and it is developed fully in the next chapter.

$$X_{net} = X_L - X_C$$

$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$

## Worked examples

Work each example with a pencil before reading the solution. Sketch the circuit, mark the reference node, and write the governing law before substituting numbers.

EXAMPLE 7.1 Impedance of a series resistive-inductive circuit

**Given.** A coil of 0.15 H and 40 Ω winding resistance is supplied at 230 V, 50 Hz.

**Find.** The reactance, impedance, current, phase angle and power factor.

Inductive reactance

$$X_L = 2\pi \times 50 \times 0.15 = 47.1\ \Omega$$

Impedance magnitude

$$Z = \sqrt{(40^2 + 47.1^2)} = 61.8\ \Omega$$

Current

$$I = 230/61.8 = 3.72\ A$$

Phase angle

$$\phi = \arctan(47.1/40) = 49.7^\circ \text{ lagging}$$

Power factor

$$\cos 49.7^\circ = 0.647 \text{ lagging}$$

Real power

$$P = 230 \times 3.72 \times 0.647 = 553\ W$$

**Check.** Check through the resistance: $P = I^2R = 3.72^2 \times 40 = 553$ W. The two agree, which confirms that only the resistance dissipates power.

EXAMPLE 7.2 Apparent, real and reactive power

**Given.** A motor draws 12 A at 400 V single phase with a power factor of 0.82 lagging.

**Find.** The apparent, real and reactive power.

Apparent power

$$S = V\,I = 400 \times 12 = 4800\ VA$$

Real power

$$P = S\cos\phi = 4800 \times 0.82 = 3936\ W$$

Phase angle

$$\phi = \arccos 0.82 = 34.9^\circ$$

Reactive power

$$Q = S\sin\phi = 4800 \times 0.572 = 2746\ var$$

Verify the triangle

$$\sqrt{(3936^2 + 2746^2)} = 4799\ VA$$

**Check.** The supply must carry 4800 VA to deliver 3936 W of useful power. The 2746 var is borrowed and returned each cycle but still occupies conductor capacity.

EXAMPLE 7.3 Sizing a correction capacitor

**Given.** The motor above is to be corrected from 0.82 to 0.95 lagging at 400 V, 50 Hz.

**Find.** The required reactive compensation, the capacitance and the new supply current.

Existing reactive power

$Q_1 = 2746 \text{ var, as calculated}$

Target angle

$\phi_2 = \arccos 0.95 = 18.2\ ^\circ$

Target reactive power

$Q_2 = 3936 \times \tan 18.2^\circ = 1294 \text{ var}$

Capacitor reactive power

$Q_c = 2746 - 1294 = 1452 \text{ var}$

Capacitance

$C = 1452 / (2\pi \times 50 \times 400^2) = 28.9\ \mu F$

New current

$S_2 = 3936/0.95 = 4143 \text{ VA, so } I = 4143/400 = 10.36 \text{ A}$

**Check.** The current falls from 12 A to 10.36 A, a 13.7 % reduction, while the real power stays at 3936 W. The conductor and the supply transformer both see less loading.

# Practical activity: impedance and power factor of a resistive-inductive load

AIM

Measure the voltage, current and power of an inductive load, compute its power factor from the measurements, then demonstrate that a parallel capacitor reduces the supply current without changing the real power.

## Equipment

- Low-voltage AC supply, 24 V, from an isolating transformer, fused at 2 A.
- Inductor or transformer winding of roughly 0.1 H with measurable resistance.
- Resistor, 22 Ω 10 W.
- Capacitors, 10 µF, 22 µF and 47 µF, AC rated for at least 100 V.
- True rms multimeter, clamp meter or series ammeter, and a wattmeter if available.

## Procedure

1. Measure the inductor's DC resistance and record it; this is the resistive part of the coil.
2. Connect the coil in series with the 22 Ω resistor across the 24 V AC supply.
3. Measure the supply voltage, the circuit current and the voltage across each element.
4. Note that the two element voltages do not add to the supply voltage arithmetically; record all three.
5. Compute the impedance from V over I, then the phase angle from the measured resistance.
6. Compute the real power as $I^2$ times the total resistance, and the apparent power as V times I.
7. Connect the 10 µF capacitor in parallel with the whole load and re-measure the supply current.
8. Repeat with 22 µF and 47 µF, recording the supply current each time.
9. Identify which capacitor gave the lowest supply current and whether any value overcorrected.

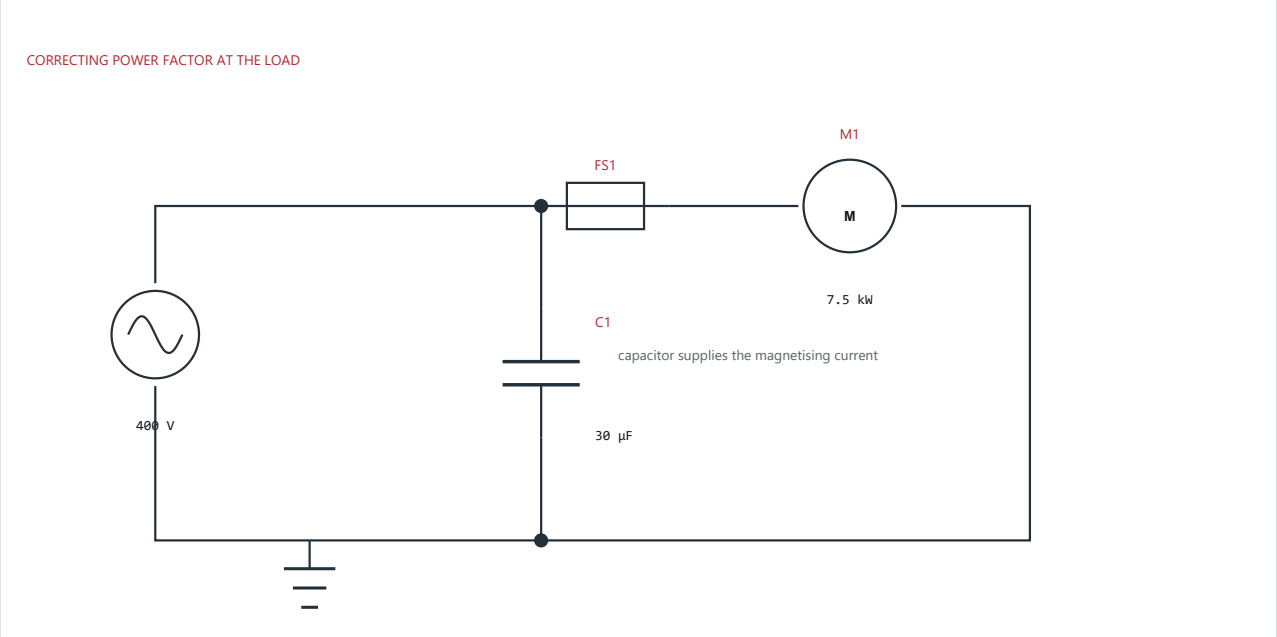



Fig. 7.L The test arrangement. The capacitor is connected across the supply terminals of the load, and the ammeter reads the total supply current.

### Results table

| Capacitor | Supply V | Supply I (A) | S = VI (VA) | Computed pf |
|---|---|---|---|---|
| none | 24.0 | | | |
| 10 µF | 24.0 | | | |
| 22 µF | 24.0 | | | |
| 47 µF | 24.0 | | | |
| repeat none | 24.0 | | | |
| best value | 24.0 | | | |

### Discussion

1. Did the two element voltages add to the supply voltage arithmetically? Explain using the voltage triangle.
2. Which capacitor produced the minimum supply current? Compare it with the value you would calculate for unity power factor.
3. Did the real power change as capacitors were added? Explain why it should not.
4. At the largest capacitor, was the current rising again? If so, explain overcorrection.
5. Compute the percentage current reduction achieved at your best value and state what that would mean for conductor sizing.

> SAFE-WORK REMINDERS
> Use an isolating transformer and keep the supply at 24 V; do not perform this activity on mains voltage.
> Capacitors must be AC rated. DC-rated or electrolytic capacitors must not be used across an AC supply.
> Discharge capacitors before handling and verify zero volts.
> Check the resistor and inductor power ratings against the measured current before energising.

## Troubleshooting guide

Faults are found by narrowing the circuit, not by replacing parts. Each row below pairs an observed symptom with the measurement that separates the likely causes.

| Symptom | Likely cause | Decisive test | Correct action |
|---|---|---|---|
| Element voltages do not add to the supply voltage | Normal: they are 90 ° apart and add vectorially | Compute $\sqrt{(V_R^2 + V_X^2)}$ and compare with the supply | No fault; record the measurement and use the triangle |
| Supply current higher than the real power suggests | Poor power factor from an inductive load | Measure watts and volt-amperes separately | Apply parallel capacitive correction sized to the reactive power |
| Current rises after adding more correction capacitance | Overcorrection, now a leading power factor | Compare supply current across several capacitor values | Reduce the capacitance to the minimum-current value |
| Correction capacitor runs hot or fails early | Harmonic current flowing into a low capacitive reactance | Measure the capacitor current and look at its waveform | Fit a detuning reactor or a harmonic-rated capacitor assembly |
| Measured power factor differs from the nameplate figure | Load not at rated output; power factor varies with loading | Measure at several load points | Use the power factor at the actual operating load for correction sizing |
| Voltage rises at light load after correction was installed | Fixed capacitors remain connected when the inductive load falls away | Log voltage against load over a cycle of operation | Use switched or automatically staged correction |

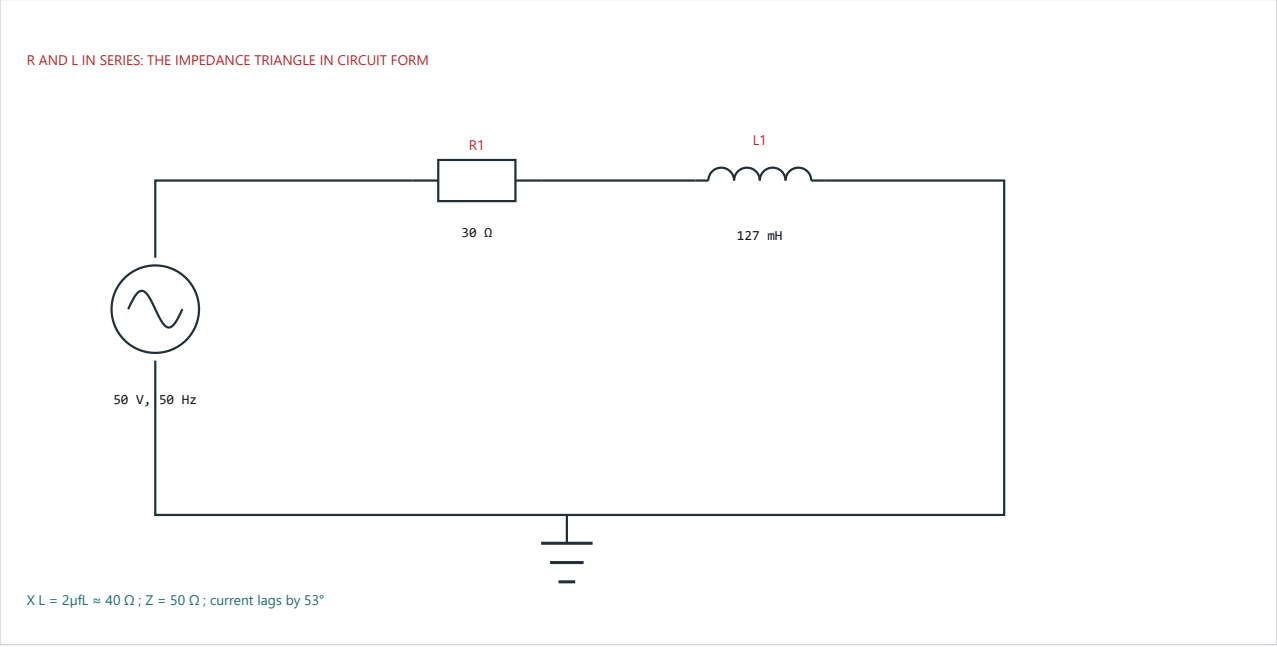



Fig. 7.T Separating resistance from reactance by measurement. The DC resistance and the AC impedance together give the reactance and the phase angle.

FIELD HABITS THAT PREVENT CALLBACKS

Measure watts and volt-amperes separately; a current reading alone cannot reveal power factor.
Record the load point with any power factor figure, because power factor varies with loading.
Size correction from the reactive power to be cancelled, never from the kilowatts.
Expect reactive element voltages to exceed what arithmetic suggests; the triangle, not addition, governs.

## Exercises

Answer from first principles. Selected answers follow; the reasoning matters more than the number.

1. Find the reactance of a 0.2 H inductor at 50 Hz and at 60 Hz.
2. Find the reactance of a 10 µF capacitor at 50 Hz.
3. A circuit has 30 Ω resistance and 40 Ω inductive reactance. Find the impedance and the phase angle.
4. The circuit above is supplied at 120 V. Find the current, the real power and the apparent power.
5. A load draws 8 A at 230 V with a power factor of 0.75. Find the real, reactive and apparent power.
6. Explain why reactive power occupies conductor capacity despite doing no work.
7. A series circuit has 25 Ω resistance, 60 Ω inductive and 20 Ω capacitive reactance. Find the impedance.
8. A 5 kW load at 0.8 power factor is to be corrected to 0.95 at 400 V, 50 Hz. Find the required capacitance.
9. State what happens to the supply current if correction capacitance is doubled beyond the unity-power-factor value.
10. Explain why a capacitor sized for 50 Hz correction presents a much lower reactance to the fifth harmonic, and why that matters.

Selected answers

| # | Answer | Reason |
|---|---|---|
| 1 | 62.8 Ω at 50 Hz; 75.4 Ω at 60 Hz | $X_L = 2\pi fL$ |
| 2 | 318 Ω | $X_C = 1/(2\pi fC)$ |
| 3 | 50 Ω; 53.1 ° lagging | $Z = \sqrt{(30^2+40^2)}$; $\varphi = \arctan(40/30)$ |
| 4 | 2.4 A; 173 W; 288 VA | $I = V/Z$; $P = I^2R$; $S = V\,I$ |
| 5 | 1380 W; 1217 var; 1840 VA | $S = V\,I$, then $P = S\cos\varphi$ and $Q = S\sin\varphi$ |
| 6 | Current is set by apparent power | Conductors and transformers are rated by current, not by watts |
| 7 | 47.2 Ω | Net reactance 40 Ω, so $Z = \sqrt{(25^2+40^2)}$ |
| 8 | About 42 µF | $Q_c = 5000(\tan 36.9° - \tan 18.2°) = 2106$ var, then $C = Q_c/(2\pi fV^2)$ |
| 9 | It rises again, now leading | The circuit becomes capacitive and the reactive current returns, opposite in sign |
| 10 | $X_C$ falls by a factor of five | Harmonic currents concentrate in the capacitor, overheating it and risking resonance |

## Chapter summary

- Reactance is frequency-dependent opposition that stores and returns energy rather than dissipating it.
- Resistance and reactance combine as a right triangle, giving an impedance magnitude and a phase angle.
- Real power is in watts, reactive power in var and apparent power in volt-amperes, and they form the same triangle.
- Power factor is the ratio of real to apparent power, and a poor value raises current without raising useful output.
- Parallel capacitors correct a lagging power factor by supplying the reactive component locally.
- Correction must be sized from the reactive power and must not be excessive, or the circuit becomes leading.

### Equations introduced in this chapter

| Relationship | Expression | When it applies |
|---|---|---|
| Inductive reactance | $X_L = 2\pi fL$ | Sinusoidal supply at frequency f |
| Capacitive reactance | $X_C = 1/(2\pi fC)$ | Sinusoidal supply at frequency f |
| Impedance | $Z = \sqrt{(R^2+X^2)}$ | Series combination |
| Phase angle | $\varphi = \arctan(X/R)$ | Series combination |
| Real power | $P = V\,I\cos\varphi$ | Sinusoidal quantities |
| Reactive power | $Q = V\,I\sin\varphi$ | Sinusoidal quantities |
| Apparent power | $S = V\,I$ | Always the product of rms values |
| Correction | $C = Q_c/(2\pi fV^2)$ | Parallel capacitor at the load voltage |

### Vocabulary check

| Term | Meaning in this chapter |
|---|---|
| Reactance | Frequency-dependent opposition that stores rather than dissipates energy |
| Impedance | The total opposition to alternating current, with magnitude and phase |
| Power factor | The ratio of real to apparent power, $\cos\varphi$ for sinusoidal supplies |
| Reactive power | Energy exchanged each cycle with reactive elements, measured in var |
| Overcorrection | Adding more capacitance than needed, producing a leading power factor |

LOOKING AHEAD

Impedance varies with frequency, so there is generally a frequency at which a circuit's behaviour changes character. The next chapter examines resonance, where inductive and capacitive reactances cancel, and the filters built from that effect.

CHAPTER 08

# Resonance and filters

The frequency where reactances cancel, and the circuits built from that effect

WHY THIS CHAPTER MATTERS

At one particular frequency an inductive and a capacitive reactance are equal and cancel. That single fact produces the sharpest voltage rises you will ever measure in a passive circuit, and it is the basis of every tuned circuit and passive filter.

## What you will be able to do

- Compute the resonant frequency of a series or parallel combination.
- Describe what happens to impedance, current and element voltages at series resonance.
- Compute the quality factor and the bandwidth of a resonant circuit.
- Identify low-pass, high-pass, band-pass and band-stop arrangements from a schematic.
- Compute the cut-off frequency of a single-stage resistive-capacitive filter.

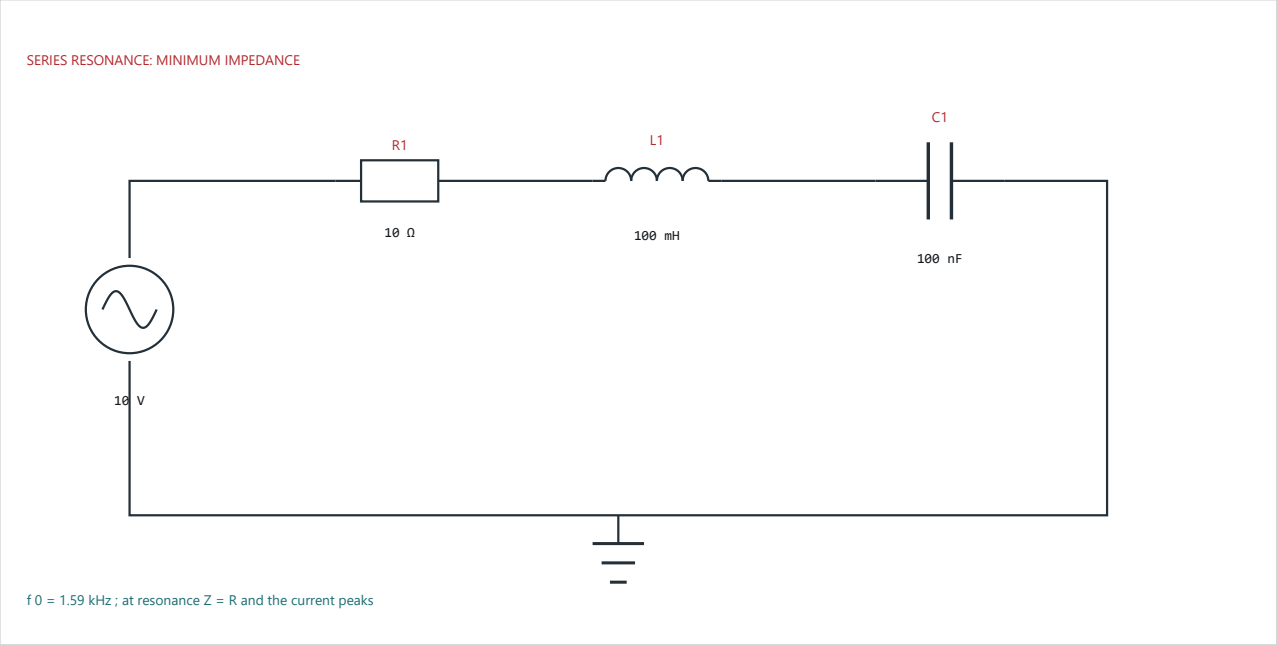



Fig. 8.1 A series resonant circuit. At resonance the inductive and capacitive reactances cancel, leaving only the resistance to limit the current.

## The resonant frequency

Inductive reactance rises with frequency while capacitive reactance falls, so there is exactly one frequency at which the two are equal. Setting them equal and solving gives the resonant frequency, which depends only on the inductance and the capacitance, not on the resistance present.

That independence from resistance is worth noting: resistance determines how sharp the resonance is and how large the element voltages become, but it does not move the resonant frequency itself.

```
At resonance:  X_L = X_C

2π f L = 1/(2π f C)

f_0 = 1 / (2π √(L C))
```

## Series resonance: minimum impedance, maximum current

In a series circuit the two reactances cancel at resonance, so the impedance falls to the resistance alone. This is the minimum impedance the circuit can present, so the current reaches its maximum and the voltage and current come into phase.

The element voltages at resonance can greatly exceed the supply voltage. Each reactance carries the resonant current and its full reactance, which may be many times the resistance. The two large voltages are in opposition and cancel as far as the supply is concerned, but each is individually real and will stress insulation and instruments.

```
Z = R      I = V / R      ϕ = 0

V_L = I X_L = Q V      V_C = I X_C = Q V
```

SERIES RESONANCE PRODUCES GENUINE VOLTAGE MAGNIFICATION

A circuit with Q = 20 supplied at 24 V develops 480 V across each reactive element. The two cancel at the supply terminals, so a supply-side meter shows nothing unusual. Measure across the individual elements before assuming a circuit is low voltage.

## Quality factor and bandwidth

The quality factor expresses how sharp a resonance is: it is the ratio of reactance at resonance to the series resistance. A high quality factor means a narrow response and large voltage magnification; a low one means a broad response and little magnification.

The bandwidth is the span of frequencies over which the response stays within the half-power points, and it equals the resonant frequency divided by the quality factor. A tuned circuit must therefore trade selectivity against bandwidth, which is the central compromise in filter design.

```
Q = X_L / R = (1/R)√(L/C)

BW = f_0 / Q

Half-power points at 0.707 of maximum
```

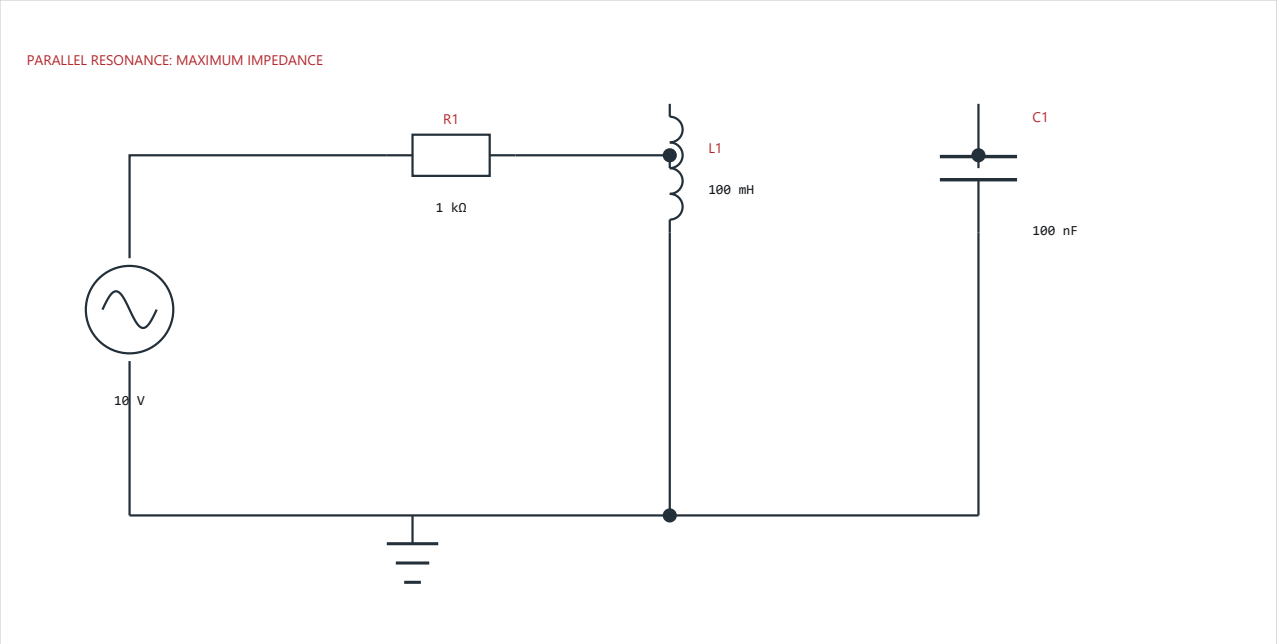



Fig. 8.2 A parallel resonant circuit. At resonance it presents maximum impedance, so it blocks rather than passes the resonant frequency.

## Parallel resonance: maximum impedance

A parallel inductor and capacitor behave in the opposite way. At resonance the two branch currents are equal and opposite, so they circulate between the elements and the supply sees almost no current. The impedance is therefore at a maximum.

This is why a parallel tuned circuit is used to reject a frequency rather than to pass it, and why the circulating current inside a parallel resonant loop can be far larger than the current drawn from the supply. In power systems an accidental parallel resonance between correction capacitors and system inductance can amplify a harmonic severely.

| Property | Series resonance | Parallel resonance |
|---|---|---|
| Impedance at $f_0$ | Minimum, equal to R | Maximum |
| Current from supply | Maximum | Minimum |
| Magnified quantity | Element voltages | Circulating branch currents |
| Typical use | Accept a frequency | Reject a frequency |
| Phase at $f_0$ | Zero | Zero |

## Passive filters

A filter is a circuit whose response depends deliberately on frequency. A single resistor and capacitor make a low-pass filter if the output is taken across the capacitor, and a high-pass filter if it is taken across the resistor. The cut-off frequency is where the reactance equals the resistance, at which point the output has fallen to 0.707 of the input and the power to half.

Combining a low-pass and a high-pass stage produces a band-pass response; a series or parallel resonant circuit does the same job more sharply. Band-stop or notch responses are built the same way with the elements interchanged.

### Identifying a filter from the schematic

- Output across the capacitor in a series resistor-capacitor pair: low pass.
- Output across the resistor in the same pair: high pass.
- Series resonant branch feeding the output: band pass.
- Parallel resonant branch across the output: band stop, or notch.
- Inductor in the series path with a capacitor to the reference: low pass with a steeper slope.

```
f_c = 1 / (2π R C)

At f_c the output is 0.707 of input, that is −3 dB
```

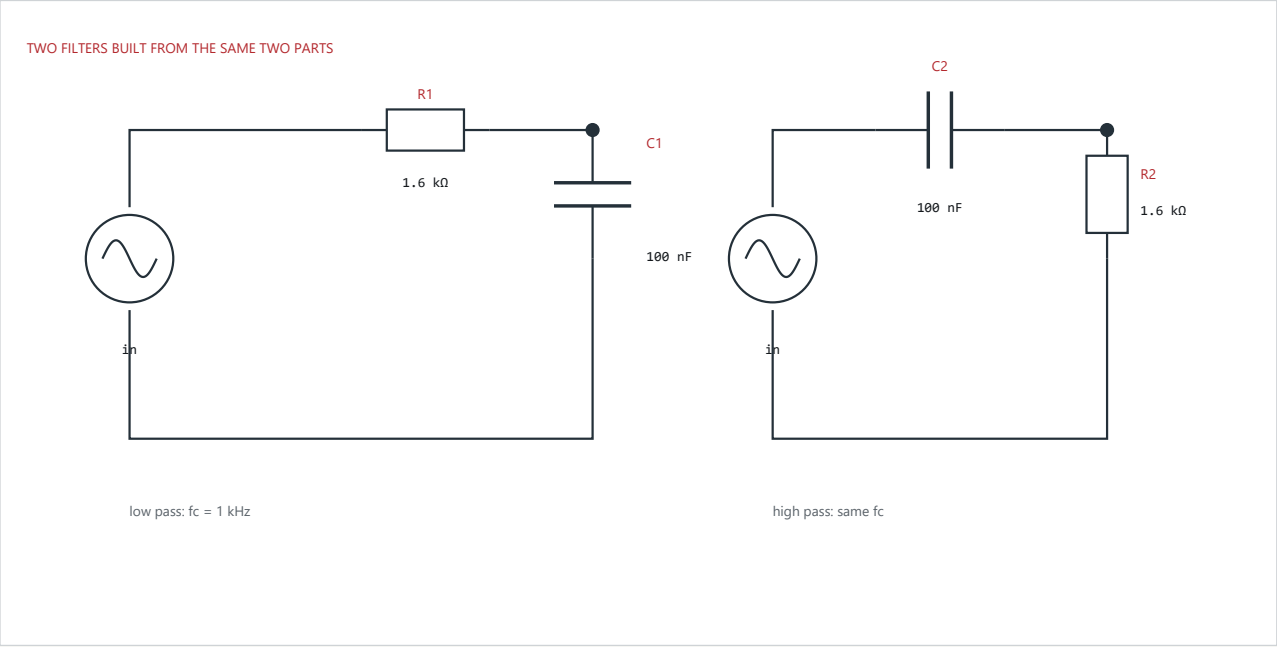



Fig. 8.3 Low-pass and high-pass arrangements from the same two components. The output terminals determine which response you get.

## Worked examples

Work each example with a pencil before reading the solution. Sketch the circuit, mark the reference node, and write the governing law before substituting numbers.

EXAMPLE 8.1 Resonant frequency and quality factor

**Given.** A series circuit has L = 100 mH, C = 100 nF and R = 50 Ω.

**Find.** The resonant frequency, the reactance at resonance, the quality factor and the bandwidth.

Resonant frequency

$$f_0 = 1/(2\pi\sqrt{(0.1 \times 100\times10^{-9})}) = 1592 \text{ Hz}$$

Reactance at resonance

$$X_L = 2\pi \times 1592 \times 0.1 = 1000\ \Omega$$

Check the capacitive reactance

$$X_C = 1/(2\pi \times 1592 \times 100\times10^{-9}) = 1000\ \Omega$$

Quality factor

$$Q = 1000/50 = 20$$

Bandwidth

$$BW = 1592/20 = 79.6 \text{ Hz}$$

**Check.** Both reactances come to 1000 Ω at the computed frequency, which confirms the resonant frequency. A quality factor of 20 is a sharp resonance.

EXAMPLE 8.2 Voltage magnification at series resonance

**Given.** The circuit above is supplied at 10 V at its resonant frequency.

**Find.** The current and the voltage across the inductor and the capacitor.

Impedance at resonance

$$Z = R = 50\ \Omega$$

Current

$$I = 10/50 = 0.2 \text{ A}$$

Inductor voltage

$$V_L = 0.2 \times 1000 = 200 \text{ V}$$

Capacitor voltage

$$V_C = 0.2 \times 1000 = 200 \text{ V}$$

Supply-side sum

$$200 - 200 = 0, \text{ leaving } 10 \text{ V across } R$$

**Check.** The 200 V across each element is twenty times the supply, equal to Q × V. A meter at the supply terminals would show only 10 V, which is why this hazard is easy to miss.

EXAMPLE 8.3 Cut-off frequency of a resistive-capacitive filter

**Given.** A filter uses R = 1.6 kΩ and C = 100 nF, with the output taken across the capacitor.

**Find.** The filter type, the cut-off frequency and the output at one tenth and ten times that frequency.

Filter type

Output across the capacitor, so low pass

Cut-off frequency

$f_c = 1/(2\pi \times 1600 \times 100\times10^{-9}) = 995$ Hz

At 99.5 Hz, a decade below

X_C  R, so output ≈ input, near 0 dB

At 9.95 kHz, a decade above

X_C  R, so output ≈ input/10, near −20 dB

At the cut-off itself

Output = 0.707 of input, that is −3 dB

**Check.** A single stage rolls off at 20 dB per decade, so the response a decade above cut-off is one tenth of the input. Taking the output across the resistor instead would give a high-pass response with the same cut-off frequency.

## Practical activity: locating resonance and plotting a filter response

AIM

Find the resonant frequency of a series combination by measurement, observe the voltage magnification across the reactive elements, then plot the response of a single-stage filter and locate its cut-off frequency from the data.

### Equipment

- Signal generator, sine output, 100 Hz to 100 kHz, with a stated output amplitude.
- Inductor, 100 mH, with its resistance measured.
- Capacitors, 100 nF and 10 nF, film type.
- Resistors, 50 Ω, 1.6 kΩ, 0.25 W.
- Oscilloscope and a true rms multimeter.

### Procedure

1. Measure and record the inductor's resistance and the actual component values.
2. Compute the expected resonant frequency before building the circuit.
3. Build the series combination of L, C and the 50 Ω resistor across the generator.
4. Set the generator to 2 V rms and keep that amplitude constant throughout the sweep.
5. Sweep the frequency and find the frequency at which the voltage across the resistor is maximum. That is resonance.
6. At resonance, measure the voltage across the inductor and across the capacitor separately.
7. Find the two frequencies at which the resistor voltage falls to 0.707 of its peak, and compute the bandwidth.
8. Rebuild as a low-pass filter with the 1.6 kΩ resistor and the 100 nF capacitor.
9. Measure the output across the capacitor at frequencies from 100 Hz to 20 kHz and tabulate.
10. Identify from your data the frequency at which the output is 0.707 of the input.

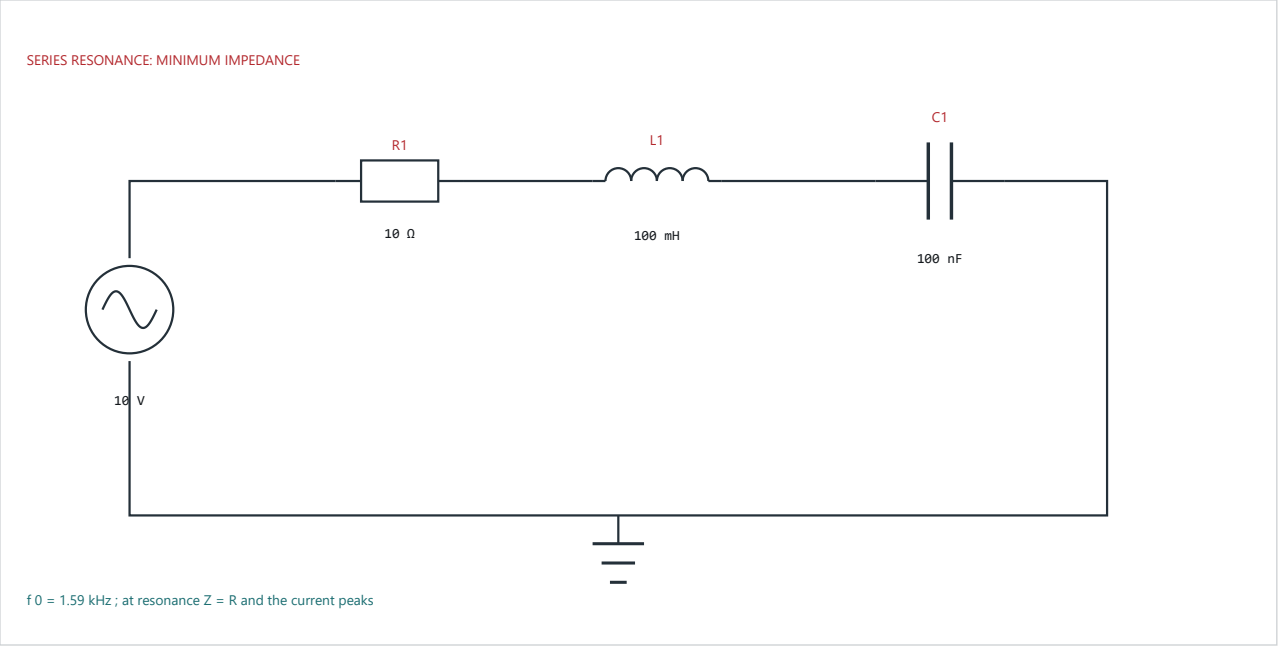



Fig. 8.L The resonance test circuit. The resistor voltage is proportional to the current, so its maximum locates resonance.

## Results table

| Frequency (Hz) | V across R | V across L | V across C | Comment |
|---|---|---|---|---|
| 500 | | | | |
| 1000 | | | | |
| 1592 (predicted $f_0$) | | | | |
| 2000 | | | | |
| 4000 | | | | |
| 8000 | | | | |

## Discussion

1. How close was the measured resonant frequency to your prediction? Account for the difference using the measured component values.
2. What was the ratio of the inductor voltage at resonance to the supply voltage? Compare it with your computed quality factor.
3. Compute the bandwidth from your half-power frequencies and compare with $f_0$ divided by Q.
4. Replace the 100 nF with 10 nF on paper and predict the new resonant frequency. By what factor should it change?
5. From your filter data, at what rate does the output fall above cut-off? Express it in decibels per decade.

SAFE-WORK REMINDERS

Keep the generator output at or below 5 V rms.

Element voltages at resonance can be many times the supply voltage. Set the oscilloscope range accordingly before probing the inductor or capacitor.

Check resistor dissipation at the resonant current before sweeping.

Reduce the generator amplitude to minimum before changing any connection.

## Troubleshooting guide

Faults are found by narrowing the circuit, not by replacing parts. Each row below pairs an observed symptom with the measurement that separates the likely causes.

| Symptom | Likely cause | Decisive test | Correct action |
|---|---|---|---|
| Measured resonant frequency differs from prediction | Component tolerance, especially the inductor | Measure L and C and recompute | Use measured values; inductors are often 10–20 % out |
| Resonance peak is broad and shallow | Low quality factor from high series resistance | Compute Q from the measured reactance and resistance | Reduce the series resistance or use a lower-loss inductor |
| Instrument damaged while probing a resonant circuit | Element voltage magnification exceeded the instrument range | Compute Q V before probing the reactive elements | Set the range for Q times the supply voltage, not the supply voltage |
| Correction capacitors overheat on a distorted supply | Parallel resonance between the capacitors and system inductance near a harmonic | Measure the capacitor current and its waveform | Fit detuning reactors to shift the resonance away from a harmonic |
| Filter passes more high frequency than expected | Stray capacitance bypassing the series element, or the output taken at the wrong node | Check the output node against the drawing and shorten leads | Correct the connection; reduce lead length and loop area |
| Notch filter does not reject the target frequency | Resonant frequency has drifted with temperature or component ageing | Sweep and locate the actual notch frequency | Retune, or specify components with tighter temperature coefficients |

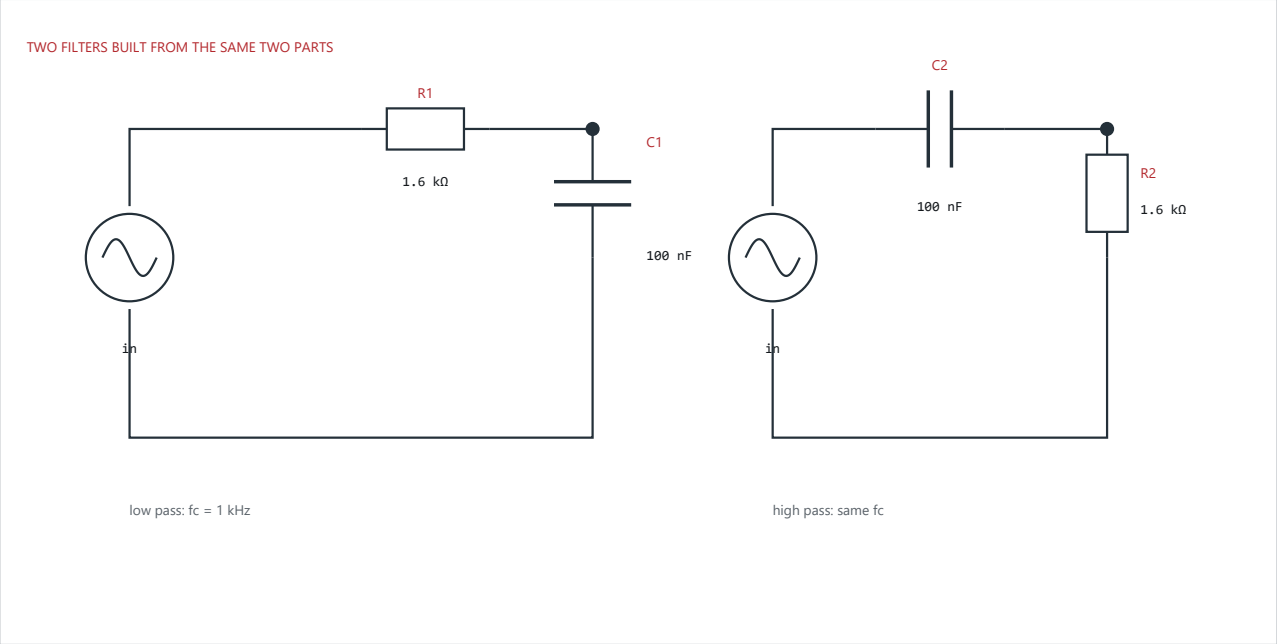



Fig. 8.T Measuring a filter response. The output is recorded against frequency at constant input amplitude, and the cut-off is read at the 0.707 point.

FIELD HABITS THAT PREVENT CALLBACKS

Compute Q V before probing a resonant circuit; the element voltages, not the supply voltage, set the instrument range you need.
Use measured inductance and capacitance for any resonance prediction, because tolerances are wide.
Hold the input amplitude constant across a frequency sweep, or the response curve is meaningless.
Treat an unexplained capacitor failure in a corrected installation as a possible harmonic resonance.

## Exercises

Answer from first principles. Selected answers follow; the reasoning matters more than the number.

1 Find the resonant frequency of 50 mH with 200 nF.
2 Find the resonant frequency of 10 mH with 1 µF.
3 A series circuit resonates at 2 kHz with X_L = 800 Ω and R = 40 Ω. Find Q and the bandwidth.
4 The circuit above is supplied at 12 V at resonance. Find the inductor voltage.
5 State what happens to the impedance, current and phase angle of a series circuit at resonance.
6 State what happens to the impedance and supply current of a parallel resonant circuit at resonance.
7 Find the cut-off frequency of a 10 kΩ resistor with a 10 nF capacitor.
8 A filter takes its output across the resistor of a series resistor-capacitor pair. State the filter type.
9 Explain why resonance frequency is independent of the series resistance.
10 A correction capacitor bank resonates with system inductance at 250 Hz on a 50 Hz system. State which harmonic that is and why it matters.

Selected answers

| # | Answer | Reason |
|---|---|---|
| 1 | 1592 Hz | $f_0 = 1/(2\pi\sqrt{(LC)})$ |
| 2 | 1592 Hz | The LC product is identical to question 1 |
| 3 | Q = 20; BW = 100 Hz | Q = X_L/R, then BW = $f_0$/Q |
| 4 | 240 V | V_L = Q V = 20 × 12 |
| 5 | Impedance minimum, current maximum, phase zero | The reactances cancel, leaving R alone |
| 6 | Impedance maximum, supply current minimum | Branch currents are equal and opposite and circulate internally |
| 7 | 1592 Hz | f_c = 1/(2πRC) |
| 8 | High pass | The resistor voltage rises as capacitive reactance falls with frequency |
| 9 | Resonance requires only X_L = X_C | Neither reactance depends on R, so R affects sharpness but not the frequency |
| 10 | The fifth harmonic; amplification | Harmonic current from electronic loads is amplified by the resonance, overheating the capacitors |

## Chapter summary

- Resonance occurs at the single frequency where inductive and capacitive reactances are equal, and depends only on L and C.
- Series resonance gives minimum impedance and maximum current; parallel resonance gives maximum impedance and minimum supply current.
- The quality factor sets both the sharpness and the voltage magnification, and bandwidth is $f_0$ divided by Q.
- Element voltages at series resonance can be many times the supply voltage and are a genuine hazard.
- A single resistor-capacitor stage gives a low-pass or high-pass response depending on which element the output is taken across, with cut-off at $1/(2\pi RC)$.
- Accidental resonance between correction capacitors and system inductance amplifies harmonics and destroys capacitors.

### Equations introduced in this chapter

| Relationship | Expression | When it applies |
|---|---|---|
| Resonant frequency | $f_0 = 1/(2\pi\sqrt{(LC)})$ | Series or parallel LC |
| Resonance condition | $X_L = X_C$ | Defines $f_0$ |
| Quality factor | $Q = X_L/R$ | Series resonant circuit |
| Bandwidth | $BW = f_0/Q$ | Between half-power points |
| Voltage magnification | $V_L = V_C = Q\,V$ | At series resonance |
| Filter cut-off | $f_c = 1/(2\pi RC)$ | Single-stage RC filter |

### Vocabulary check

| Term | Meaning in this chapter |
|---|---|
| Resonance | The condition where inductive and capacitive reactances cancel |
| Quality factor | The ratio of reactance to resistance at resonance, setting sharpness |
| Bandwidth | The frequency span between the half-power points |
| Cut-off frequency | Where a filter's output has fallen to 0.707 of its input |
| Detuning reactor | An inductor added to shift a capacitor bank's resonance away from a harmonic |

LOOKING AHEAD

Resonance and filtering depend on inductance, which is produced by magnetic fields. The next chapter examines magnetism directly: how fields are produced, how they produce force and voltage, and how magnetic circuits are analysed.

CHAPTER 09

# Magnetism and electromagnetism

Fields, flux, force and induced voltage

WHY THIS CHAPTER MATTERS

Every transformer, motor, relay, contactor and solenoid in an installation works because a current produces a magnetic field and a changing field produces a voltage. Those two statements, taken together, account for most of the equipment an electrician ever touches.

## What you will be able to do

- Describe a magnetic field in terms of flux and flux density.
- Apply the right-hand rule to find field direction around a conductor and inside a coil.
- Compute magnetomotive force, field strength and reluctance in a simple magnetic circuit.
- Compute the force on a current-carrying conductor in a field.
- State Faraday's and Lenz's laws and compute an induced voltage.
- Explain hysteresis and eddy-current losses and how laminated cores reduce them.

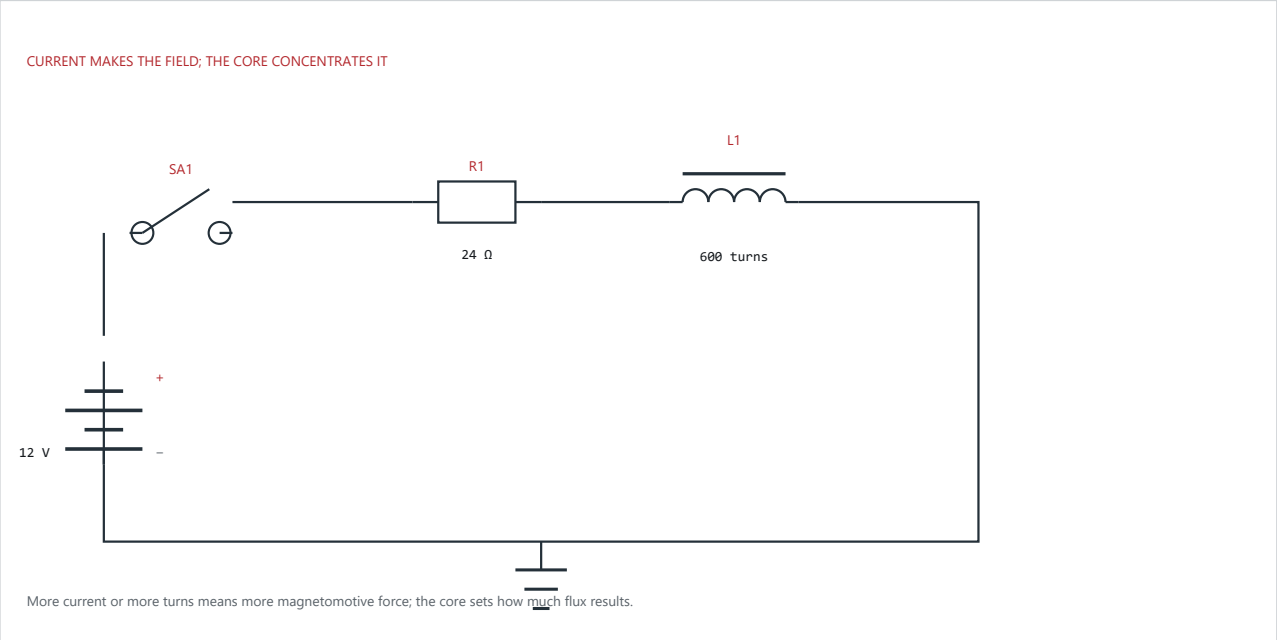



Fig. 9.1 A coil wound on a core. The current in the winding drives flux around the magnetic circuit, and the core's permeability determines how much flux a given current produces.

## Flux and flux density

A magnetic field is described by its flux, measured in webers, and by how concentrated that flux is, measured as flux density in teslas. Flux density is simply flux divided by the area it passes through at right angles, so squeezing the same flux into a smaller cross-section raises the density.

Flux density is the quantity that matters in practice, because force, induced voltage and core saturation all depend on it. A core driven beyond about 1.6 to 1.8 teslas saturates: further increases in current produce almost no further flux, and the equipment draws heavy current while doing no useful magnetic work.

```
B = Φ / A

Φ in webers, A in square metres, B in teslas

1 T = 1 Wb/m^2
```

## Field direction and the right-hand rules

Field direction is fixed by the current direction and found by a rule rather than by memory. Grip a straight conductor with the right hand, thumb pointing along conventional current, and the fingers wrap around it in the same sense as the circular field.

For a coil, let the right-hand fingers follow the direction the current circulates in the turns and the thumb points to the north pole of the resulting field. Reversing the supply polarity reverses the poles, which is why a direct-current solenoid pulls in the same direction regardless of polarity only if its construction is deliberately polarity-independent.

### Working with field direction

- Thumb along current, fingers wrapping the conductor: straight conductor.
- Fingers following the turns, thumb to north: coil or solenoid.
- Reversing current reverses the poles of an electromagnet.
- Two conductors carrying current the same way attract; opposite ways repel.
- A permanent magnet has fixed poles, so a relay containing one will respond to polarity.

## The magnetic circuit

Flux around a core behaves much like current around an electric circuit, and the analogy is worth using. The driving quantity is magnetomotive force, the product of current and turns. The opposition is reluctance, which depends on path length, cross-sectional area and the permeability of the material.

The crucial practical point is that air has a permeability thousands of times lower than iron, so even a small air gap dominates the reluctance of an otherwise iron path. This is why a contactor that fails to close fully draws heavy current: the residual gap keeps the reluctance high, the flux low, and the inrush current never falls to its sealed value.

```
F = N I      (ampere-turns)

H = N I / l      (ampere-turns per metre)

Φ = F / S      S = l / (μ0 μ_r A)

B = μ0 μ_r H      μ0 = 4π × 10^-7 H/m
```

WHY A SMALL AIR GAP MATTERS SO MUCH

Relative permeability of transformer steel may be 4000; of air it is 1.
A 1 mm gap in a 400 mm iron path can contribute more reluctance than all the iron.
A contactor held partly open therefore keeps drawing its inrush current and overheats.
Always investigate a buzzing or hot contactor coil as a mechanical problem first.

## Force on a conductor

A conductor carrying current across a magnetic field experiences a force at right angles to both. The force is proportional to flux density, current and the length of conductor in the field, and it is the mechanism by which every motor develops torque.

The same effect explains why heavy fault currents are mechanically violent. Busbars carrying tens of kiloamperes exert large forces on each other, which is why switchgear bracing and bus supports are rated for short-circuit withstand and not merely for weight.

```
F = B I l sinθ

Maximum when the conductor is perpendicular to the field

F in newtons, B in teslas, l in metres
```

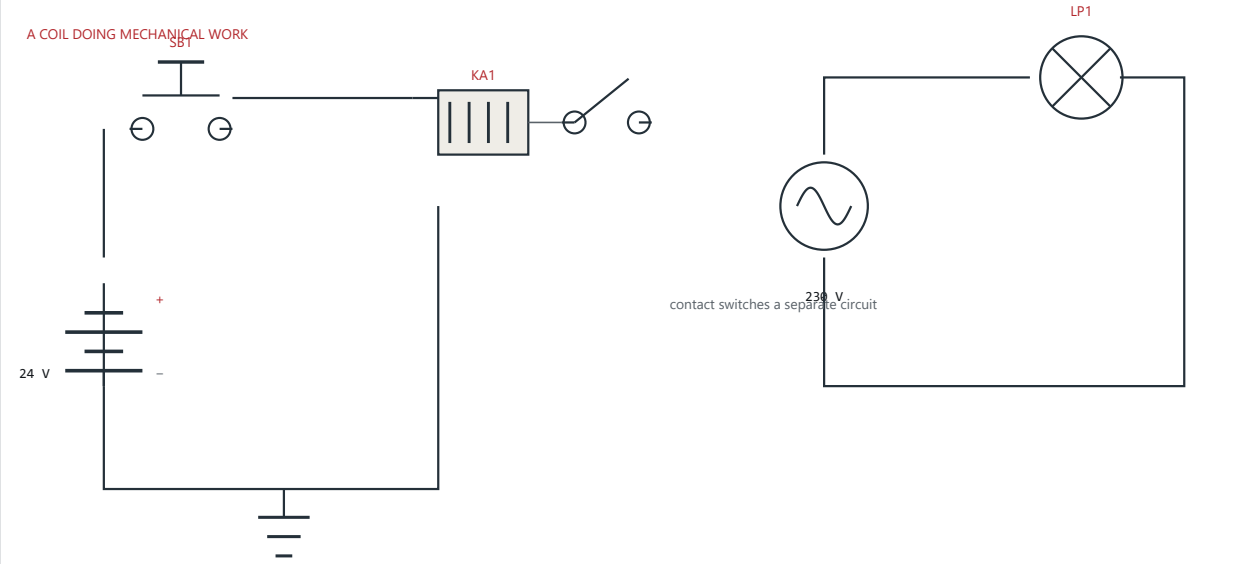



Fig. 9.2 A relay: a coil drives flux through a magnetic circuit, the armature is attracted, and the contacts change state. Coil and contact circuits are electrically separate.

## Induced voltage: Faraday and Lenz

Faraday's law states that a voltage is induced in a circuit whenever the flux linking it changes, and that the magnitude is proportional to the rate of change and the number of turns. Nothing is induced by a steady field; only change produces voltage.

Lenz's law fixes the polarity: the induced voltage always acts to oppose the change that produced it. This is not an arbitrary sign convention but a consequence of energy conservation, and it is the reason a generator becomes harder to turn as load is applied and the reason an inductor produces a large reverse voltage when its current is interrupted.

```
e = −N dΦ/dt

For a conductor moving through a field:  e = B l v sinθ

The minus sign is Lenz's law: opposition to the change
```

Both forms describe the same physics. The first is used for transformers and coils, where flux changes while the conductor stays still; the second is used for machines, where the conductor moves through a steady field.

## Core losses: hysteresis and eddy currents

Driving a core through repeated magnetisation cycles costs energy, because the material resists reversing its internal domains. The energy lost per cycle is proportional to the area of the hysteresis loop, so loss rises with frequency and with peak flux density.

A solid core also behaves as a short-circuited turn: the changing flux induces circulating eddy currents in the iron itself, which dissipate heat. Laminating the core into thin insulated sheets interrupts those current paths while leaving the magnetic path intact, which is why transformer and motor cores are built from stacked laminations rather than solid blocks.

| Loss mechanism | Cause | Varies with | Reduced by |
|---|---|---|---|
| Hysteresis | Energy to reverse magnetic domains | Frequency and peak flux density | Grain-oriented silicon steel |
| Eddy current | Circulating current induced in the core | Square of frequency and of flux density | Thin insulated laminations |
| Copper loss | Resistance of the windings | Square of the load current | Larger conductor cross-section |
| Stray loss | Flux leaking into tanks and clamps | Load current | Magnetic shielding and layout |

CORE LOSS IS PRESENT WHENEVER THE EQUIPMENT IS ENERGISED

Hysteresis and eddy-current loss depend on voltage and frequency, not on load.
A lightly loaded transformer therefore still dissipates its full core loss.
Copper loss varies with the square of the load current and falls away at light load.
This is why leaving transformers energised to serve no load wastes energy continuously.

## Worked examples

Work each example with a pencil before reading the solution. Sketch the circuit, mark the reference node, and write the governing law before substituting numbers.

EXAMPLE 9.1 Flux density and saturation

**Given.** A core of cross-section 25 mm × 40 mm carries a flux of 1.4 mWb.

**Find.** The flux density, and whether the core is likely to be saturated.

Cross-sectional area

$$A = 0.025 \times 0.040 = 1.0 \times 10^{-3}\ \text{m}^2$$

Flux density

$$B = \Phi / A = 1.4 \times 10^{-3} / 1.0 \times 10^{-3} = 1.4\ \text{T}$$

Compare with typical saturation

Silicon steel saturates near 1.6 to 1.8 T

Conclusion

1.4 T is below saturation but with little margin

**Check.** A design at 1.4 T is normal practice. A 15 % supply overvoltage would push the flux to about 1.6 T and the magnetising current would rise sharply, which is why transformers hum louder and run hotter on a high supply.

EXAMPLE 9.2 Ampere-turns and the effect of an air gap

**Given.** A coil of 600 turns carries 0.5 A on an iron path 300 mm long, area $1.0 \times 10^{-3}$ m$^2$, relative permeability 3000. A 0.8 mm air gap is then introduced.

**Find.** The flux before and after the gap is introduced.

Magnetomotive force

$$F = N I = 600 \times 0.5 = 300\ \text{ampere-turns}$$

Iron reluctance

$$S_{Fe} = 0.300 / (4\pi \times 10^{-7} \times 3000 \times 1 \times 10^{-3}) = 79\,600\ \text{A/Wb}$$

Flux with no gap

$$\Phi = 300 / 79\,600 = 3.77\ \text{mWb}$$

Gap reluctance

$$S_{air} = 0.0008 / (4\pi \times 10^{-7} \times 1 \times 1 \times 10^{-3}) = 636\,600\ \text{A/Wb}$$

Total reluctance

$$S = 79\,600 + 636\,600 = 716\,200\ \text{A/Wb}$$

Flux with the gap

$$\Phi = 300 / 716\,200 = 0.42\ \text{mWb}$$

**Check.** A gap less than three thousandths of the path length cut the flux to about one ninth, because its reluctance is eight times that of all the iron. This is the quantitative reason a contactor must seal fully.

EXAMPLE 9.3 Induced voltage in a moving conductor

**Given.** A conductor 250 mm long moves at 12 m/s perpendicular to a field of 0.8 T.

**Find.** The induced voltage, and the value if the conductor moves at 30° to the field.

Perpendicular motion

```
e = B l v = 0.8 × 0.25 × 12 = 2.4 V
```

At 30° to the field

```
e = 2.4 × sin30° = 1.2 V
```

At 0°, moving along the field

```
e = 2.4 × sin0° = 0 V
```

**Check.** Only motion across the flux generates voltage. A conductor sliding along the field lines cuts no flux and generates nothing, which is why machine geometry places conductors across the field at the air gap.

EXAMPLE 9.4 Force on a busbar during a fault

**Given.** Two parallel busbars 1.2 m long sit in a stray field of 0.35 T and carry 18 kA during a fault.

**Find.** The force on one bar.

Force equation

```
F = B I l
```

Substitute

```
F = 0.35 × 18 000 × 1.2
```

Result

```
F = 7560 N, about 770 kg equivalent
```

Duration

```
Present only for the clearing time, perhaps 100 ms
```

**Check.** Nearly a tonne of force applied for a tenth of a second explains why bus supports are rated for short-circuit withstand and why damaged bracing must be replaced after a major fault.

# Practical activity: magnetic field, induced voltage and a relay's operating points

AIM

Observe field direction around a coil, demonstrate that only changing flux induces a voltage, measure the pull-in and drop-out voltages of a relay, and show the effect of preventing the armature from sealing.

## Equipment

- Air-cored and iron-cored coils, 600 turns.
- Bar magnet, plotting compass and iron filings with a covered tray.
- Direct-current supply, 0 to 24 V, current limited to 1 A.
- Centre-zero galvanometer or a multimeter on its most sensitive direct-voltage range.
- Relay, 12 V coil, with accessible contacts; a 24 V lamp as the contact load.
- Oscilloscope with a single channel.

## Procedure

1. With the coil de-energised, map the field of the bar magnet using the plotting compass and sketch the pattern.
2. Energise the iron-cored coil at 6 V and use the compass to identify its north pole. Confirm the result against the right-hand rule and the winding direction.
3. Reverse the supply polarity and confirm that the poles exchange.
4. Connect the galvanometer across a second coil placed near the first. Switch the first coil on and off and record the direction of deflection in each case.
5. Hold the first coil energised steadily and confirm that the galvanometer reads zero.
6. Move the bar magnet into the second coil, hold it still, then withdraw it. Record the deflection for each of the three actions.
7. Repeat the insertion at two clearly different speeds and compare the deflections.
8. Wire the relay coil to the variable supply and the lamp through the relay contacts.
9. Raise the coil voltage slowly and record the voltage at which the contacts close.
10. Lower the voltage slowly and record the voltage at which they reopen. Repeat three times.
11. Insert a 0.5 mm shim to prevent the armature sealing, re-energise at rated coil voltage and record the coil current. Compare with the sealed current.

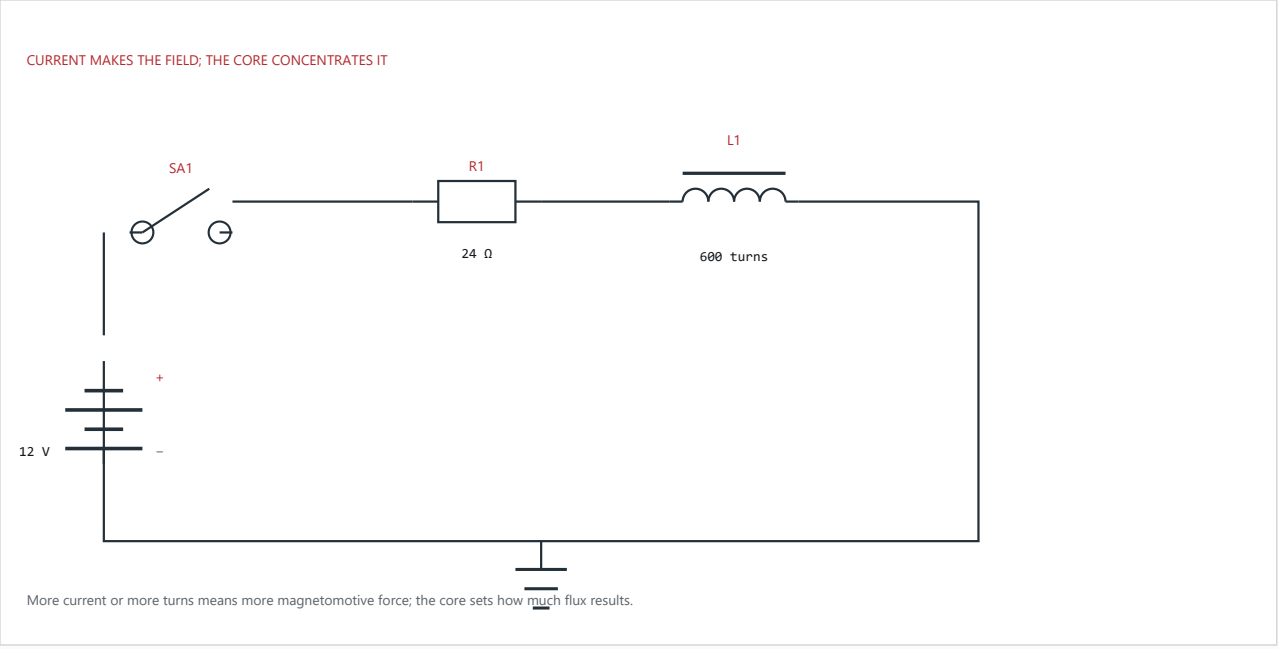



Fig. 9.L The coil and core used for the field and induction tests. The second coil is placed to share the first coil's flux.

## Results table

| Test | Observation | Deflection or reading | Explanation |
|---|---|---|---|
| Coil switched on | | | |
| Coil switched off | | | |
| Coil held energised | | | |
| Magnet inserted slowly | | | |
| Magnet inserted quickly | | | |
| Magnet held stationary | | | |
| Relay pull-in voltage | | | |
| Relay drop-out voltage | | | |
| Coil current, sealed | | | |
| Coil current, shimmed open | | | |

## Discussion

1. Why did the galvanometer read zero while the coil was held energised, even though a strong field was present?
2. Why were the switch-on and switch-off deflections in opposite directions? Relate your answer to Lenz's law.
3. How did deflection depend on the speed of the magnet, and which term in Faraday's law does that confirm?

4 Why is the drop-out voltage lower than the pull-in voltage? Explain using reluctance and the air gap.

5 By what factor did the coil current rise when the armature was prevented from sealing, and what would that do to the coil over time?

6 A relay in service chatters on and off rapidly. Using your pull-in and drop-out figures, explain how a sagging supply could produce this.

SAFE-WORK REMINDERS

Limit the supply current to 1 A and do not exceed the coil's rated voltage for more than a few seconds.

Do not leave the relay coil shimmed and energised for longer than the measurement requires; the coil overheats quickly.

Interrupting an inductive coil produces a high reverse voltage. Do not hold bare coil terminals while switching.

Keep iron filings in a sealed tray and clear of eyes and of the equipment.

Energised coils and cores become hot. Allow them to cool before handling.

## Troubleshooting guide

Faults are found by narrowing the circuit, not by replacing parts. Each row below pairs an observed symptom with the measurement that separates the likely causes.

| Symptom | Likely cause | Decisive test | Correct action |
|---|---|---|---|
| Contactor buzzes loudly and the coil overheats | Armature not sealing, so the air gap keeps reluctance and current high | Measure coil current and compare with the sealed value; inspect the pole faces | Clean or dress the pole faces, free the mechanism, replace a worn armature |
| Transformer hums louder than usual and runs hot at no load | Supply voltage high, driving the core toward saturation | Measure supply voltage and compare with the nameplate | Correct the supply voltage or change the tap setting |
| Relay drops out on motor starting | Supply sag takes the coil below its drop-out voltage | Record coil voltage during the start with a logging meter | Supply the control circuit from a stiffer source, or fit a relay with a wider voltage window |
| Solenoid valve will not pull in but the coil is energised | Mechanical obstruction, or coil voltage below pull-in | Measure coil voltage at the terminals and check free movement | Clear the obstruction; verify the coil voltage rating matches the supply |
| Core of a repaired transformer runs much hotter than before | Laminations shorted together during repair, allowing eddy currents | Compare no-load loss with the original test figure | Restore the interlaminar insulation or replace the core assembly |
| Permanent magnet motor or relay has weakened over time | Magnet partly demagnetised by heat or by an opposing field | Measure flux density at the pole face against the original value | Replace the magnet assembly and remove the source of overheating |




Fig. 9.T A thermocouple: two dissimilar conductors produce a small voltage that depends on the junction temperature, used to monitor magnetic equipment for overheating.

FIELD HABITS THAT PREVENT CALLBACKS

Treat a buzzing contactor as a mechanical fault, not an electrical one, and check the pole faces before condemning the coil.

Record pull-in and drop-out voltages when commissioning control gear, so later nuisance operation can be diagnosed against known figures.

Fit a suppression diode or snubber across any coil switched by a semiconductor or a delicate contact.

Investigate a transformer that has become noisier, because increased hum usually means flux density has risen.

## Exercises

Answer from first principles. Selected answers follow; the reasoning matters more than the number.

1. A core of area $8 \times 10^{-4}\ \text{m}^2$ carries 1.0 mWb. Find the flux density.
2. Find the flux needed to produce 1.5 T in a core of area $1.2 \times 10^{-3}\ \text{m}^2$.
3. A coil of 450 turns carries 0.8 A. Find the magnetomotive force.
4. For the coil above on a 250 mm path, find the field strength H.
5. State the two laws that together determine the magnitude and the polarity of an induced voltage.
6. A conductor 0.4 m long moves at 15 m/s perpendicular to a 0.6 T field. Find the induced voltage.
7. Explain why a steady direct current in a coil induces no voltage in a nearby coil.
8. A conductor carrying 250 A lies 0.3 m long in a 0.5 T field. Find the force on it.
9. Explain why transformer cores are laminated rather than solid.
10. State which transformer losses depend on load current and which do not, and why that matters for a lightly loaded unit.
11. Explain why a 1 mm air gap can dominate the reluctance of a 400 mm iron magnetic circuit.
12. A relay pulls in at 9.2 V and drops out at 4.1 V. Explain the difference in terms of the air gap.

## Selected answers

| # | Answer | Reason |
|---|---|---|
| 1 | 1.25 T | $B = \Phi / A$ |
| 2 | 1.8 mWb | $\Phi = B\,A$ |
| 3 | 360 ampere-turns | $F = N\,I$ |
| 4 | 1440 A/m | $H = N\,I / l$ |
| 5 | Faraday's law and Lenz's law | Faraday gives magnitude from rate of change; Lenz gives the opposing polarity |
| 6 | 3.6 V | $e = B\,l\,v$ |
| 7 | Flux is not changing | Faraday's law depends on $d\Phi/dt$, which is zero for steady current |
| 8 | 37.5 N | $F = B\,I\,l$ |
| 9 | To interrupt eddy-current paths | Thin insulated sheets break the circulating current while keeping the magnetic path |
| 10 | Core loss is load-independent; copper loss varies with current squared | A lightly loaded transformer still dissipates full core loss, so energising unused units wastes energy |
| 11 | Air's permeability is thousands of times lower | Reluctance is inversely proportional to permeability, so a short air path can outweigh a long iron one |
| 12 | Once sealed, the gap is gone and reluctance falls | Less current is needed to hold the armature than to pull it in, so drop-out is well below pull-in |

## Chapter summary

- Flux density, not total flux, determines force, induced voltage and saturation.
- Right-hand rules fix field direction from current direction for both straight conductors and coils.
- The magnetic circuit analogy relates ampere-turns, reluctance and flux, and shows why a small air gap dominates an iron path.
- Force on a conductor is B I l, which produces motor torque and makes heavy fault currents mechanically destructive.
- Only changing flux induces voltage; Faraday sets the magnitude and Lenz sets the opposing polarity.
- Hysteresis and eddy-current losses depend on voltage and frequency rather than load, and lamination addresses the eddy-current component.

### Equations introduced in this chapter

| Relationship | Expression | When it applies |
|---|---|---|
| Flux density | $B = \Phi / A$ | Uniform flux through a known area |
| Magnetomotive force | $F = N I$ | Driving quantity of a magnetic circuit |
| Field strength | $H = N I / l$ | Along a uniform magnetic path |
| Reluctance | $S = l / (\mu_0 \mu_r A)$ | Opposition of a magnetic path to flux |
| Magnetic Ohm's law | $\Phi = F / S$ | Series magnetic circuit |
| Force on a conductor | $F = B I l \sin\theta$ | Conductor crossing a field |
| Faraday's law | $e = -N \, d\Phi/dt$ | Changing flux linking a coil |
| Motional voltage | $e = B l v \sin\theta$ | Conductor moving through a field |

### Vocabulary check

| Term | Meaning in this chapter |
|---|---|
| Flux | The total magnetic field passing through a surface, in webers |
| Flux density | Flux per unit area, in teslas; the quantity that sets force and saturation |
| Magnetomotive force | The ampere-turn product that drives flux around a magnetic circuit |
| Reluctance | The opposition a magnetic path presents to flux |
| Saturation | The condition where further magnetomotive force produces little further flux |
| Hysteresis | Energy lost per magnetisation cycle in reversing magnetic domains |
| Eddy current | Circulating current induced in a core, suppressed by lamination |

LOOKING AHEAD

Faraday's law applied to two windings on a shared core gives the transformer, the single most important application of electromagnetism in power distribution. The next chapter develops it.

CHAPTER 10

# Transformers

Changing voltage without moving parts, and what that costs

> WHY THIS CHAPTER MATTERS
>
> Transformers make the whole distribution system possible. Generating at one voltage, transmitting at a much higher one and using at a safe one is only practical because a transformer converts between them efficiently and with no moving parts.

### What you will be able to do

- Derive and apply the turns, voltage and current ratios.
- Explain why a transformer cannot operate on direct current.
- Distinguish step-up, step-down, isolating and auto-transformer connections.
- Compute transformer rating in volt-amperes and relate it to load current.
- Account for no-load and load losses and compute efficiency and regulation.
- Interpret a nameplate, select tap positions and explain inrush current.

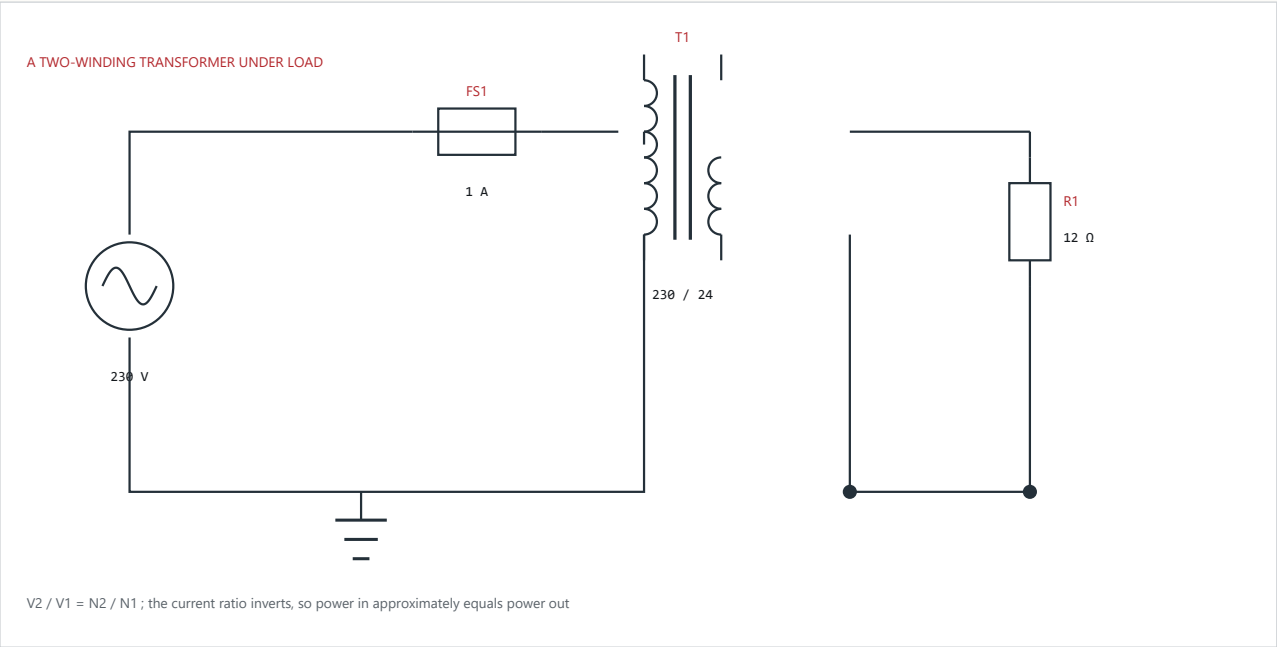



Fig. 10.1 A two-winding transformer on a shared iron core. Alternating flux produced by the primary links the secondary, inducing a voltage set by the turns ratio.

## How a transformer works

Two windings share a magnetic core. Alternating voltage applied to the primary drives an alternating flux around that core, and because the same flux links the secondary, a voltage is induced there. No electrical connection between the windings is needed; the coupling is entirely magnetic.

Because the induced voltage depends on the rate of change of flux, a transformer needs alternating supply. Applied to direct current it produces a brief output at switch-on and then nothing, while the primary draws a current limited only by its own winding resistance. This is why connecting a transformer to a direct supply destroys it.

```
e = −N dΦ/dt      so steady flux induces nothing

V_p / V_s = N_p / N_s = a      (turns ratio)
```

> A TRANSFORMER IS AN ALTERNATING-CURRENT DEVICE ONLY
> Output voltage depends on flux changing, not on flux existing.
> On direct current the primary becomes a low-resistance coil across the supply.
> The winding overheats within seconds and the transformer is lost.
> Confirm the supply type before energising any transformer.

## Voltage, current and impedance ratios

The voltage ratio follows the turns ratio directly. The current ratio is the inverse, because an ideal transformer neither creates nor absorbs power: whatever volt-amperes enter the primary leave the secondary. A step-down transformer therefore raises current in the same proportion as it lowers voltage.

This inversion has an important safety consequence. The secondary of a step-down transformer can deliver very large currents from a modest primary supply, so secondary conductors and protection must be sized for the stepped-up current, not for the primary current.

```
V_p / V_s = N_p / N_s
I_s / I_p = N_p / N_s      so V_p I_p = V_s I_s (ideal)
Z_p / Z_s = (N_p / N_s)^2 = a^2
```

The impedance ratio is the square of the turns ratio, which is why a transformer can match a load to a source and why a secondary short circuit appears to the primary as a fault reduced by the square of the ratio.

## Rating in volt-amperes

A transformer is rated in volt-amperes rather than watts because its limits are set by current in the windings and by flux in the core, neither of which depends on the load's power factor. A winding heated by 20 A is equally heated whether that current is in phase with the voltage or not.

Load current is therefore found from the apparent power and the secondary voltage. Where the load power factor is poor, the transformer must be sized on apparent power even though the useful power delivered is lower.

```
S = V I      (single phase, in volt-amperes)

I_s = S / V_s

S = √3 V_L I_L      (three phase)
```

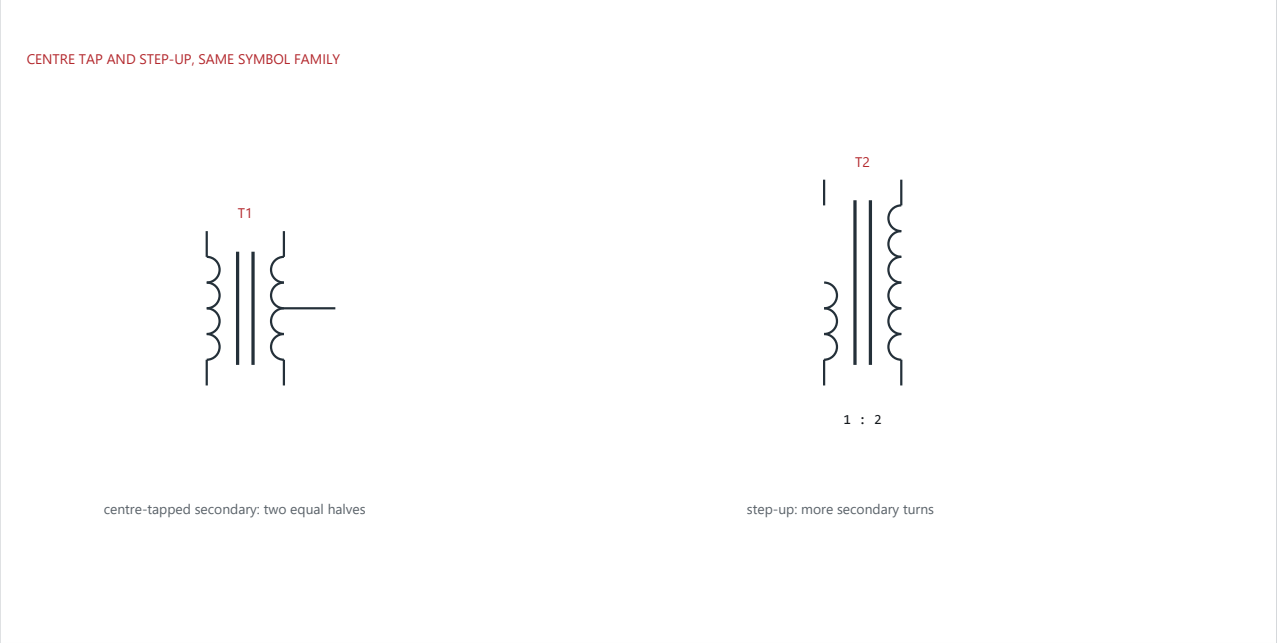



Fig. 10.2 Tap connections on the primary winding. Selecting a different tap changes the effective turns ratio to compensate for a high or low incoming supply.

## Losses, efficiency and regulation

Real transformers lose power in two distinct ways. Core loss, comprising hysteresis and eddy-current loss, depends on voltage and frequency and is therefore almost constant whenever the unit is energised. Copper loss arises from winding resistance and varies with the square of the load current.

Efficiency is highest at the load where these two losses are equal, which for distribution transformers is usually designed to fall somewhere between half and three-quarters of full load. Regulation is a separate measure: the fall in secondary voltage from no load to full load, expressed as a percentage of the no-load value.

```
η = output / (output + P_core + P_cu)

P_cu at any load = P_cu(full) × (I / I_full)^2

Regulation % = (V_noload − V_load) / V_noload × 100
```

| Quantity | Determined by | Varies with load? | Measured by |
|---|---|---|---|
| Core loss | Applied voltage and frequency | No | Open-circuit test |
| Copper loss | Winding resistance and current | Yes, as current squared | Short-circuit test |

| Quantity | Determined by | Varies with load? | Measured by |
|---|---|---|---|
| Efficiency | Ratio of output to input | Yes, peaks where losses are equal | Computed from both tests |
| Regulation | Winding impedance and load power factor | Yes | Loaded voltage measurement |

## Isolation, autotransformers and inrush

A two-winding transformer provides galvanic isolation: there is no conductive path between primary and secondary, so a fault on one side does not reference the other to earth. This is the property exploited by isolating transformers used for sensitive equipment and for certain tool supplies.

An autotransformer uses a single tapped winding instead, which makes it smaller and cheaper for small ratio changes but provides no isolation whatever. The primary and secondary share conductors, so an earth fault or an open point in the common section can place full input voltage on the output.

All transformers draw a heavy inrush current at energisation, because the core may be left magnetised from the previous switch-off and can be driven briefly into deep saturation. Inrush may reach ten or more times rated current for a few cycles, which is why transformer protection must tolerate it without tripping.

### Choosing and energising transformers

- Specify a two-winding type wherever isolation is required; an autotransformer cannot provide it.
- Size secondary conductors for the stepped-up secondary current.
- Select protection that rides through inrush but still clears a genuine fault.
- Record the tap position at commissioning, so later voltage complaints can be checked.
- De-energise unused transformers, because core loss continues regardless of load.

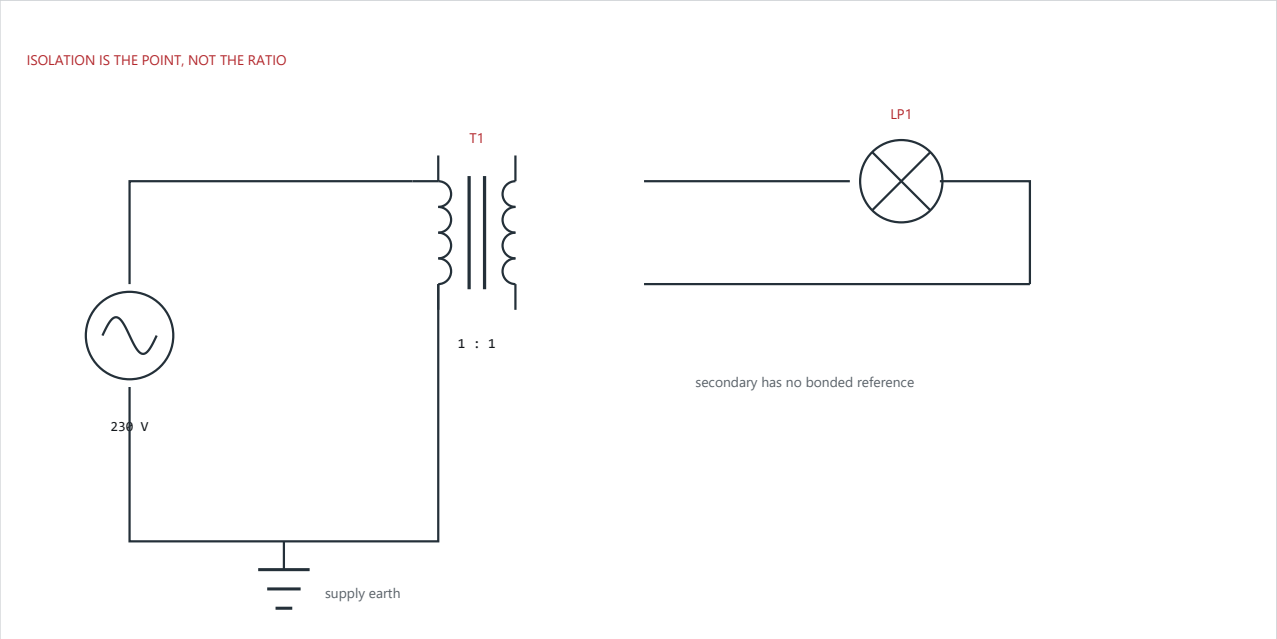



Fig. 10.3 An isolating transformer. With no conductive path between the windings, the secondary has no fixed relationship to the primary reference.

## Worked examples

Work each example with a pencil before reading the solution. Sketch the circuit, mark the reference node, and write the governing law before substituting numbers.

EXAMPLE 10.1 Turns ratio, secondary voltage and currents

**Given.** A transformer has 960 primary turns and 80 secondary turns, supplied at 240 V, delivering 15 A to its load.

**Find.** The turns ratio, secondary voltage, primary current and the apparent power.

Turns ratio

$a = 960/80 = 12 : 1$, step down

Secondary voltage

$V_s = 240/12 = 20\ V$

Apparent power

$S = 20 \times 15 = 300\ VA$

Primary current

$I_p = 15/12 = 1.25\ A$

Confirm the power balance

$240 \times 1.25 = 300\ VA$

**Check.** Voltage fell by twelve times and current rose by twelve times, leaving the apparent power unchanged. A 1.25 A primary circuit is feeding a 15 A secondary, which is why the secondary cable must be sized on 15 A.

EXAMPLE 10.2 Efficiency at full and half load

**Given.** A 5 kVA transformer has a core loss of 60 W and a full-load copper loss of 110 W. The load power factor is 0.9.

**Find.** The efficiency at full load and at half load.

Full-load output power

$P = 5000 \times 0.9 = 4500\ W$

Full-load total loss

$60 + 110 = 170\ W$

Full-load efficiency

$\eta = 4500/(4500 + 170) = 0.964$, that is 96.4 %

Half-load output

$P = 2250\ W$

Half-load copper loss

$110 \times 0.5^2 = 27.5\ W$

Half-load efficiency

$\eta = 2250/(2250 + 60 + 27.5) = 0.963$, that is 96.3 %

**Check.** Efficiency barely changes between half and full load because copper loss falls as the square of current while core loss stays fixed. The peak lies where the two losses are equal, near 73 % of full load here.

EXAMPLE 10.3 Voltage regulation

**Given.** A transformer measures 230 V at its secondary with no load and 219 V at full load.

**Find.** The regulation, and whether it is acceptable for a lighting circuit.

Voltage drop

$$230 - 219 = 11 \text{ V}$$

Regulation

$$11/230 \times 100 = 4.8 \%$$

Typical expectation

Distribution transformers commonly give 2 to 5 %

Assessment

Acceptable, but near the upper end

**Check.** A 4.8 % regulation leaves little margin once cable drop is added downstream. If terminal voltage at the far end of the final circuit matters, this transformer should be tapped up or a lower-impedance unit chosen.

EXAMPLE 10.4 Secondary fault current reflected to the primary

**Given.** A 12 : 1 step-down transformer suffers a secondary short circuit drawing 480 A.

**Find.** The resulting primary current, and why the primary protection may not see it as alarming.

Current ratio

$$I_p = I_s / a$$

Substitute

$$I_p = 480/12 = 40 \text{ A}$$

Impedance seen at the primary

$$Z_p = a^2 Z_s = 144 \times Z_s$$

Consequence

A severe secondary fault appears as a modest primary overload

**Check.** The transformer transforms faults as well as load. A 480 A secondary fault looks like 40 A at the primary, which may sit within the primary device's tolerance long enough to burn the secondary winding, so secondary-side protection is required.

## Practical activity: transformer ratio, loss separation and regulation

AIM

Verify the turns and current ratios by measurement, separate core loss from copper loss using open-circuit and short-circuit tests, then load the transformer progressively and determine its regulation and efficiency curve.

### Equipment

- Two-winding transformer, 240 V to 24 V, 150 VA or similar, on a bench mount.
- Variable alternating supply with an isolating transformer ahead of it.
- Two true rms multimeters and a power meter or wattmeter.
- Resistive load bank, switchable in steps, rated for the secondary current.
- Clamp meter suitable for small alternating currents.
- Insulation resistance tester.

### Procedure

1. With the transformer de-energised and disconnected, measure and record the resistance of each winding and the insulation resistance between them.
2. Identify primary and secondary from the nameplate and compute the expected turns ratio.
3. Open-circuit test: apply rated voltage to the primary with the secondary open. Record primary voltage, current and power. This power is essentially the core loss.
4. Record the secondary open-circuit voltage and compute the measured voltage ratio.
5. Short-circuit test: with the secondary shorted through an ammeter, raise the primary voltage from zero only until rated secondary current flows. Record primary voltage, current and power. This power is essentially the full-load copper loss.
6. Return the supply to zero before changing any connection.
7. Reconnect the load bank and apply rated primary voltage.
8. Record secondary voltage and current at no load and at four increasing load steps up to rated current.
9. Measure primary current at each step and compare the measured current ratio with the turns ratio.
10. Compute efficiency at each step from the two loss figures, and compute regulation from your no-load and full-load secondary voltages.

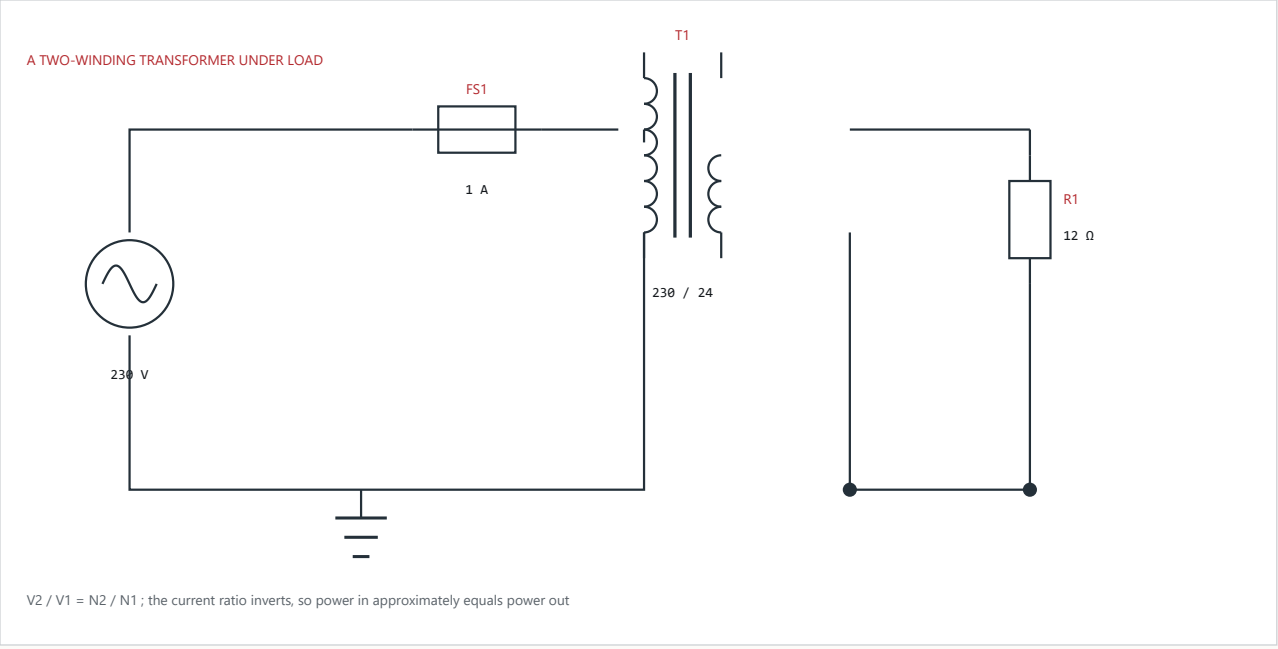



Fig. 10.L The test arrangement. The same transformer serves the open-circuit, short-circuit and loading tests with only the secondary connection changed.

## Results table

| Load step | $V_s$ (V) | $I_s$ (A) | $I_p$ (A) | Efficiency and comment |
|---|---|---|---|---|
| No load | | | | |
| 25 % load | | | | |
| 50 % load | | | | |
| 75 % load | | | | |
| 100 % load | | | | |
| Short-circuit test | | | | |

## Discussion

1. How closely did your measured voltage ratio match the nameplate turns ratio, and what accounts for the difference?
2. Why is the open-circuit power essentially all core loss, and the short-circuit power essentially all copper loss?
3. At which load step was your computed efficiency highest? Compare that load with the point where your two losses are equal.
4. Compute the regulation. What would that drop do to a load at the end of a long secondary cable?
5. Why must the short-circuit test be started from zero volts and raised only to rated current?
6. Your measured current ratio probably departs from the turns ratio slightly more than the voltage ratio does. Suggest why.

SAFE-WORK REMINDERS

Always supply the bench through an isolating transformer and a residual current device.

During the short-circuit test, raise the voltage from zero slowly. Applying rated voltage to a shorted secondary will destroy the transformer and is dangerous.

Secondary current in a step-down transformer is much higher than primary current. Size the load leads and the ammeter accordingly.

Return the variable supply to zero and prove dead before altering any connection.

Windings and the load bank become hot. Allow them to cool before handling.

Do not apply any direct-current supply to the transformer.

## Troubleshooting guide

Faults are found by narrowing the circuit, not by replacing parts. Each row below pairs an observed symptom with the measurement that separates the likely causes.

| Symptom | Likely cause | Decisive test | Correct action |
|---|---|---|---|
| Transformer hums loudly and runs hot with no load | Supply voltage high or tap set too low, pushing the core toward saturation | Measure the incoming voltage and check the selected tap | Move to the correct tap or correct the supply voltage |
| Secondary voltage low only when loaded | Excessive regulation, undersized transformer or undersized secondary cable | Measure secondary voltage at no load and at full load, then at the load terminals | Tap up, fit a larger transformer, or increase cable size |
| Primary protection trips at every energisation | Inrush current exceeding the device's instantaneous setting | Compare the device characteristic with the expected inrush | Fit a device with a suitable inrush-tolerant characteristic |
| Transformer destroyed shortly after installation | Connected to a direct-current supply, or primary and secondary reversed | Check the supply type and compare terminal markings with the nameplate | Replace the unit and correct the supply or the connection |
| Shock risk present on an output believed to be isolated | Autotransformer fitted where an isolating transformer was specified | Test for continuity between input and output terminals with the unit de-energised | Replace with a genuine two-winding isolating transformer |
| Insulation resistance between windings has fallen | Moisture ingress or winding insulation degradation | Insulation resistance test between windings and to earth, de-energised | Dry out and retest, or withdraw the transformer from service |

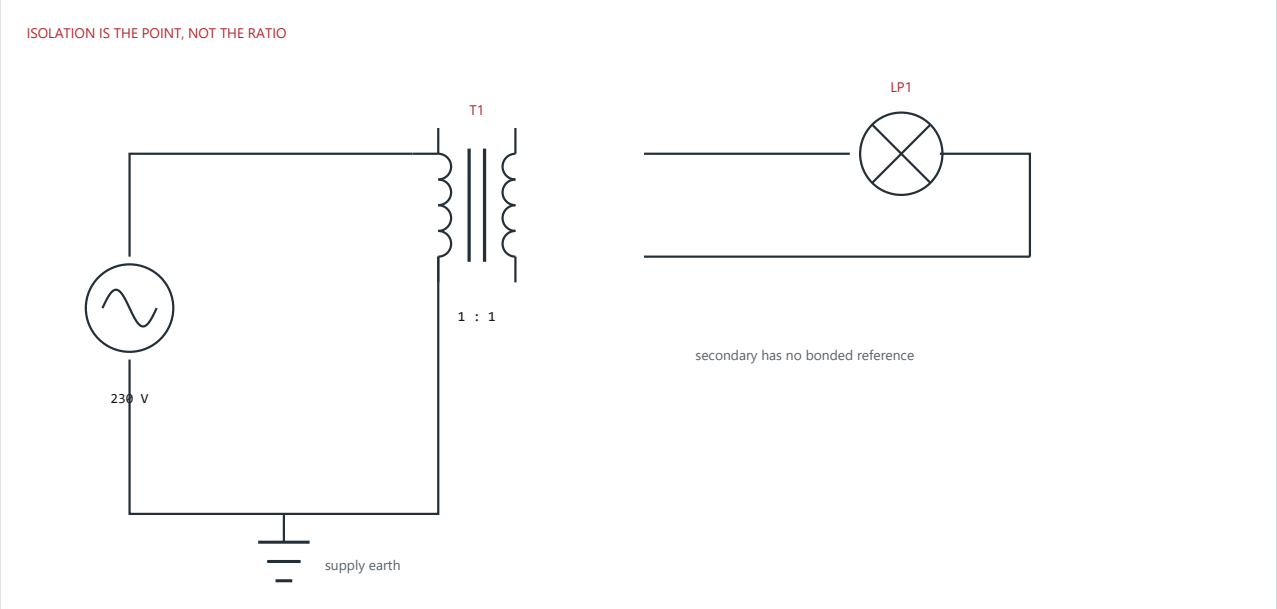



Fig. 10.T Verifying isolation. With the unit de-energised, a continuity test between primary and secondary distinguishes a true isolating transformer from an autotransformer.

FIELD HABITS THAT PREVENT CALLBACKS

Record the tap position and the no-load secondary voltage at commissioning, so later voltage complaints have a baseline.
Size secondary conductors and protection on the secondary current, never on the primary current.
Test for continuity between windings before relying on any transformer for isolation.
De-energise transformers serving no load, because core loss continues the whole time they are energised.
Treat a change in transformer noise as a real symptom and investigate the supply voltage.

## Exercises

Answer from first principles. Selected answers follow; the reasoning matters more than the number.

1. A transformer has 1200 primary and 100 secondary turns on 230 V. Find the secondary voltage.
2. The transformer above delivers 8 A. Find the primary current.
3. Find the apparent power of a transformer delivering 24 V at 12 A.
4. A 3 kVA transformer has a 48 V secondary. Find the rated secondary current.
5. State why transformers are rated in volt-amperes rather than watts.
6. Explain why a transformer cannot work on a direct-current supply.
7. A transformer has 40 W core loss and 90 W full-load copper loss. Find the copper loss at half load.
8. For the transformer above delivering 2 kW at full load, find the full-load efficiency.
9. A secondary measures 48 V unloaded and 45.5 V at full load. Find the regulation.
10. State the single most important difference between an autotransformer and a two-winding transformer.
11. Explain why inrush current can reach many times rated current at energisation.
12. A 20 : 1 step-down transformer has a secondary fault of 600 A. Find the primary current and state the protection implication.

## Selected answers

| # | Answer | Reason |
|---|---|---|
| 1 | 19.2 V | $V_s = V_p N_s/N_p$ |
| 2 | 0.667 A | $I_p = I_s N_s/N_p$ |
| 3 | 288 VA | S = V I |
| 4 | 62.5 A | I = S/V |
| 5 | Limits are set by current and flux, not power factor | A winding is heated by its current whatever the phase angle |
| 6 | Induced voltage needs changing flux | Steady current gives constant flux, so nothing is induced and the primary acts as a low-resistance coil |
| 7 | 22.5 W | P_cu scales with current squared, so 90 × 0.25 |
| 8 | 93.9 % | η = 2000/(2000 + 40 + 90) |
| 9 | 5.2 % | (48 − 45.5)/48 × 100 |
| 10 | The autotransformer gives no galvanic isolation | Its windings share conductors, so input and output are conductively connected |
| 11 | Residual core magnetism plus switching instant | The core can be driven into deep saturation for a few cycles, briefly collapsing the limiting impedance |
| 12 | 30 A; secondary protection is required | A severe secondary fault looks like a modest primary overload, so the primary device may not clear it in time |

## Chapter summary

- A transformer couples two windings magnetically, so it changes voltage with no electrical connection and no moving parts, but only on alternating supply.
- Voltage follows the turns ratio, current follows its inverse, and impedance follows the square of the ratio.
- Rating is in volt-amperes because winding current and core flux, not load power factor, set the limits.
- Core loss is constant with load while copper loss rises as current squared, so efficiency peaks where the two are equal.
- Regulation measures the voltage fall from no load to full load and must be allowed for in downstream design.
- Only a two-winding transformer provides isolation; an autotransformer does not.
- Inrush current at energisation can be many times rated current and protection must tolerate it.

Equations introduced in this chapter

| Relationship | Expression | When it applies |
|---|---|---|
| Turns ratio | $V_p/V_s = N_p/N_s$ | Two-winding transformer |
| Current ratio | $I_s/I_p = N_p/N_s$ | Ideal transformer, power balance |
| Impedance ratio | $Z_p/Z_s = (N_p/N_s)^2$ | Referring a load to the primary |
| Apparent power | $S = V\,I$ | Single-phase rating |
| Copper loss at part load | $P_cu = P_cu(full)(I/I_full)^2$ | Any load current |
| Efficiency | $\eta = P_out/(P_out + P_core + P_cu)$ | Any load |
| Regulation | $(V_nl - V_fl)/V_nl \times 100$ | No load to full load |

Vocabulary check

| Term | Meaning in this chapter |
|---|---|
| Turns ratio | The ratio of primary to secondary turns, which fixes the voltage ratio |
| Galvanic isolation | Absence of any conductive path between two circuits |
| Autotransformer | A single tapped winding serving as both primary and secondary, with no isolation |
| Core loss | Hysteresis and eddy-current loss, present whenever the unit is energised |
| Regulation | The percentage fall in secondary voltage from no load to full load |
| Inrush current | The brief heavy current drawn at energisation as the core is driven toward saturation |
| Tap | A connection point on a winding used to adjust the effective turns ratio |

LOOKING AHEAD

A transformer converts voltage but produces no motion. Adding a rotating magnetic field to the same principles gives the induction motor, which the next chapter examines along with generators and machine control.

CHAPTER 11

# Rotating machines

Motors, generators, slip, torque and starting

WHY THIS CHAPTER MATTERS

Motors consume the majority of the electrical energy used in industry, and a motor that is wrongly started, wrongly protected or wrongly loaded is the most common cause of both energy waste and unplanned downtime. Understanding slip, torque and starting current explains most of what goes wrong.

## What you will be able to do

- Explain how a three-phase supply produces a rotating magnetic field.
- Compute synchronous speed, slip and rotor frequency.
- Relate torque, speed and mechanical power output.
- Explain why starting current is high and describe methods of reducing it.
- Interpret a motor nameplate and use it to select protection.
- Describe generator action and the effect of applying load.

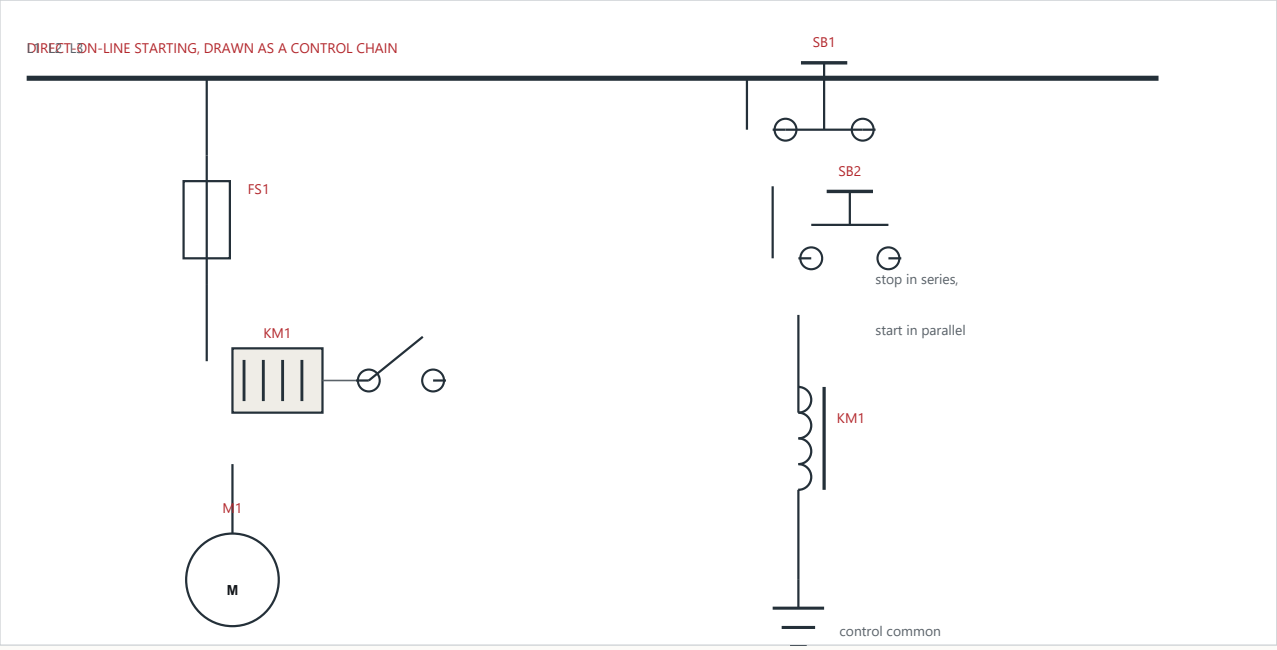



Fig. 11.1 A direct-on-line starter. The contactor seals in through its own auxiliary contact, and the overload relay opens the control circuit if the motor draws excessive current.

## The rotating magnetic field

Three windings spaced 120 mechanical degrees apart and fed from a three-phase supply produce a magnetic field of constant magnitude that rotates at a speed fixed by the supply frequency and the number of poles. Nothing mechanical rotates to produce it; the rotation comes from the sequential rise and fall of the three phase currents.

This is the central elegance of the three-phase induction motor. The rotating field sweeps past the rotor conductors, inducing voltage and current in them, and the resulting force drags the rotor around after the field. Reversing any two supply phases reverses the direction of field rotation and therefore of the motor.

$$n_s = 120 f / p \qquad \text{(revolutions per minute)}$$

$f$ in hertz, $p$ the number of poles

At 50 Hz: 4 poles gives 1500 rev/min; 60 Hz gives 1800 rev/min

REVERSING A THREE-PHASE MOTOR

Exchanging any two of the three supply conductors reverses rotation.
The third conductor and the protective conductor are left undisturbed.
Confirm direction with the coupling removed or the load safely free to turn.
Never reverse a motor by altering the protective earthing connection.

## Slip and rotor frequency

The rotor can never quite reach synchronous speed. If it did, there would be no relative motion between the field and the rotor conductors, no flux would be cut, no current would be induced, and no torque would be produced. The rotor therefore settles at a speed slightly below synchronous, and that difference is the slip.

Slip is small at light load and grows as load increases, because more relative motion is needed to induce the larger rotor current that larger torque requires. Typical full-load slip for a general-purpose induction motor is 2 to 5 per cent. Rotor frequency is the supply frequency multiplied by the fractional slip, so at standstill the rotor sees full supply frequency and at running speed it sees only a few hertz.

$$s = (n_s - n_r) / n_s \qquad \text{(fractional slip)}$$

$$\text{Slip \%} = s \times 100$$

$$f_r = s f \qquad \text{(rotor frequency)}$$

$$n_r = n_s (1 - s)$$

The nameplate speed is the full-load speed, not the synchronous speed. A nameplate reading 1440 rev/min on a 50 Hz supply indicates a four-pole machine with 4 per cent slip.

## Torque, speed and power

Mechanical output power is the product of torque and angular speed. For the same output power, a slower machine must develop proportionally more torque, which is why low-speed motors and geared drives are physically larger for a given rating.

Torque is not constant across the speed range. Starting torque is typically one and a half to two and a half times full-load torque, torque then rises to a pull-up and breakdown maximum and falls away as the machine approaches synchronous speed. A load whose torque demand exceeds the available torque at any speed will stall the motor there.

```
P = 2π n T / 60      (n in rev/min, T in newton metres, P in watts)

T = 9.55 P / n

Efficiency η = mechanical output / electrical input
```

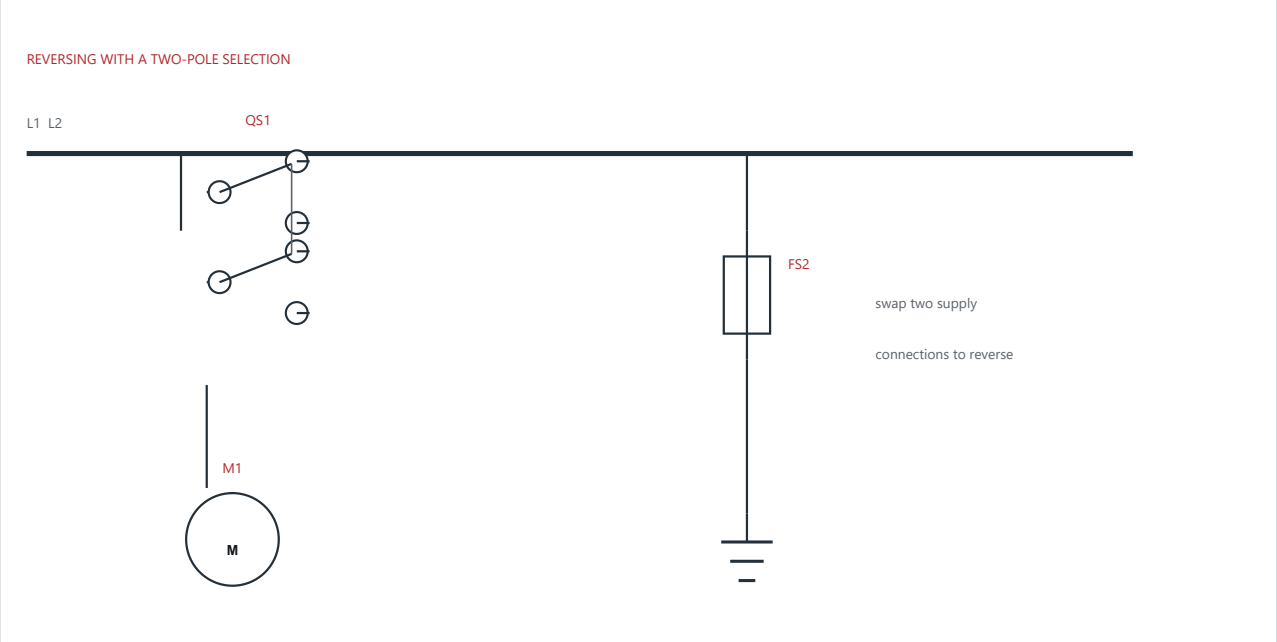



Fig. 11.2 A reversing arrangement. Two contactors supply the motor with two phases exchanged, and they are interlocked so that both can never close together.

## Starting current and starting methods

At the instant of starting, the rotor is stationary, slip is unity and the machine behaves much like a transformer with a short-circuited secondary. The current drawn is typically five to eight times full-load current, and it persists until the rotor accelerates.

Direct-on-line starting is simplest and gives full starting torque, but the current step may cause supply voltage dips that disturb other equipment. Star-delta starting reconnects the windings to reduce the starting current to about one third, but reduces starting torque in the same proportion, so it suits only loads that start lightly. Soft starters and variable frequency drives control the applied voltage or frequency electronically and give the smoothest start, at greater cost and complexity.

| Starting method | Starting current | Starting torque | Suited to |
|---|---|---|---|
| Direct on line | 5 to 8 times full load | Full, 150 to 250 % | Small motors, stiff supplies, loads needing full torque |

| Starting method | Starting current | Starting torque | Suited to |
|---|---|---|---|
| Star-delta | About one third of direct on line | About one third | Fans, pumps, unloaded compressors |
| Autotransformer | Selectable by tap | Reduced as the square of voltage | Larger machines needing a chosen compromise |
| Soft starter | Ramped and adjustable | Ramped and adjustable | Belt drives and loads sensitive to shock |
| Variable frequency drive | Near full-load current | Full torque from zero speed | Variable-speed duty and high-inertia loads |

STARTING CURRENT IS A DESIGN CONSTRAINT, NOT A NUISANCE

A six-times starting current on a 40 A motor draws 240 A for several seconds.
Protection must tolerate that without tripping, yet still clear a genuine fault.
Supply impedance determines how far the resulting voltage dip propagates.
Frequent starting accumulates heating; respect the stated starts-per-hour limit.

## Reading a nameplate

The nameplate is the authoritative source for protection and connection decisions. It states rated voltage and the connection it applies to, full-load current, output power, full-load speed, frequency, power factor, efficiency class, insulation class, enclosure rating and duty type.

Two entries cause the most confusion. The stated power is mechanical output at the shaft, not electrical input, so input power is always higher by the efficiency. And where two voltages are shown, each corresponds to a particular winding connection, so a motor marked for both 400 V in star and 230 V in delta must be connected to suit the actual supply.

Nameplate entries that drive decisions

- Full-load current sets the overload relay setting and the cable size.
- Rated voltage must be read together with the stated connection.
- Full-load speed reveals the pole count and the slip.
- Insulation class fixes the permitted temperature rise.
- Duty type states whether continuous running is permitted.
- Enclosure rating must suit the installed environment.

## Generator action

The same machine run as a generator reverses the energy flow. Driving the rotor mechanically while its field is excited induces voltage in the stator windings, and the frequency produced depends on the driven speed and the pole count.

Applying electrical load makes the machine harder to turn. The load current produces its own field, which opposes the rotation by Lenz's law, so the prime mover must supply more mechanical power. This is why a generator's engine note changes audibly as load is applied and why governor response determines how well frequency is held.

```
f = p n / 120      (generated frequency)

Mechanical input = electrical output + losses
```

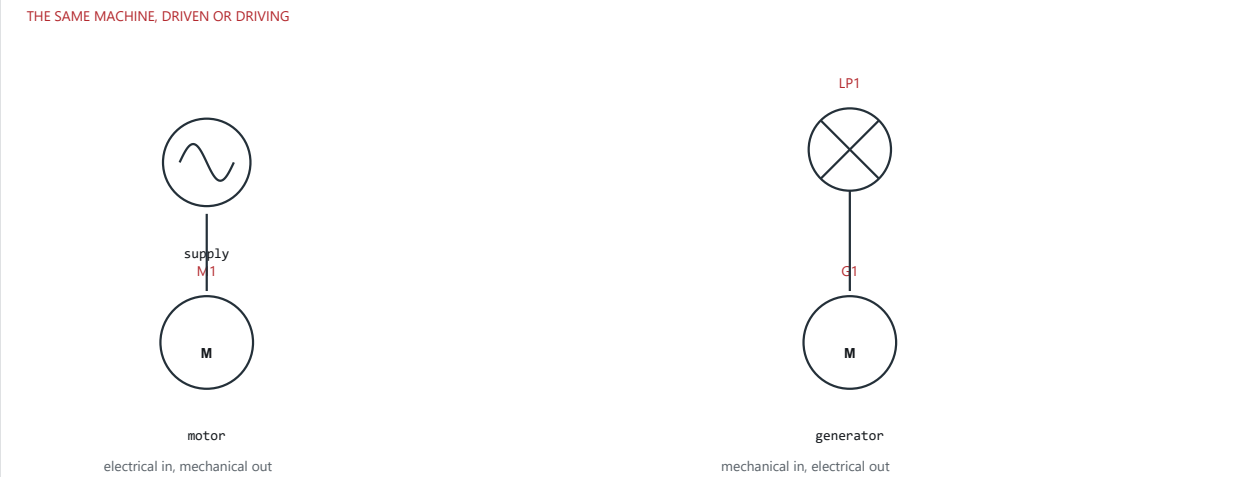



Fig. 11.3 A generator supplying a load. Mechanical drive turns the excited rotor, and the load current produces an opposing field that the prime mover must work against.

## Worked examples

Work each example with a pencil before reading the solution. Sketch the circuit, mark the reference node, and write the governing law before substituting numbers.

EXAMPLE 11.1 Synchronous speed, slip and rotor frequency

**Given.** A four-pole induction motor on a 50 Hz supply runs at 1440 rev/min at full load.

**Find.** The synchronous speed, the slip, and the rotor frequency.

Synchronous speed

$n_s = 120 \times 50/4 = 1500$ rev/min

Slip speed

$1500 - 1440 = 60$ rev/min

Fractional slip

$s = 60/1500 = 0.04$, that is 4 %

Rotor frequency

$f_r = 0.04 \times 50 = 2$ Hz

At standstill

$s = 1$, so $f_r = 50$ Hz

**Check.** Four per cent slip is normal for a general-purpose machine. Note that the rotor sees only 2 Hz when running but full supply frequency at the instant of starting, which is why starting conditions are so much more severe.

EXAMPLE 11.2 Torque from rated output and speed

**Given.** A motor is rated 7.5 kW at 1440 rev/min.

**Find.** The full-load torque, and the torque if the same power were produced at 720 rev/min.

Torque equation

$T = 9.55\ P / n$

At 1440 rev/min

$T = 9.55 \times 7500/1440 = 49.7$ N·m

At 720 rev/min

$T = 9.55 \times 7500/720 = 99.5$ N·m

Observation

Halving the speed doubles the torque for the same power

**Check.** The low-speed machine must develop twice the torque, so its shaft, bearings and frame are larger for the same rating. This is the reason an eight-pole motor is physically bigger than a four-pole motor of equal power.

EXAMPLE 11.3 Starting current and overload setting

**Given.** A motor has a nameplate full-load current of 18 A and a starting current six times full load.

**Find.** The starting current, a suitable overload relay setting, and the implication for cable sizing.

Starting current

```
6 × 18 = 108 A
```

Duration

```
Typically a few seconds until the rotor accelerates
```

Overload relay setting

```
Set at the nameplate full-load current, 18 A
```

Cable sizing

```
Sized on 18 A continuous, with voltage drop checked at 108 A
```

Protective device

```
Must tolerate 108 A for the acceleration time without tripping
```

**Check.** The overload relay is set on full-load current because it protects against sustained overloading; its time characteristic is what allows the starting surge through. The cable carries 18 A continuously, but the momentary 108 A must not cause an unacceptable voltage dip.

EXAMPLE 11.4 Input power from mechanical output

**Given.** A 5.5 kW motor has an efficiency of 0.88 and a power factor of 0.85 on a 400 V three-phase supply.

**Find.** The electrical input power, the apparent power and the line current.

Input power

```
P_in = 5500/0.88 = 6250 W
```

Apparent power

```
S = 6250/0.85 = 7353 VA
```

Line current

```
I = 7353/(√3 × 400) = 10.6 A
```

Compare

```
A nameplate would show roughly 10.6 A full-load current
```

**Check.** The nameplate kilowatt figure is shaft output, so input is always higher. Both efficiency and power factor must be applied to arrive at the current the cable and protection must handle.

## Practical activity: motor nameplate, starting current, slip and direction

> AIM
> Interpret a nameplate and predict the machine's behaviour, then measure starting current, running current, slip at two loads and verify the effect of exchanging two supply phases.

### Equipment

- Three-phase cage induction motor, 0.75 to 1.5 kW, on a guarded test bed.
- Direct-on-line starter with an adjustable overload relay, and a star-delta starter if available.
- Clamp meter with inrush or peak-hold capability.
- Non-contact tachometer.
- Mechanical brake or dynamometer load, with a guard.
- True rms multimeter and a three-phase power meter.
- Lock-off devices and a voltage proving unit.

### Procedure

1. Record every entry from the nameplate before energising anything.
2. From the nameplate speed and frequency, compute the pole count, the synchronous speed and the full-load slip, and write the predictions down.
3. From the rated output, efficiency and power factor, predict the full-load line current and compare it with the nameplate figure.
4. Prove the supply dead, lock off, and confirm the motor terminal connection matches the supply voltage and the nameplate.
5. Set the overload relay to the nameplate full-load current.
6. With the coupling guarded and the load disconnected, start the motor direct on line and capture the starting current with the clamp meter on inrush.
7. Measure the no-load running current and the no-load speed.
8. Apply the brake to roughly half of rated torque. Record current, speed and input power.
9. Increase to full-load torque, then record current, speed and input power again.
10. Compute slip at each load condition from your measured speeds.
11. Stop the motor, prove dead, lock off, and exchange two supply conductors.
12. Restart and confirm the direction of rotation has reversed.
13. If a star-delta starter is available, repeat the starting-current measurement and compare.

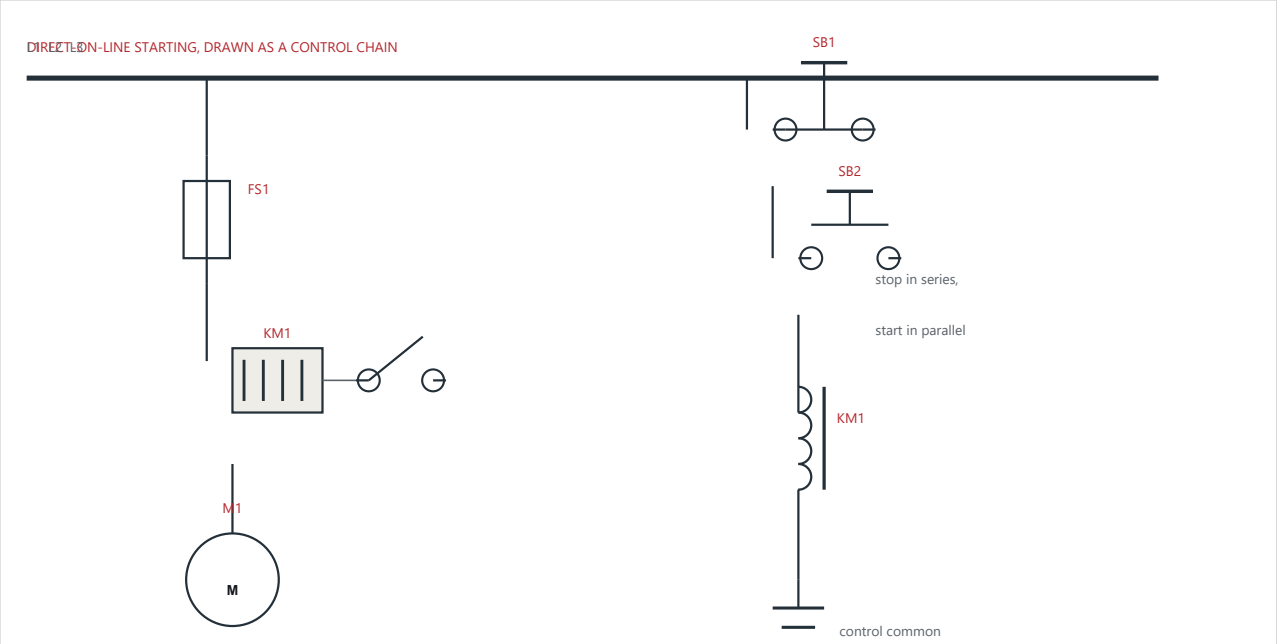



Fig. 11.L The direct-on-line test arrangement, with the overload relay in the control circuit and the clamp meter positioned on one supply conductor.

## Results table

| Condition | Current (A) | Speed (rev/min) | Slip (%) | Input power and comment |
|---|---|---|---|---|
| Starting, direct on line | | | | |
| No load, running | | | | |
| Half load | | | | |
| Full load | | | | |
| Starting, star-delta | | | | |
| Reversed rotation check | | | | |

## Discussion

1. What was the ratio of your measured starting current to full-load current, and how does that compare with the usual five to eight times?
2. How did slip change between no load, half load and full load? Explain the trend using induced rotor current.
3. Why does the motor draw appreciable current at no load, and what is that current doing?
4. Compare your predicted full-load current with the measured value and account for any difference.
5. If you used the star-delta starter, by what factor did starting current fall, and what happened to starting torque?
6. The nameplate speed is below synchronous speed. Explain why a cage induction motor can never run at synchronous speed.

7 Compute the full-load torque from your measured output and speed, and compare it with the brake reading.

SAFE-WORK REMINDERS

Rotating machinery must be fully guarded, with couplings and shaft ends enclosed before energising.
Prove dead and lock off before touching any motor terminal, and before exchanging phases.
Confirm the direction of rotation with the load disconnected or safely free to turn.
Tie back long hair, remove loose clothing, jewellery and lanyards near the shaft.
Starting current is many times running current. Confirm the clamp meter range before capturing it.
Respect the stated starts-per-hour limit; repeated starting overheats the windings.
Allow the brake and the motor frame to cool before handling.

## Troubleshooting guide

Faults are found by narrowing the circuit, not by replacing parts. Each row below pairs an observed symptom with the measurement that separates the likely causes.

| Symptom | Likely cause | Decisive test | Correct action |
|---|---|---|---|
| Motor hums but will not start, and draws heavy current | Single phasing from one lost supply conductor or one open winding | Measure current in all three conductors; they should be nearly equal | Restore the lost phase or rewind; investigate why the conductor opened |
| Overload relay trips a few seconds after each start | Relay set below full-load current, or the starting time exceeds the relay class | Compare the setting with the nameplate and time the acceleration | Set the relay to nameplate full-load current and select a suitable trip class |
| Motor runs hot at normal current | Blocked ventilation, high ambient temperature, or excessive starts per hour | Check airflow and measure frame temperature; count the starts | Clear the cooling path, reduce start frequency, or fit a machine rated for the duty |
| Motor runs in the wrong direction after rewiring | Two supply conductors exchanged relative to the original connection | Confirm rotation with the load uncoupled | Exchange any two supply conductors, leaving the protective conductor alone |
| Speed is well below the nameplate figure under load | Mechanical overload, or reduced supply voltage lowering available torque | Measure terminal voltage under load and compare current with full load | Reduce the load, correct the voltage, or fit a larger machine |
| Motor stalls partway through acceleration | Load torque exceeds the machine's torque at that speed, often with reduced-voltage starting | Compare the load torque curve with the motor torque curve | Start direct on line, use a soft starter, or fit a machine with higher pull-up torque |
| Bearing noise and rising vibration over weeks | Bearing wear, shaft misalignment or an unbalanced coupling | Measure vibration and check alignment at the coupling | Replace the bearings and realign; correct the cause of the misalignment |

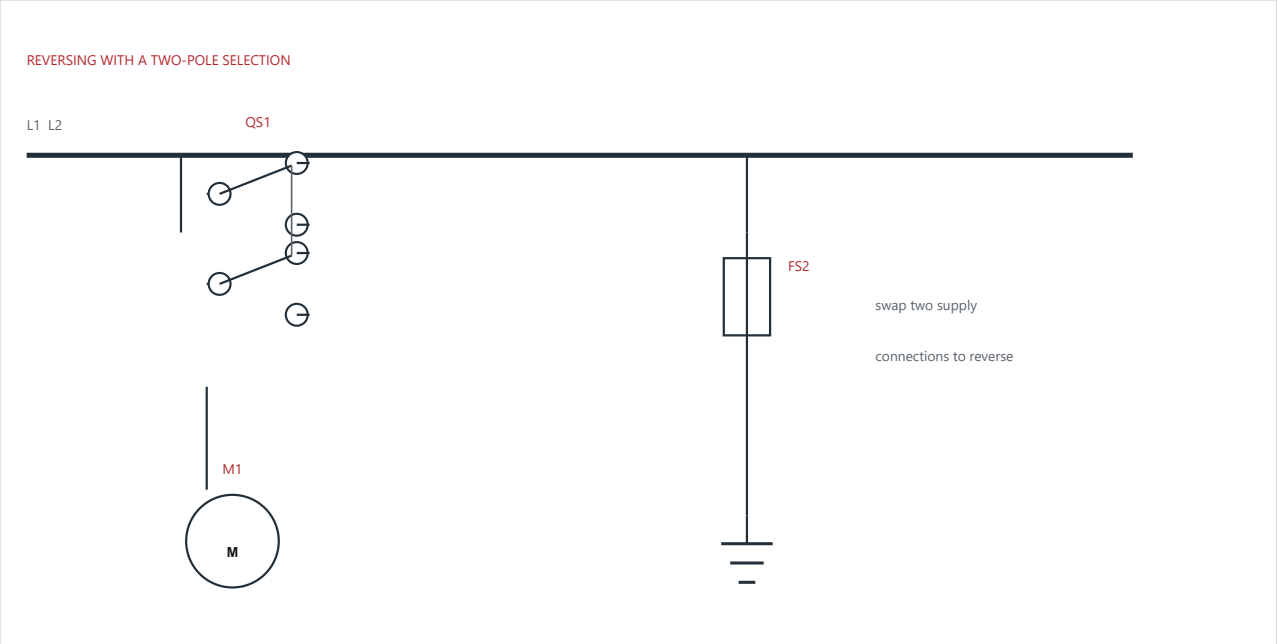



Fig. 11.T Interlocked reversing contactors. The interlock prevents both contactors closing together, which would short the supply between two phases.

FIELD HABITS THAT PREVENT CALLBACKS

Set overload relays to the nameplate full-load current, not to a convenient round number.

Measure current in all three conductors when a motor misbehaves; an imbalance points straight to single phasing.

Confirm rotation direction with the load uncoupled before coupling a pump or compressor.

Photograph the nameplate at commissioning, because it becomes illegible long before the motor fails.

Count starts per hour on cyclic duty, since cumulative starting heat destroys more windings than running load does.

## Exercises

Answer from first principles. Selected answers follow; the reasoning matters more than the number.

1 Find the synchronous speed of a six-pole motor on a 50 Hz supply.

2 A four-pole 60 Hz motor runs at 1740 rev/min. Find the slip as a percentage.

3 Find the rotor frequency for 5 % slip on a 50 Hz supply.

4 State why a cage induction motor cannot run at synchronous speed.

5 Find the full-load torque of a 4 kW motor running at 1455 rev/min.

6 Find the output power of a motor developing 65 N·m at 960 rev/min.

7 A motor has a full-load current of 24 A and starts at seven times that. Find the starting current.

8 State the setting you would apply to the overload relay for the motor above, and why.

9 State the approximate effect of star-delta starting on starting current and on starting torque.

10 A nameplate reads 400 V, 11 kW, 0.86 power factor, efficiency 0.9. Find the full-load line current on a three-phase supply.

11 Explain why a generator becomes mechanically harder to turn as electrical load is applied.

12 A motor nameplate reads 1455 rev/min at 50 Hz. Determine the pole count and the full-load slip.

13 Explain why exchanging two supply conductors reverses a three-phase motor.

14 State two nameplate entries that must be checked before selecting the supply cable, and why.

### Selected answers

| # | Answer | Reason |
|---|---|---|
| 1 | 1000 rev/min | $n_s = 120f/p$ |
| 2 | 3.3 % | $n_s = 1800$; slip = 60/1800 |
| 3 | 2.5 Hz | $f_r = s\,f$ |
| 4 | No relative motion means no induced current | Without slip no flux is cut, so no rotor current and no torque exist |
| 5 | 26.3 N·m | $T = 9.55\,P/n$ |
| 6 | 6.53 kW | $P = 2\pi nT/60$ |
| 7 | 168 A | Starting current is a multiple of full-load current |
| 8 | 24 A | The relay guards against sustained overload; its time characteristic passes the starting surge |
| 9 | Both fall to about one third | Reduced winding voltage reduces current, and torque falls with the square of voltage |
| 10 | 20.5 A | P_in = 11000/0.9 = 12222 W; $I = 12222/(\sqrt{3} \times 400 \times 0.86)$ |
| 11 | Load current produces an opposing field | By Lenz's law the reaction opposes rotation, so the prime mover must supply more mechanical power |
| 12 | Four poles, 3 % slip | $n_s = 1500$, so slip = 45/1500 |
| 13 | It reverses the direction of field rotation | The phase sequence determines which way the rotating field sweeps |
| 14 | Full-load current and rated voltage with its connection | Current sets the cross-section and the voltage with connection confirms the machine suits the supply |

## Chapter summary

- A three-phase supply in distributed windings produces a magnetic field that rotates at a speed fixed by frequency and pole count.
- Slip is essential: without relative motion between field and rotor there is no induced current and no torque.
- Mechanical power is the product of torque and speed, so low-speed machines are larger for the same rating.
- Starting current is five to eight times full load because the stationary rotor behaves as a shorted secondary.
- Reduced-voltage starting cuts current but cuts torque by the square of the voltage reduction.
- The nameplate states shaft output, not input, and its voltage entry must be read with the stated connection.
- Loading a generator produces an opposing field, so electrical output must be matched by mechanical input.

### Equations introduced in this chapter

| Relationship | Expression | When it applies |
|---|---|---|
| Synchronous speed | $n_s = 120\,f / p$ | Three-phase machine |
| Fractional slip | $s = (n_s - n_r)/n_s$ | Induction machine |
| Rotor speed | $n_r = n_s(1 - s)$ | Induction machine |
| Rotor frequency | $f_r = s\,f$ | Induction machine |
| Mechanical power | $P = 2\pi\,n\,T / 60$ | n in rev/min |
| Torque from power | $T = 9.55\,P / n$ | n in rev/min, P in watts |
| Three-phase power | $P = \sqrt{3}\,V_L\,I_L \cos\varphi$ | Balanced load |
| Generated frequency | $f = p\,n / 120$ | Synchronous generator |

### Vocabulary check

| Term | Meaning in this chapter |
|---|---|
| Synchronous speed | The rotational speed of the magnetic field, set by frequency and poles |
| Slip | The fractional difference between synchronous and rotor speed |
| Breakdown torque | The maximum torque a machine can develop before stalling |
| Direct on line | Starting by applying full supply voltage through a single contactor |
| Star-delta | A starting method that reconnects the windings to reduce starting current |
| Duty type | The nameplate classification of permitted running and starting pattern |
| Single phasing | Operation with one supply conductor lost, producing severe imbalance |

LOOKING AHEAD

Machines and transformers rely on magnetic coupling. The next chapter turns to semiconductors, where conduction is controlled within a solid material rather than by a field, beginning with the diode and the rectifier circuits built from it.

CHAPTER 12

# Diodes and rectification

One-way conduction, and converting alternating supply to direct

WHY THIS CHAPTER MATTERS

Almost every piece of modern equipment contains a rectifier, because electronics needs direct supply while distribution provides alternating. The diode is the component that makes the conversion possible, and understanding its forward drop, its reverse limit and its heat explains most rectifier failures.

## What you will be able to do

- Describe forward and reverse behaviour of a junction diode and its forward voltage drop.
- Identify half-wave, full-wave centre-tapped and bridge rectifier circuits from a schematic.
- Compute mean and root-mean-square output values for rectifier circuits.
- Select a diode on the basis of forward current and peak reverse voltage.
- Explain how a reservoir capacitor reduces ripple and what it does to peak current.
- Apply Zener diodes for reference and simple regulation, and compute the series resistor.

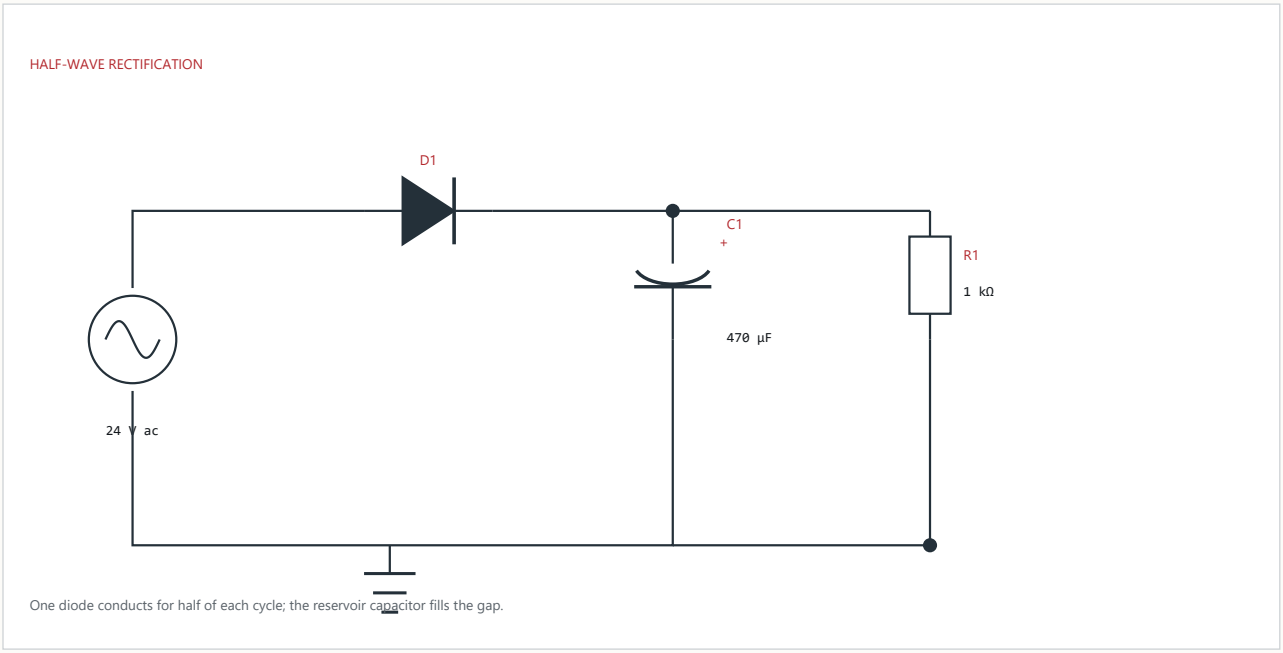



Fig. 12.1 A half-wave rectifier. The diode conducts on one polarity only, so the load receives current during alternate half-cycles.

## The junction diode

A diode conducts readily in one direction and blocks in the other. In the forward direction, conduction begins once the applied voltage exceeds the junction's threshold, about 0.7 volts for silicon and about 0.3 volts for germanium or a Schottky type. Beyond that point the forward drop changes only slightly as current rises, so it is usually treated as a constant.

In the reverse direction only a negligible leakage current flows, until the reverse voltage reaches the breakdown value. For an ordinary rectifier diode, breakdown is destructive, so the peak reverse voltage the circuit will impose must be known and the diode rated above it with a margin.

```
Forward drop, silicon:  V_F ≈ 0.7 V

Power dissipated in the diode:  P = V_F I_F

Reverse condition:  V_R must stay below the rated peak reverse voltage
```

THE FORWARD DROP IS A HEAT PROBLEM, NOT JUST A VOLTAGE LOSS

A diode carrying 8 A at 0.8 V dissipates 6.4 W as heat.
That heat must leave through the package and its mounting.
A rectifier without adequate heatsinking fails thermally, not electrically.
Schottky types drop less and run cooler, at the cost of higher leakage.

## Half-wave rectification

A single diode in series with the load passes current on one polarity only. The load therefore receives half-cycles separated by gaps, which gives a low mean output and a large ripple content. The diode must withstand the full peak supply voltage in reverse during the blocked half-cycles.

Half-wave rectification is cheap and occasionally adequate, but it wastes half of each cycle, it draws current from the supply in one direction only, and it needs a much larger reservoir capacitor for the same ripple. It is used mainly in very low-power circuits.

```
V_mean = 0.318 V_pk      V_rms = 0.5 V_pk

Ripple frequency = supply frequency

Peak reverse voltage across the diode = V_pk
```

## Full-wave and bridge rectification

A full-wave rectifier uses both half-cycles. With a centre-tapped secondary and two diodes, each diode handles one half-cycle; with four diodes in a bridge, no centre tap is needed and the whole secondary winding is used on every half-cycle.

The bridge is almost always preferred. It makes better use of the transformer, and although two diodes conduct in series so the total forward drop doubles, the output is higher than the centre-tapped arrangement gives from the same winding. The ripple frequency is twice the supply frequency, which makes smoothing easier.

```
V_mean = 0.637 V_pk      V_rms = 0.707 V_pk

Ripple frequency = 2 × supply frequency

Bridge output ≈ V_pk − 2 V_F      Centre-tap output ≈ V_pk − V_F
```

| Property | Half wave | Centre-tapped full wave | Bridge |
|---|---|---|---|
| Diodes required | 1 | 2 | 4 |
| Mean output | 0.318 V_pk | 0.637 V_pk | 0.637 V_pk |
| Ripple frequency | Supply frequency | Twice supply | Twice supply |
| Forward drops in path | One | One | Two |
| Transformer utilisation | Poor | Half winding per half-cycle | Full winding |
| Peak reverse voltage | V_pk | 2 V_pk per diode | V_pk |

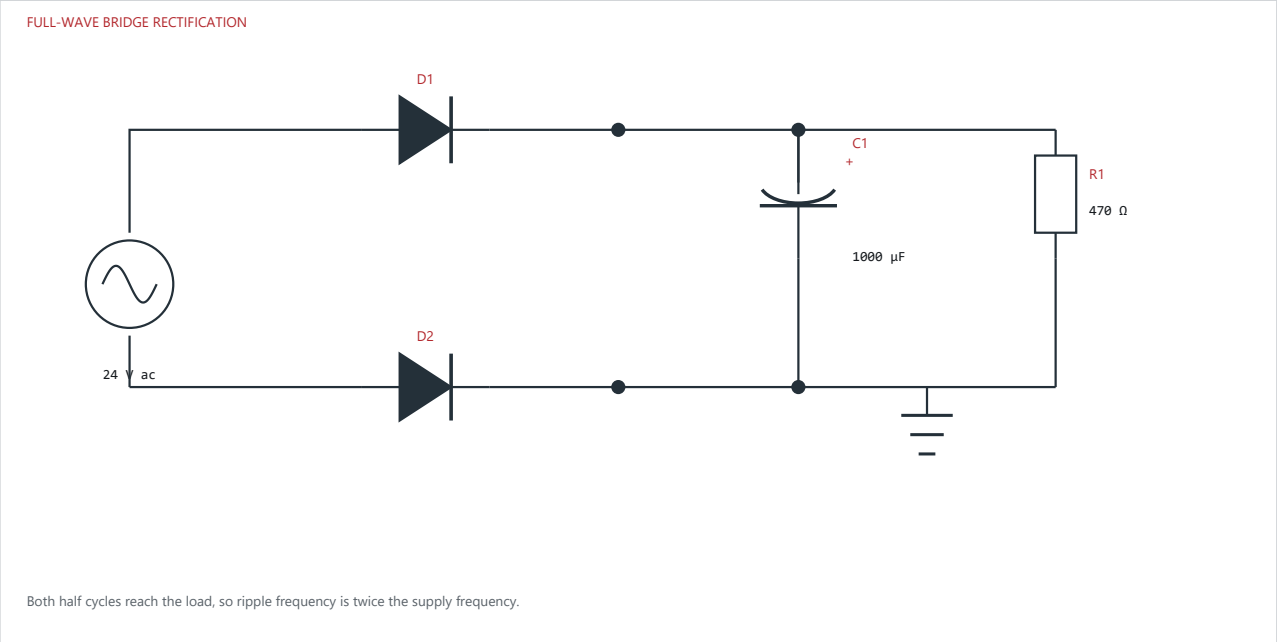



Fig. 12.2 A bridge rectifier. Two diodes conduct on each half-cycle, so current through the load is always in the same direction.

## Smoothing and the reservoir capacitor

Rectified output is unidirectional but far from steady. A capacitor across the load charges to near the peak and then supplies the load during the intervals when the rectifier output falls below the capacitor voltage, so the output becomes a nearly steady voltage with a small sawtooth ripple superimposed.

The price of smoothing is current. The capacitor recharges only during a short period near each peak, so the peak repetitive current through the diodes and the transformer is many times the mean load current. Larger capacitors give less ripple but shorter, higher charging pulses, which is why oversizing a reservoir capacitor can destroy the rectifier it was meant to protect.

```
Ripple voltage ≈ I_load / (f_ripple C)

Larger C or higher f_ripple gives less ripple

Peak diode current   mean load current
```

WHY A BIGGER CAPACITOR IS NOT ALWAYS BETTER

Smoothing works by recharging the capacitor in a brief pulse near each peak.
Doubling the capacitance roughly halves the ripple but shortens the charging window.
Peak repetitive diode current and transformer heating both rise.
Check the diode's repetitive peak current rating, not only its mean rating.

## Zener diodes and simple regulation

A Zener diode is designed to break down at a specified reverse voltage without damage, and to hold that voltage nearly constant over a wide current range. Operated in reverse with a series resistor to limit the current, it provides a stable reference or a simple regulated supply for a light load.

The series resistor must be chosen so that enough current flows to keep the Zener in breakdown at the minimum input voltage and maximum load current, while the Zener's dissipation stays within rating at the maximum input voltage and minimum load current. Those two conditions bracket the acceptable resistor values.

### Selecting a Zener regulator

- Operate the Zener in reverse; it is not a forward conduction device.
- Always include the series resistor, or the Zener will be destroyed.
- Check dissipation at maximum input with the load disconnected.
- Keep the Zener current above a few milliamperes for a stable voltage.
- Use a Zener as a reference for larger loads, not as the regulator itself.

```
R = (V_in − V_Z) / (I_Z + I_load)

P_Z = V_Z I_Z      P_R = (V_in − V_Z) I_total

Worst case for the Zener: maximum V_in with no load
```

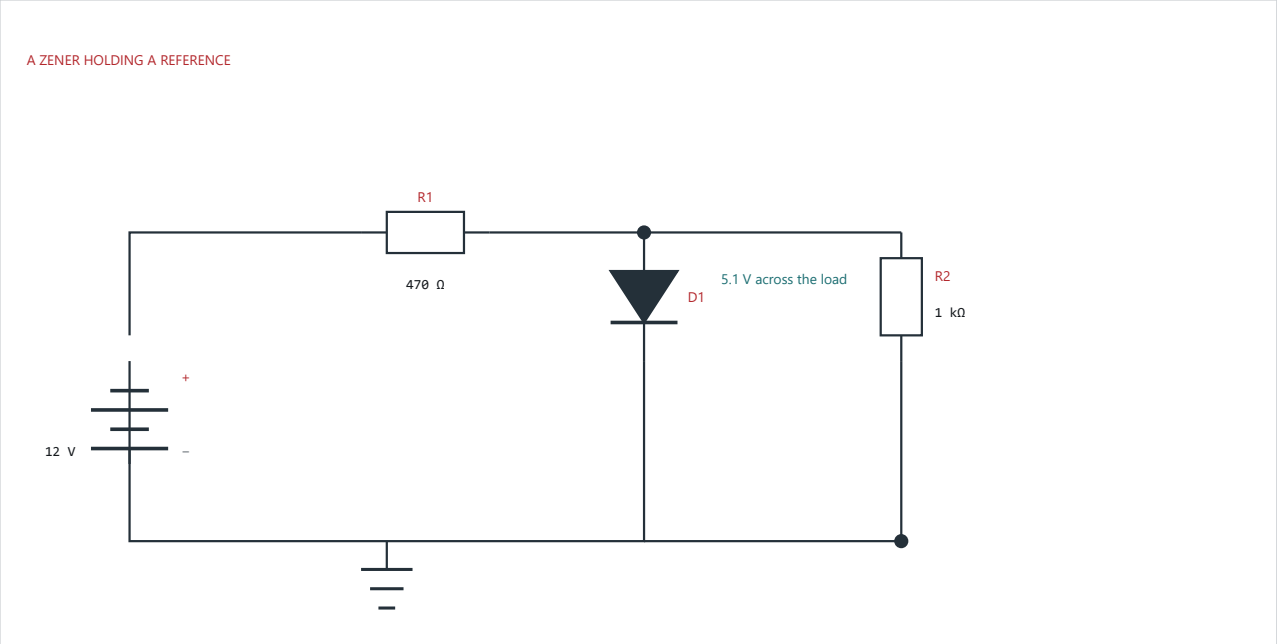



Fig. 12.3 A Zener diode as a shunt regulator. The series resistor sets the total current and the Zener absorbs whatever the load does not take.

## Light-emitting diodes and other variants

A light-emitting diode is a forward-conducting junction that emits light, with a forward drop between about 1.8 and 3.4 volts depending on colour. Like any diode it has no internal current limiting, so a series resistor or a constant-current supply is mandatory; connecting one directly across a supply destroys it immediately.

Other variants adapt the same junction to specific tasks. Schottky diodes switch fast and drop less. Photodiodes convert light to current. Varactors present a voltage-dependent capacitance. Each appears in the symbol set with a distinguishing mark on the standard diode outline.

| Device | Operated | Characteristic exploited | Typical use |
|---|---|---|---|
| Rectifier diode | Forward | One-way conduction | Mains rectification |
| Schottky | Forward | Low drop, fast recovery | Switching supplies |
| Zener | Reverse breakdown | Constant breakdown voltage | Reference and regulation |
| Light-emitting | Forward | Light emission | Indication and illumination |
| Photodiode | Reverse | Light-dependent current | Sensing and detection |
| Varactor | Reverse | Voltage-dependent capacitance | Tuning circuits |

## Worked examples

Work each example with a pencil before reading the solution. Sketch the circuit, mark the reference node, and write the governing law before substituting numbers.

EXAMPLE 12.1 Half-wave and bridge outputs from the same supply

**Given.** A transformer secondary gives 24 V root mean square.

**Find.** The peak value, and the mean output as a half-wave rectifier and as a bridge, allowing for silicon forward drops.

Peak voltage

```
V_pk = 24 × 1.414 = 33.9 V
```

Half-wave mean, ideal

```
0.318 × 33.9 = 10.8 V
```

Bridge mean, ideal

```
0.637 × 33.9 = 21.6 V
```

Bridge smoothed output

```
33.9 − 1.4 = 32.5 V across the capacitor
```

Peak reverse voltage, bridge

```
About 33.9 V per diode
```

**Check.** The bridge gives twice the mean output of the half-wave circuit from the same winding. Note that the smoothed output approaches the peak, not the mean, which is why a 24 V winding produces about 32 V of direct supply.

EXAMPLE 12.2 Diode selection for a rectifier

**Given.** A bridge rectifier supplies 5 A at 30 V direct from a 24 V root-mean-square winding.

**Find.** The minimum diode ratings and the heat each diode must dissipate.

Peak reverse voltage present

```
V_pk = 33.9 V
```

Rating with margin

```
Select at least 100 V peak reverse voltage
```

Mean current per diode

```
Each conducts half the time, so about 2.5 A mean
```

Current rating with margin

```
Select at least 6 A continuous
```

Dissipation per diode

```
P = 0.8 × 2.5 = 2 W
```

Total bridge dissipation

```
About 8 W, requiring a heatsink
```

**Check.** Eight watts in a small bridge package needs real heatsinking. Rating a rectifier only on voltage and mean current, without checking the thermal path, is the usual reason bridges fail in service.

EXAMPLE 12.3 Reservoir capacitor for a specified ripple

**Given.** A bridge rectifier on a 50 Hz supply must hold ripple below 1 V peak to peak while supplying 0.8 A.

**Find.** The minimum reservoir capacitance.

Ripple frequency

```
Bridge on 50 Hz gives 100 Hz
```

Rearrange the ripple relation

```
C = I / (f_ripple V_ripple)
```

Substitute

```
C = 0.8/(100 × 1)
```

Result

```
C = 8000 µF
```

Practical choice

```
Fit 10 000 µF rated above 50 V
```

**Check.** Note how large smoothing capacitors must be for appreciable load current at mains frequency. A half-wave circuit would need twice this value for the same ripple, because its ripple frequency is half as high.

EXAMPLE 12.4 Series resistor for a Zener regulator

**Given.** A 5.6 V Zener is to supply a load of up to 20 mA from an input that varies between 12 V and 15 V. A minimum Zener current of 5 mA is required.

**Find.** The series resistor, and the Zener dissipation in the worst case.

Worst case for resistor sizing

```
Minimum input, maximum load
```

Current required

```
I = 20 + 5 = 25 mA
```

Resistor value

```
R = (12 − 5.6)/0.025 = 256 Ω, so use 240 Ω
```

Worst case for the Zener

```
Maximum input, no load
```

Total current then

```
I = (15 − 5.6)/240 = 39 mA, all through the Zener
```

Zener dissipation

```
P = 5.6 × 0.039 = 0.22 W, so use a 0.5 W device
```

**Check.** The two worst cases are different conditions and both must be checked. The Zener is most stressed when the load is disconnected, which is exactly the condition a casual bench test might overlook.

EXAMPLE 12.5 Series resistor for a light-emitting diode

**Given.** A red light-emitting diode with a 2.0 V forward drop is to operate at 15 mA from a 24 V supply.

**Find.** The series resistor and its dissipation.

Voltage across the resistor

```
24 − 2.0 = 22 V
```

Resistor value

```
R = 22/0.015 = 1467 Ω, so use 1.5 kΩ
```

Resulting current

```
I = 22/1500 = 14.7 mA
```

Resistor dissipation

```
P = 22 × 0.0147 = 0.32 W, so use a 0.5 W part
```

**Check.** Most of the supply voltage is dropped across the resistor, which is why the resistor dissipates far more than the diode. Running indicators from a high supply voltage this way is wasteful but simple.

# Practical activity: diode characteristic, rectifier waveforms and a Zener regulator

AIM

Measure the forward characteristic of a silicon diode, display and measure half-wave and bridge rectifier waveforms with and without smoothing, then test the line and load regulation of a Zener shunt regulator.

## Equipment

- Silicon diodes, 1 A, four required for the bridge.
- Schottky diode and a 5.6 V Zener diode, 0.5 W.
- Transformer, 230 V to 24 V, supplied through an isolating transformer.
- Resistors: 100 Ω, 240 Ω, 1.5 kΩ and a 0 to 1 kΩ decade box.
- Electrolytic capacitors, 100 µF and 1000 µF, rated above 50 V.
- Oscilloscope with a differential or isolated input, and two multimeters.
- Variable direct supply, 0 to 20 V, current limited.

## Procedure

1. With the variable direct supply and a 100 Ω series resistor, raise the voltage across the silicon diode in small steps and record forward voltage against forward current up to 100 mA.
2. Repeat for the Schottky diode and note the difference in threshold.
3. Reverse the silicon diode, apply 10 V, and record the leakage current.
4. Plot the forward characteristic and identify the threshold voltage from your data.
5. Build the half-wave rectifier from the transformer secondary into a 1 kΩ load.
6. Display the load waveform and measure its peak and mean values. Record the ripple frequency.
7. Add the 100 µF capacitor across the load and record the new mean output and the ripple.
8. Replace the single diode with the four-diode bridge and repeat both measurements.
9. Change the capacitor to 1000 µF and record the ripple again.
10. Reduce the load resistance in steps and record how ripple varies with load current.
11. Build the Zener regulator with the 240 Ω series resistor from the variable direct supply.
12. Hold the load constant and vary the input from 8 V to 18 V, recording the output each time.
13. Hold the input at 15 V and vary the load from open circuit to 25 mA, recording the output.

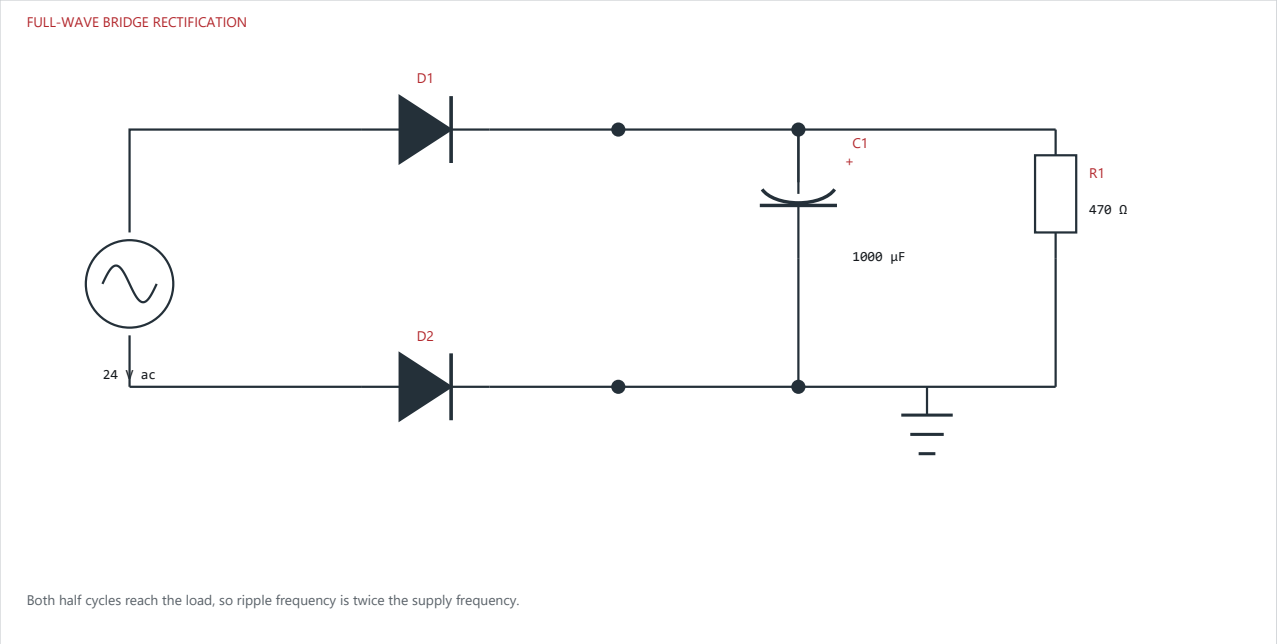



Fig. 12.L The bridge rectifier under test. The oscilloscope is connected across the load through an isolated input, and the capacitor is switched in and out.

## Results table

| Configuration | Mean output (V) | Ripple (V pk-pk) | Ripple frequency (Hz) | Comment |
|---|---|---|---|---|
| Half wave, no capacitor | | | | |
| Half wave, 100 µF | | | | |
| Bridge, no capacitor | | | | |
| Bridge, 100 µF | | | | |
| Bridge, 1000 µF | | | | |
| Bridge, 1000 µF, heavy load | | | | |
| Zener output, 8 V input | | | | |
| Zener output, 18 V input | | | | |

## Discussion

1. From your forward characteristic, at what voltage did appreciable conduction begin? Compare silicon with the Schottky device.
2. Why was the bridge ripple frequency twice that of the half-wave circuit, and why does that make smoothing easier?
3. Compare your smoothed bridge output with the peak of the secondary waveform. Why is it close to the peak rather than to the unsmoothed mean?

4 How did ripple change when you increased the capacitance tenfold? Compare the measured change with the prediction from the ripple relation.

5 How did ripple change as the load current rose, and why?

6 Over your input range from 8 V to 18 V, how much did the Zener output vary? Express that as a percentage.

7 What happened to the Zener output when the input fell to 8 V, and can you explain it using the minimum Zener current?

8 Compute the Zener dissipation at 18 V input with no load and compare it with the device rating.

SAFE-WORK REMINDERS

Supply the bench through an isolating transformer and a residual current device.

Do not connect an earthed oscilloscope ground to any point of a mains-referenced rectifier; use an isolated or differential input.

Observe electrolytic capacitor polarity. A reversed electrolytic can rupture.

A smoothing capacitor holds dangerous charge after the supply is removed. Discharge it through a resistor and prove it dead before touching the circuit.

Check resistor and Zener dissipation before applying power, not after.

Diodes and heatsinks become hot. Allow them to cool before handling.

## Troubleshooting guide

Faults are found by narrowing the circuit, not by replacing parts. Each row below pairs an observed symptom with the measurement that separates the likely causes.

| Symptom | Likely cause | Decisive test | Correct action |
|---|---|---|---|
| Rectifier diodes fail repeatedly in service | Peak reverse voltage rating too low, or supply transients exceeding it | Compute the peak reverse voltage present and compare with the rating | Fit diodes rated well above the peak, and add transient suppression |
| Bridge rectifier runs very hot and eventually fails | Inadequate heatsinking for the forward drop at the operating current | Compute V_F I_F per diode and measure the package temperature | Fit an adequate heatsink, improve airflow, or use Schottky devices |
| Output voltage sags badly under load | Reservoir capacitance too small, or the transformer is undersized | Measure the ripple and the output at load against the no-load figure | Increase the capacitance and verify the transformer volt-ampere rating |
| Smoothing capacitor fails after a capacitance upgrade | Peak repetitive diode and capacitor current rose with the larger capacitor | Measure the charging pulse current with a current probe | Return to a suitable capacitance or fit components rated for the ripple current |
| Zener regulator output falls below its rated voltage | Input too low or load too high, so Zener current fell below the breakdown minimum | Measure the current through the series resistor at the worst case | Reduce the series resistor or raise the input voltage |
| Zener diode overheats with the load disconnected | All the series current diverted through the Zener at maximum input | Compute V_Z I_Z at maximum input with no load | Select a higher-wattage Zener or a larger series resistor |
| Light-emitting diode fails instantly on connection | Connected without a series current-limiting resistor, or reverse connected | Check for a series resistor and confirm polarity | Fit the correct series resistor and observe polarity |

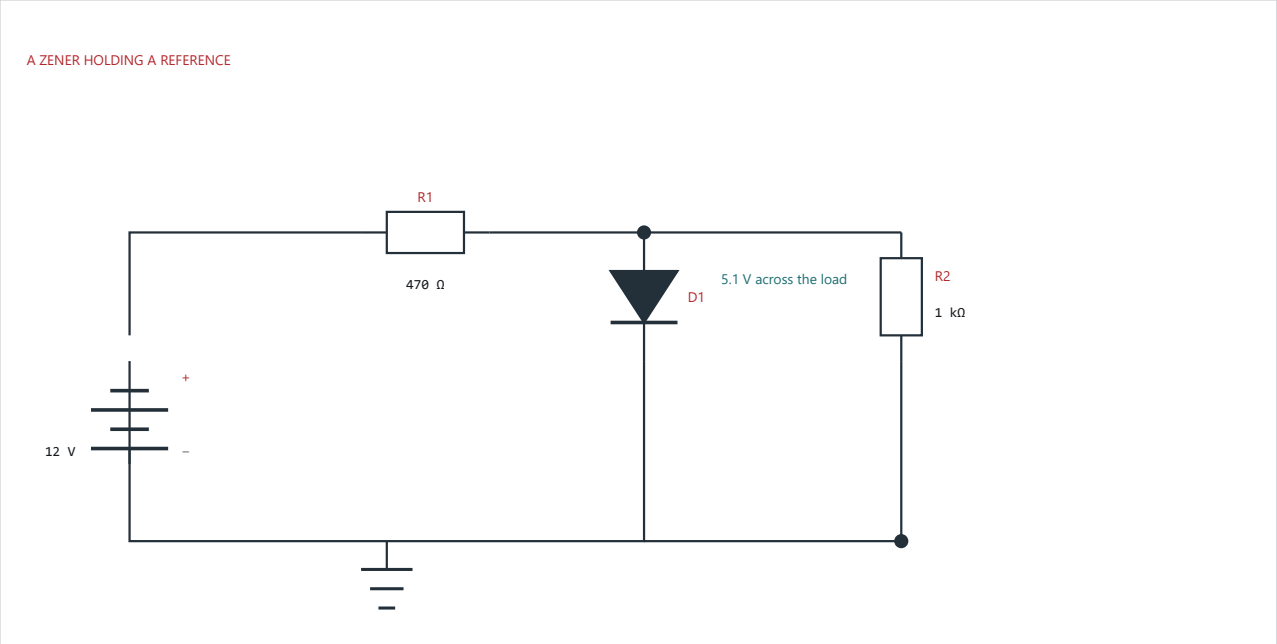



Fig. 12.T Testing the Zener regulator. Line regulation is checked by varying the input; load regulation by varying the load while the input is held constant.

FIELD HABITS THAT PREVENT CALLBACKS

Compute diode dissipation and provide for the heat before selecting a rectifier package.
Rate the peak reverse voltage with generous margin, because supply transients are not optional events.
Discharge smoothing capacitors through a resistor and prove dead before working on any rectifier.
Check the Zener worst case with the load disconnected, which is the condition that destroys it.
Never connect a light-emitting diode without a current-limiting resistor or a constant-current supply.

## Exercises

Answer from first principles. Selected answers follow; the reasoning matters more than the number.

1 A transformer secondary gives 18 V root mean square. Find the peak value.
2 Find the ideal mean output of a half-wave rectifier from that secondary.
3 Find the ideal mean output of a bridge rectifier from the same secondary.
4 State the ripple frequency of a bridge rectifier on a 60 Hz supply.
5 Explain why a smoothed bridge output approaches the peak rather than the mean value.
6 A diode carries 4 A with a 0.75 V forward drop. Find the power it dissipates.
7 State the number of diodes in a bridge and how many conduct at any instant.
8 Find the reservoir capacitance for 2 V ripple at 1 A from a bridge on a 50 Hz supply.
9 Explain why increasing a reservoir capacitor can shorten the life of the rectifier diodes.
10 A 9.1 V Zener feeds a 25 mA load from 18 V, with 6 mA minimum Zener current. Find the series resistor.
11 Find the Zener dissipation above at 18 V with the load disconnected.
12 State the operating region of a Zener diode and why a series resistor is mandatory.
13 A 3.0 V light-emitting diode is to run at 20 mA from 12 V. Find the series resistor.
14 State one advantage and one disadvantage of a Schottky diode compared with a silicon rectifier diode.

## Selected answers

| # | Answer | Reason |
|---|---|---|
| 1 | 25.5 V | V_pk = 1.414 V_rms |
| 2 | 8.1 V | 0.318 V_pk |
| 3 | 16.2 V | 0.637 V_pk |
| 4 | 120 Hz | A bridge gives twice the supply frequency |
| 5 | The capacitor charges to near the peak | It then holds that level between peaks, so the output follows the peak rather than the average of the unsmoothed waveform |
| 6 | 3 W | P = V_F I_F |
| 7 | Four diodes; two conduct at a time | Each half-cycle uses a diagonally opposite pair |
| 8 | 5000 µF | C = I/(f_ripple V_ripple) with f_ripple = 100 Hz |
| 9 | Peak repetitive current rises | The capacitor recharges in a shorter, higher pulse, heating the diodes and the transformer |
| 10 | 287 Ω, so use 270 Ω | R = (18 − 9.1)/0.031 |
| 11 | About 0.3 W | All 33 mA flows through the Zener at 9.1 V |
| 12 | Reverse breakdown; the resistor limits current | Without it the current is unlimited and the device is destroyed |
| 13 | 450 Ω | R = (12 − 3.0)/0.020 |
| 14 | Lower forward drop; higher reverse leakage | Less heat in conduction but poorer blocking performance |

## Chapter summary

- A diode conducts above its forward threshold and blocks until its reverse breakdown voltage is reached, where an ordinary rectifier fails.
- The forward drop multiplied by the current is heat, and thermal design determines rectifier reliability.
- Half-wave rectification uses one half-cycle; full-wave and bridge circuits use both and double the mean output.
- A bridge needs no centre tap, uses the whole winding, and gives twice the supply ripple frequency.
- A reservoir capacitor raises the output to near the peak and reduces ripple, but raises peak repetitive diode current.
- A Zener diode operated in reverse breakdown with a series resistor provides a reference or simple regulation, with its worst case at maximum input and no load.
- Every light-emitting diode requires external current limiting.

### Equations introduced in this chapter

| Relationship | Expression | When it applies |
|---|---|---|
| Silicon forward drop | $V_F \approx 0.7\ V$ | Conducting junction diode |
| Diode dissipation | $P = V_F I_F$ | Forward conduction |
| Half-wave mean | $V_{mean} = 0.318\ V_{pk}$ | Resistive load, no smoothing |
| Full-wave mean | $V_{mean} = 0.637\ V_{pk}$ | Resistive load, no smoothing |
| Smoothed output | $V \approx V_{pk} - n V_F$ | With reservoir capacitor, n diodes in path |
| Ripple voltage | $V_{ripple} \approx I/(f_{ripple} C)$ | Capacitor-smoothed rectifier |
| Zener series resistor | $R = (V_{in} - V_Z)/(I_Z + I_{load})$ | Shunt regulator |
| Indicator resistor | $R = (V_{supply} - V_F)/I_F$ | Light-emitting diode circuit |

### Vocabulary check

| Term | Meaning in this chapter |
|---|---|
| Forward drop | The nearly constant voltage across a conducting diode |
| Peak reverse voltage | The highest reverse voltage a diode must withstand without breakdown |
| Ripple | The alternating component remaining on a rectified and smoothed output |
| Reservoir capacitor | The capacitor that holds the output near the peak between rectifier pulses |
| Zener voltage | The specified reverse breakdown voltage a Zener diode maintains |
| Line regulation | The change in output caused by a change in input voltage |
| Load regulation | The change in output caused by a change in load current |

LOOKING AHEAD

A diode conducts or blocks according to the polarity applied to it, with no third terminal to intervene. Adding that control terminal gives the transistor, which can switch and amplify, and which the next chapter examines.

CHAPTER 13

# Transistors

Switching and amplifying with a controlled third terminal

WHY THIS CHAPTER MATTERS

A diode's conduction is decided entirely by the polarity across it. A transistor adds a third terminal that controls conduction between the other two, which makes electronic switching and amplification possible. Every logic gate, every solid-state relay and every variable-speed drive rests on this one capability.

## What you will be able to do

- Identify bipolar and field-effect transistors and their terminals from the symbols.
- Describe the three operating regions and state which are used for switching.
- Compute base, collector and emitter currents using current gain.
- Design a transistor switch to drive a relay or lamp, including the base resistor.
- Explain why a field-effect transistor is voltage controlled and what that implies.
- Compute switching and conduction losses and explain why saturation matters.

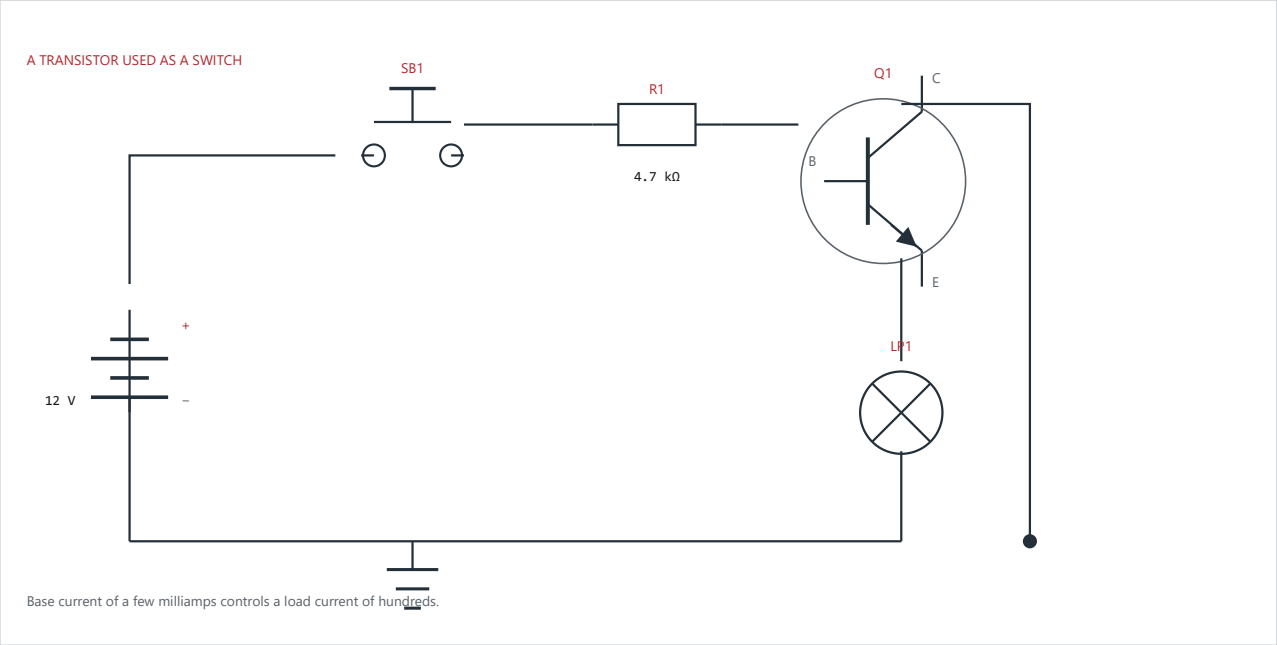



Fig. 13.1 A transistor as a low-side switch driving a relay coil. The base resistor sets the drive current and the suppression diode handles the coil's reverse voltage.

## The bipolar junction transistor

A bipolar transistor has three terminals: emitter, base and collector. A small current into the base allows a much larger current to flow from collector to emitter, and the ratio between them is the current gain. Because a current controls a current, the bipolar device always consumes drive current while it is on.

Two polarities exist. In an n-p-n device the collector sits positive and base current flows into the base; in a p-n-p device the polarities are reversed throughout. The symbols are distinguished only by the direction of the emitter arrow, which always points in the direction of conventional current through the emitter.

Reading a bipolar transistor symbol

- The arrow is always on the emitter, never on the collector.
- Arrow pointing out of the device: n-p-n.
- Arrow pointing into the device: p-n-p.
- The base is the terminal on the flat bar.
- A second arrow on the base region indicates a phototransistor.
- Two transistors drawn in cascade within one outline indicates a Darlington.

```
I_E = I_B + I_C      (currents always balance)

β = I_C / I_B      (current gain, often 50 to 400)

I_C = β I_B      in the active region only
```

## The three operating regions

With no base current the device is cut off: essentially no collector current flows and the full supply appears across the device. With a large base current the device is saturated: the collector current is limited by the external circuit rather than by the gain, and the collector-to-emitter voltage falls to a few tenths of a volt.

Between those two states lies the active region, where collector current is proportional to base current. Amplifiers work in the active region; switches deliberately avoid it, because a device carrying substantial current with substantial voltage across it dissipates substantial heat. A switch should pass through the active region quickly and settle in saturation or cut-off.

| Region | Base drive | Collector-emitter voltage | Dissipation | Used for |
|---|---|---|---|---|
| Cut-off | None | Full supply | Negligible | Switch off state |
| Active | Proportional | Intermediate | High | Amplification |
| Saturation | Generous excess | 0.2 to 0.4 V | Low | Switch on state |

A SWITCH MUST BE DRIVEN HARD INTO SATURATION

In saturation the device drops a few tenths of a volt and dissipates little.
Partial drive leaves it in the active region, where it dissipates heavily.
Provide base current well above I_C divided by the minimum gain.
A transistor switch that runs hot is almost always underdriven.

## Designing a transistor switch

The design sequence is short. Establish the load current the collector must carry, take the worst-case minimum gain from the data sheet, and compute the base current needed. Then multiply that by a factor of three to five to guarantee saturation across temperature and device spread.

The base resistor follows from the drive voltage less the base-emitter drop of about 0.7 volts, divided by the chosen base current. Where the load is a coil or any other inductive device, a suppression diode across the load is essential: interrupting coil current produces a reverse voltage that will otherwise exceed the transistor's rating and destroy it.

```
I_B(min) = I_C / β(min)

I_B(design) = 3 to 5 × I_B(min)

R_B = (V_drive − 0.7) / I_B(design)

P = V_CE(sat) × I_C      in the on state
```

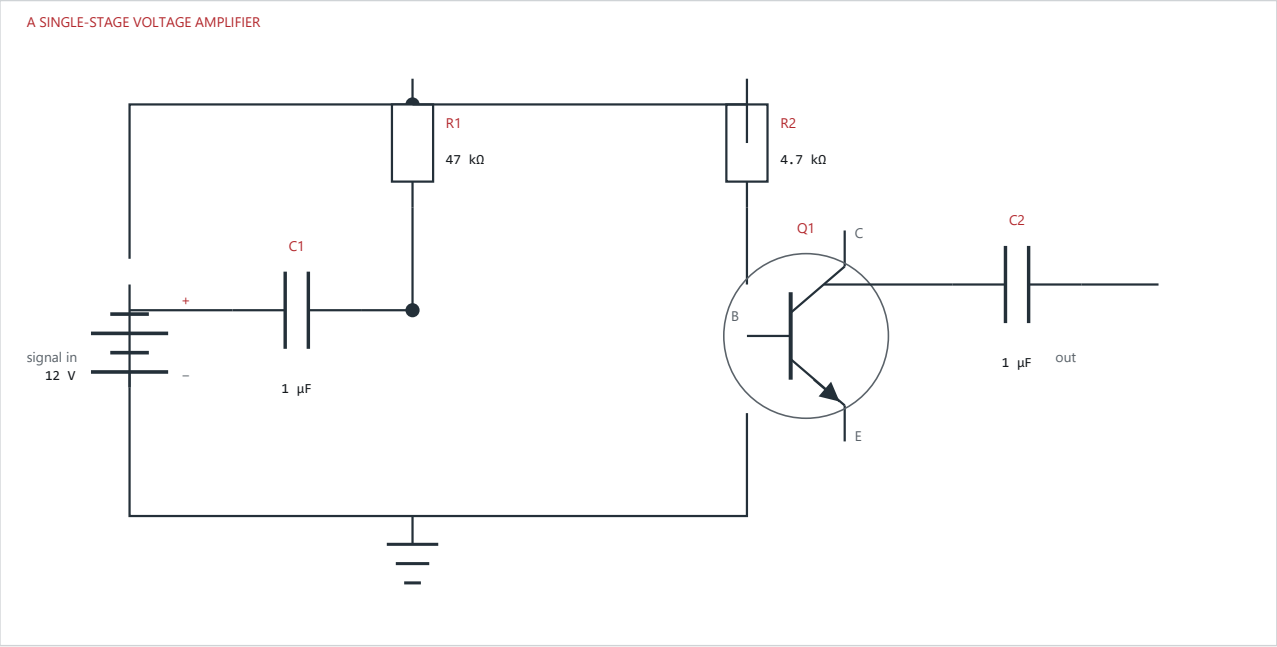



Fig. 13.2 A single-stage amplifier. The bias network holds the device in the active region and the signal is superimposed on that operating point.

## Field-effect transistors

A field-effect transistor controls current between drain and source by means of a voltage on an insulated or reverse-biased gate, rather than by a current. Once switched, the gate draws almost no current, so the drive circuit need supply only the brief charge that the gate capacitance requires.

In the metal-oxide-semiconductor type, the gate is insulated by a thin oxide layer that is easily punctured by static discharge or by gate overvoltage, so handling precautions and gate protection matter. Junction types use a reverse-biased junction instead and are more robust but offer less switching performance.

```
Gate voltage controls drain current; gate current ≈ 0

On-state loss:  P = I_D² R_DS(on)
```

```
Gate charge:  Q = C_gate V_gate
```

| Property | Bipolar transistor | Field-effect transistor |
|---|---|---|
| Controlled by | Base current | Gate voltage |
| Steady drive power | Continuous base current | Almost none once switched |
| On-state loss | V_CE(sat) × I_C | $I_D^2 \times R_{DS(on)}$ |
| Parallel operation | Difficult, current hogging | Straightforward, resistance rises with temperature |
| Static sensitivity | Low | High for insulated-gate types |
| Typical switching speed | Moderate | Fast |

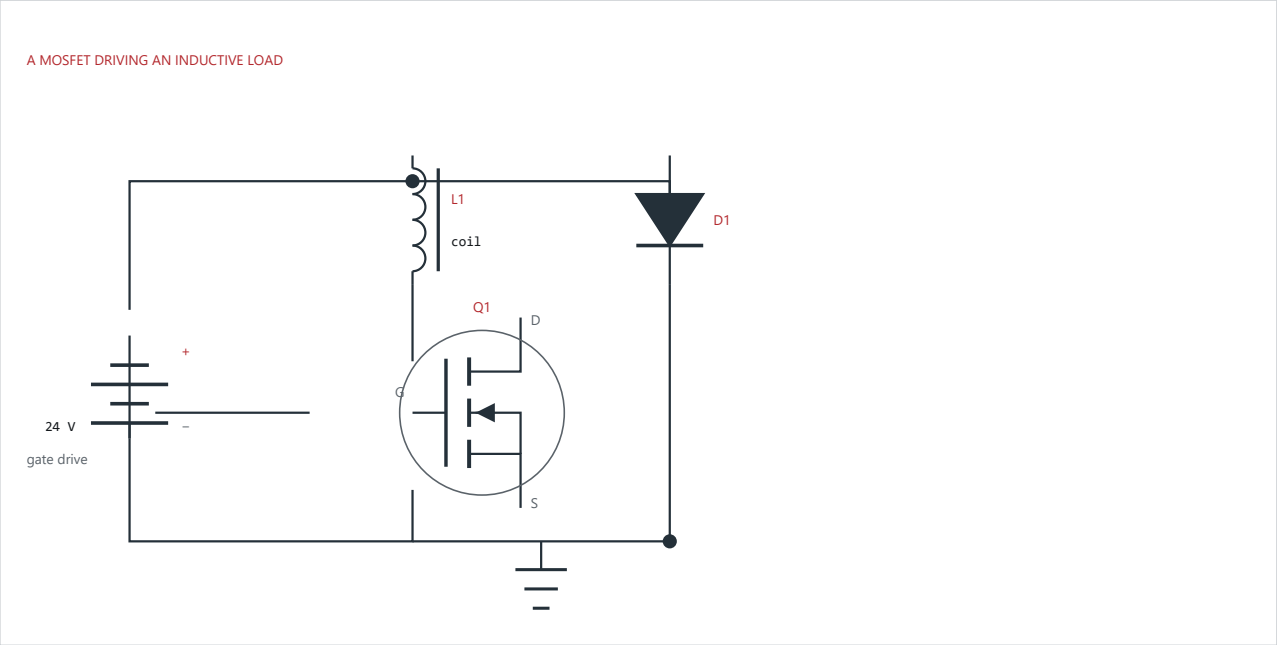



Fig. 13.3 A metal-oxide-semiconductor field-effect transistor switching a load. The gate is driven by voltage and draws current only while the gate capacitance charges.

## Heat, saturation and parallel devices

On-state loss differs fundamentally between the two families. A bipolar device drops a roughly fixed saturation voltage, so its loss rises in proportion to current. A field-effect device behaves as a resistance, so its loss rises as the square of current, which makes it excellent at low current and progressively worse at high current.

That resistance rises with temperature, which has a useful consequence: paralleled field-effect devices share current automatically, because the hottest device becomes the most resistive and sheds load to the others. Bipolar devices do the opposite and must be ballasted with emitter resistors if paralleled at all.

### Keeping a solid-state switch alive

- Drive the gate or base hard enough to reach full saturation or full enhancement.

- Fit a suppression diode across every inductive load.
- Provide a thermal path sized on the computed on-state loss.
- Include a gate resistor and a gate-to-source resistor on insulated-gate devices.
- Observe static-handling precautions for insulated-gate parts.
- Check the device rating against the load's inrush, not only its running current.

SWITCHING LOSS IS SEPARATE FROM CONDUCTION LOSS

Conduction loss occurs while the device is fully on.
Switching loss occurs during each transition through the active region.
Switching loss is proportional to switching frequency.
A device cool at low frequency may overheat at high frequency with the same current.

## Worked examples

Work each example with a pencil before reading the solution. Sketch the circuit, mark the reference node, and write the governing law before substituting numbers.

EXAMPLE 13.1 Base current and base resistor for a relay driver

**Given.** A relay coil draws 90 mA at 24 V. The transistor has a minimum current gain of 60 and is driven from a 5 V logic output.

**Find.** The minimum base current, a suitable design base current, and the base resistor.

Collector current

```
I_C = 90 mA
```

Minimum base current

```
I_B = 90/60 = 1.5 mA
```

Design base current, factor of 4

```
I_B = 6 mA
```

Voltage across the base resistor

```
5 − 0.7 = 4.3 V
```

Base resistor

```
R_B = 4.3/0.006 = 717 Ω, so use 680 Ω
```

Resulting base current

```
4.3/680 = 6.3 mA
```

**Check.** The factor of four guarantees saturation despite gain spread and temperature. Note the suppression diode across the coil is not optional: without it the collapsing field will exceed the transistor's collector rating.

EXAMPLE 13.2 On-state dissipation compared across device types

**Given.** A load draws 3 A. A bipolar device saturates at 0.3 V; a field-effect device has an on resistance of 40 mΩ.

**Find.** The on-state loss of each, and which is preferable if the current rises to 10 A.

Bipolar loss at 3 A

$P = 0.3 \times 3 = 0.9\ W$

Field-effect loss at 3 A

$P = 3^2 \times 0.04 = 0.36\ W$

Bipolar loss at 10 A

$P = 0.3 \times 10 = 3.0\ W$

Field-effect loss at 10 A

$P = 10^2 \times 0.04 = 4.0\ W$

Conclusion

The field-effect device wins at 3 A, loses at 10 A with this resistance

**Check.** Because one loss is linear in current and the other is quadratic, there is a crossover current. For high-current duty a field-effect device must have a correspondingly lower on resistance to stay ahead.

EXAMPLE 13.3 Underdriven switch running in the active region

**Given.** A transistor with a gain of 100 is given only 0.2 mA of base current while its load demands 60 mA from a 24 V supply through a 400 Ω load.

**Find.** What the device actually does, and the dissipation.

Collector current the gain permits

$I_C = 100 \times 0.0002 = 20\ mA$

Load current demanded at saturation

$24/400 = 60\ mA$

Actual state

Active region, limited to 20 mA by insufficient drive

Voltage across the load

$0.020 \times 400 = 8\ V$

Voltage across the device

$24 - 8 = 16\ V$

Dissipation

$P = 16 \times 0.020 = 0.32\ W$

**Check.** Compare that 0.32 W with a properly saturated device, which would dissipate about 0.3 V times 60 mA, or 18 mW. Underdriving multiplied the heat by nearly twenty times while delivering a third of the current.

EXAMPLE 13.4 Gate drive charge and switching frequency

**Given.** A field-effect transistor has a gate charge of 50 nC and is driven to 10 V at a switching frequency of 40 kHz.

**Find.** The average gate drive current and the drive power.

Charge per switching cycle

$$Q = 50\ \text{nC, delivered twice per cycle}$$

Average gate current

$$I = Q\,f = 50\times10^{-9} \times 40\,000 = 2\ \text{mA}$$

Gate drive power

$$P = Q\,f\,V = 50\times10^{-9} \times 40\,000 \times 10 = 20\ \text{mW}$$

Steady-state gate current

$$\text{Essentially zero between transitions}$$

**Check.** The drive requirement is almost entirely dynamic. This is why a field-effect device needs negligible drive at low switching rates but a substantial gate driver at high frequency.

## Practical activity: transistor regions, a relay driver and comparing two switch devices

AIM

Map the three operating regions of a bipolar transistor, build and verify a relay driver including its suppression diode, then compare the on-state dissipation of a bipolar and a field-effect switch at the same load current.

### Equipment

- Bipolar transistors, n-p-n, medium power, with a data sheet.
- Field-effect transistor, n-channel enhancement type, logic-level gate.
- Relay, 24 V coil, and a suppression diode.
- Resistors: 220 Ω, 680 Ω, 4.7 kΩ, 100 kΩ and a 0 to 100 kΩ decade box.
- Two variable direct supplies, 0 to 24 V, current limited.
- Two multimeters, an oscilloscope, and a clip-on or infrared thermometer.
- Antistatic wrist strap and mat for handling insulated-gate devices.

### Procedure

1. Identify the terminals of each device from the data sheet before wiring anything.
2. Build the bipolar device with a 400 Ω collector load from a 12 V supply and a decade box feeding the base from a separate 5 V supply.
3. Set the decade box to its maximum and record collector current and collector-emitter voltage. Note which region this is.
4. Reduce the base resistance in steps, recording base current, collector current and collector-emitter voltage at each step.
5. Continue until collector-emitter voltage stops falling. Record the base current at which saturation was reached.
6. Compute the current gain from a mid-range pair of readings and compare with the data sheet.
7. Plot collector current against base current and mark the three regions on your graph.
8. At one clearly active-region point, compute the device dissipation and measure its temperature after two minutes.
9. Repeat the temperature measurement in saturation at the same collector current and compare.
10. Build the relay driver with the 680 Ω base resistor and the suppression diode fitted.
11. Confirm the relay operates, then observe the collector waveform on switch-off.
12. Remove the suppression diode, switch off once, and observe the collector voltage spike. Refit the diode immediately.
13. Wearing the wrist strap, build the field-effect device switching the same relay from a gate voltage, with a gate resistor and a gate-to-source resistor fitted.
14. Measure the on-state voltage across each device at the same load current and compute the dissipation of each.

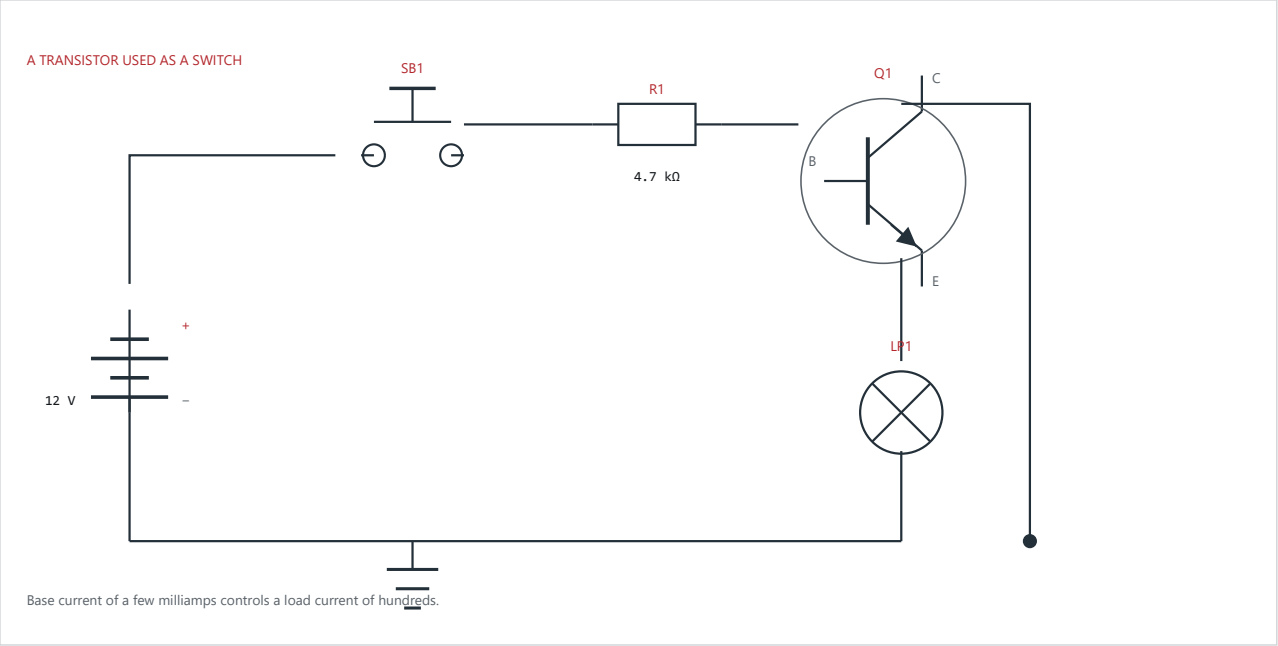



Fig. 13.L The switch test arrangement. The suppression diode across the coil is fitted for all tests except the single deliberate observation of the switch-off spike.

## Results table

| Base current (mA) | Collector current (mA) | V_CE (V) | Region | Dissipation and comment |
|---|---|---|---|---|
| 0 | | | | |
| 0.05 | | | | |
| 0.2 | | | | |
| 0.5 | | | | |
| 1.0 | | | | |
| 3.0 | | | | |
| Field-effect device, on | | | | |

## Discussion

1. From your graph, over what range of base current was collector current proportional to base current? What limited it beyond that?
2. How did your computed current gain compare with the data sheet figure, and why would a designer use the minimum rather than the typical value?
3. Compare the measured device temperature in the active region with that in saturation at the same collector current. Explain the difference.
4. What collector voltage did you observe on switch-off with the suppression diode removed? Compare it with the supply voltage and with the device rating.
5. Explain, using Lenz's law, why the spike appears and why its polarity is such that the diode can conduct it.

6 Compare the on-state dissipation of the two devices at the same load current. At what current would the comparison reverse?

7 Why does the field-effect device need almost no steady drive current, and what does it need instead?

8 Why were a gate resistor and a gate-to-source resistor fitted to the field-effect device?

SAFE-WORK REMINDERS

Wear an antistatic wrist strap when handling insulated-gate devices; their gate oxide is destroyed by static discharge.

Limit both supplies in current before connecting any device.

Interrupting the relay coil without the suppression diode produces a voltage spike well above the supply. Do this once, briefly, and do not touch the circuit while switching.

Devices in the active region become hot quickly. Use the thermometer rather than a finger.

Check the computed dissipation against the device rating before applying power.

Allow devices and heatsinks to cool before handling or rewiring.

## Troubleshooting guide

Faults are found by narrowing the circuit, not by replacing parts. Each row below pairs an observed symptom with the measurement that separates the likely causes.

| Symptom | Likely cause | Decisive test | Correct action |
|---|---|---|---|
| Transistor switch runs hot but the load works | Insufficient base or gate drive, so the device sits in the active region | Measure the voltage across the device while it is on; it should be a few tenths of a volt | Reduce the base resistor or raise the gate drive voltage |
| Transistor fails whenever an inductive load is switched off | No suppression diode, so the coil's reverse voltage exceeds the device rating | Observe the device voltage at switch-off on an oscilloscope | Fit a suppression diode across the coil, cathode to the positive supply |
| Insulated-gate device fails on the bench but not in service | Static discharge during handling, or gate overvoltage | Inspect for gate-to-source short with a meter | Observe antistatic precautions and fit a gate resistor and gate-to-source resistor |
| Paralleled bipolar devices share current unevenly and one fails | Current hogging, because forward drop falls as temperature rises | Measure current in each device branch separately | Fit emitter ballast resistors, or use field-effect devices instead |
| Switch works at low frequency but overheats at high frequency | Switching loss rising in proportion to frequency | Compute conduction and switching loss separately | Improve gate drive to shorten transitions, or reduce the switching frequency |
| Field-effect device turns on by itself | Gate left floating and charged by leakage or capacitive coupling | Check for a gate-to-source resistor | Fit a gate-to-source resistor to define the off state |
| Logic output cannot drive the transistor into saturation | Logic source current too low for the required base current | Compare the required base current with the logic output rating | Use a Darlington, a field-effect device, or a buffer stage |

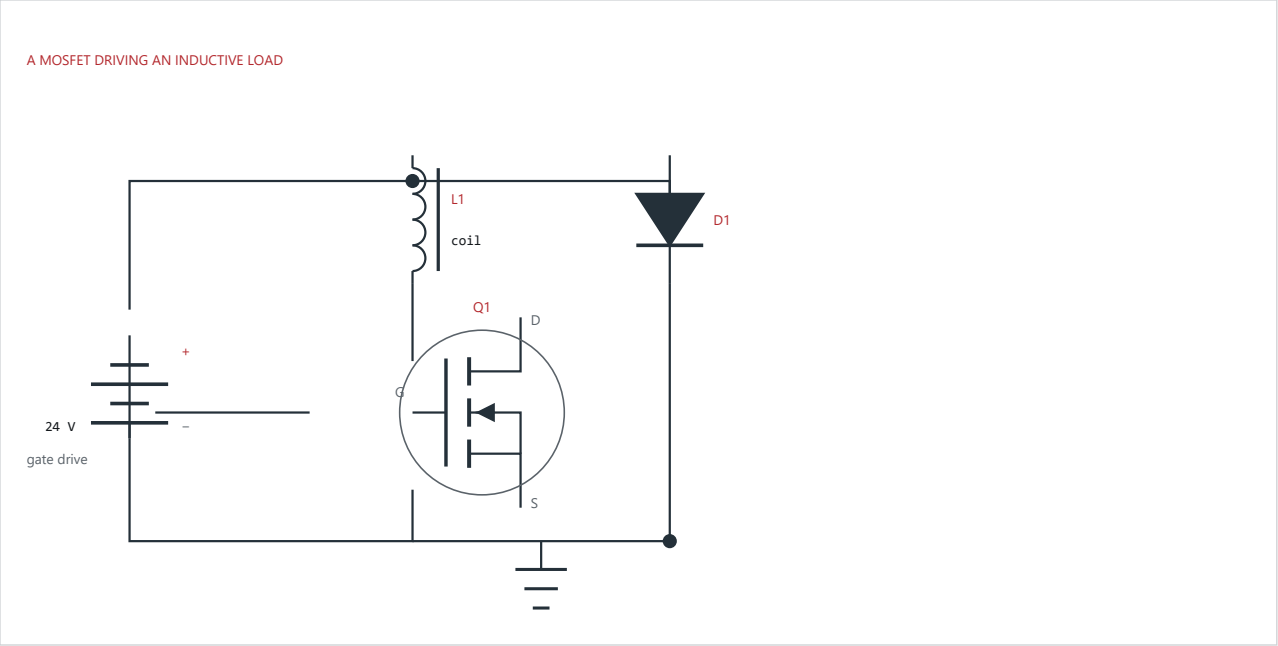



Fig. 13.T Measuring on-state loss. The voltage across the conducting device at a known load current gives its dissipation directly.

FIELD HABITS THAT PREVENT CALLBACKS

Drive a switch to full saturation; a transistor switch running hot is usually underdriven, not undersized.
Fit a suppression diode across every inductive load as a matter of routine.
Use the minimum current gain from the data sheet, never the typical value, when sizing base drive.
Treat insulated-gate devices as static sensitive from the moment they leave the packaging.
Compute conduction and switching loss separately when the switching frequency is appreciable.

## Exercises

Answer from first principles. Selected answers follow; the reasoning matters more than the number.

1 State which terminal of a bipolar transistor symbol carries the arrow, and what its direction indicates.
2 A transistor has a base current of 0.4 mA and a collector current of 48 mA. Find the gain.
3 For the device above, find the emitter current.
4 Name the three operating regions and state which two a switch should use.
5 Explain why a switching transistor must be driven into saturation rather than the active region.
6 A load draws 120 mA and the minimum gain is 40. Find the minimum base current.
7 Using a factor of four on the answer above, find the base resistor for a 5 V drive.
8 State what a suppression diode across a relay coil does and why it is necessary.
9 State the fundamental difference between how a bipolar and a field-effect transistor are controlled.
10 A field-effect device has 25 mΩ on resistance and carries 8 A. Find its on-state loss.
11 A bipolar device saturating at 0.35 V carries the same 8 A. Find its loss and compare.
12 Explain why paralleled field-effect devices share current but paralleled bipolar devices do not.
13 State two handling precautions required for insulated-gate devices.
14 Explain why a device that is cool at 1 kHz switching may overheat at 100 kHz with the same current.

## Selected answers

| # | Answer | Reason |
|---|---|---|
| 1 | The emitter; its direction shows conventional current | Pointing out is n-p-n, pointing in is p-n-p |
| 2 | 120 | $\beta = I_C / I_B$ |
| 3 | 48.4 mA | $I_E = I_B + I_C$ |
| 4 | Cut-off, active, saturation; a switch uses cut-off and saturation | The active region dissipates heavily |
| 5 | To minimise dissipation | In saturation the device drops a few tenths of a volt rather than a large fraction of the supply |
| 6 | 3 mA | $I_B = I_C / \beta(\min)$ |
| 7 | 358 Ω, so use 330 Ω | $R_B = (5 - 0.7)/0.012$ |
| 8 | It conducts the coil's collapsing current | Without it the induced reverse voltage exceeds the transistor's rating |
| 9 | Current control against voltage control | The bipolar needs continuous base current; the field-effect device needs only gate charge |
| 10 | 1.6 W | $P = I^2 R_{DS(on)}$ |
| 11 | 2.8 W; the field-effect device is better here | $P = V_{CE(sat)} I$, which is linear in current |
| 12 | On resistance rises with temperature | The hottest field-effect device sheds current; a bipolar device's drop falls with temperature, so it takes more |
| 13 | Antistatic strap and a gate-to-source resistor | The gate oxide is destroyed by static and a floating gate can self-trigger |
| 14 | Switching loss is proportional to frequency | Each transition dissipates energy, so a hundredfold frequency increase multiplies that component accordingly |

## Chapter summary

- A transistor's third terminal controls conduction between the other two, which makes electronic switching and amplification possible.
- Bipolar devices are current controlled and consume base current continuously; field-effect devices are voltage controlled and need only gate charge.
- Cut-off and saturation are the switching states; the active region is for amplification and dissipates heavily.
- Base drive for a switch is computed from minimum gain and then multiplied by three to five to guarantee saturation.
- Every inductive load switched by a semiconductor needs a suppression diode.
- Bipolar on-state loss is linear in current; field-effect loss is quadratic, so each wins in a different current range.
- Switching loss rises with frequency and is separate from conduction loss.

### Equations introduced in this chapter

| Relationship | Expression | When it applies |
|---|---|---|
| Current balance | $I_E = I_B + I_C$ | Any bipolar transistor |
| Current gain | $\beta = I_C / I_B$ | Active region |
| Minimum base drive | $I_B = I_C / \beta(\text{min})$ | Switch design |
| Base resistor | $R_B = (V_{drive} - 0.7)/I_B$ | Bipolar switch |
| Bipolar on-state loss | $P = V_{CE(sat)}\, I_C$ | Saturated switch |
| Field-effect on-state loss | $P = I_D^2\, R_{DS(on)}$ | Conducting channel |
| Gate drive power | $P = Q\, f\, V$ | Switching at frequency f |

### Vocabulary check

| Term | Meaning in this chapter |
|---|---|
| Current gain | The ratio of collector current to base current in the active region |
| Saturation | The fully on state, where collector current is set by the external circuit |
| Cut-off | The fully off state, with negligible collector current |
| Active region | The proportional region used for amplification |
| On resistance | The effective resistance of a conducting field-effect channel |
| Suppression diode | A diode across an inductive load that conducts its collapsing current |
| Current hogging | The tendency of paralleled bipolar devices to share current unequally |

LOOKING AHEAD

A single transistor amplifies, but its gain varies between devices and with temperature. Building a feedback amplifier from many transistors in one package gives the operational amplifier, whose behaviour depends on external components instead. The next chapter develops it.

CHAPTER 14

# Operational amplifiers

Very high gain tamed by feedback

WHY THIS CHAPTER MATTERS

A single transistor's gain varies between samples and with temperature, which makes precise analogue work difficult. An operational amplifier has so much gain that feedback can be used to make the circuit's behaviour depend on external resistors instead of on the device, which is why almost all analogue signal conditioning is built this way.

## What you will be able to do

- Identify the operational amplifier symbol and its inverting and non-inverting inputs.
- State the ideal assumptions and the virtual-earth principle.
- Compute the gain of inverting and non-inverting amplifier circuits.
- Analyse summing, difference and buffer configurations.
- Explain comparator operation and the purpose of hysteresis.
- Recognise the practical limits: supply rails, offset, bandwidth and slew rate.

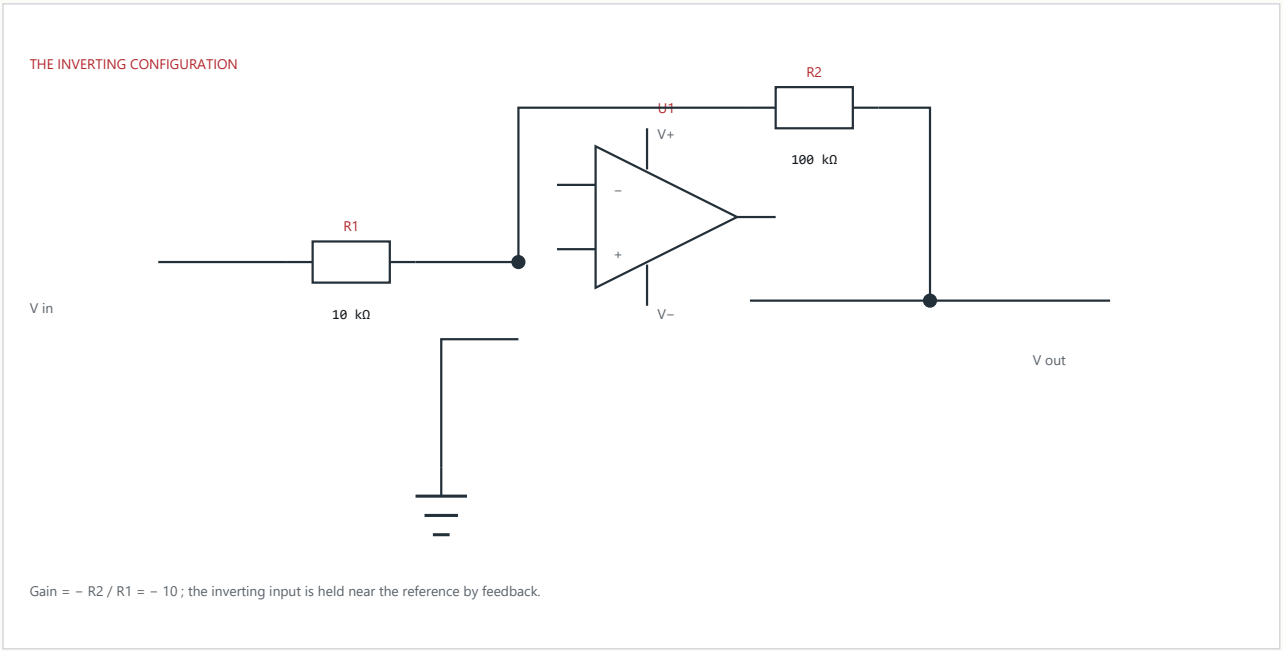



Fig. 14.1 An inverting amplifier. The feedback resistor and the input resistor set the gain, and the inverting input is held at the reference potential by feedback.

## The device and the ideal assumptions

An operational amplifier is a differential amplifier with two inputs and one output. It amplifies the difference between the inputs by an enormous factor, typically a hundred thousand or more at low frequency. That raw gain is almost never used directly; it is traded away through feedback in exchange for predictability.

Analysis is made simple by two assumptions that real devices approach closely. First, the input terminals draw negligible current. Second, when negative feedback is present, the amplifier drives its output to whatever value makes the two input voltages equal. Together these let every standard configuration be analysed with Ohm's law alone.

```
V_out = A(V⁺ − V⁻)      with A very large

Ideal assumption 1:  input currents ≈ 0

Ideal assumption 2:  with feedback, V⁺ ≈ V⁻
```

WHY FEEDBACK IS WORTH THE LOSS OF GAIN

Open-loop gain varies widely between devices and with temperature.
Closed-loop gain depends on a ratio of two resistors, which is stable and known.
Feedback also widens bandwidth and lowers output impedance.
The design question becomes which resistors to fit, not which device to select.

## The inverting amplifier and virtual earth

In the inverting configuration the signal enters through a resistor to the inverting input, and the feedback resistor returns from the output to the same point. The non-inverting input is tied to the reference, so feedback holds the inverting input at the reference potential too. That point is called a virtual earth: it sits at reference potential without being connected to it.

The consequence is that the input current is set entirely by the input resistor, and because no current enters the amplifier, all of it continues through the feedback resistor. The gain is therefore the ratio of the two resistors, with a sign inversion, and the input impedance of the stage equals the input resistor.

```
Gain = −R_f / R_in

Input impedance = R_in

At the virtual earth:  I_in = V_in / R_in = I_f
```

The inversion is a genuine phase reversal, not a sign convention. A positive input produces a negative output, which matters whenever the stage feeds something polarity sensitive.

## The non-inverting amplifier and the buffer

Taking the signal directly to the non-inverting input and returning feedback through a divider to the inverting input gives a non-inverting amplifier. Its gain is one plus the ratio of the feedback resistor to the grounded resistor, so the minimum gain of this configuration is unity and it cannot attenuate.

Setting the feedback resistor to zero and omitting the grounded resistor gives the voltage follower, or buffer, with a gain of exactly one. It is used not for gain but for impedance transformation: it presents a very high impedance to the source and a very low impedance to the load, so a weak high-impedance signal can drive a substantial load without being loaded down.

```
Gain = 1 + R_f / R_g

Buffer:  gain = 1, very high input impedance

Input impedance ≈ that of the amplifier input itself
```

| Configuration | Gain | Input impedance | Phase | Typical use |
|---|---|---|---|---|
| Inverting | −R_f/R_in | R_in | Inverted | Gain, summing, virtual-earth circuits |
| Non-inverting | 1 + R_f/R_g | Very high | Preserved | Amplifying a weak source |
| Buffer | 1 | Very high | Preserved | Impedance transformation |
| Summing | −R_f per input resistor | Each input resistor | Inverted | Mixing signals |
| Difference | Set by matched pairs | Moderate | Differential | Rejecting common-mode noise |

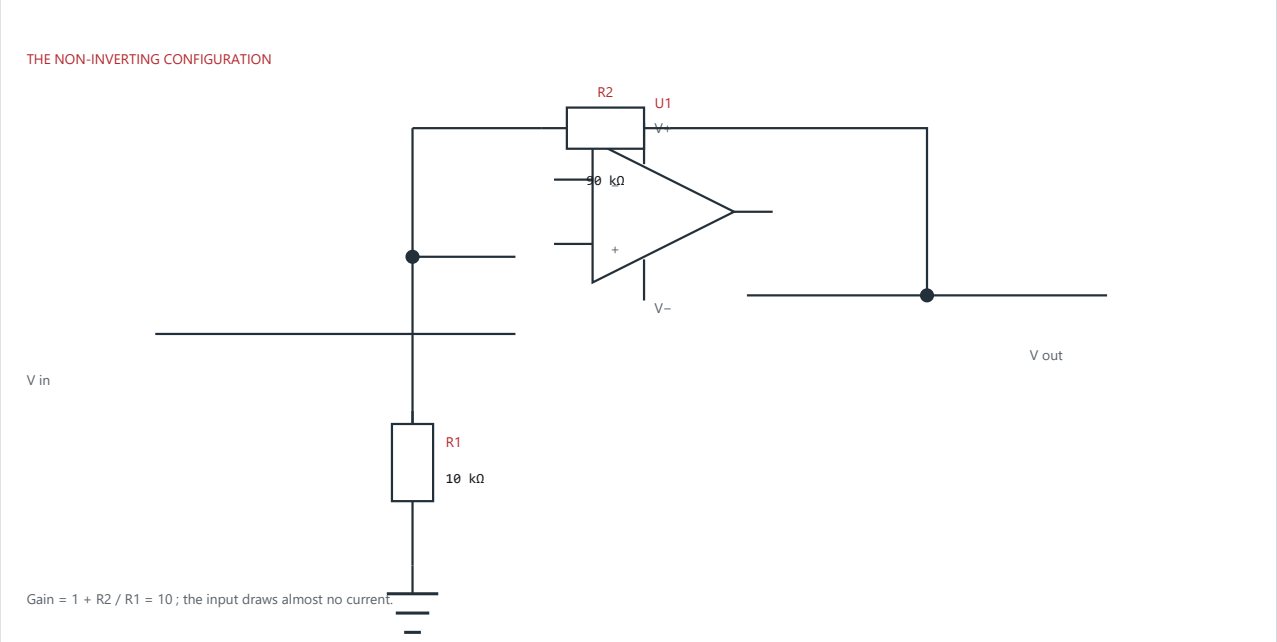



Fig. 14.2 A non-inverting amplifier. The signal drives the high-impedance input directly and the divider sets the gain.

## Summing and difference amplifiers

Because the virtual earth holds its potential regardless of what is connected to it, several input resistors can feed the same point without interacting. Each contributes a current proportional to its own input voltage, the currents add, and the output reflects their sum. Giving each input a different resistor applies a different weight to it.

A difference amplifier uses both inputs with matched resistor pairs and responds only to the difference between its two inputs, rejecting any voltage common to both. This is the standard approach to recovering a small signal from a sensor at the end of a long cable, where interference appears almost equally on both conductors.

```
Summing:  V_out = −R_f (V_1/R_1 + V_2/R_2 + V_3/R_3)

Difference:  V_out = (R_f/R_in)(V_2 − V_1)   with matched pairs

Common-mode rejection depends on resistor matching
```

COMMON-MODE REJECTION DEPENDS ON MATCHED COMPONENTS

Interference on a twisted pair appears almost equally on both conductors.
A difference amplifier subtracts it out, leaving the wanted signal.
Rejection is limited by how closely the two resistor pairs match.
Use close-tolerance resistors, or an instrumentation amplifier, for demanding work.

## The comparator and hysteresis

Without feedback, the enormous open-loop gain drives the output to one supply rail or the other according to which input is higher. This is comparator operation, and it converts an analogue input into a two-state output, which is how an analogue measurement becomes a control decision.

A plain comparator chatters when the input hovers near the threshold, because noise repeatedly crosses it. Adding positive feedback moves the threshold slightly whenever the output changes, so the input must travel further to switch back. That deliberate gap is hysteresis, and it is why a thermostat has a dead band rather than buzzing at setpoint.

```
V_out → positive rail when V+ > V−

V_out → negative rail when V+ < V−

Hysteresis gap set by the positive feedback ratio
```

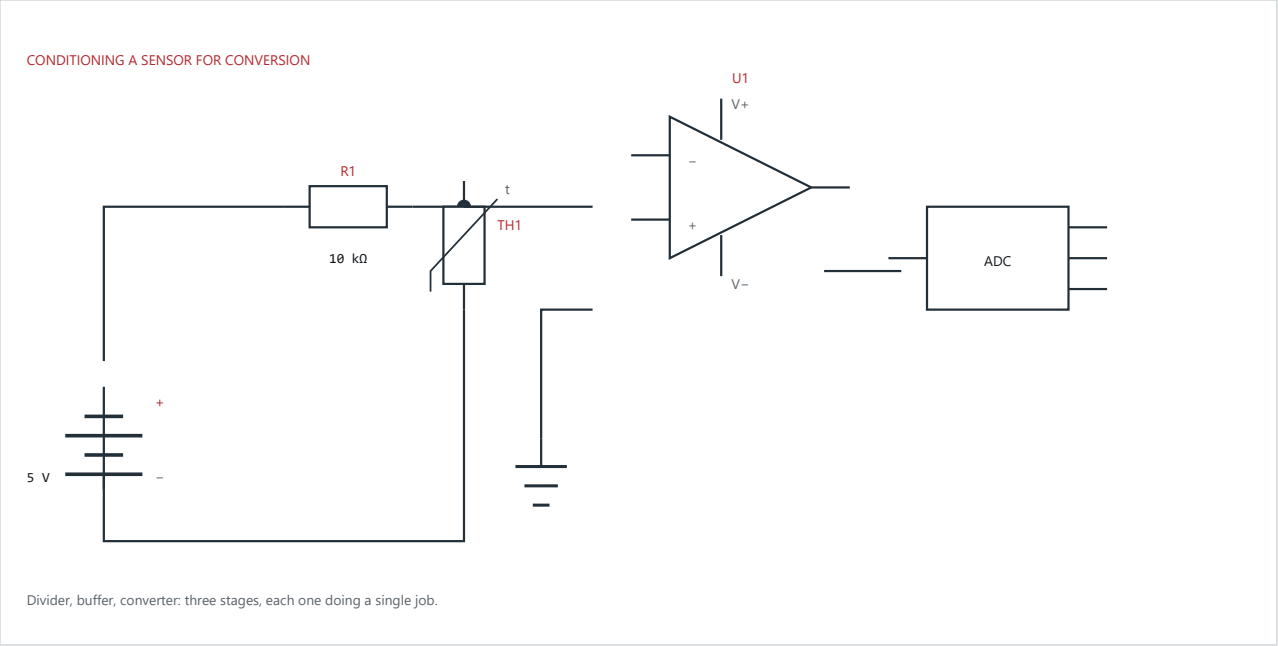



Fig. 14.3 A signal conditioning stage. The sensor output is amplified and compared against a reference to produce a two-state control output.

## Practical limits

Real devices depart from the ideal in ways that matter. The output cannot reach beyond the supply rails, and many types stop short of them, so a circuit designed for a gain that would demand more output than the supply allows will simply clip. Input offset voltage means the output is not exactly zero for a zero input, which matters in precision direct-voltage measurement.

Gain also falls with frequency, so a device with a gain-bandwidth product of one megahertz gives a gain of a hundred only up to ten kilohertz. Slew rate limits how fast the output can change regardless of gain, which distorts large fast signals. Each of these is a data-sheet figure to be checked against the application rather than a defect.

### Checking a design against real limits

- Confirm the required output swing fits within the supply rails.
- Divide the gain-bandwidth product by the gain to find the usable bandwidth.
- Check slew rate against the fastest large output change required.
- Allow for input offset in precision direct-voltage work.
- Provide a path for input bias current at both inputs.
- Decouple the supply pins close to the device.

## Worked examples

Work each example with a pencil before reading the solution. Sketch the circuit, mark the reference node, and write the governing law before substituting numbers.

EXAMPLE 14.1 Inverting amplifier gain and output

**Given.** An inverting amplifier has an input resistor of 10 kΩ and a feedback resistor of 220 kΩ, with an input of 45 mV.

**Find.** The gain, the output voltage, and the stage input impedance.

Gain

```
−220/10 = −22
```

Output voltage

```
−22 × 0.045 = −0.99 V
```

Input impedance

```
Equal to the input resistor, 10 kΩ
```

Current at the virtual earth

```
0.045/10 000 = 4.5 µA
```

**Check.** The output is just under one volt and inverted. Note that the input impedance is only 10 kΩ, so this configuration loads the source; a non-inverting stage would not.

EXAMPLE 14.2 Non-inverting amplifier for a sensor

**Given.** A sensor produces 0 to 120 mV and must drive a 0 to 5 V input. The feedback resistor is to be a convenient value.

**Find.** The required gain and a suitable resistor pair.

Required gain

```
5/0.120 = 41.7
```

Non-inverting relation

```
Gain = 1 + R_f/R_g, so R_f/R_g = 40.7
```

Choose R_g

```
1 kΩ
```

Required R_f

```
40.7 kΩ, so use 39 kΩ with a trimmer, or 40.2 kΩ close tolerance
```

Resulting gain with 39 kΩ

```
1 + 39 = 40, giving 4.8 V full scale
```

**Check.** The non-inverting configuration suits a sensor because its input impedance is very high and it does not load the source. The slight gain shortfall is normally trimmed rather than accepted.

EXAMPLE 14.3 Summing amplifier with unequal weights

**Given.** A summing amplifier has a 100 kΩ feedback resistor and three inputs through 10 kΩ, 20 kΩ and 50 kΩ. The inputs are 0.4 V, 0.8 V and 1.0 V.

**Find.** The output voltage.

Contribution from the first input

$-(100/10) \times 0.4 = -4.0$ V

Contribution from the second

$-(100/20) \times 0.8 = -4.0$ V

Contribution from the third

$-(100/50) \times 1.0 = -2.0$ V

Total output

$-4.0 - 4.0 - 2.0 = -10.0$ V

Supply check

Requires rails beyond −10 V

**Check.** Each input is weighted by its own resistor and the three contributions simply add. The output demand of ten volts must be checked against the available supply, or the stage will clip.

EXAMPLE 14.4 Bandwidth available at a chosen gain

**Given.** A device has a gain-bandwidth product of 1.5 MHz and is used at a closed-loop gain of 50.

**Find.** The usable bandwidth, and the gain available at 100 kHz.

Bandwidth at a gain of 50

$1.5\times10^{6}/50 = 30$ kHz

Gain available at 100 kHz

$1.5\times10^{6}/100\times10^{3} = 15$

Implication

A gain of 50 cannot be achieved at 100 kHz with this device

**Check.** Gain and bandwidth trade directly against each other. A design needing both must use a faster device or distribute the gain across two cascaded stages.

EXAMPLE 14.5 Comparator with hysteresis

**Given.** A comparator operates from 0 to 10 V rails with a reference of 5 V, and positive feedback set to one fiftieth of the output.

**Find.** The two thresholds and the hysteresis band.

Feedback contribution

```
10/50 = 0.2 V
```

Upper threshold, output low

```
5 + 0.1 = 5.1 V approximately
```

Lower threshold, output high

```
5 − 0.1 = 4.9 V approximately
```

Hysteresis band

```
0.2 V
```

Effect

```
Input noise below 0.2 V peak to peak cannot cause chatter
```

**Check.** A 0.2 V band is enough to reject typical sensor noise while keeping the control threshold close to the intended 5 V. Too little hysteresis gives chatter; too much gives a sloppy control point.

## Practical activity: inverting and non-inverting gain, saturation, bandwidth and hysteresis

AIM

Verify the predicted gain of inverting and non-inverting configurations, observe clipping against the supply rails, measure the gain-bandwidth trade-off, then demonstrate comparator chatter and cure it with hysteresis.

### Equipment

- General-purpose operational amplifier, dual in-line package, with its data sheet.
- Split supply, plus and minus 12 V, or a single 12 V supply with a mid-rail reference.
- Resistors: 1 kΩ, 10 kΩ, 22 kΩ, 100 kΩ, 220 kΩ, 1 MΩ, close tolerance where available.
- Signal generator with sine and square outputs and a calibrated amplitude.
- Two-channel oscilloscope and two multimeters.
- Potentiometer, 10 kΩ, to provide a slowly varying input.
- Decoupling capacitors, 100 nF, for the supply pins.

### Procedure

1. Fit the decoupling capacitors across the supply pins before any measurement.
2. Build the inverting amplifier with a 10 kΩ input resistor and a 100 kΩ feedback resistor.
3. Apply a 1 kHz sine input of 100 mV peak and measure the output amplitude. Compute the measured gain and compare with the prediction.
4. Confirm the phase relationship between input and output on the two oscilloscope channels.
5. Measure the voltage at the inverting input with respect to the reference and confirm it is near zero. Record the value.
6. Increase the input amplitude in steps until the output stops growing. Record the amplitude at which clipping begins and the clipped output level.
7. Compare the clipped level with the supply rail voltage.
8. Change the feedback resistor to 220 kΩ and repeat the gain measurement.
9. Rebuild as a non-inverting amplifier with 1 kΩ and 22 kΩ, measure the gain and confirm the phase is preserved.
10. Reconfigure as a buffer and confirm unity gain.
11. Return to a gain of 100 and sweep the input frequency upward, recording the frequency at which the output has fallen to 0.707 of its low-frequency value.
12. Compute the gain-bandwidth product from that result and compare with the data sheet.
13. Build a comparator with a mid-rail reference and drive the input from the potentiometer.
14. Turn the potentiometer slowly through the threshold and observe the output on the oscilloscope. Record whether chatter occurs.
15. Add a 1 MΩ positive feedback resistor from output to the non-inverting input and repeat. Measure the two thresholds and compute the hysteresis band.

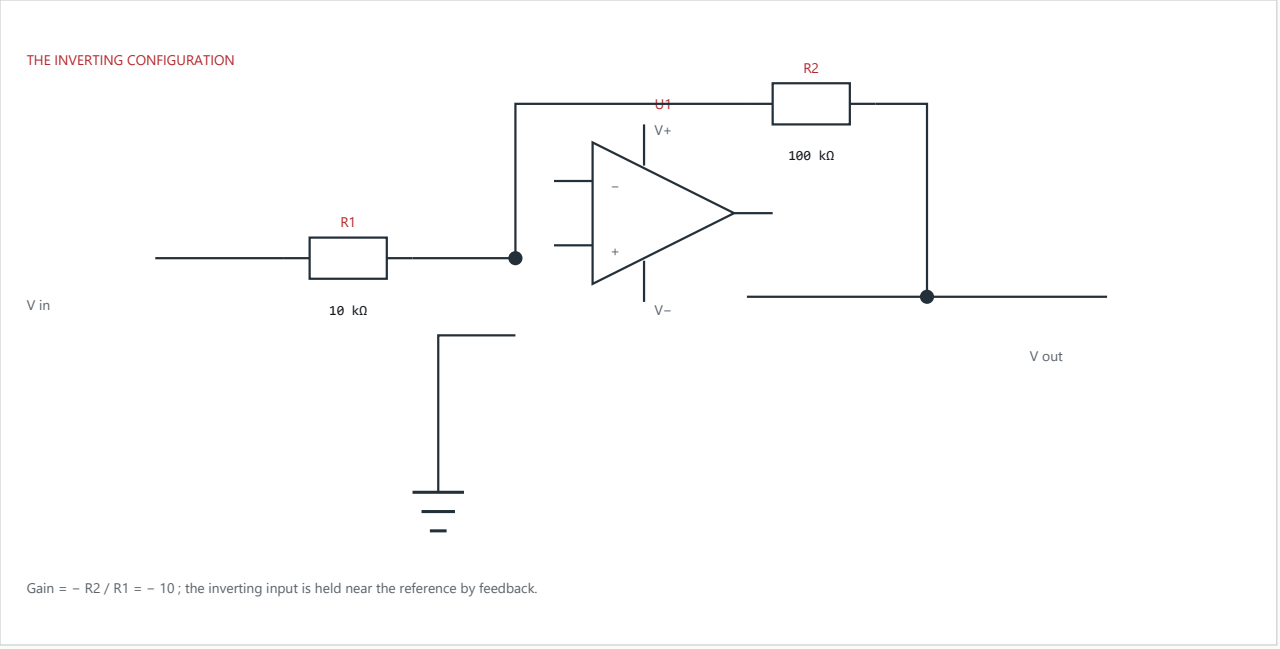



Fig. 14.L The inverting configuration under test. The virtual-earth node is monitored directly to confirm it remains near reference potential.

## Results table

| Configuration | Predicted gain | Measured gain | Phase | Observation |
|---|---|---|---|---|
| Inverting, 10 k / 100 k | | | | |
| Inverting, 10 k / 220 k | | | | |
| Non-inverting, 1 k / 22 k | | | | |
| Buffer | | | | |
| Clipping onset | | | | |
| Bandwidth at gain of 100 | | | | |
| Comparator, no hysteresis | | | | |
| Comparator with hysteresis | | | | |

## Discussion

1. How closely did your measured gains match the resistor ratios, and what accounts for the difference?
2. What voltage did you measure at the inverting input during normal operation? Explain why it stays there without being connected to the reference.
3. At what output level did clipping begin, and how did that compare with the supply rails? What does that imply for a design needing a large output swing?

4 From your bandwidth measurement at a gain of 100, compute the gain-bandwidth product. What bandwidth would the same device give at a gain of 10?

5 Why does the non-inverting configuration not load the signal source, while the inverting one does?

6 Describe what the comparator output did as the input crossed the threshold without hysteresis, and explain the cause.

7 Compute the hysteresis band from your two measured thresholds and compare it with the prediction from the feedback ratio.

8 In what application would you deliberately choose a wider hysteresis band, and what is the penalty?

SAFE-WORK REMINDERS

Observe supply polarity carefully; a reversed supply destroys the device immediately.
Keep the input signal within the supply rails at all times.
Fit the supply decoupling capacitors before applying power, or the stage may oscillate.
Do not short the output to either supply rail; many devices tolerate it only briefly.
Return the signal generator amplitude to minimum before changing any connection.
Check that the computed output swing fits the supply before increasing gain.

## Troubleshooting guide

Faults are found by narrowing the circuit, not by replacing parts. Each row below pairs an observed symptom with the measurement that separates the likely causes.

| Symptom | Likely cause | Decisive test | Correct action |
|---|---|---|---|
| Output sits hard at one supply rail | No negative feedback path, or an input outside the permitted range | Check the feedback resistor is fitted and measure both input voltages | Restore the feedback path and bring the inputs within range |
| Output waveform is flat topped | Required output swing exceeds the supply rails | Compare the predicted output amplitude with the rail voltages | Reduce the gain or raise the supply voltage |
| Gain falls off well below the expected frequency | Gain-bandwidth product exceeded at the chosen gain | Divide the data-sheet product by the closed-loop gain | Use a faster device or split the gain across two stages |
| Large fast signals are distorted while small ones are clean | Slew-rate limiting | Compare the required rate of output change with the data-sheet slew rate | Select a device with a higher slew rate or reduce the signal amplitude |
| Stage oscillates with no signal applied | Inadequate supply decoupling or stray feedback through the layout | Observe the output with the input grounded | Fit decoupling capacitors at the supply pins and shorten feedback wiring |
| Comparator output chatters near the threshold | No hysteresis, so input noise crosses the threshold repeatedly | Observe the input and output together on an oscilloscope | Add a positive feedback resistor to establish a hysteresis band |
| Precision direct-voltage reading has a constant error | Input offset voltage amplified by the stage gain | Short the input and measure the residual output | Use an offset trim, or select a low-offset device |
| Difference stage passes interference as well as signal | Resistor pairs not matched closely enough | Measure the output with both inputs tied together and driven | Fit close-tolerance matched resistors or an instrumentation amplifier |

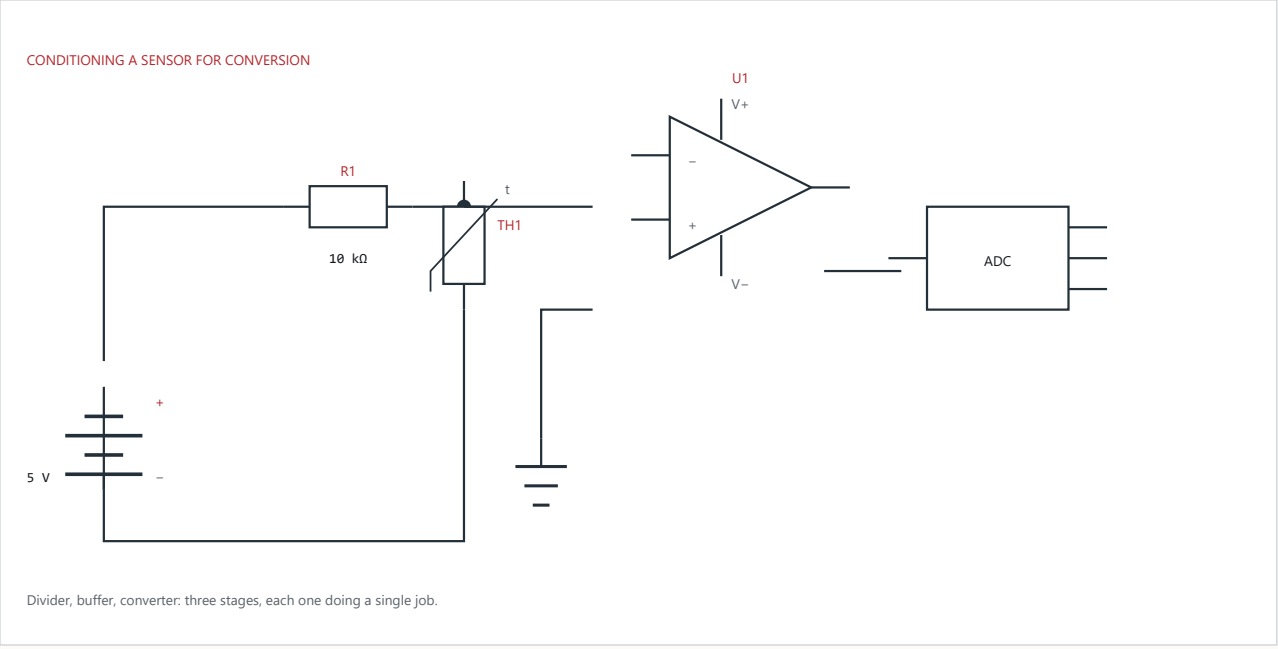



Fig. 14.T Signal conditioning under test. The amplified sensor signal and the comparator output are observed together to confirm the switching thresholds.

FIELD HABITS THAT PREVENT CALLBACKS

Fit supply decoupling capacitors at the device pins as a matter of routine, not as a cure after oscillation appears.
Check the required output swing against the supply rails before selecting a gain.
Divide the gain-bandwidth product by the intended gain early in the design, not after testing.
Add deliberate hysteresis to every comparator that drives a contactor or a relay.
Use close-tolerance matched resistors wherever common-mode rejection matters.

## Exercises

Answer from first principles. Selected answers follow; the reasoning matters more than the number.

1 State which input of an operational amplifier produces a phase-inverted output.
2 State the two ideal assumptions used to analyse feedback amplifier circuits.
3 Find the gain of an inverting amplifier with a 4.7 kΩ input and a 47 kΩ feedback resistor.
4 Find the output voltage of the stage above for a 60 mV input.
5 Find the gain of a non-inverting amplifier with 2.2 kΩ and 68 kΩ.
6 State the gain of a voltage follower and its purpose.
7 Explain what a virtual earth is and why it occurs.
8 A summing amplifier has a 68 kΩ feedback resistor and inputs through 10 kΩ and 22 kΩ at 0.3 V and 0.5 V. Find the output.
9 State the minimum gain achievable with the non-inverting configuration and explain why.
10 A device has a gain-bandwidth product of 4 MHz. Find the bandwidth at a gain of 80.
11 Explain why an amplifier output clips, and what determines the clipping level.
12 State the purpose of hysteresis in a comparator and one consequence of omitting it.

13 Explain why a difference amplifier rejects interference picked up on a long sensor cable.

14 State which configuration you would choose to amplify a high-impedance sensor, and why.

15 Explain why input offset voltage matters more in direct-voltage measurement than in audio work.

## Selected answers

| # | Answer | Reason |
|---|---|---|
| 1 | The inverting input | Signal applied there appears inverted at the output |
| 2 | Inputs draw no current; feedback equalises the input voltages | These two permit analysis with Ohm's law alone |
| 3 | −10 | Gain = −R_f/R_in |
| 4 | −0.6 V | Output = gain × input |
| 5 | 31.9 | Gain = 1 + R_f/R_g |
| 6 | Unity; impedance transformation | It presents high impedance to the source and low impedance to the load |
| 7 | A node held at reference potential by feedback | Feedback drives the output until the inverting input matches the non-inverting input |
| 8 | −3.59 V | −(68/10)(0.3) − (68/22)(0.5) |
| 9 | Unity | The relation 1 + R_f/R_g cannot fall below one, so it cannot attenuate |
| 10 | 50 kHz | Bandwidth = product divided by gain |
| 11 | The demanded output exceeds the rails | The supply voltage, less the device's rail shortfall, sets the limit |
| 12 | To prevent chatter near the threshold | Without it, input noise repeatedly crosses the threshold and the output oscillates |
| 13 | Interference appears equally on both conductors | The stage responds only to the difference, so the common part is subtracted out |
| 14 | Non-inverting or buffer | Its very high input impedance does not load the source |
| 15 | Offset is indistinguishable from a small direct signal | Audio coupling removes the steady component, but a direct measurement carries the offset as an error |

## Chapter summary

- An operational amplifier has very high open-loop gain, which feedback trades for predictable closed-loop behaviour set by external resistors.
- The two ideal assumptions, negligible input current and equal input voltages under feedback, allow analysis with Ohm's law.
- Inverting gain is the resistor ratio with inversion, and the input impedance equals the input resistor.
- Non-inverting gain is one plus the resistor ratio, cannot fall below unity, and presents a very high input impedance.
- The virtual earth lets several weighted inputs be summed without interaction.
- A difference amplifier rejects common-mode interference to the extent that its resistor pairs match.
- Without feedback the device acts as a comparator, and hysteresis is needed to prevent chatter.
- Supply rails, offset, gain-bandwidth product and slew rate are the practical limits to check.

### Equations introduced in this chapter

| Relationship | Expression | When it applies |
|---|---|---|
| Differential gain | $V_{out} = A(V^+ - V^-)$ | Open loop |
| Inverting gain | $\text{Gain} = -R_f/R_{in}$ | Inverting configuration |
| Non-inverting gain | $\text{Gain} = 1 + R_f/R_g$ | Non-inverting configuration |
| Summing output | $V_{out} = -R_f \Sigma(V_n/R_n)$ | Virtual-earth summing node |
| Difference output | $V_{out} = (R_f/R_{in})(V_2 - V_1)$ | Matched resistor pairs |
| Usable bandwidth | $BW = GBP / \text{gain}$ | Closed-loop operation |
| Input impedance, inverting | $Z_{in} = R_{in}$ | Inverting configuration |

### Vocabulary check

| Term | Meaning in this chapter |
|---|---|
| Open-loop gain | The device's own gain before feedback is applied |
| Closed-loop gain | The gain of the complete circuit, set by external components |
| Virtual earth | A node held at reference potential by feedback rather than by connection |
| Buffer | A unity-gain stage used to transform impedance |
| Common-mode rejection | The ability to ignore a voltage present equally on both inputs |
| Hysteresis | A deliberate gap between a comparator's rising and falling thresholds |
| Slew rate | The maximum rate at which the output voltage can change |
| Gain-bandwidth product | The constant relating achievable gain to available bandwidth |

LOOKING AHEAD

A comparator turns an analogue quantity into a two-state output. Once signals are reduced to two states, they can be combined by logic rules rather than by Ohm's law, which is the subject of the next chapter.

CHAPTER 15

# Digital logic

Two-state signals, gates, and the circuits that remember

WHY THIS CHAPTER MATTERS

Once a signal is reduced to two states, it can be combined by logical rules rather than by Ohm's law. Every control panel built in the last forty years uses this idea, whether in discrete gates, in a programmable controller, or in the interlocking of a motor starter.

### What you will be able to do

- State the truth tables of the basic gates and recognise their symbols.
- Convert between binary, decimal and hexadecimal.
- Build and simplify a combinational function from a specification.
- Explain how a latch stores a state and why a seal-in contact is the same idea.
- Describe flip-flop operation and clocked behaviour.
- Interpret a control interlock drawn as either a ladder or a logic diagram.

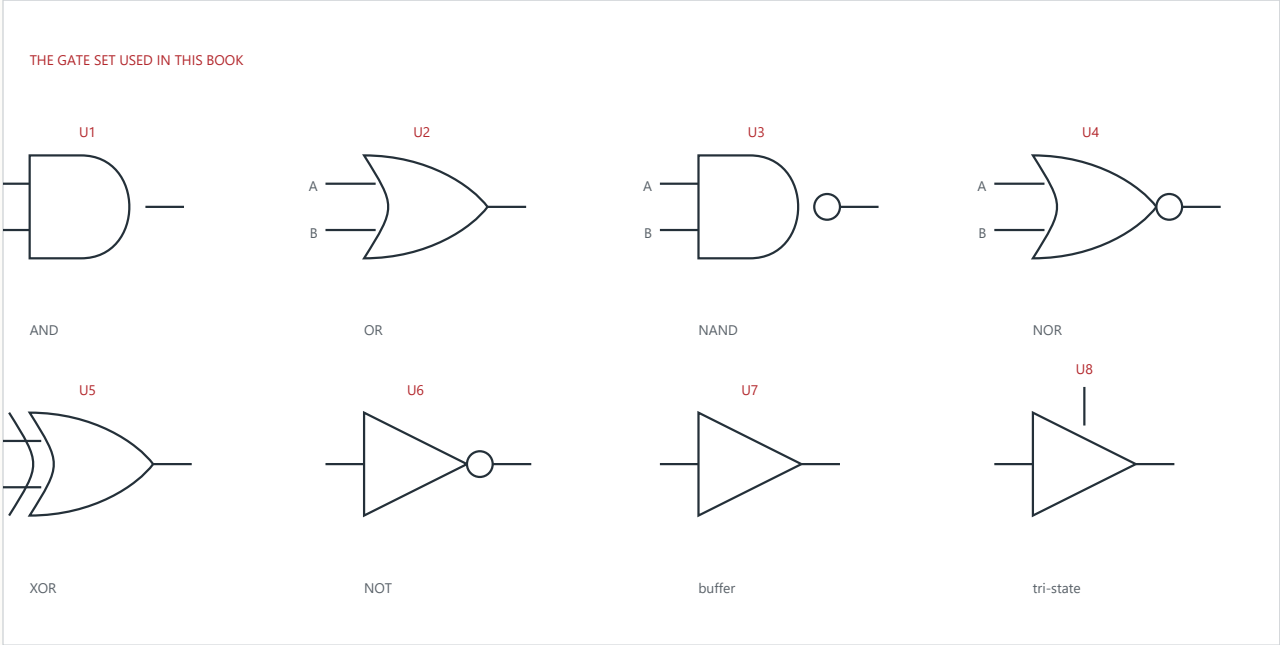



Fig. 15.1 The basic logic gates. Each symbol's outline identifies its function, and a small circle on the output indicates inversion.

## Two states and number systems

A digital signal is interpreted as one of two states. Any voltage above a defined threshold reads as one state and any voltage below a lower threshold reads as the other, with a forbidden band between them. That tolerance is the reason digital signals survive noise and long cable runs so much better than analogue ones.

Because there are two states per digit, counting is naturally binary. Hexadecimal is used as a shorthand, because one hexadecimal digit represents exactly four binary digits, which makes long binary values readable without arithmetic.

```
Each bit position carries 2^n:  ... 16, 8, 4, 2, 1

One hexadecimal digit = four bits

n bits represent 2^n distinct values
```

| Decimal | Binary | Hexadecimal | Decimal | Binary | Hexadecimal |
|---|---|---|---|---|---|
| 0 | 0000 | 0 | 8 | 1000 | 8 |
| 1 | 0001 | 1 | 9 | 1001 | 9 |
| 2 | 0010 | 2 | 10 | 1010 | A |
| 3 | 0011 | 3 | 11 | 1011 | B |
| 4 | 0100 | 4 | 12 | 1100 | C |
| 5 | 0101 | 5 | 13 | 1101 | D |
| 6 | 0110 | 6 | 14 | 1110 | E |
| 7 | 0111 | 7 | 15 | 1111 | F |

## The basic gates

Five functions account for almost everything. An AND output is active only when all inputs are active. An OR output is active when any input is active. A NOT output is the inverse of its single input. NAND and NOR are the inverted forms of AND and OR, and the exclusive-OR output is active when its inputs differ.

The small circle on a symbol's output always means inversion, so recognising NAND as an AND outline with an output circle is all that is needed to read it. NAND and NOR are particularly important because either one alone can construct any other function, which is why real integrated circuits are built predominantly from them.

### Reading a logic symbol

- Flat back with a curved front: AND.
- Curved back with a pointed front: OR.
- Triangle with an output circle: NOT, or inverter.
- A circle on any output means that output is inverted.
- A double curve on the input side of an OR outline: exclusive-OR.
- A triangle without a circle: buffer, which drives rather than inverts.

| A | B | AND | OR | NAND | NOR | XOR |
|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 1 | 1 | 0 |
| 0 | 1 | 0 | 1 | 1 | 0 | 1 |
| 1 | 0 | 0 | 1 | 1 | 0 | 1 |
| 1 | 1 | 1 | 1 | 0 | 0 | 0 |

## Combining gates into a function

A control requirement stated in words translates directly into gates. The phrase ‘run when the start button is pressed and no trip is active’ is an AND of two conditions, one of them inverted. Building the function is a matter of writing the statement precisely enough that each condition becomes a single input.

Two simplifications are worth knowing, because they appear constantly. An inverted AND is the same as an OR of inverted inputs, and an inverted OR is the same as an AND of inverted inputs. These let a function be rearranged into whichever form uses fewer gates or matches the available devices.

```
NOT(A AND B) = (NOT A) OR (NOT B)

NOT(A OR B) = (NOT A) AND (NOT B)

A AND A = A       A OR NOT A = 1
```

SAFETY CONDITIONS ARE WIRED AS NORMALLY CLOSED, NOT AS LOGIC

A logic NOT depends on the device working and being powered.
A physically normally closed emergency-stop contact opens on demand or on a broken wire.
Safety functions must therefore be hard-wired, not merely programmed.
Use logic to coordinate a process; use hard wiring to make it safe.

## Latches and the seal-in contact

A latch holds a state after the input that set it has gone away. Cross-coupling two NAND or two NOR gates so that each drives the other produces exactly this: a brief input sets the output, and it stays set until a separate input resets it.

An electrician has already met this circuit. A motor starter in which the contactor's own auxiliary contact maintains the coil after the start button is released is a latch built from a relay. The start button sets it, the stop button resets it, and the seal-in contact is the feedback path. Recognising the two as the same circuit makes both easier to troubleshoot.

### The starter circuit as a latch

- Start button: the set input, acting momentarily.
- Seal-in auxiliary contact: the feedback that maintains the state.
- Stop button: the reset input, wired normally closed.
- Overload contact: a second reset, also wired normally closed.
- Loss of supply resets the latch, which is why the motor does not restart by itself.

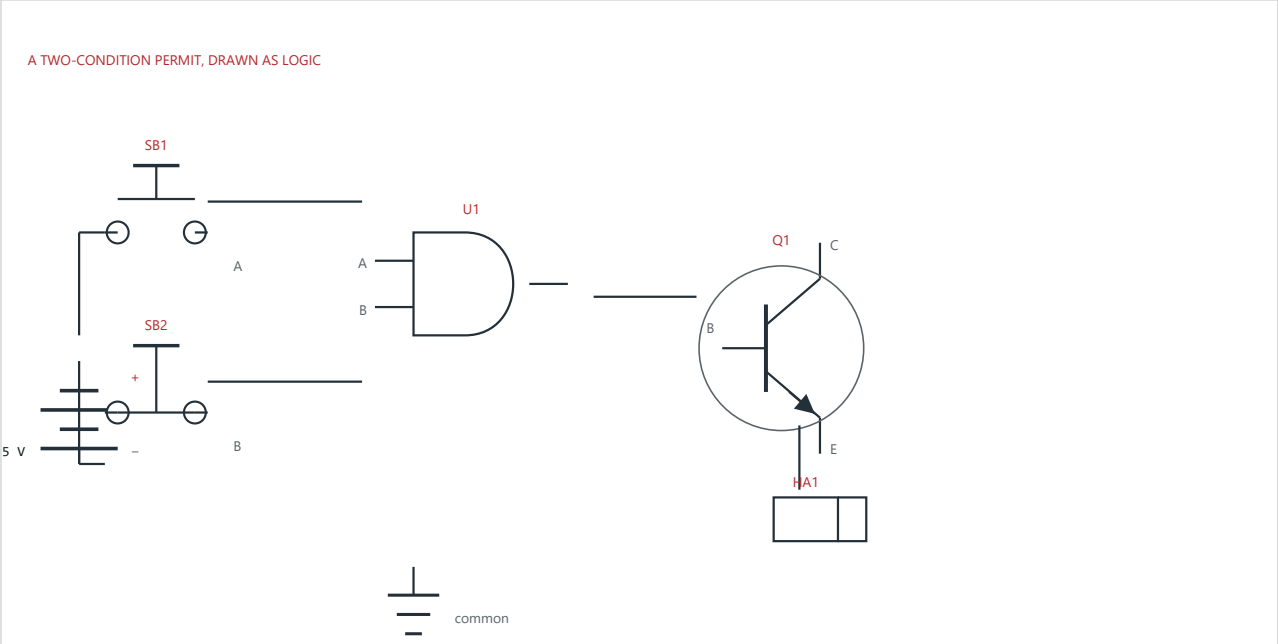



Fig. 15.2 A control interlock. The seal-in path maintains the coil after the start button is released, and the stop contact breaks that path.

## Flip-flops and clocked logic

A flip-flop is a latch whose inputs are only examined at a defined instant, usually the edge of a clock signal. That discipline matters because it stops the circuit responding to transient conditions while signals are still settling, and it allows many stages to advance together in step.

From clocked flip-flops come counters, shift registers and timing chains. A counter is a cascade in which each stage changes state when the previous one completes a cycle, which is how a digital timer measures intervals and how a frequency is divided down.

```
Edge triggered: inputs sampled at the clock transition

A chain of n flip-flops divides frequency by 2^n

Counter range with n stages: 0 to 2^n – 1
```

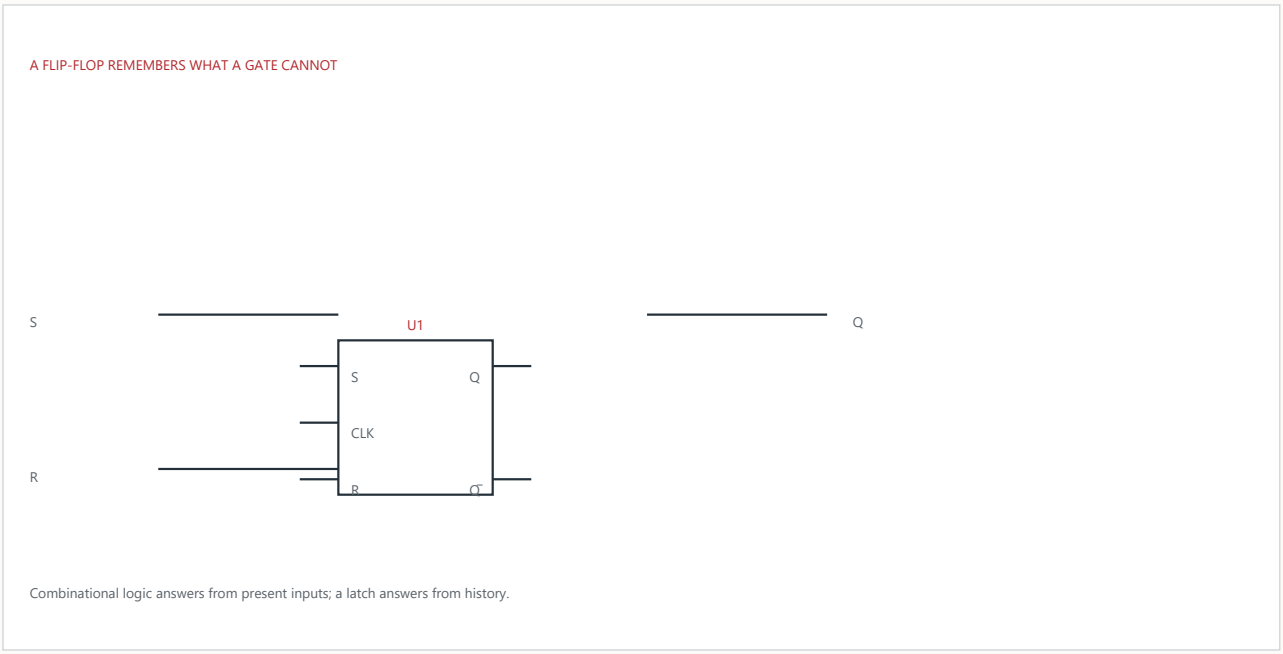



Fig. 15.3 A flip-flop. Its stored state changes only at the clock edge, so the output ignores input activity between edges.

## Logic families and interfacing

Different logic families use different thresholds and different supply voltages, and they differ widely in how much current an output can source or sink. Mixing families without checking those figures is a common source of intermittent faults, because a signal may be within the forbidden band for the receiving device even though it looks correct on a meter.

Interfacing logic to real loads needs an intervening stage. A logic output cannot drive a contactor coil directly, so a transistor or an optically coupled driver is used, and the coil receives its suppression diode. In the other direction, field inputs are conditioned and often optically isolated before they reach the logic.

| Concern | What to check | Consequence if ignored |
|---|---|---|
| Supply voltage | Both devices rated for the same rail | Device damage or no output |
| Threshold levels | Output levels fall outside the receiver's forbidden band | Intermittent and temperature-dependent faults |
| Output drive current | Fan-out against the receiving inputs | Degraded levels and slow edges |
| Unused inputs | Tied to a defined level, never left floating | Random switching and noise pickup |
| Load driving | A transistor or coupler stage with suppression | Logic output destroyed by the coil |
| Field inputs | Conditioning and isolation | Induced transients reaching the logic |

NEVER LEAVE A LOGIC INPUT FLOATING
A floating input has no defined state and picks up noise capacitively.
The device may switch randomly or oscillate at high frequency.
Tie every unused input to the supply or to the reference through a resistor.
This is among the most common causes of unrepeatable digital faults.

## Worked examples

Work each example with a pencil before reading the solution. Sketch the circuit, mark the reference node, and write the governing law before substituting numbers.

EXAMPLE 15.1 Converting between number systems

**Given.** The value 1011 0110 in binary.

**Find.** Its decimal and hexadecimal equivalents.

Split into four-bit groups

```
1011 0110
```

First group

```
1011 = 8 + 2 + 1 = 11, which is B
```

Second group

```
0110 = 4 + 2 = 6
```

Hexadecimal

```
B6
```

Decimal

```
(11 × 16) + 6 = 182
```

**Check.** Grouping into four bits makes the hexadecimal conversion immediate with no arithmetic. The decimal value then follows from the place values of the hexadecimal digits.

EXAMPLE 15.2 Translating a control requirement into logic

**Given.** A pump must run when the level switch calls for it and the selector is in automatic, or whenever the manual button is pressed, but never when the trip is active.

**Find.** The logic expression and the gates required.

First condition

```
Level AND Automatic
```

Second condition

```
OR Manual
```

Trip condition

```
AND NOT Trip
```

Complete expression

```
Run = ((Level AND Auto) OR Manual) AND NOT Trip
```

Gates needed

```
One AND for the pair, one OR, one inverter, one final AND
```

Hard-wiring note

```
The trip must also break the coil circuit physically
```

**Check.** The expression follows the wording directly. Note the final condition is implemented twice: once in logic for orderly operation, and once in hard wiring so that it still works if the logic fails.

EXAMPLE 15.3 Applying the inversion rules

**Given.** The expression NOT(A AND B AND C) is to be built, but only OR gates and inverters are available.

**Find.** An equivalent form.

Original

```
NOT(A AND B AND C)
```

Apply the rule

```
= (NOT A) OR (NOT B) OR (NOT C)
```

Gates needed

```
Three inverters and one three-input OR
```

Verification, all inputs active

```
Original gives 0; new form gives 0 OR 0 OR 0 = 0
```

Verification, A inactive

```
Original gives 1; new form gives 1 OR 0 OR 0 = 1
```

**Check.** The two forms are identical in every case, which is what makes the substitution safe. Checking two rows, one where the result differs from the others, is a quick confidence test.

EXAMPLE 15.4 Counter range and frequency division

**Given.** A chain of eight flip-flops is clocked at 32 768 Hz.

**Find.** The count range and the output frequency of the final stage.

Count range

$0$ to $2^{8} - 1$, that is $0$ to $255$

Division per stage

Each stage divides by two

Total division

$2^{8} = 256$

Output frequency

$32\,768/256 = 128$ Hz

For a 1 Hz output

A division of 32 768 is needed, requiring fifteen stages

**Check.** The relationship between stage count and division is why 32 768 Hz crystals are standard in timekeeping: fifteen divisions produce exactly one hertz.

EXAMPLE 15.5 Driving a contactor from a logic output

**Given.** A logic output can source 20 mA at 5 V. A contactor coil draws 150 mA at 24 V.

**Find.** Why the output cannot drive the coil, and what is required.

Coil current required

150 mA

Logic output capability

20 mA

Voltage mismatch

Coil needs 24 V; logic provides 5 V

Required interface

A transistor switch with a base resistor

Base current needed

With a minimum gain of 50, $I_B$ = 3 mA, driven to 12 mA

Additional component

A suppression diode across the coil

**Check.** Both the current and the voltage are wrong, so an interface stage is unavoidable. The suppression diode protects the transistor, and an optical coupler would be added if isolation between field and logic were required.

## Practical activity: gate truth tables, a latch, and the starter circuit as logic

AIM

Verify the truth table of each basic gate by measurement, confirm the inversion rules experimentally, build a cross-coupled latch and relate it to a motor starter seal-in circuit, then observe clocked flip-flop behaviour.

### Equipment

- Logic integrated circuits: quad NAND, quad NOR, hex inverter, quad AND, quad OR, and a dual flip-flop.
- Breadboard, 5 V regulated supply, and 100 nF decoupling capacitors.
- Light-emitting diodes with 1.5 kΩ series resistors for output indication.
- Switches or links with 10 kΩ pull-down resistors for inputs.
- Two push buttons, one normally open and one normally closed.
- Relay with an auxiliary contact, 24 V coil, with a transistor driver and suppression diode.
- Signal generator for a slow clock, and a two-channel oscilloscope.

### Procedure

1. Fit a decoupling capacitor across the supply pins of every integrated circuit before applying power.
2. Tie all unused inputs to a defined level through resistors and note why this is necessary.
3. Wire the AND gate with switched inputs and an indicator output. Record the output for all four input combinations.
4. Repeat for the OR, NAND, NOR and exclusive-OR gates, tabulating each full truth table.
5. Build NOT(A AND B) using a NAND gate, and separately build (NOT A) OR (NOT B) using inverters and an OR gate.
6. Drive both circuits from the same two switches and confirm their outputs agree for all four combinations.
7. Cross-couple two NAND gates so that each output feeds the other's input, with the remaining input of each brought out as set and reset.
8. Pulse the set input briefly and confirm the output stays set after the input is released.
9. Pulse the reset input and confirm the output clears.
10. Apply both inputs simultaneously and record what happens. Note why this condition is avoided.
11. Build the relay starter: a normally open start button, a normally closed stop button, the coil through its transistor driver, and the auxiliary contact wired as a seal-in.
12. Operate the start button momentarily and confirm the relay stays energised.
13. Operate the stop button and confirm it drops out.
14. Interrupt the supply briefly and confirm the relay does not re-energise by itself.
15. Apply a slow clock to the flip-flop, change its input between clock edges, and observe on the oscilloscope when the output actually responds.

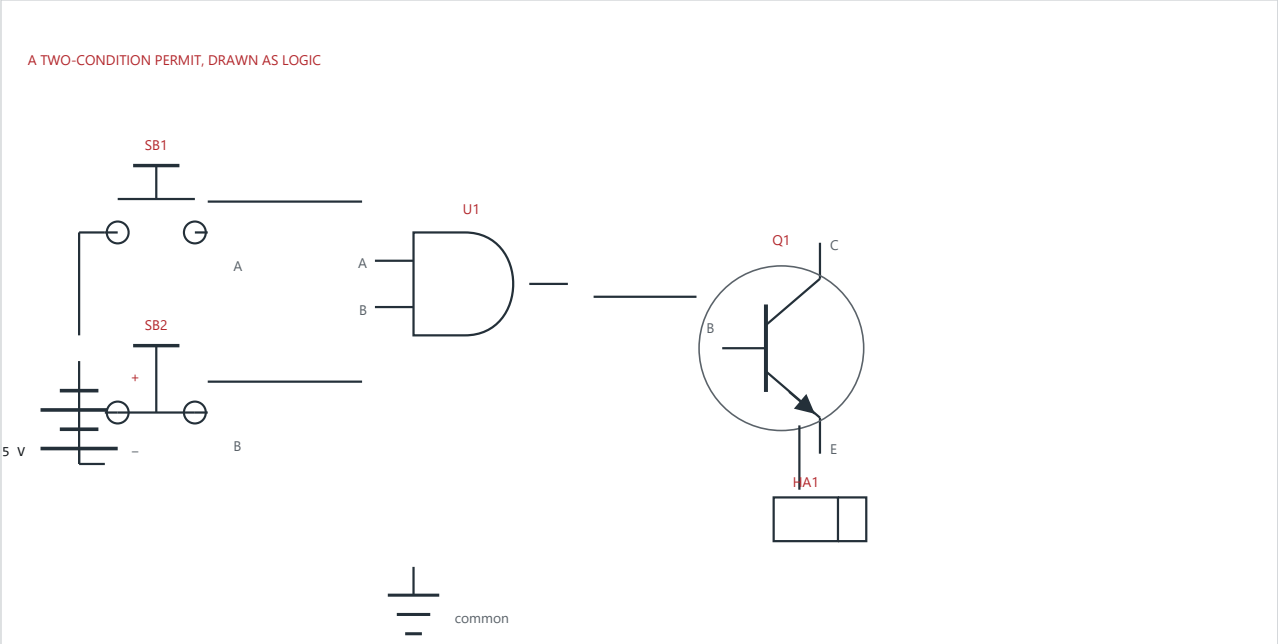



Fig. 15.L The starter circuit built as a latch. The seal-in contact provides the feedback path that corresponds to the cross-coupling in the gate version.

## Results table

| Circuit | Input combination | Output | Matches prediction? | Observation |
|---|---|---|---|---|
| AND gate | All four | | | |
| OR gate | All four | | | |
| NAND gate | All four | | | |
| NOR gate | All four | | | |
| Exclusive-OR | All four | | | |
| Inversion rule check | All four | | | |
| Latch, set then release | | | | |
| Latch, reset | | | | |
| Relay seal-in circuit | | | | |
| Flip-flop at clock edge | | | | |

## Discussion

1. Did every measured truth table match the predicted one? Note any discrepancy and its cause.
2. Did your two circuits for NOT(A AND B) agree in all four cases? What does that confirm?
3. Explain in your own words how the cross-coupled latch retains its state after the set input is released.
4. What happened when both latch inputs were applied together, and why is that condition designed out of real circuits?

5 Identify which part of the relay starter corresponds to the set input, which to the reset, and which to the cross-coupling.

6 Why did the relay not re-energise after the supply was restored? What safety purpose does that serve?

7 When did the flip-flop output actually change relative to your input change, and what does that tell you about edge triggering?

8 Deliberately leave one unused gate input floating and describe what the output does. Then tie it off again.

SAFE-WORK REMINDERS

Confirm supply polarity and voltage before inserting any integrated circuit.
Fit decoupling capacitors before applying power.
Never connect a logic output directly to a relay coil; use the transistor driver and its suppression diode.
Observe the polarity of every light-emitting diode and fit its series resistor.
Remove power before inserting or removing an integrated circuit.
Keep the relay coil circuit at low voltage and separate from any mains wiring on the bench.

## Troubleshooting guide

Faults are found by narrowing the circuit, not by replacing parts. Each row below pairs an observed symptom with the measurement that separates the likely causes.

| Symptom | Likely cause | Decisive test | Correct action |
|---|---|---|---|
| Logic output switches randomly with no input change | An unused input left floating and picking up noise | Check every input pin for a defined level | Tie all unused inputs to the supply or the reference through a resistor |
| Circuit works intermittently and worse when warm | Output levels falling within the receiver's forbidden band, often from mixed families | Measure the actual output levels and compare with the receiving device's thresholds | Match the logic families or insert a proper level-translating buffer |
| Logic output destroyed when it was connected to a coil | Coil current and voltage far beyond the output rating | Compare coil current with the output drive rating | Fit a transistor or optically coupled driver with a suppression diode |
| Outputs glitch whenever a nearby contactor operates | Inadequate decoupling and induced transients on field wiring | Observe the supply rail on an oscilloscope during switching | Fit decoupling at every device, and isolate and condition field inputs |
| Latch sets itself at power-up | Both inputs momentarily active, or no defined reset at start-up | Observe the inputs during the power-up transition | Add a power-up reset and ensure the inputs cannot both be active |
| Counter occasionally skips or doubles a count | Input contact bouncing, producing multiple edges from one operation | Observe the input on an oscilloscope while operating the contact | Debounce the input with a filter or a latch stage |
| Programmed interlock did not stop the machine on emergency stop | Safety function implemented in logic rather than hard wired | Trace the emergency-stop circuit through to the coil | Hard wire the emergency-stop contacts in series with the coil, normally closed |
| Flip-flop output changes at unexpected times | Input changing too close to the clock edge, so the sampled value is uncertain | Observe input and clock together on two channels | Ensure the input is stable for the required interval around the clock edge |

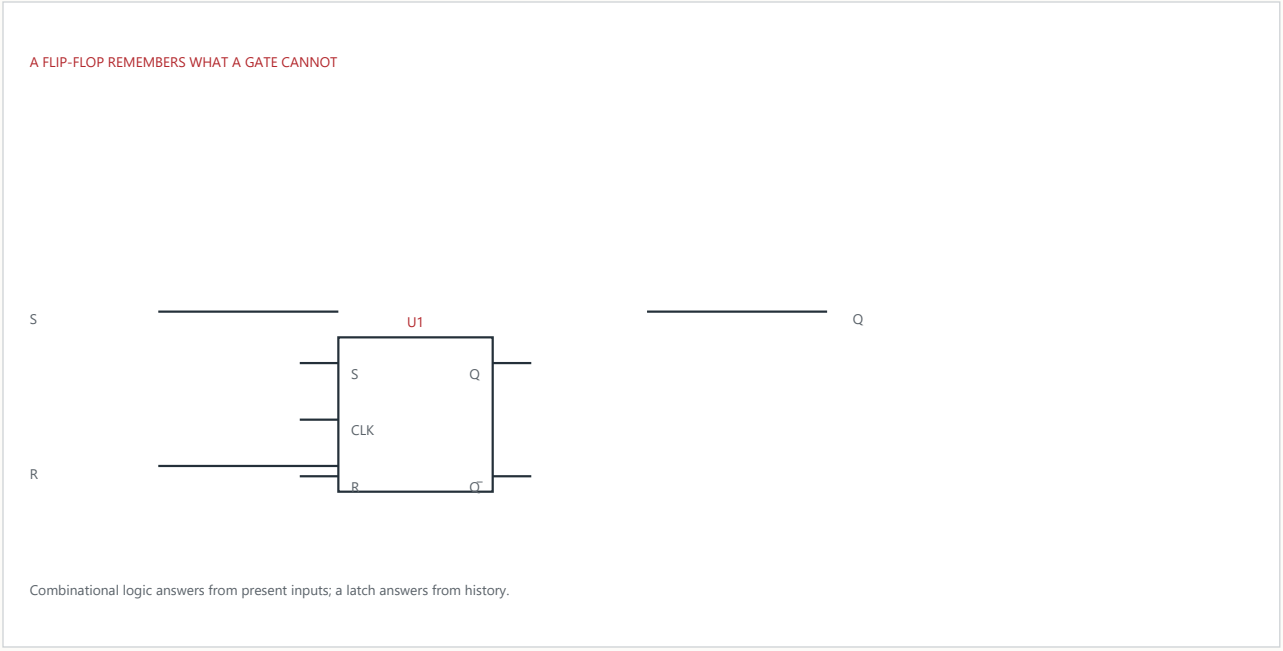



Fig. 15.T Observing clocked behaviour. The input is changed between clock edges and the output is seen to respond only at the edge itself.

FIELD HABITS THAT PREVENT CALLBACKS

Tie off every unused logic input; floating inputs cause the least repeatable faults you will ever chase.
Hard wire every safety function and treat the programmed version as coordination only.
Fit decoupling capacitors at each device as part of the build, not as a later remedy.
Always interpose a driver and a suppression diode between logic and any coil.
Check output levels against the receiving device's thresholds when mixing logic families.

## Exercises

Answer from first principles. Selected answers follow; the reasoning matters more than the number.

1 State the truth table of a two-input NAND gate.
2 Convert 1101 0011 to hexadecimal and to decimal.
3 Convert decimal 205 to binary and to hexadecimal.
4 State how many distinct values twelve bits can represent.
5 State what a small circle on a gate output indicates.
6 Write the equivalent of NOT(A OR B) using AND and inverters.
7 Express in logic: a fan runs when the thermostat calls and the system is enabled, unless the door is open.
8 Explain why an emergency stop must be hard wired rather than implemented in logic.
9 Explain how a cross-coupled pair of gates retains its state.
10 Identify the three parts of a motor starter circuit that correspond to set, reset and cross-coupling.
11 Explain why a motor does not restart by itself after a supply interruption.
12 A chain of ten flip-flops is clocked at 1.024 MHz. Find the final stage frequency.
13 State the count range of a twelve-stage counter.
14 Explain why an unused logic input must not be left floating.

15 State two reasons a 5 V logic output cannot drive a 24 V contactor coil.

### Selected answers

| # | Answer | Reason |
|---|---|---|
| 1 | Output is 0 only when both inputs are 1 | It is an AND with an inverted output |
| 2 | D3; 211 | 1101 = D, 0011 = 3; decimal is 13×16 + 3 |
| 3 | 1100 1101; CD | 128 + 64 + 8 + 4 + 1 = 205 |
| 4 | 4096 | $2^{12}$ distinct values |
| 5 | That the output is inverted | It converts AND to NAND and OR to NOR |
| 6 | (NOT A) AND (NOT B) | The inverted OR equals the AND of the inverted inputs |
| 7 | Run = Thermostat AND Enable AND NOT Door | Each stated condition becomes one input |
| 8 | It must work when the logic has failed or lost supply | A normally closed contact also opens on a broken wire, which logic does not |
| 9 | Each output drives the other's input | The state is self-maintaining once established |
| 10 | Start button, stop button, seal-in auxiliary contact | They are the set, reset and feedback path respectively |
| 11 | Loss of supply resets the latch | The seal-in path is broken, so the coil needs a fresh start command |
| 12 | 1 kHz | 1.024 MHz divided by $2^{10}$, that is 1024 |
| 13 | 0 to 4095 | $2^{12} - 1$ |
| 14 | It has no defined state and picks up noise | The device may switch randomly or oscillate |
| 15 | Insufficient current and wrong voltage | The coil needs far more current than the output can source and a higher supply rail |

## Chapter summary

- A digital signal is interpreted against thresholds with a forbidden band between, which is why it tolerates noise so well.
- Five functions, AND, OR, NOT, and the inverted and exclusive forms, account for almost all combinational logic.
- A circle on a symbol output always means inversion, and NAND or NOR alone can build any function.
- The two inversion rules allow a function to be rearranged into whichever form suits the available devices.
- A latch retains state through cross-coupled feedback, and a motor starter seal-in circuit is the same circuit built from a relay.
- A flip-flop samples its inputs only at a clock edge, which makes counters and timing chains possible.
- Safety functions must be hard wired, because logic depends on the device working and powered.
- Floating inputs, mismatched thresholds and direct coil driving are the three commonest practical failures.

### Equations introduced in this chapter

| Relationship | Expression | When it applies |
|---|---|---|
| Bit place value | Position n carries $2^n$ | Binary number |
| Values from n bits | $2^n$ distinct values | Any binary field |
| Hexadecimal grouping | One hex digit = four bits | Conversion shorthand |
| Inverted AND | NOT(A AND B) = NOT A OR NOT B | Any number of inputs |
| Inverted OR | NOT(A OR B) = NOT A AND NOT B | Any number of inputs |
| Frequency division | f_out = f_in / $2^n$ | Chain of n flip-flops |
| Counter range | 0 to $2^n - 1$ | n-stage counter |

### Vocabulary check

| Term | Meaning in this chapter |
|---|---|
| Forbidden band | The voltage range between the two valid logic thresholds |
| Truth table | A complete listing of a function's output for every input combination |
| Latch | A circuit that retains its state after the setting input is removed |
| Seal-in contact | The auxiliary contact that maintains a contactor coil after the start button is released |
| Flip-flop | A latch whose inputs are sampled only at a clock edge |
| Fan-out | The number of inputs a single output can drive within its ratings |
| Debounce | Conditioning applied to remove multiple edges from one contact operation |

LOOKING AHEAD

Logic and analogue circuits both need a stable supply, and the quality of that supply determines whether either works reliably. The next chapter examines power supplies, regulation and the protection built around them.

CHAPTER 16

# Power supplies and protection

Delivering a stable supply, and surviving what the installation does to it

WHY THIS CHAPTER MATTERS

Every circuit in the previous four chapters assumes a clean, steady supply of the right voltage. Producing that from an alternating mains input, holding it steady as load and input vary, and protecting both the supply and its load are what a power supply does. Most electronic equipment failures trace back to this part of the circuit.

## What you will be able to do

- Describe the stages of a linear power supply and the function of each.
- Compute the dissipation of a series regulator and size its heatsink requirement.
- Explain how a switching supply differs and why its efficiency is so much higher.
- Compute line and load regulation from measurements.
- Select overcurrent and overvoltage protection appropriate to a supply.
- Explain earthing, isolation and the purpose of a protective conductor in equipment supplies.

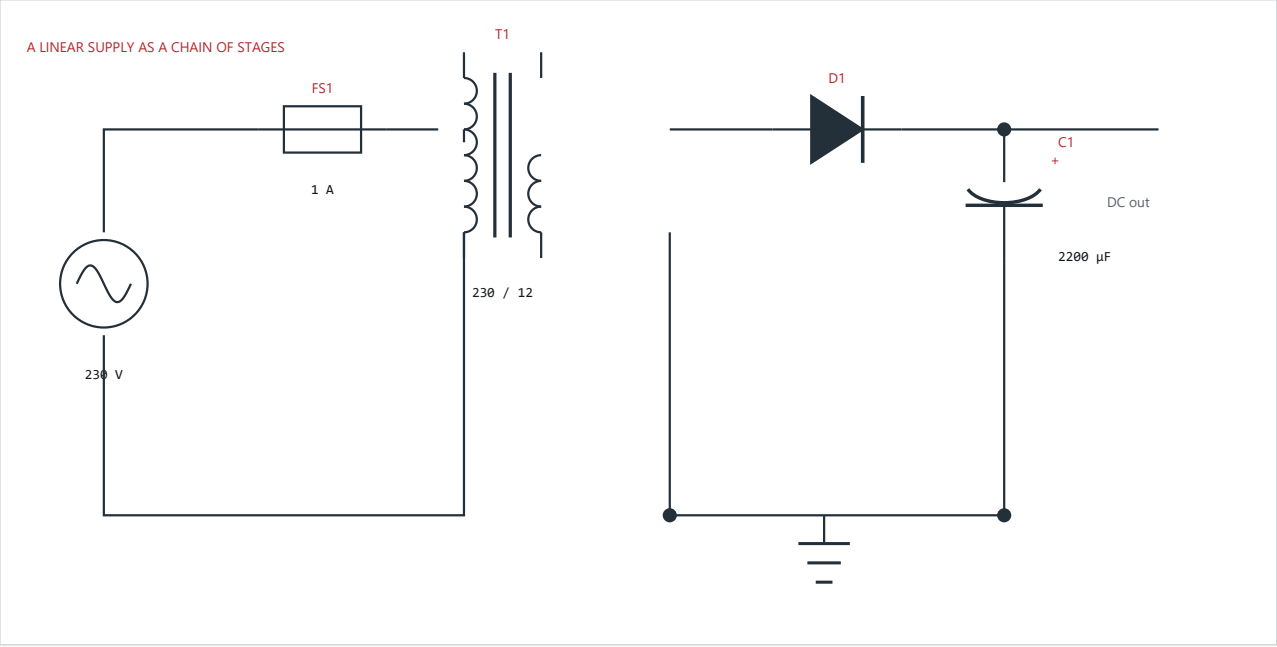



Fig. 16.1 A complete linear supply. The transformer isolates and steps down, the bridge rectifies, the reservoir capacitor smooths, and the regulator holds the output steady.

## Stages of a linear supply

A linear supply performs four distinct jobs in sequence. The transformer isolates the output from the mains and steps the voltage down. The rectifier converts the alternating waveform to unidirectional. The reservoir capacitor smooths that into a roughly steady voltage with ripple on it. The regulator then removes the remaining ripple and holds the output constant despite changes in input voltage and load current.

Each stage must be sized for what the next one demands. The regulator needs a minimum input above its output, called the dropout voltage, maintained even at the bottom of the ripple and at the lowest expected mains voltage. Designing the reservoir capacitor without accounting for that minimum is why some supplies work on the bench and fail on a low mains.

```
V_in(min) at the ripple trough > V_out + V_dropout

Headroom = V_in − V_out

Stages: isolate → rectify → smooth → regulate
```

HEADROOM MUST SURVIVE THE WORST CASE, NOT THE TYPICAL CASE

The regulator sees the lowest input at the ripple trough, on a low mains, at full load.
Too little headroom and the output follows the input down, with ripple appearing on it.
Too much headroom and the regulator dissipates the surplus as heat.
Design for the minimum case and then check dissipation at the maximum case.

## The series regulator and its heat

A linear regulator works by dropping the surplus voltage across a series device, adjusting that drop continuously so the output stays constant. It is simple, quiet and has excellent output purity, which is why it remains the choice for sensitive analogue work.

The cost is heat. Every volt of surplus multiplied by every ampere of load becomes heat in the regulator, and none of it reaches the load. A supply dropping twelve volts to five at two amperes delivers ten watts and dissipates fourteen. Efficiency falls as the difference between input and output grows, so a linear supply is poorly suited to large voltage reductions or high currents.

```
P_reg = (V_in − V_out) × I_load

η ≈ V_out / V_in

Required thermal path from P_reg and the permitted temperature rise
```

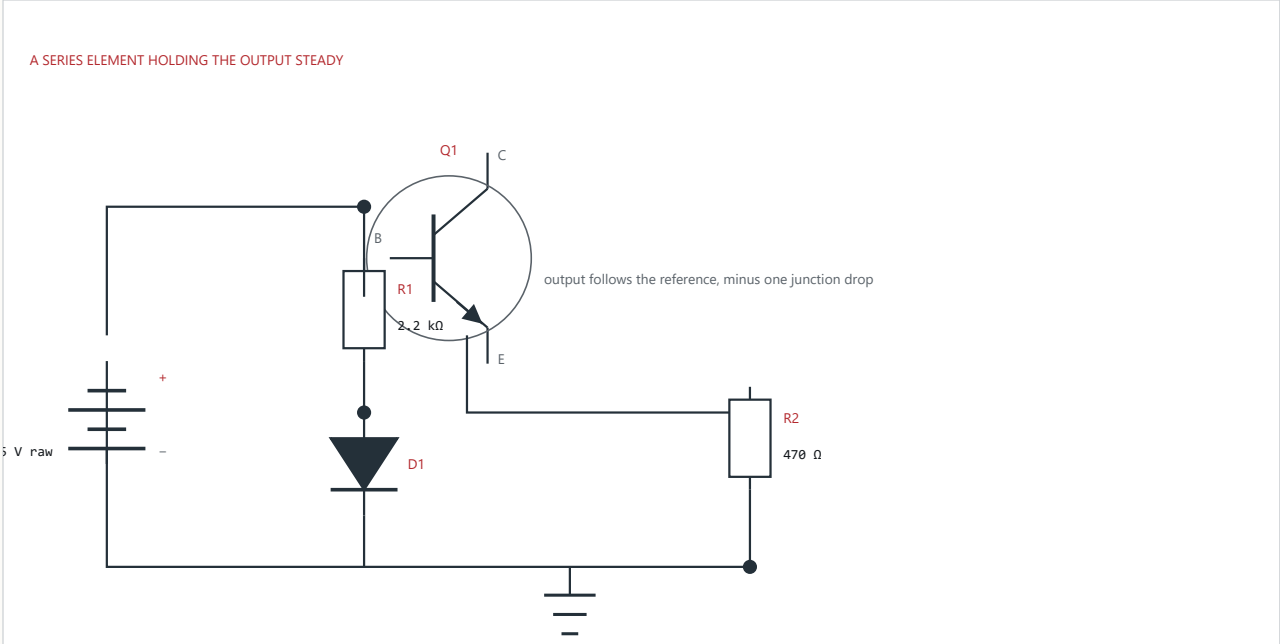



Fig. 16.2 A series regulator. The pass device carries the full load current and drops the difference between input and output, dissipating that product as heat.

## Switching supplies

A switching supply avoids the dissipation by not dropping voltage continuously. Instead it switches the input on and off rapidly into an inductor and capacitor, and varies the proportion of time it is on to set the average output. Because the switch is either fully on or fully off, it dissipates little in either state.

The result is efficiency typically between eighty-five and ninety-five per cent, almost independent of the voltage ratio, and much smaller magnetic components because the switching frequency is far above mains frequency. The penalties are electrical noise at the switching frequency and its harmonics, greater circuit complexity, and a failure mode that can place the input voltage on the output if the switch fails short.

```
Duty ratio D = on time / period

Step-down output:  V_out ≈ D V_in

Efficiency largely independent of the voltage ratio
```

| Property | Linear supply | Switching supply |
|---|---|---|
| Efficiency | Low, falls as the ratio grows | High, 85 to 95 % |
| Heat produced | Large, proportional to voltage dropped | Small |
| Output noise | Very low | Switching frequency and harmonics present |
| Size and mass | Large mains-frequency transformer | Small high-frequency magnetics |
| Complexity | Low | High |

| Property | Linear supply | Switching supply |
|---|---|---|
| Best suited to | Sensitive analogue, small voltage drops | General supply, large drops, high current |

## Regulation and measurement

A supply's quality is expressed in two figures. Line regulation is how much the output changes when the input voltage varies across its specified range. Load regulation is how much it changes from no load to full load. Both are quoted as a percentage of the nominal output and both are measured by deliberately varying one condition while holding the other fixed.

Output impedance is the underlying cause of load regulation: a supply with low output impedance holds its voltage as current is drawn. Ripple rejection describes how much of the input ripple survives to the output, and is usually quoted in decibels.

```
Line regulation % = ΔV_out / V_out(nom) × 100  for the input range

Load regulation % = (V_noload − V_fullload)/V_nominal × 100

Output impedance = ΔV_out / ΔI_load
```

## Protecting the supply and the load

Protection is needed in both directions. The supply must be protected against an overload or short circuit in the load, which is the job of a fuse or a current limit; and the load must be protected against the supply failing in a way that raises the output voltage, which is the job of overvoltage protection.

A fuse interrupts a sustained fault but is slow against a short-lived surge, so electronic current limiting is usually added to act immediately. Overvoltage protection matters because a shorted pass device in a linear supply, or a shorted switch in a switching supply, places the unregulated input directly on the load, which may destroy everything connected.

### Layers of protection in a supply

- Input fuse, rated for inrush but clearing a genuine fault.
- Transformer thermal cut-out, where fitted.
- Electronic current limiting, acting immediately on an output short.
- Output fuse where several loads share one supply.
- Overvoltage protection against pass-device failure.
- Transient suppression at the input against mains surges.
- Thermal shutdown in the regulator against loss of cooling.

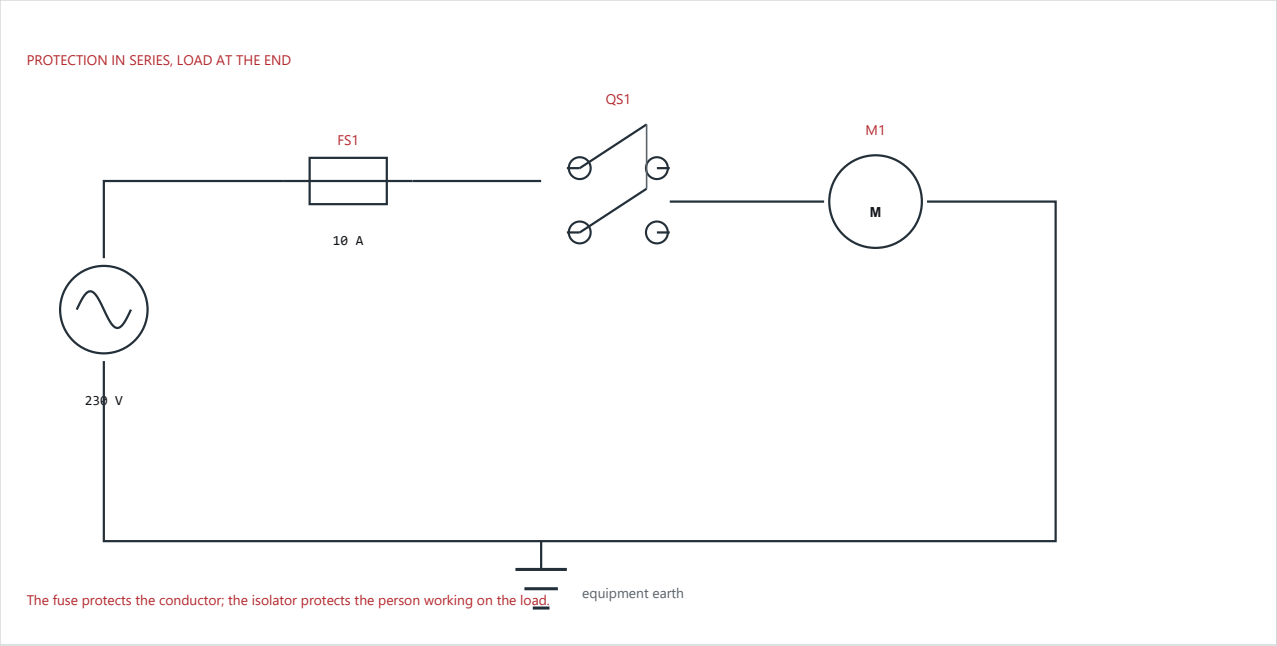



Fig. 16.3 Protection arrangements. The fuse clears a sustained fault, while a voltage limiting device across the output absorbs transients and limits overvoltage.

## Isolation, earthing and the protective conductor

A mains-supplied power supply must separate its output from the mains, and the transformer provides that separation. The quality of the separation is what distinguishes a supply suitable for accessible low-voltage circuits from one that is not, and it is defined by the insulation between the windings and the spacing within the unit.

Where exposed conductive parts of equipment could become live through an insulation failure, they are connected to a protective conductor so that a fault produces enough current to operate the protective device. A supply whose output is referenced to the protective conductor cannot be connected in series with another similarly referenced supply without creating a parallel path, which is a frequent cause of unexplained circulating currents in multi-supply installations.

National wiring rules set the detailed requirements for conductor sizing, protective device selection and isolation in these circuits. Where a specific national code is being applied, the edition in force must be identified before relying on any particular article.

> WHY TWO EARTHED SUPPLIES CANNOT SIMPLY BE STACKED
> If both outputs reference the protective conductor, that conductor bridges them.
> The intended series connection is shorted out through the protective path.
> Circulating current flows and the voltages are not what was expected.
> Use one supply with a floating output, or a genuinely isolated second supply.

## Worked examples

Work each example with a pencil before reading the solution. Sketch the circuit, mark the reference node, and write the governing law before substituting numbers.

EXAMPLE 16.1 Regulator dissipation and heatsink requirement

**Given.** A linear regulator produces 5 V at 1.5 A from an unregulated input of 14 V.

**Find.** The dissipation, the efficiency, and the implication for cooling.

Voltage dropped

```
14 − 5 = 9 V
```

Dissipation

```
P = 9 × 1.5 = 13.5 W
```

Power delivered

```
5 × 1.5 = 7.5 W
```

Efficiency

```
7.5/21 = 36 %
```

Cooling

```
13.5 W demands a substantial heatsink
```

**Check.** Nearly twice as much power is wasted as delivered. A switching supply at ninety per cent efficiency would dissipate under a watt for the same output, which is why linear regulators are confined to small drops or small currents.

EXAMPLE 16.2 Reservoir capacitor sized for regulator headroom

**Given.** A 12 V regulator with a 2 V dropout requirement draws 1 A from a bridge rectifier on a 50 Hz supply. The rectified peak is 19 V.

**Find.** The maximum permissible ripple and the required capacitance.

Minimum acceptable input

```
12 + 2 = 14 V
```

Available ripple allowance

```
19 − 14 = 5 V peak to peak
```

Ripple frequency

```
100 Hz for a bridge
```

Required capacitance

```
C = I/(f V) = 1/(100 × 5) = 2000 µF
```

With margin for low mains

```
Fit 4700 µF
```

**Check.** The capacitor is sized by the regulator's headroom requirement, not by any ripple target of its own. The margin covers a low mains condition, where the peak falls and the headroom shrinks.

EXAMPLE 16.3 Line and load regulation from measurements

**Given.** A 12 V supply measures 12.04 V at no load and 11.88 V at full load. Varying the input across its full range changes the output from 11.96 V to 12.02 V.

**Find.** The load regulation, the line regulation, and the output impedance if full load is 2 A.

Load regulation

```
(12.04 − 11.88)/12 × 100 = 1.33 %
```

Line regulation

```
(12.02 − 11.96)/12 × 100 = 0.5 %
```

Output impedance

```
ΔV/ΔI = 0.16/2 = 0.08 Ω
```

Assessment

```
Both figures are typical of a competent linear supply
```

**Check.** The output impedance of 80 milliohms is what produces the load regulation. A supply feeding a pulsed load would need a lower figure, or local decoupling at the load to supply the pulses.

EXAMPLE 16.4 Switching supply duty ratio and efficiency comparison

**Given.** A step-down switching supply produces 5 V at 4 A from a 24 V input at 88 % efficiency.

**Find.** The duty ratio, the input power, the dissipation, and the dissipation a linear supply would have had.

Duty ratio

```
D ≈ V_out/V_in = 5/24 = 0.21
```

Output power

```
5 × 4 = 20 W
```

Input power

```
20/0.88 = 22.7 W
```

Switching supply dissipation

```
22.7 − 20 = 2.7 W
```

Linear equivalent dissipation

```
(24 − 5) × 4 = 76 W
```

Ratio

```
The linear supply would dissipate 28 times as much
```

**Check.** Seventy-six watts in a pass device is impractical to cool in most enclosures. This comparison is why nearly all equipment supplies with large voltage ratios are switching types.

EXAMPLE 16.5 Fuse selection for a supply

**Given.** A supply draws 0.9 A from the mains in normal operation, with an inrush of 12 A for 20 ms at switch-on.

**Find.** A suitable fuse rating and type.

Continuous requirement

```
Must carry 0.9 A indefinitely
```

Inrush requirement

```
Must not clear on 12 A for 20 ms
```

Rating

```
Choose 1.25 A or 1.6 A, above the continuous current
```

Type

```
Time-delay, to ride through the inrush
```

Fault requirement

```
Must clear a genuine sustained overload
```

**Check.** A fast-acting fuse of the same rating would clear on the inrush at every switch-on. The selection must satisfy three separate conditions, and only the type distinguishes them.

## Practical activity: supply stages, regulation, dissipation and protection

AIM

Build a linear supply stage by stage and measure the waveform at each point, then measure line and load regulation, compute regulator dissipation against measured temperature rise, and verify current limiting and overvoltage protection.

### Equipment

- Transformer, 230 V to 18 V, supplied through an isolating transformer and a residual current device.
- Bridge rectifier, 2 A, on a small heatsink.
- Electrolytic capacitors, 470 µF and 4700 µF, rated above 50 V.
- Three-terminal regulator, 12 V, with a heatsink and mounting hardware.
- Resistive load bank, switchable, and a decade resistance box.
- Variable alternating supply to simulate a varying mains input.
- Two multimeters, an oscilloscope with an isolated input, and an infrared thermometer.
- Fuses of assorted ratings and types, and a voltage limiting device for the output.

### Procedure

1. Confirm the bench supply is fed through an isolating transformer and a residual current device before beginning.
2. Connect the transformer alone and measure its secondary open-circuit voltage. Compute the expected rectified peak.
3. Add the bridge rectifier with a 1 kΩ load and observe the unsmoothed waveform. Record its peak, mean and ripple frequency.
4. Add the 470 µF capacitor and record the new mean output and the ripple at 0.2 A load.
5. Replace it with the 4700 µF capacitor and record the ripple again at the same load.
6. Add the regulator with its heatsink. Measure the regulated output and the ripple remaining on it.
7. Compare the ripple before and after the regulator and express the reduction in decibels.
8. Load regulation: record the output at no load and at four increasing load steps up to 1 A.
9. At each step, measure the voltage across the regulator and compute its dissipation.
10. Measure the regulator case temperature at the highest load after five minutes.
11. Line regulation: hold the load at 0.5 A and vary the input from the variable supply across a range of plus and minus 10 per cent, recording the output at each setting.
12. Reduce the input further until the output begins to fall, and record the input at which regulation is lost. Compare with the stated dropout voltage.
13. Discharge the capacitors through a resistor and prove dead.
14. Fit a fuse one size above the measured continuous current and confirm the supply starts without clearing it.
15. Apply a short circuit to the output briefly through an ammeter and record the current the supply allows. Remove it immediately.
16. Discharge, prove dead, and fit the voltage limiting device across the output. Note the voltage at which it would act.

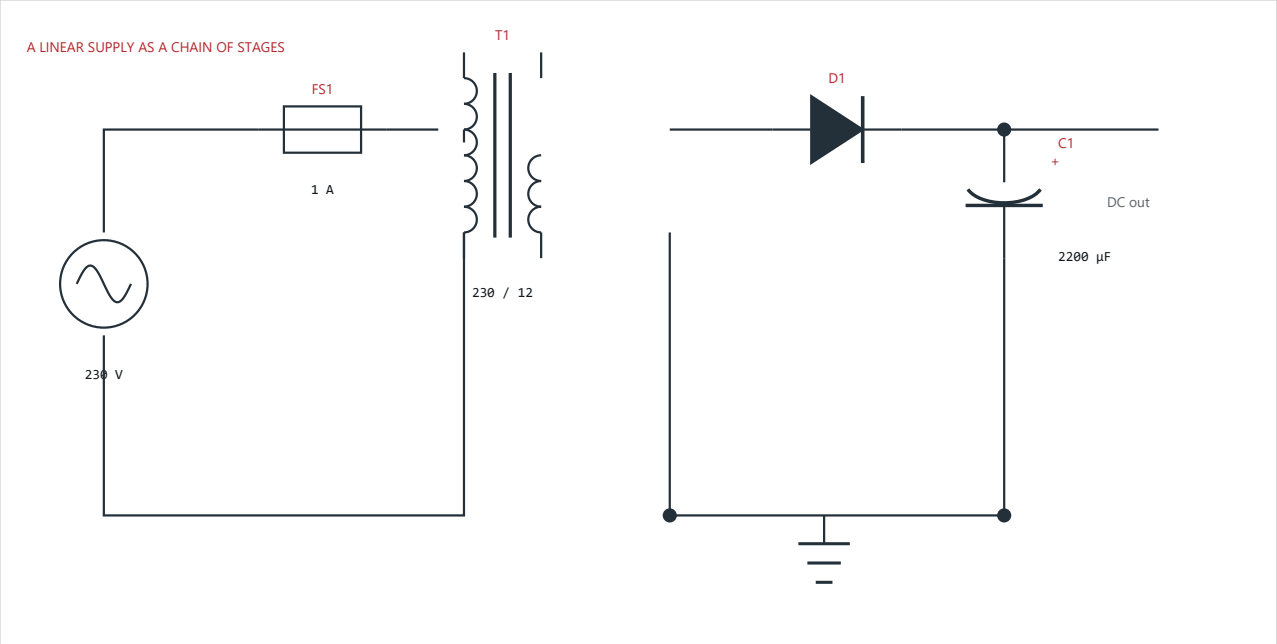



Fig. 16.L The supply under test. Measurements are taken at the transformer secondary, after the rectifier, after the reservoir capacitor and at the regulated output.

## Results table

| Measurement point or condition | Voltage (V) | Ripple (V pk-pk) | Dissipation (W) | Comment |
|---|---|---|---|---|
| Secondary, open circuit | | | | |
| After rectifier, unsmoothed | | | | |
| After 470 µF at 0.2 A | | | | |
| After 4700 µF at 0.2 A | | | | |
| Regulated output, no load | | | | |
| Regulated output, 0.25 A | | | | |
| Regulated output, 0.5 A | | | | |
| Regulated output, 1.0 A | | | | |
| Input at loss of regulation | | | | |
| Short-circuit current | | | | |

## Discussion

1. How did your smoothed voltage compare with the rectified peak and with the unsmoothed mean? Explain which it approached and why.
2. By what factor did ripple fall when you increased the capacitance tenfold? Compare with the prediction.
3. Express the regulator's ripple reduction in decibels from your measurements before and after it.
4. Compute your load regulation as a percentage. What output impedance does that imply at your full-load current?
5. Compute your line regulation across the input range you tested.
6. At what input voltage was regulation lost, and how does the difference between that and the output compare with the stated dropout voltage?
7. Compute the regulator dissipation at 1 A and compare the measured temperature rise with what that dissipation implies for your heatsink.
8. Compute the efficiency of your supply at 1 A. How would a switching supply of the same output compare?
9. What short-circuit current did the supply allow, and is that consistent with an internal current limit?
10. Why must a supply have both a fuse and electronic current limiting rather than only one of them?

> SAFE-WORK REMINDERS
> Supply the bench through an isolating transformer and a residual current device at all times.
> Do not connect an earthed oscilloscope ground to a mains-referenced point; use an isolated or differential input.
> Reservoir capacitors hold a dangerous charge after the supply is removed. Discharge through a resistor and prove dead before touching the circuit.
> Observe electrolytic capacitor polarity; a reversed electrolytic can rupture violently.
> Apply the output short circuit only momentarily and through an ammeter, never by hand.
> The regulator and its heatsink reach high temperatures. Use the thermometer, not a finger.
> Prove dead and discharge before fitting or removing any component.

## Troubleshooting guide

Faults are found by narrowing the circuit, not by replacing parts. Each row below pairs an observed symptom with the measurement that separates the likely causes.

| Symptom | Likely cause | Decisive test | Correct action |
|---|---|---|---|
| Output voltage correct at no load but falls under load | Regulator losing headroom at the ripple trough, or transformer undersized | Observe the regulator input on an oscilloscope while loaded | Increase the reservoir capacitance or use a higher secondary voltage |
| Ripple appears on the regulated output only on a low mains | Dropout voltage reached at the trough of the ripple | Measure the minimum instantaneous input at full load | Raise the input voltage or increase the capacitance to reduce the ripple depth |
| Regulator shuts down intermittently when warm | Thermal protection operating because dissipation exceeds the heatsink capability | Compute dissipation and measure case temperature | Fit a larger heatsink, improve airflow, or change to a switching supply |
| Equipment downstream destroyed by overvoltage | Pass device or switch failed short, placing the unregulated input on the output | Measure the output with the load disconnected | Replace the supply and fit overvoltage protection across the output |
| Input fuse clears at every switch-on | Fast-acting type fitted where the inrush requires a time-delay type | Measure the inrush current and duration | Fit a time-delay fuse of suitable rating |
| Noise from a switching supply disturbs analogue measurements | Switching frequency and harmonics coupling into the signal circuit | Observe the analogue signal with the supply running and inhibited | Add filtering, improve layout and separation, or use a linear supply for that circuit |
| Unexplained circulating current between two supplies | Both outputs referenced to the protective conductor, creating a parallel path | Test for continuity between the two output reference points | Use one floating output or a genuinely isolated supply |
| Electrolytic capacitor has bulged or vented | Exceeded ripple current rating, excessive temperature, or reverse polarity | Inspect, and check the ripple current against the component rating | Replace with a component rated for the ripple current and verify polarity |

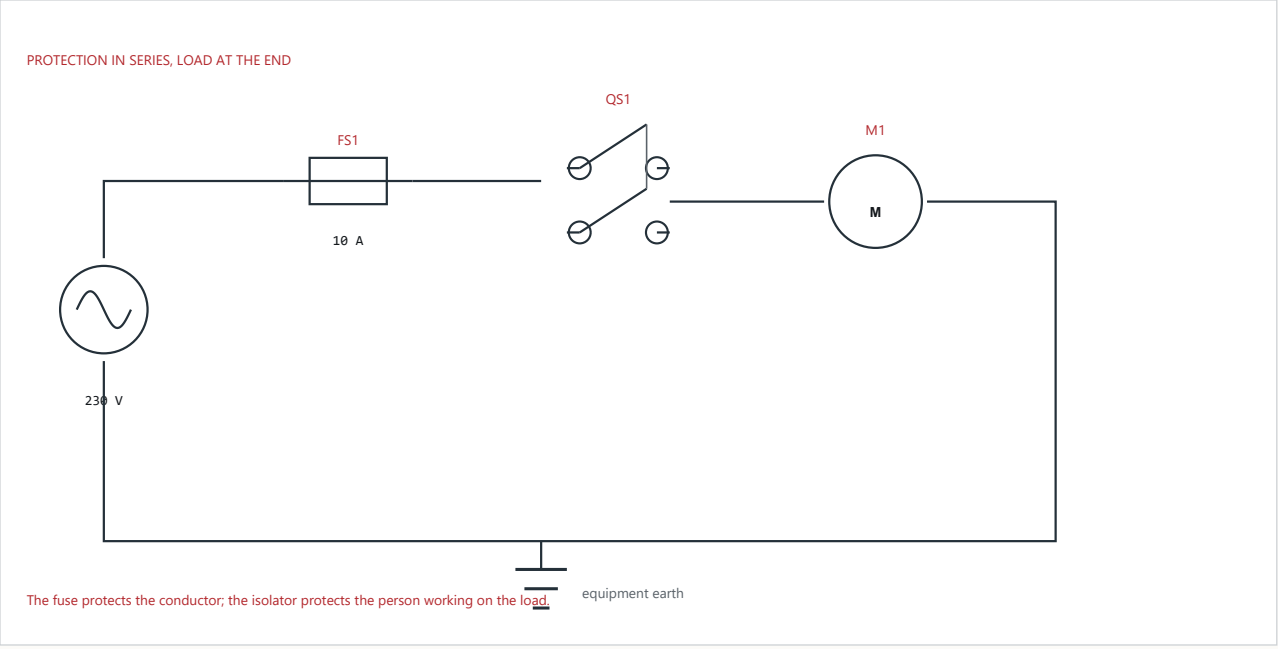



Fig. 16.T Verifying protection. The current limit is checked through an ammeter and the overvoltage device is selected to act below the downstream equipment's limit.

FIELD HABITS THAT PREVENT CALLBACKS

Design headroom for the worst case: ripple trough, low mains and full load together.
Compute regulator dissipation before selecting a package, and provide the thermal path to match.
Discharge reservoir capacitors through a resistor and prove dead as a routine step.
Fit overvoltage protection wherever a supply failure could destroy expensive downstream equipment.
Check whether a supply output is referenced to the protective conductor before connecting two supplies together.
Match the fuse type, not only its rating, to the inrush the supply actually draws.

## Exercises

Answer from first principles. Selected answers follow; the reasoning matters more than the number.

1 Name the four stages of a linear power supply in order and state the function of each.
2 A regulator drops 15 V to 9 V at 0.8 A. Find its dissipation.
3 Find the efficiency of the supply above.
4 Explain why a linear supply becomes impractical for large voltage reductions at high current.
5 A 5 V regulator has a 2.5 V dropout. State the minimum input it requires.
6 A bridge rectifier on 50 Hz feeds a regulator needing 3 V of ripple allowance at 0.75 A. Find the capacitance required.
7 Explain how a switching supply achieves high efficiency.
8 A step-down switching supply gives 12 V from 48 V. Find the approximate duty ratio.
9 A supply gives 24.1 V at no load and 23.7 V at 3 A. Find the load regulation and the output impedance.
10 State the difference between line regulation and load regulation.
11 State one advantage and one disadvantage of a switching supply compared with a linear supply.

12 Explain why a supply needs both a fuse and electronic current limiting.

13 Explain what overvoltage protection guards against and why it is needed.

14 Explain why two supplies whose outputs are both referenced to the protective conductor cannot be connected in series.

15 State why the edition of a national wiring code must be identified before relying on a particular requirement.

## Selected answers

| # | Answer | Reason |
|---|---|---|
| 1 | Transformer, rectifier, reservoir capacitor, regulator | Isolate and step down, convert to unidirectional, smooth, then hold steady |
| 2 | 4.8 W | P = (V_in − V_out) I |
| 3 | 60 % | η ≈ V_out/V_in |
| 4 | Dissipation is the dropped voltage times the current | Both the heat and the cooling requirement become impractical |
| 5 | 7.5 V | Output plus dropout |
| 6 | 2500 µF | C = I/(f_ripple V_ripple) with f_ripple = 100 Hz |
| 7 | The switch is either fully on or fully off | It drops little voltage while conducting and carries none while off, so it dissipates little |
| 8 | 0.25 | D ≈ V_out/V_in |
| 9 | 1.67 %; 0.133 Ω | Regulation from the voltage change; impedance from ΔV/ΔI |
| 10 | Line regulation responds to input change; load regulation to load change | They are measured by varying one while holding the other fixed |
| 11 | High efficiency; switching noise | Less heat and smaller magnetics, but noise at the switching frequency |
| 12 | The fuse is too slow for a transient fault | Current limiting acts immediately, while the fuse clears a sustained fault |
| 13 | The pass device or switch failing short | That places the unregulated input directly on the load, destroying downstream equipment |
| 14 | The protective conductor bridges both outputs | It shorts out the intended series connection and circulating current flows |
| 15 | Requirements change between editions | A citation is only meaningful with its edition identified |

## Chapter summary

- A linear supply isolates, rectifies, smooths and regulates, and each stage must be sized for the demands of the next.
- Regulator headroom must be maintained at the ripple trough, on a low mains, at full load.
- A series regulator dissipates the dropped voltage times the load current, so efficiency falls as the voltage ratio grows.
- A switching supply varies duty ratio instead of dropping voltage, giving high efficiency and small magnetics at the cost of noise and complexity.
- Line and load regulation quantify supply quality, and output impedance is the underlying cause of load regulation.
- Protection is needed in both directions: current limiting and fusing for the supply, overvoltage protection for the load.
- Isolation and the protective conductor arrangement determine whether supplies can be combined safely, and national code requirements must be applied by edition.

### Equations introduced in this chapter

| Relationship | Expression | When it applies |
|---|---|---|
| Regulator dissipation | $P = (V_{in} - V_{out})\, I_{load}$ | Linear series regulator |
| Linear efficiency | $\eta \approx V_{out} / V_{in}$ | Linear supply |
| Minimum input | $V_{in} > V_{out} + V_{dropout}$ | At the ripple trough, full load |
| Reservoir capacitance | $C = I/(f_{ripple}\, V_{ripple})$ | Capacitor-smoothed rectifier |
| Switching output | $V_{out} \approx D\, V_{in}$ | Step-down switching converter |
| Load regulation | $(V_{nl} - V_{fl})/V_{nom} \times 100$ | No load to full load |
| Output impedance | $Z_{out} = \Delta V_{out}/\Delta I_{load}$ | From load regulation measurements |

### Vocabulary check

| Term | Meaning in this chapter |
|---|---|
| Dropout voltage | The minimum excess of input over output a regulator requires |
| Headroom | The margin between the input and the output of a regulator |
| Duty ratio | The proportion of each switching period for which the switch conducts |
| Line regulation | Output change caused by a change in input voltage |
| Load regulation | Output change caused by a change in load current |
| Current limiting | Electronic restriction of output current during a fault |
| Overvoltage protection | A device that limits the output if the regulator fails short |
| Protective conductor | The conductor connecting exposed conductive parts for fault protection |

LOOKING AHEAD

Single-phase supplies serve most electronic equipment, but industrial power is distributed and consumed in three phases. The next chapter develops three-phase systems, their connections and their measurement.

CHAPTER 17

# Three-phase systems

Star and delta, line and phase quantities, and how power is measured

WHY THIS CHAPTER MATTERS

Industrial power is generated, transmitted and consumed in three phases, because three-phase transmission carries more power for a given weight of conductor and because three-phase motors are self-starting and smooth-running. Nearly every mistake made with three-phase systems comes from confusing line quantities with phase quantities.

## What you will be able to do

- Describe a balanced three-phase supply and the 120-degree relationship between phases.
- Distinguish line and phase voltages and currents in star and delta connections.
- Apply the root-three relationships correctly in each connection.
- Compute three-phase power, apparent power and power factor.
- Explain the function of the neutral conductor and the effect of unbalance.
- Describe the two-wattmeter method and interpret its readings.

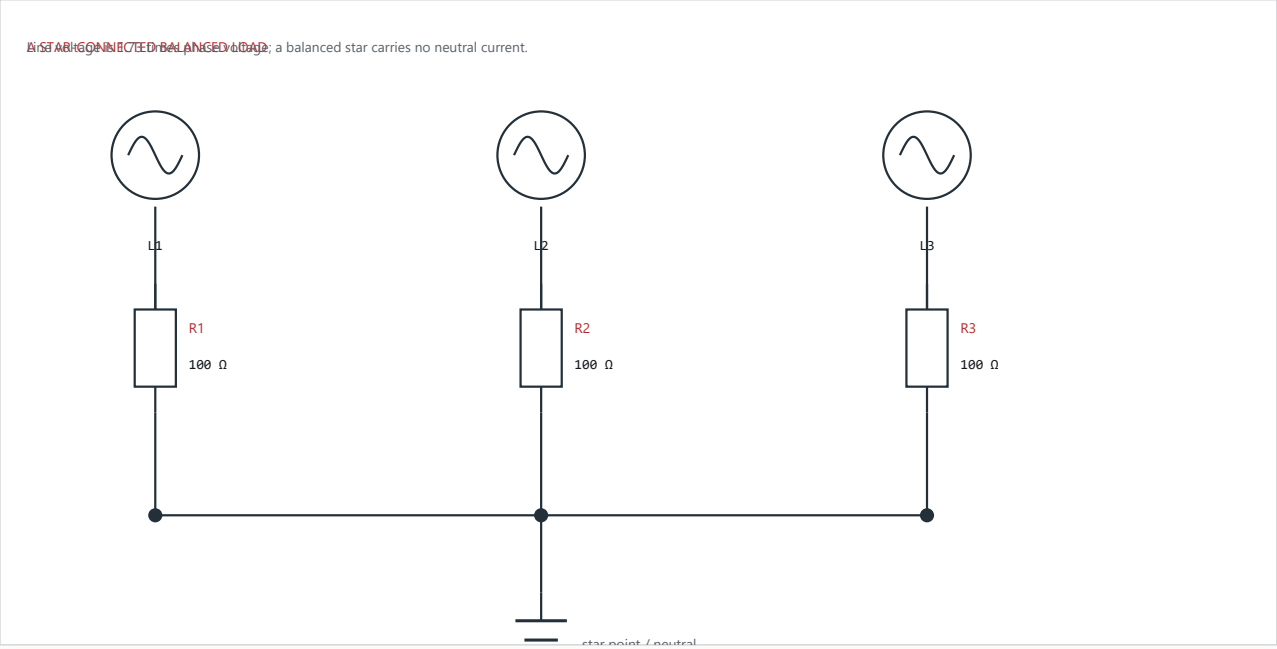



Fig. 17.1 A star-connected load. The three phase windings meet at a common star point, which may be brought out as a neutral conductor.

## The balanced three-phase supply

A three-phase supply consists of three alternating voltages of equal magnitude, separated in phase by 120 degrees. Taken together they sum to zero at every instant, which is the property that makes the whole system efficient: in a balanced system the return currents cancel and no neutral current flows at all.

The order in which the three phases reach their maxima is the phase sequence, and it determines the direction in which a connected motor will rotate. Reversing any two conductors reverses the sequence, which is why phase sequence must be verified before connecting rotating machinery.

```
Three voltages, equal magnitude, 120° apart

Instantaneous sum of a balanced set = 0

Phase sequence determines rotation direction
```

THE NEUTRAL CARRIES ONLY THE UNBALANCE

In a perfectly balanced star-connected load the neutral current is zero.
The neutral exists to carry the difference when phases are unequally loaded.
It also holds each phase voltage fixed relative to the star point.
Losing the neutral on an unbalanced load lets phase voltages shift severely.

## Star connection

In a star connection, one end of each phase winding is joined at a common point. The voltage measured between any two line conductors is root three times the voltage across any one phase, because the two phase voltages involved are 120 degrees apart rather than in line. The current in each line conductor is the same as the current through the phase winding it feeds.

This is the connection used for distribution to mixed loads, because the star point provides a neutral. A 400-volt line voltage gives 230 volts to neutral, so three-phase machines and single-phase loads can be served from the same supply.

```
V_L = √3 V_ph      so V_ph = V_L / √3

I_L = I_ph

Example: 400 V line gives 400/1.732 = 231 V to neutral
```

## Delta connection

In a delta connection the three windings are joined end to end in a closed loop, with the line conductors taken from the three junctions. Here the relationship is reversed: each phase winding sees the full line voltage, while the line current is root three times the current in any one phase winding.

Delta has no star point and therefore no neutral, so it suits balanced three-phase loads such as motors rather than mixed single-phase loads. The absence of a neutral also means a delta load is unaffected by the kind of voltage shift that a lost neutral produces in a star system.

```
V_ph = V_L
```

```
I_L = √3 I_ph      so I_ph = I_L / √3

No star point, so no neutral available
```

| Quantity | Star connection | Delta connection |
|---|---|---|
| Phase voltage | V_L / √3 | Equal to V_L |
| Line voltage | √3 V_ph | Equal to V_ph |
| Phase current | Equal to I_L | I_L / √3 |
| Line current | Equal to I_ph | √3 I_ph |
| Neutral available | Yes, at the star point | No |
| Typical use | Distribution to mixed loads | Balanced loads, motor windings |

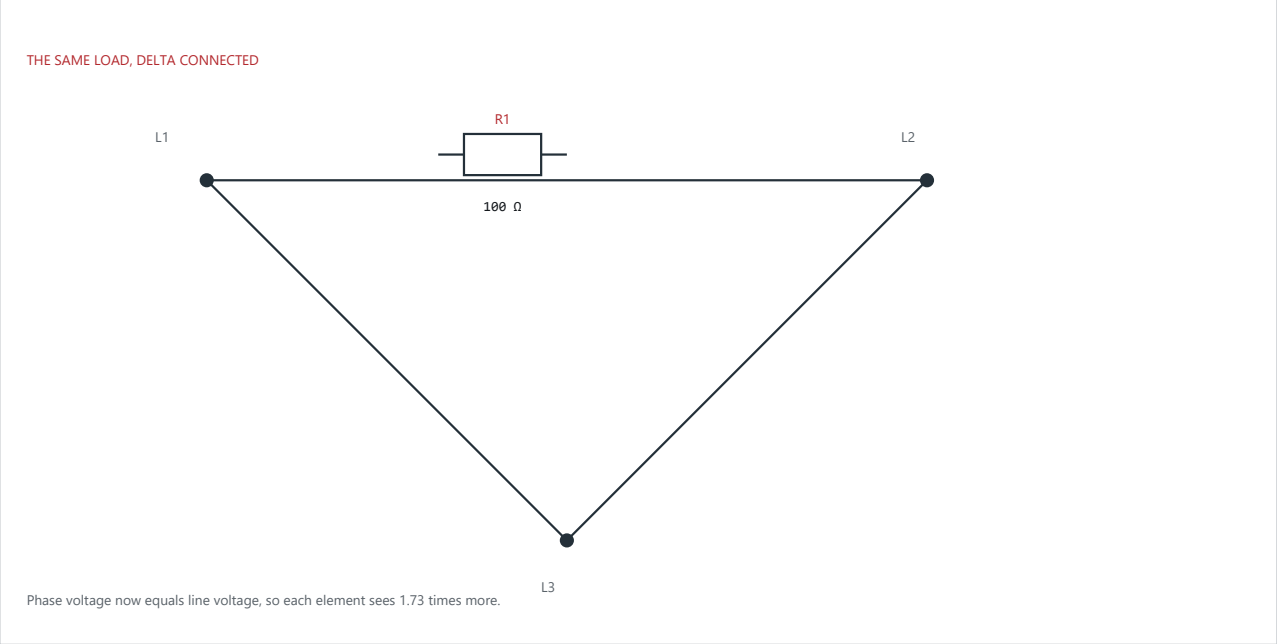



Fig. 17.2 A delta-connected load. Each winding sees the full line voltage, and the line current divides between two windings at each junction.

## Three-phase power

Total power in a balanced three-phase system is three times the power in one phase, and when expressed in terms of line quantities this reduces to a single formula that holds for both star and delta. The root-three factor appears once, and the power factor is the cosine of the angle between the phase voltage and the phase current.

The convenience of this is that total power can be found from line measurements alone, without opening the load to find out whether it is star or delta connected. The same formula applies either way, which is why it is worth memorising in the line-quantity form.

```
P = √3 V_L I_L cosϕ      (watts, either connection)
```

```
S = √3 V_L I_L      (volt-amperes)

Q = √3 V_L I_L sinϕ      (reactive volt-amperes)

Power factor = P / S
```

Note that a load reconnected from star to delta on the same supply draws three times the power, because each winding then sees root three times the voltage and the current rises by the same factor. This is exactly the relationship exploited by star-delta starting.

## Unbalance and the neutral

Real installations are rarely balanced, because single-phase loads are connected between each line and neutral and cannot be matched exactly. The neutral conductor then carries the resultant unbalance current, which in normal circumstances is modest.

Two situations change that. A lost or high-resistance neutral on an unbalanced star load allows the star point to shift, so lightly loaded phases rise well above their nominal voltage while heavily loaded phases fall; equipment is damaged on both sides. And harmonic currents from electronic loads, particularly the third harmonic, add arithmetically in the neutral rather than cancelling, so the neutral can carry more current than any line conductor despite an apparently balanced installation.

```
Balanced load:  I_N = 0

Unbalanced load:  I_N = phasor sum of the three line currents

Triplen harmonics add in the neutral rather than cancelling
```

A LOST NEUTRAL IS A DESTRUCTIVE FAULT, NOT AN INCONVENIENCE

With the neutral intact, each phase voltage is fixed at the nominal value.
With it lost, the star point floats and divides the line voltage by load impedance.
A lightly loaded phase can rise toward the line voltage and destroy its equipment.
Suspect a lost neutral whenever several phases show simultaneous and opposite errors.

## Measuring three-phase power

A single wattmeter measures total power only in a balanced four-wire system, where one phase can be measured and the result multiplied by three. For a three-wire system, balanced or not, the two-wattmeter method gives the total power from two instruments connected between the three conductors.

The two readings are added to obtain total power, and their ratio reveals the power factor. At unity power factor the readings are equal; as power factor falls one reading diminishes and at a power factor of 0.5 it reads zero. Below that, one wattmeter reads negative and its reading must be subtracted, which is the step most often got wrong.

### Interpreting two-wattmeter readings

- Equal readings: unity power factor.
- One reading zero: power factor of 0.5.
- One reading negative: power factor below 0.5, so subtract it.
- Both readings negative: check the connections, not the load.

- Sum is always the total power, whether the load is balanced or not.

```
P_total = P1 + P2
tanϕ = √3 (P1 − P2)/(P1 + P2)
Below 0.5 power factor, one reading is negative
```

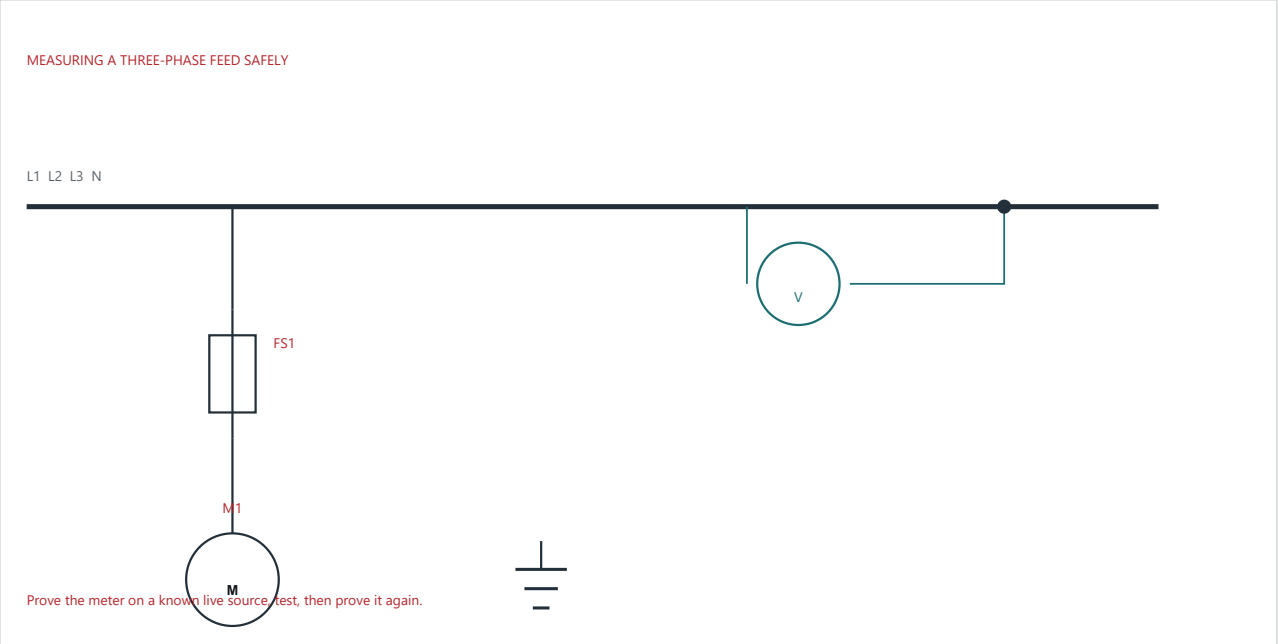



Fig. 17.3 Two-wattmeter measurement on a three-wire supply. The sum of the two readings is the total power regardless of balance.

## Worked examples

Work each example with a pencil before reading the solution. Sketch the circuit, mark the reference node, and write the governing law before substituting numbers.

EXAMPLE 17.1 Star-connected load: phase and line quantities

**Given.** A balanced star-connected load of 25 Ω per phase is supplied at 400 V line voltage.

**Find.** The phase voltage, the phase and line currents, and the total power.

Phase voltage

$V_{ph} = 400/\sqrt{3} = 231\ V$

Phase current

$I_{ph} = 231/25 = 9.24\ A$

Line current

$I_L = I_{ph} = 9.24\ A$ in star

Total power, phase quantities

$3 \times 231 \times 9.24 = 6.40\ kW$

Total power, line quantities

$\sqrt{3} \times 400 \times 9.24 \times 1 = 6.40\ kW$

**Check.** Both routes give the same answer, which is a useful check. With a purely resistive load the power factor is unity, so the real and apparent powers are equal.

EXAMPLE 17.2 The same load reconnected in delta

**Given.** The 25 Ω per phase load above is reconnected in delta on the same 400 V supply.

**Find.** The new phase and line currents and the total power, compared with the star case.

Phase voltage now

$V_{ph} = V_L = 400\ V$

Phase current

$I_{ph} = 400/25 = 16\ A$

Line current

$I_L = \sqrt{3} \times 16 = 27.7\ A$

Total power

$\sqrt{3} \times 400 \times 27.7 = 19.2\ kW$

Ratio to the star case

$19.2/6.40 = 3$ exactly

**Check.** Delta draws three times the power of star from the same supply. This is the basis of star-delta starting: starting in star applies a reduced voltage to each winding and draws a third of the current and develops a third of the torque.

EXAMPLE 17.3 Three-phase motor current from its rating

**Given.** A motor is rated 15 kW output at 400 V, with an efficiency of 0.91 and a power factor of 0.86.

**Find.** The input power, the apparent power and the line current.

Input power

$$P_{in} = 15\,000/0.91 = 16\,484 \text{ W}$$

Apparent power

$$S = 16\,484/0.86 = 19\,168 \text{ VA}$$

Line current

$$I_L = 19\,168/(\sqrt{3} \times 400) = 27.7 \text{ A}$$

Alternative route

$$I_L = 16\,484/(\sqrt{3} \times 400 \times 0.86) = 27.7 \text{ A}$$

**Check.** Both efficiency and power factor must be applied: efficiency converts shaft output to electrical input, and power factor converts real power to the apparent power that determines the current.

EXAMPLE 17.4 Two-wattmeter readings and power factor

**Given.** Two wattmeters on a three-wire supply read 7.2 kW and 2.4 kW.

**Find.** The total power and the power factor.

Total power

$$7.2 + 2.4 = 9.6 \text{ kW}$$

Difference

$$7.2 - 2.4 = 4.8 \text{ kW}$$

Tangent of the angle

$$\tan\phi = \sqrt{3} \times 4.8/9.6 = 0.866$$

Phase angle

$$\phi = 40.9^\circ$$

Power factor

$$\cos 40.9^\circ = 0.756$$

**Check.** The unequal readings indicate a lagging load. Had one reading been negative, it would have been subtracted in the total, and the power factor would have been below 0.5.

EXAMPLE 17.5 Neutral current in an unbalanced installation

**Given.** Three single-phase loads draw 32 A, 26 A and 18 A from the three phases of a four-wire supply, at unity power factor.

**Find.** An estimate of the neutral current.

Resultant of a balanced portion

$18\ \text{A in each phase cancels in the neutral}$

Remaining unbalance

$14\ \text{A and}\ 8\ \text{A on two phases}$

Phasor sum of the remainder

$$\sqrt{(14^2 + 8^2 - 14\times 8)} = \sqrt{(196 + 64 - 112)}$$

Neutral current

$$\sqrt{148} = 12.2\ \text{A}$$

Check against the simple difference

$32 - 18 = 14\ \text{A, so}\ 12.2\ \text{A is plausible}$

**Check.** The neutral carries only the unbalance, and because the remaining currents are 120 degrees apart they partly cancel. Note this calculation assumes no harmonic content; with electronic loads the neutral current could exceed any line current.

## Practical activity: star and delta relationships, power measurement and the effect of a lost neutral

AIM

Verify the root-three relationships in both connections, measure three-phase power by the two-wattmeter method, confirm the three-to-one power ratio between delta and star, and observe safely the voltage shift caused by an interrupted neutral on an unbalanced load.

### Equipment

- Three-phase supply, reduced voltage, through an isolating transformer with a residual current device and lockable isolator.
- Three-phase resistive load bank with individually switchable phases and accessible star and delta links.
- Two wattmeters, or a three-phase power analyser.
- Three true rms voltmeters and three true rms ammeters, or a clamp meter.
- Phase rotation indicator.
- Low-voltage lamp loads of unequal rating for the unbalance test.
- Lock-off devices, voltage proving unit and insulated tools.

### Procedure

1. Prove the supply dead, lock off, and confirm the isolation before making any connection.
2. Verify the phase sequence with the rotation indicator and record it.
3. Connect the load bank in star with all three phases equal. Prove dead again before energising.
4. Measure all three line voltages and all three phase voltages. Compute the ratio and compare with root three.
5. Measure line and phase currents in one phase and confirm they are equal in star.
6. Measure the neutral current with the load balanced and record it.
7. Connect the two wattmeters and record both readings. Compute total power and power factor.
8. Compute total power independently from your line voltage and current measurements and compare.
9. Prove dead, lock off, and reconnect the same load bank in delta.
10. Energise and measure line and phase voltages, confirming they are now equal.
11. Measure line and phase currents and compute their ratio, comparing with root three.
12. Record the total power in delta and compare it with the star figure. Confirm the ratio.
13. Prove dead, lock off, and reconnect in star with deliberately unequal lamp loads on the three phases, with the neutral connected.
14. Energise and record all three phase voltages and the neutral current.
15. With the supply reduced to the lowest usable voltage, and under direct supervision, open the neutral link and immediately record the three phase voltages.
16. Restore the neutral at once, then prove dead and lock off.

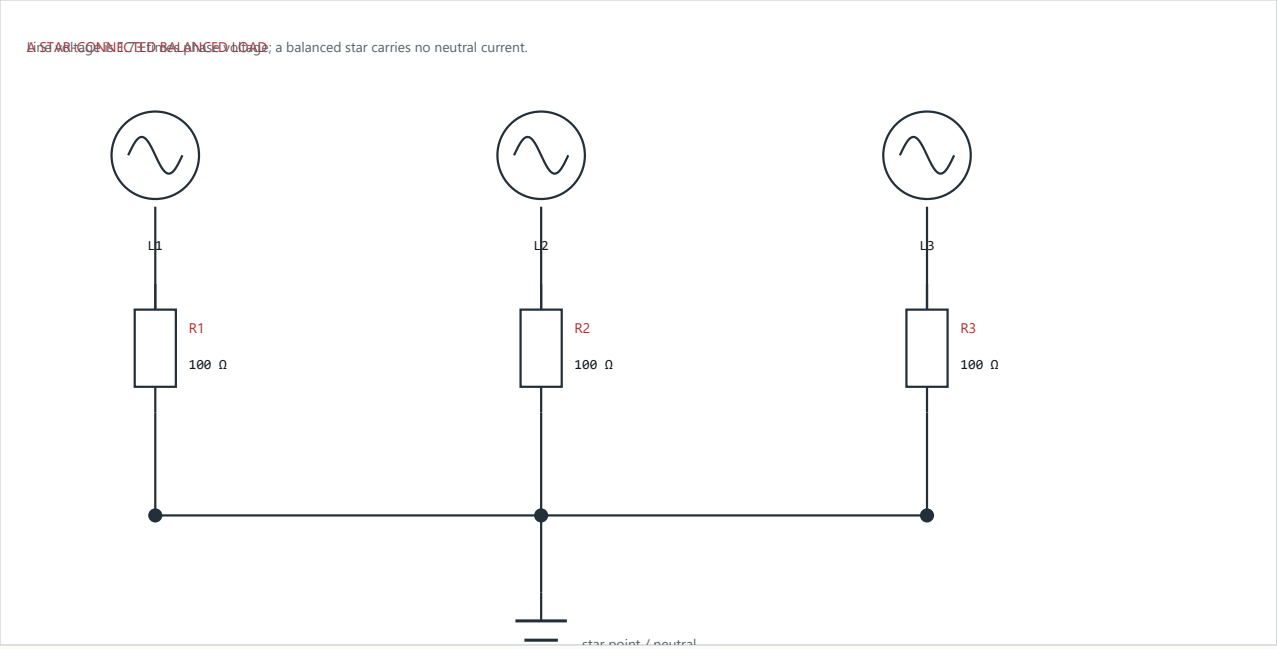



Fig. 17.L The star-connected test arrangement, with the neutral link accessible and all three phase and line measurement points brought out.

## Results table

| Connection and condition | Line voltage (V) | Phase voltage (V) | Current (A) | Power and comment |
|---|---|---|---|---|
| Star, balanced | | | | |
| Star, phase current | | | | |
| Star, neutral current | | | | |
| Delta, balanced | | | | |
| Delta, phase current | | | | |
| Wattmeter 1 and 2 readings | | | | |
| Star, unbalanced, neutral intact | | | | |
| Star, unbalanced, neutral open | | | | |

## Discussion

1. What was the measured ratio of line to phase voltage in star? Compare it with root three and account for the difference.
2. What was the measured ratio of line to phase current in delta? Compare with root three.
3. Confirm from your measurements that line and phase currents are equal in star, and that line and phase voltages are equal in delta.
4. What was the neutral current with the load balanced, and why was it not exactly zero?

5 Compare the total power in delta with that in star. How close was the ratio to three, and why does that ratio arise?

6 Compute the power factor from your two wattmeter readings and compare it with the value implied by your load.

7 How did the three phase voltages behave when the neutral was opened on the unbalanced load? Which rose and which fell, and why?

8 Explain in terms of the star point what happens when the neutral is lost.

9 What would the consequence of that voltage shift be for equipment connected to the lightly loaded phase in a real installation?

10 Why does the two-wattmeter method work regardless of whether the load is balanced?

SAFE-WORK REMINDERS

This work involves three-phase voltages. It must be carried out at the lowest practicable voltage, on a supply fed through an isolating transformer and a residual current device, and under direct supervision.

Prove dead and lock off before every reconnection, and confirm isolation with a proving unit.

Line voltage is root three times phase voltage. A circuit that seems safe phase to neutral is not safe line to line.

The neutral-interruption test must be performed only at reduced voltage, momentarily, under supervision, and never on a live installation.

Never open a neutral conductor on an energised unbalanced load in service; the resulting voltage shift destroys equipment and is dangerous.

Confirm instrument ranges before connecting; three-phase currents and voltages exceed single-phase bench values.

Use insulated tools and keep one hand clear when working near energised terminals.

Load banks become hot. Allow them to cool before handling.

## Troubleshooting guide

Faults are found by narrowing the circuit, not by replacing parts. Each row below pairs an observed symptom with the measurement that separates the likely causes.

| Symptom | Likely cause | Decisive test | Correct action |
|---|---|---|---|
| One phase voltage high and another low at the same time | Lost or high-resistance neutral on an unbalanced star load | Measure all three phase voltages and the neutral-to-earth voltage | Locate and repair the neutral connection; inspect all equipment on the high phase |
| Motor rotates in the wrong direction after a supply alteration | Phase sequence reversed | Check with a phase rotation indicator | Exchange any two line conductors, leaving the protective conductor undisturbed |
| Motor runs hot and noisy and current is unequal between phases | Single phasing or supply voltage unbalance | Measure current in all three conductors and all three line voltages | Restore the lost phase or correct the unbalance at source |
| Neutral conductor hotter than the line conductors | Triplen harmonic currents adding in the neutral rather than cancelling | Measure neutral current with a true rms instrument and compare with the line currents | Increase the neutral cross-section or fit harmonic mitigation |
| Load draws three times the expected current | Connected in delta where star was intended | Inspect the terminal links against the nameplate and supply voltage | Reconnect to the correct configuration for the supply voltage |
| Two-wattmeter total disagrees with a computed figure | One wattmeter reading negative and added instead of subtracted | Note the sign of each reading and recompute | Subtract the negative reading; verify the instrument connections |
| Star-delta starter trips during the changeover | Open transition causing a current surge, or incorrect changeover timing | Record current through the changeover with a logging instrument | Adjust the changeover timing or fit a closed-transition starter |
| Single-phase loads on one phase cause flicker on another | Severe unbalance combined with a high-impedance neutral or supply | Measure phase voltages while the offending load switches | Redistribute single-phase loads across the three phases and verify neutral integrity |

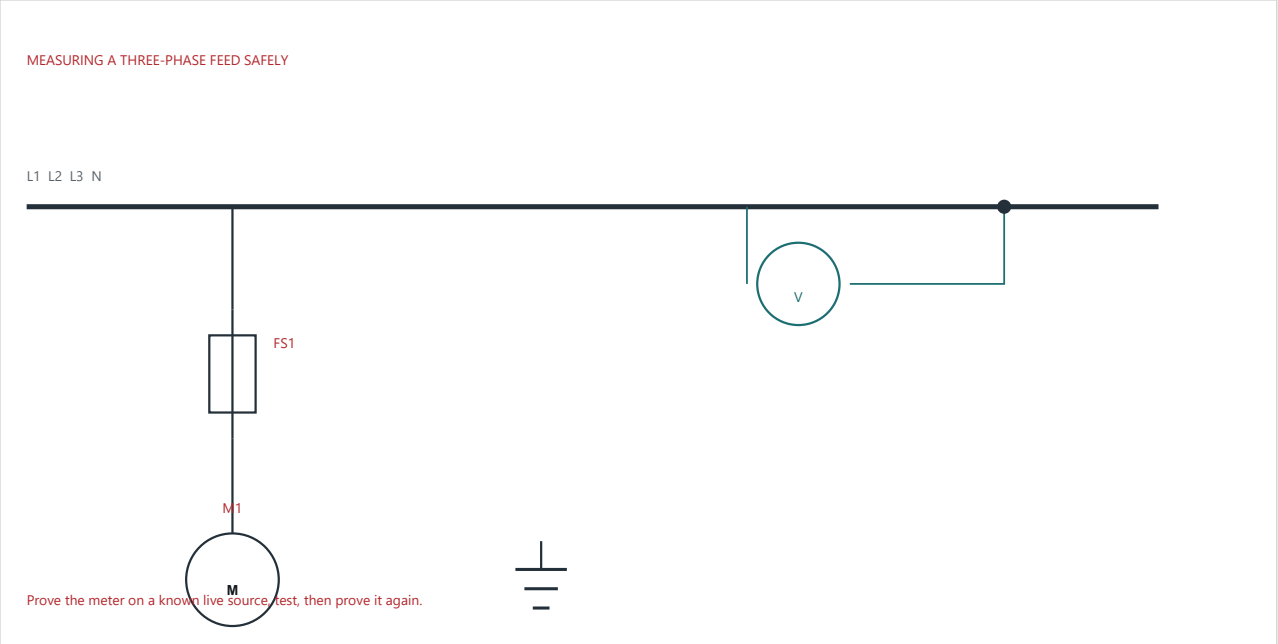



Fig. 17.T The two-wattmeter connection. Both readings are recorded with their signs, and their sum gives total power for any degree of unbalance.

FIELD HABITS THAT PREVENT CALLBACKS

State explicitly whether a quoted voltage or current is a line or a phase quantity; most three-phase errors begin here.
Verify phase sequence before connecting any rotating machine.
Treat a lost neutral as an urgent and destructive fault, and inspect equipment on the phases that rose.
Measure neutral current with a true rms instrument on installations with electronic loads.
Check terminal links against the nameplate and the supply voltage before energising a motor.
Record the sign of each wattmeter reading, because a negative reading must be subtracted.
Distribute single-phase loads across all three phases as evenly as the design allows.

## Exercises

Answer from first principles. Selected answers follow; the reasoning matters more than the number.

1 State the phase displacement between the three voltages of a three-phase supply.
2 A star-connected supply has a 415 V line voltage. Find the phase voltage.
3 A delta-connected load draws 12 A per phase. Find the line current.
4 State the relationship between line and phase current in a star connection.
5 State the relationship between line and phase voltage in a delta connection.
6 A balanced star load of 40 Ω per phase is supplied at 400 V line. Find the line current.
7 Find the total power of the load above.
8 The same load is reconnected in delta on the same supply. Find the new total power.
9 Explain why reconnecting from star to delta triples the power.
10 Find the line current of a 400 V three-phase load drawing 22 kW at 0.88 power factor.
11 State the neutral current of a perfectly balanced star-connected load and explain why.

12 Explain what happens to the phase voltages when the neutral is lost on an unbalanced star load.

13 Explain why the neutral of an installation with many electronic loads may carry more current than any line conductor.

14 Two wattmeters read 9.0 kW and 3.6 kW. Find the total power and the power factor.

15 Two wattmeters read 8.0 kW and −1.2 kW. Find the total power and state what the negative reading indicates.

16 State why phase sequence must be verified before connecting a three-phase motor.

## Selected answers

| # | Answer | Reason |
|---|---|---|
| 1 | 120 degrees | Three equal voltages equally spaced through the cycle |
| 2 | 240 V | $V_{ph} = V_L/\sqrt{3}$ |
| 3 | 20.8 A | $I_L = \sqrt{3}\, I_{ph}$ |
| 4 | They are equal | Each line conductor carries the current of the phase it feeds |
| 5 | They are equal | Each winding is connected directly between two lines |
| 6 | 5.77 A | $V_{ph} = 231$ V, so $I = 231/40$ |
| 7 | 4.0 kW | $P = \sqrt{3}\, V_L I_L \cos\varphi$ with unity power factor |
| 8 | 12.0 kW | Three times the star value |
| 9 | Each winding sees $\sqrt{3}$ times the voltage | Current also rises by $\sqrt{3}$, so power rises by three |
| 10 | 36.1 A | $I = P/(\sqrt{3}\, V_L \cos\varphi)$ |
| 11 | Zero | The three equal currents are 120 degrees apart and sum to zero |
| 12 | The star point shifts, so lightly loaded phases rise and loaded phases fall | Voltage divides according to load impedance instead of being fixed |
| 13 | Triplen harmonics add rather than cancel | Third-harmonic currents are in phase in all three conductors, so they sum in the neutral |
| 14 | 12.6 kW; power factor 0.76 | Sum gives power; $\tan\varphi = \sqrt{3}(P_1 - P_2)/(P_1 + P_2)$ |
| 15 | 6.8 kW; power factor below 0.5 | The negative reading is subtracted and indicates a strongly reactive load |
| 16 | It determines the direction of rotation | Connecting without checking may drive a pump or compressor backwards |

## Chapter summary

- A three-phase supply is three equal voltages 120 degrees apart, which sum to zero and so need no return current when balanced.
- In star, line voltage is root three times phase voltage and the currents are equal; in delta, the voltages are equal and line current is root three times phase current.
- Total power is root three times line voltage times line current times power factor, for either connection.
- Reconnecting a load from star to delta on the same supply triples its power, which is the basis of star-delta starting.
- The neutral carries only the unbalance, and losing it on an unbalanced star load shifts the phase voltages destructively.
- Triplen harmonics add in the neutral instead of cancelling, so neutral current can exceed line current.
- The two-wattmeter method gives total power for any three-wire load, with a negative reading subtracted and indicating a power factor below 0.5.

### Equations introduced in this chapter

| Relationship | Expression | When it applies |
|---|---|---|
| Star voltage relation | $V_L = \sqrt{3}\, V_{ph}$ | Star connection |
| Star current relation | $I_L = I_{ph}$ | Star connection |
| Delta voltage relation | $V_{ph} = V_L$ | Delta connection |
| Delta current relation | $I_L = \sqrt{3}\, I_{ph}$ | Delta connection |
| Three-phase power | $P = \sqrt{3}\, V_L I_L \cos\varphi$ | Balanced load, either connection |
| Three-phase apparent power | $S = \sqrt{3}\, V_L I_L$ | Balanced load |
| Two-wattmeter total | $P = P_1 + P_2$ | Any three-wire load |
| Power factor from wattmeters | $\tan\varphi = \sqrt{3}(P_1 - P_2)/(P_1 + P_2)$ | Two-wattmeter method |

### Vocabulary check

| Term | Meaning in this chapter |
|---|---|
| Phase voltage | The voltage across one winding or one phase of the load |
| Line voltage | The voltage between any two line conductors |
| Phase sequence | The order in which the three phases reach their maxima |
| Star point | The common junction of three windings in a star connection |
| Unbalance | Inequality between the three phase loads or voltages |
| Triplen harmonics | Harmonics at multiples of three times the fundamental, which add in the neutral |
| Two-wattmeter method | Measuring total three-wire power with two instruments |

LOOKING AHEAD

Three-phase distribution brings substantial energy to the point of use, which makes the consequences of a fault correspondingly serious. The next chapter turns to safety, fault paths and the discipline of safe isolation.

CHAPTER 18

# Safety, fault paths and protection

How electricity injures, how faults clear, and the discipline of safe isolation

WHY THIS CHAPTER MATTERS

Everything in the previous chapters describes how circuits behave when they are working. This chapter concerns what happens when they are not, and what keeps the person working on them alive. The physics is the same; only the consequences of a mistake have changed.

## What you will be able to do

- Describe how electric current injures the body and why current, not voltage, is the hazard.
- Trace the fault-current path from source to earth and identify what limits it.
- Explain how an overcurrent device achieves disconnection within the required time.
- Describe residual current protection and what it does and does not detect.
- Carry out safe isolation in the correct sequence and justify each step.
- Recognise arc-flash hazard and describe the precautions that reduce it.
- Explain the role of national wiring codes and the need to identify the edition in force.

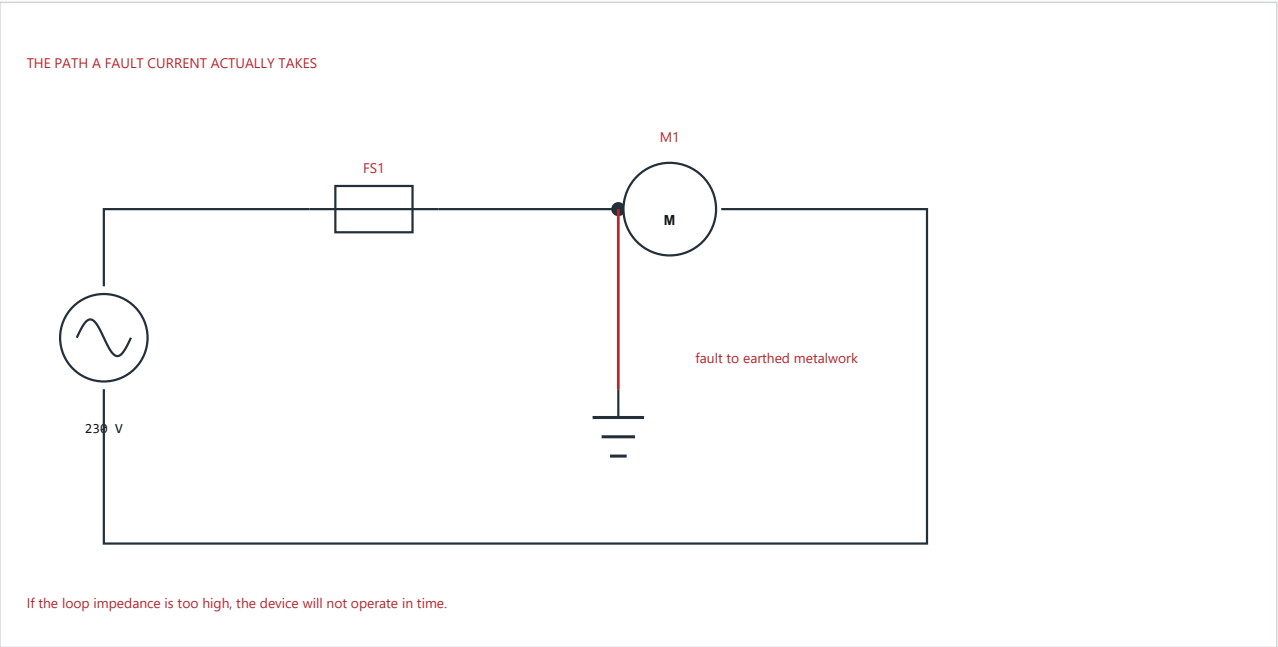



Fig. 18.1 An earth fault path. Current returns to the source through the protective conductor, and the total impedance of that loop determines whether the device will operate.

## How current injures

It is current through the body, not voltage across it, that causes injury. The body's impedance varies enormously with skin condition, contact area and contact pressure, which is why the same voltage can be harmless in one circumstance and lethal in another. Dry, calloused skin may present tens of thousands of ohms, while wet skin or a break in it may present only a few hundred.

Effects progress with current magnitude and with duration. A current of around one milliampere is perceptible; around ten milliamperes the muscles may contract so that the victim cannot let go; and in the region of tens of milliamperes through the chest, ventricular fibrillation becomes possible. Both the magnitude and the time matter, which is why protective devices are specified by disconnection time as well as by current.

| Approximate current | Typical effect | Relevance to protection |
|---|---|---|
| 1 mA | Perception threshold | Below the level of concern for injury |
| 5 to 10 mA | Painful shock, muscular reaction | Startle reaction causes secondary injury |
| 10 to 20 mA | Muscular contraction, cannot let go | Duration now becomes the critical factor |
| 30 mA | Residual current device operating level | Chosen to act before fibrillation becomes likely |
| 50 mA and above | Ventricular fibrillation possible | Disconnection must be within fractions of a second |
| Several amperes | Severe burns, cardiac arrest | Overcurrent protection territory |

> LOW VOLTAGE IS NOT THE SAME AS SAFE VOLTAGE
> Body impedance varies by two orders of magnitude with conditions.
> Wet hands, perspiration or broken skin drastically reduce it.
> A voltage harmless in dry conditions can be lethal in wet ones.
> Treat every circuit as dangerous until it has been proved dead.

## The fault path

An earth fault occurs when a live conductor contacts an exposed conductive part. Current then flows from the source, through the line conductor, across the fault, through the protective conductor and back to the source. The total impedance of that complete loop determines how much current flows, and therefore whether the protective device will operate and how quickly.

This is why the continuity and the cross-section of the protective conductor matter so much. A high-resistance protective connection limits the fault current, which may leave it below the level that operates the device while still being far above the level that is lethal. The exposed part then stays live, which is the precise condition protective bonding exists to prevent.

```
I_fault = V / Z_loop
```

```
Z_loop = source + line conductor + protective conductor

Disconnection requires I_fault above the device's operating level
```

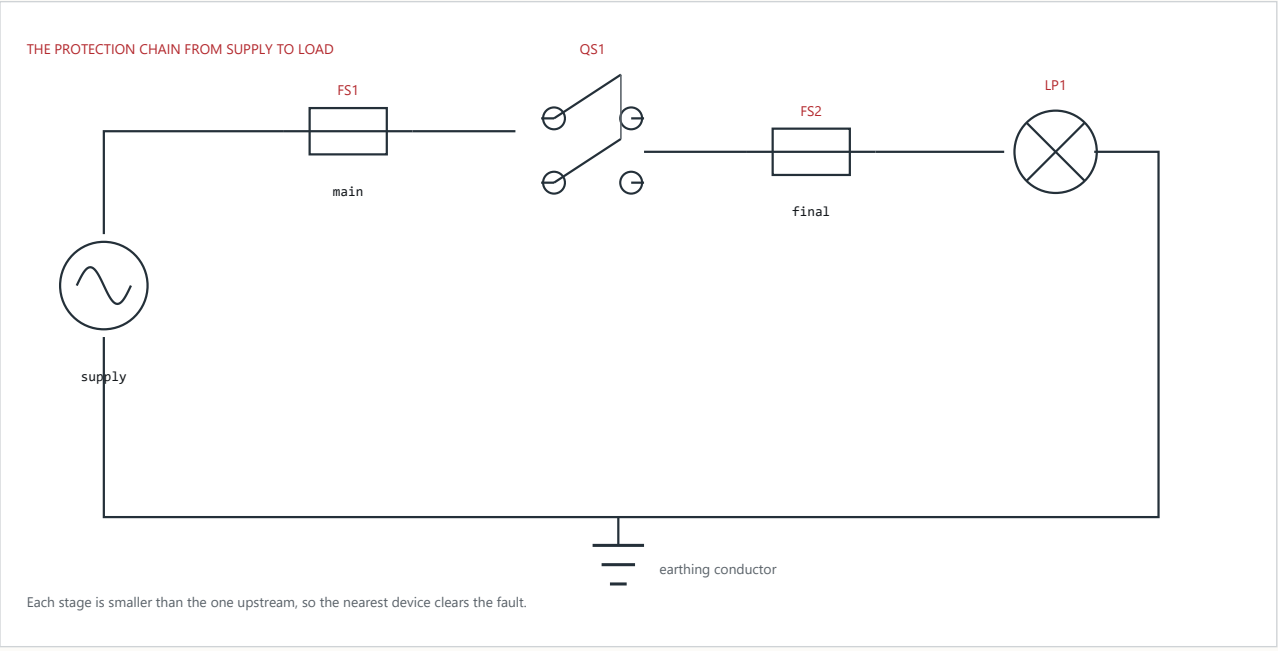



Fig. 18.2 The protective conductor path. Its continuity and cross-section determine the fault current and therefore whether disconnection occurs.

## Overcurrent protection

Overcurrent devices respond to current magnitude and duration together. A fuse or a circuit breaker will carry its rated current indefinitely, clear a modest overload after a deliberate delay, and clear a heavy fault almost instantly. That shape of response is the device's time-current characteristic, and selecting a device means matching that curve to the circuit.

Two separate requirements must be satisfied. The device must protect the conductor against thermal damage under overload, and it must disconnect an earth fault quickly enough to limit shock duration. The first depends on the conductor's current-carrying capacity; the second depends on the fault loop impedance. A device can satisfy one and fail the other, which is why both are verified.

### What an overcurrent device must achieve

- Carry the design current of the circuit continuously without operating.
- Operate before the conductor's thermal limit is reached under overload.
- Disconnect an earth fault within the required time at the measured loop impedance.
- Interrupt the maximum prospective fault current without damage.
- Discriminate with upstream devices so that only the affected circuit is lost.
- Tolerate the legitimate inrush of the connected load.

## Residual current protection

A residual current device compares the current flowing out in the line conductor with that returning in the neutral. In a healthy circuit they are equal. If some current is leaving by another route, through a person or through damaged insulation to earth, the difference is detected and the device opens.

It is important to understand what this does not cover. A residual current device cannot detect a fault between line and neutral, because that current still balances, so it provides no protection against an ordinary overload or a line-to-neutral short circuit. Nor does it protect someone who contacts line and neutral simultaneously. It supplements overcurrent protection; it does not replace it.

```
Detects:  I_line ≠ I_neutral

Typical personal protection rating: 30 mA

Does not detect line-to-neutral faults
```

TWO DEVICES FOR TWO DIFFERENT HAZARDS

Overcurrent protection guards the conductor against thermal damage.
Residual current protection guards a person against earth leakage.
Neither substitutes for the other, and most circuits need both.
A residual current device sees nothing wrong with a line-to-neutral short.

## Safe isolation

Safe isolation is a fixed sequence, not a set of preferences, and it is followed in order every time. Identify the correct circuit from the drawings and the labelling. Isolate it at an appropriate point. Secure the isolation with a lock and retain the key. Post a warning notice. Then prove the circuit dead using an approved voltage indicator that has itself been proved on a known source immediately before and immediately after the test.

The proving step before and after the test is what catches a failed indicator, and it is the step most often skipped. An indicator that has failed since its last use will show dead on a live circuit, and without the proving unit there is nothing to reveal that. Testing all combinations of conductors, not only line to earth, catches the circuit that has been isolated at the wrong pole.

The safe isolation sequence

- Identify the circuit from the drawings and confirm the labelling.
- Isolate at a point that removes all sources of supply.
- Secure with a lock and retain the key personally.
- Post a warning notice at the point of isolation.
- Prove the voltage indicator on a known live source.
- Test the circuit: all live conductors to earth and to each other.
- Re-prove the indicator on the known source.
- Only then begin work.

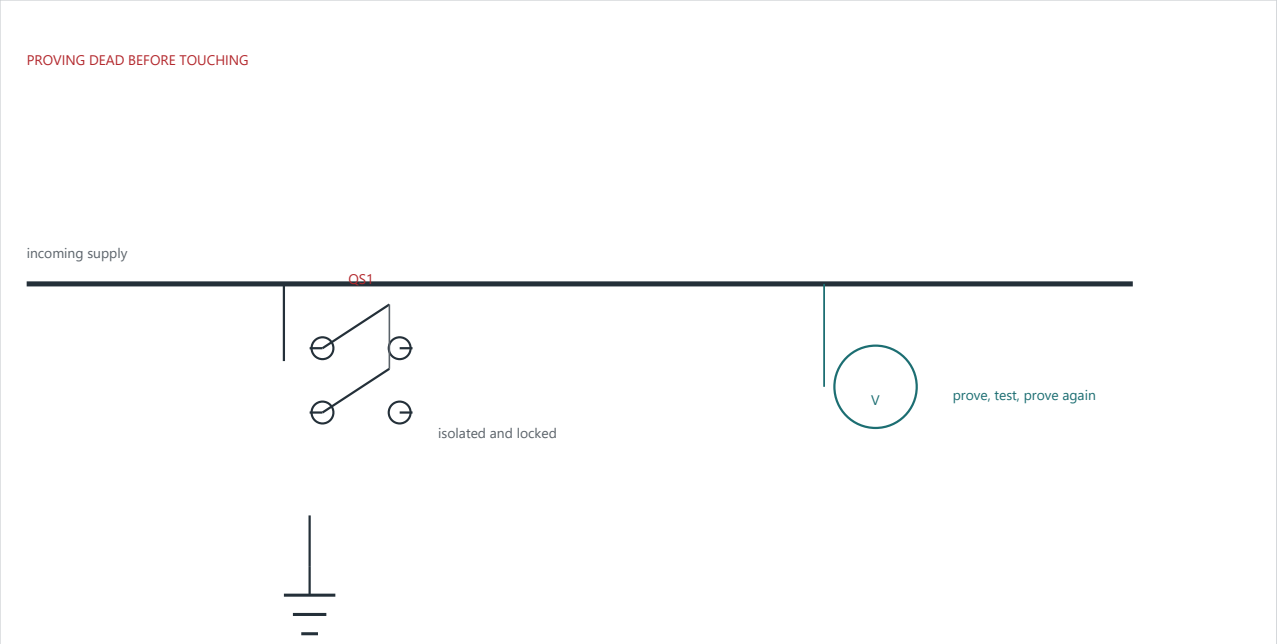



Fig. 18.3 Securing an isolation. The isolator is locked in the off position, the key is retained by the person working, and a warning notice is fitted.

## Arc flash and stored energy

Not every electrical injury is a shock. An arc fault converts fault energy into heat, light, pressure and molten metal, and the resulting burns can be severe at distances where no contact was ever made. The energy available depends on the prospective fault current and on how long the protective device takes to clear, so an arc close to a large supply is far more dangerous than the same event on a small final circuit.

Stored energy deserves the same respect. Capacitors retain charge after isolation, inductive circuits produce a voltage when interrupted, and rotating machines generate while they coast down. Each must be discharged or allowed to stop, and then proved dead, before work begins.

National wiring codes set the detailed requirements for disconnection times, conductor sizing, protective device selection, bonding and isolation. These requirements change between editions, so the edition in force for the installation must be established before any particular requirement is relied upon, and the relevant national code always takes precedence over any general guidance in a textbook.

### Sources of energy that remain after isolation

- Capacitors and reservoir capacitors hold charge after isolation; discharge and prove.
- Inductive circuits produce a reverse voltage when interrupted.
- Rotating machines generate while coasting down.
- Uninterruptible supplies and standby generators are additional sources.
- Photovoltaic arrays remain live in daylight regardless of any isolator.
- Prove dead after all stored energy has been dealt with, not before.

## Worked examples

Work each example with a pencil before reading the solution. Sketch the circuit, mark the reference node, and write the governing law before substituting numbers.

EXAMPLE 18.1 Body current from a contact voltage

**Given.** A person contacts a faulty appliance at 230 V. Their body impedance is 1300 Ω with dry hands and 600 Ω with wet hands.

**Find.** The current in each case and the likely effect.

Dry hands

```
I = 230/1300 = 177 mA
```

Wet hands

```
I = 230/600 = 383 mA
```

Comparison with the fibrillation region

```
Both far exceed 50 mA
```

Significance of duration

```
At these currents, disconnection time is critical
```

Residual current device response

```
A 30 mA device operates well below either value
```

**Check.** Both cases are potentially lethal, and the difference between them is only a matter of degree. This is why a residual current device is specified at 30 milliamperes rather than at any value related to body impedance.

EXAMPLE 18.2 Fault current and disconnection

**Given.** A circuit has a measured earth fault loop impedance of 1.15 Ω on a 230 V supply, protected by a device requiring 150 A to operate instantaneously.

**Find.** The prospective fault current and whether the device will operate instantaneously.

Fault current

```
I = 230/1.15 = 200 A
```

Device requirement

```
150 A for instantaneous operation
```

Comparison

```
200 A exceeds 150 A
```

Conclusion

```
The device operates in its instantaneous region
```

Margin

```
A 33 % margin above the requirement
```

**Check.** The verification is a simple comparison, but it depends entirely on the measured loop impedance. A deteriorated protective conductor connection raising the impedance to 1.6 ohms would reduce the fault current to 144 amperes and the device would no longer operate instantaneously.

EXAMPLE 18.3 Effect of a high-resistance protective conductor

**Given.** The circuit above develops a corroded protective conductor joint adding 2.0 Ω to the loop.

**Find.** The new fault current, the device response, and the voltage on the exposed part.

New loop impedance

```
1.15 + 2.0 = 3.15 Ω
```

New fault current

```
230/3.15 = 73 A
```

Device response

```
Below 150 A, so no instantaneous operation
```

Voltage on the exposed part

```
73 × 2.0 = 146 V
```

Hazard

```
The exposed part sits at 146 V while the device may not operate
```

**Check.** This is the precise failure that protective conductor testing exists to find. The fault current is far too small to clear the device quickly, yet the voltage on the accessible metal is lethal, and it may persist indefinitely.

EXAMPLE 18.4 What a residual current device detects

**Given.** A circuit draws 9.2 A in the line conductor. In one case 9.2 A returns in the neutral; in another 9.16 A returns and 40 mA flows to earth.

**Find.** Whether a 30 mA device operates in each case.

First case, difference

```
9.2 − 9.2 = 0 mA, no operation
```

Second case, difference

```
9.2 − 9.16 = 40 mA
```

Comparison with the rating

```
40 mA exceeds 30 mA, so the device operates
```

A line-to-neutral fault of 200 A

```
Currents still balance, so no operation
```

Protection in that case

```
Provided only by the overcurrent device
```

**Check.** The device responds only to an imbalance. A heavy line-to-neutral fault produces no imbalance at all, which is why the overcurrent device remains essential alongside it.

EXAMPLE 18.5 Stored energy in a capacitor after isolation

**Given.** A 4700 µF capacitor in an isolated supply remains charged to 45 V.

**Find.** The stored energy, and the discharge resistor to bring it to a safe level in two seconds.

Stored energy

$E = \tfrac{1}{2} C V^2 = 0.5 \times 0.0047 \times 45^2 = 4.76\ J$

Hazard assessment

Enough to produce a severe startle and an arc at the terminals

Time constant for a 2 s discharge

Allow about five time constants, so $\tau = 0.4$ s

Resistor value

$R = \tau / C = 0.4 / 0.0047 = 85\ \Omega$

Resistor dissipation at the instant of connection

$45^2 / 85 = 24$ W briefly

**Check.** Nearly five joules is a real hazard at the terminals even though the voltage is modest. The resistor must be rated for the initial pulse, and the capacitor must still be proved dead after discharging.

## Practical activity: fault loop measurement, protective conductor continuity, residual current testing and safe isolation

AIM

Practise the safe isolation sequence until it is automatic, measure protective conductor continuity and earth fault loop impedance on a training rig, verify residual current device operating current and time, and demonstrate what a residual current device does not detect.

### Equipment

- Training rig or test board with simulated final circuits, fed through an isolating transformer and a supply-side residual current device.
- Approved voltage indicator complying with the relevant national requirements, and its matching proving unit.
- Low-resistance ohmmeter for continuity testing.
- Earth fault loop impedance tester.
- Residual current device tester with ramp and timing functions.
- Lock-off devices for several isolator types, padlocks with unique keys, and warning notices.
- Insulation resistance tester.
- Introduced resistance, 2 Ω, to simulate a degraded protective conductor.

### Procedure

1. Before any measurement, identify the circuit to be worked on from the drawings and confirm it against the labelling at the distribution board.
2. Isolate the circuit at the correct device, secure it with a padlock, retain the key, and fit a warning notice.
3. Prove the voltage indicator on the proving unit and confirm it functions.
4. Test the isolated circuit between every pair of live conductors and between each live conductor and earth. Record each result.
5. Re-prove the indicator on the proving unit and record that it still functions.
6. Have the sequence checked by the supervisor before proceeding, and repeat it until it is performed without prompting.
7. With the circuit still isolated, measure the continuity of the protective conductor end to end with the low-resistance ohmmeter. Record the value.
8. Measure the insulation resistance between live conductors and earth. Record the value.
9. Restore the supply under supervision and measure the earth fault loop impedance at the furthest point of the circuit.
10. Compute the prospective fault current from your measured impedance and compare it with the protective device's requirement for instantaneous operation.
11. Isolate again following the full sequence, introduce the 2 Ω resistance into the protective conductor path, and restore the supply.
12. Re-measure the loop impedance and recompute the fault current. Compare with the device requirement and compute the voltage that would appear on the exposed part.
13. Isolate, remove the introduced resistance, and restore.
14. Test the residual current device with the ramp function and record the current at which it operates.
15. Test it at its rated residual current and record the operating time.
16. Test at five times the rated residual current and record the operating time.

17 Operate the device's test button and confirm it trips.

18 With supervision, demonstrate on the rig that a simulated line-to-neutral fault does not operate the residual current device, and identify which device does respond.

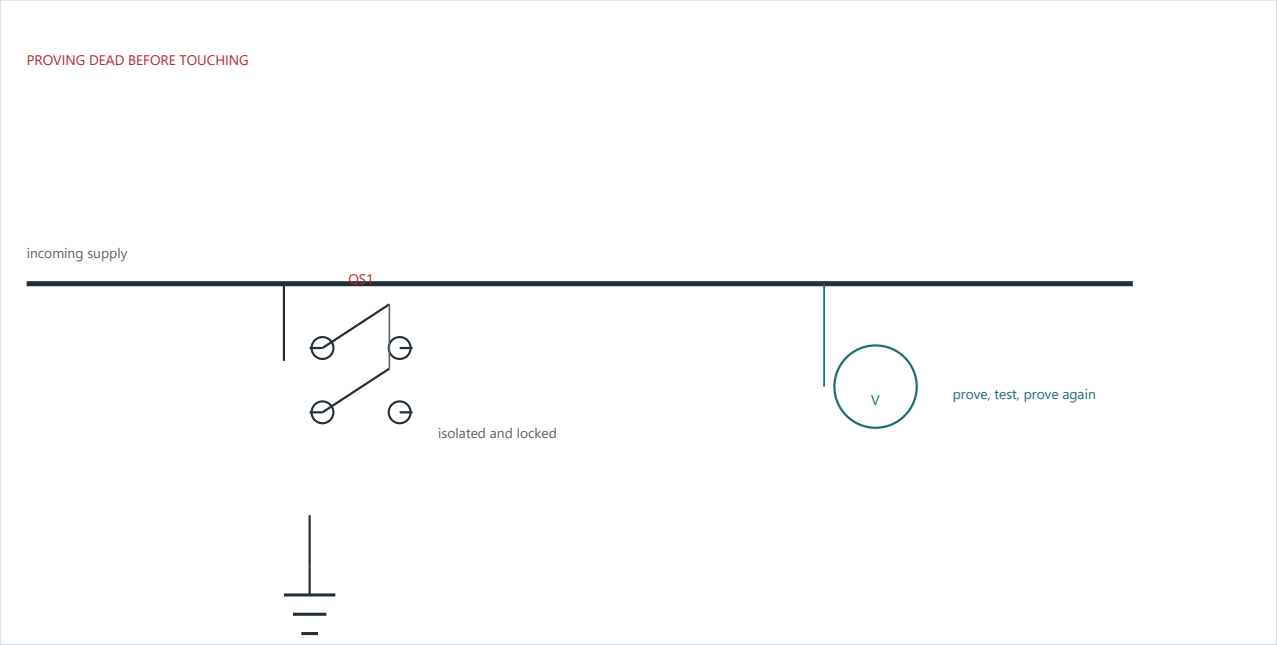



Fig. 18.L The isolation point secured with a lock and a warning notice, with the key retained by the person carrying out the work.

## Results table

| Test | Measured value | Required value | Pass or fail | Comment |
|---|---|---|---|---|
| Proving unit, before test | | | | |
| Live to live conductors | | | | |
| Live conductors to earth | | | | |
| Proving unit, after test | | | | |
| Protective conductor continuity | | | | |
| Insulation resistance | | | | |
| Loop impedance, healthy | | | | |
| Loop impedance, 2 Ω introduced | | | | |
| Residual device operating current | | | | |

| Test | Measured value | Required value | Pass or fail | Comment |
|---|---|---|---|---|
| Operating time at rated current | | | | |
| Operating time at five times rated | | | | |
| Line-to-neutral fault simulation | | | | |

### Discussion

1. Why is the voltage indicator proved both before and after the test, and what failure does the second proving catch?
2. Why is it necessary to test between all pairs of live conductors rather than only from line to earth?
3. What fault current did your healthy loop impedance imply, and did it satisfy the device's requirement for instantaneous operation?
4. How did the introduced 2 Ω change the fault current, and what voltage would have appeared on the exposed conductive part?
5. Explain why that condition is more dangerous than a dead short to earth.
6. At what current did the residual current device actually operate, and how does that compare with its rated residual current?
7. How did the operating time at five times rated current compare with the time at rated current? Why is the device designed that way?
8. Why did the simulated line-to-neutral fault not operate the residual current device, and which device protects against it?
9. What does the device's own test button prove, and what does it not prove?
10. List, in order and from memory, the full safe isolation sequence, and state the purpose of each step.

> **SAFE-WORK REMINDERS**
>
> This exercise must be carried out only on a purpose-built training rig, under direct supervision, and never on a live installation.
>
> The full safe isolation sequence must be followed every time, in order, without abbreviation.
>
> Use only an approved voltage indicator with its matching proving unit. Do not use a multimeter as the primary means of proving dead.
>
> Retain the padlock key personally. Never allow another person to hold it or to use a duplicate.
>
> Treat every circuit as live until it has been proved dead by the correct method.
>
> Discharge capacitors and prove dead before touching any conductor.
>
> Restore supplies only under supervision, after confirming all persons are clear.
>
> Do not work alone on this exercise.
>
> Apply the national wiring code in force for your installation; the values used here are for training illustration only.

## Troubleshooting guide

Faults are found by narrowing the circuit, not by replacing parts. Each row below pairs an observed symptom with the measurement that separates the likely causes.

| Symptom | Likely cause | Decisive test | Correct action |
|---|---|---|---|
| Exposed metal is live but the protective device has not operated | High-resistance protective conductor limiting fault current below the operating level | Measure protective conductor continuity and earth fault loop impedance | Locate and repair the high-resistance joint, then re-verify the loop impedance |
| Residual current device trips without apparent cause | Accumulated standing leakage from several items of equipment, or a genuine deteriorating insulation fault | Measure standing leakage with the circuit loaded; test insulation resistance | Divide the circuit between devices, or locate and repair the failing insulation |
| Residual current device did not operate on a line-to-neutral short circuit | It detects imbalance only, and such a fault remains balanced | Confirm which device operated | None required: the overcurrent device is the correct protection for that fault |
| Voltage indicator showed dead on a circuit that was live | Indicator failed since its last use and was not proved before the test | Prove the indicator on a proving unit | Replace the indicator and prove it before and after every test, without exception |
| Circuit still live after isolating the device labelled for it | Incorrect labelling, or a second source of supply such as a standby supply | Test all conductors and trace the circuit back to every possible source | Correct the labelling and isolate every source before working |
| Protective device trips as soon as the load is reconnected | Insulation failure in the load, or a conductor damaged during the work | Insulation resistance test on the load and the final circuit separately | Repair or replace the faulty item and retest before re-energising |
| Severe burns from an incident where no contact was made | Arc flash, with energy set by prospective fault current and clearing time | Assess the prospective fault current and the device clearing time at that point | Apply appropriate arc-rated protection and reduce clearing time where practicable |
| Shock received from equipment that had been isolated for some time | Stored charge in capacitors, or a second live source | Discharge capacitors through a resistor and re-prove dead | Discharge all stored energy as part of the isolation procedure and prove dead again |

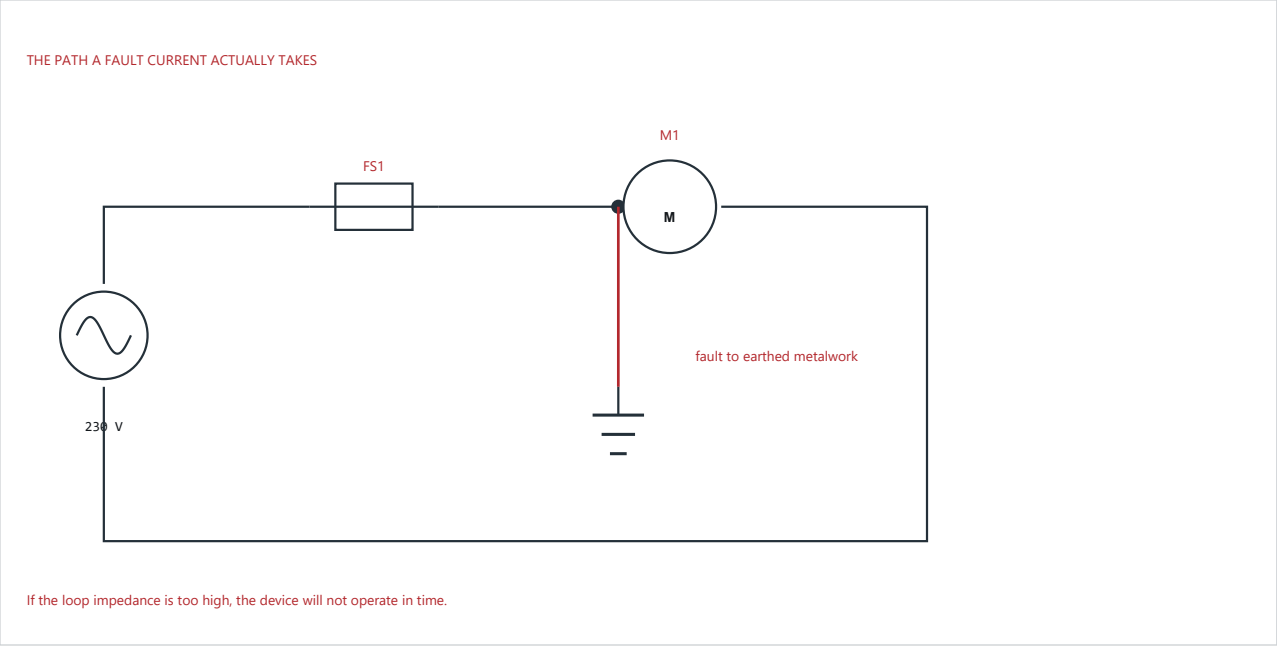



Fig. 18.T Verifying the fault path. Protective conductor continuity and loop impedance together confirm that a fault will be disconnected in the required time.

FIELD HABITS THAT PREVENT CALLBACKS

Follow the safe isolation sequence in full, in order, every single time, regardless of how familiar the circuit is.

Prove the voltage indicator before and after the test; this one step catches the failure that kills.

Test all conductor combinations, not only line to earth.

Treat a high-resistance protective conductor as an urgent fault, because it leaves metal live without operating the device.

Discharge stored energy and re-prove dead before touching any conductor.

Identify every source of supply, including standby supplies and photovoltaic arrays, before isolating.

Establish which edition of the applicable national code governs the installation before relying on any specific requirement.

## Exercises

Answer from first principles. Selected answers follow; the reasoning matters more than the number.

1. State whether current or voltage is the direct cause of electric shock injury, and explain why.
2. State approximately the current at which a person may be unable to release their grip.
3. Explain why body impedance varies so widely and give two conditions that reduce it.
4. List, in order, the components of an earth fault current loop.
5. A circuit has a loop impedance of 0.92 Ω on a 230 V supply. Find the prospective fault current.
6. The device above requires 200 A for instantaneous operation. State whether it is satisfied.
7. A corroded joint adds 1.5 Ω to that loop. Find the new fault current and the voltage on the exposed part.
8. State the two separate requirements an overcurrent device must satisfy.
9. Explain the principle by which a residual current device detects a fault.

10 State one type of fault a residual current device cannot detect, and explain why.

11 State the typical rated residual current chosen for personal protection.

12 List the safe isolation sequence in order.

13 Explain why the voltage indicator is proved both before and after the test.

14 Explain why testing only line to earth is insufficient when proving a circuit dead.

15 State what determines the severity of an arc flash incident.

16 Name three sources of energy that may remain in equipment after it has been isolated.

17 Explain why the edition of a national wiring code must be established before relying on a specific requirement.

## Selected answers

| # | Answer | Reason |
|---|---|---|
| 1 | Current, because it is what passes through tissue | Voltage only determines the current that flows for a given body impedance |
| 2 | About 10 to 20 mA | Muscular contraction prevents release at this level |
| 3 | Skin condition, contact area and pressure vary; moisture and broken skin reduce it | Impedance can fall by two orders of magnitude |
| 4 | Source, line conductor, fault, protective conductor, back to source | The total impedance of this loop sets the fault current |
| 5 | 250 A | I = V/Z_loop |
| 6 | Yes, 250 A exceeds 200 A | The device operates in its instantaneous region |
| 7 | 95 A; about 143 V on the exposed part | Fault current falls below the operating level while the metal remains live |
| 8 | Protect the conductor thermally; disconnect an earth fault in time | Both must be verified separately |
| 9 | It compares line and neutral current | Any imbalance indicates current leaving by another route |
| 10 | A line-to-neutral fault | That current still balances, so no imbalance is detected |
| 11 | 30 mA | Chosen to operate below the level at which fibrillation becomes likely |
| 12 | Identify, isolate, secure, notice, prove indicator, test, re-prove indicator | Each step removes one way the isolation could be wrong |
| 13 | To detect an indicator that has failed | A failed indicator reads dead on a live circuit |
| 14 | The circuit may be isolated at the wrong pole | Line to neutral or neutral to earth may still be live |
| 15 | Prospective fault current and the device clearing time | Together they determine the energy released |
| 16 | Charged capacitors, inductive circuits, coasting machines | Standby supplies and photovoltaic arrays are also sources |
| 17 | Requirements change between editions | A citation is meaningless without its edition, and the current edition governs the work |

## Chapter summary

- Current through the body causes injury, and body impedance varies so widely that no voltage can be assumed safe.
- Both magnitude and duration determine the severity of a shock, which is why devices are specified by disconnection time.
- Fault current is set by the impedance of the complete loop, so protective conductor continuity and cross-section are critical.
- A high-resistance protective conductor is especially dangerous because it leaves metal live without producing enough current to operate the device.
- Overcurrent devices protect conductors and clear faults; residual current devices detect earth leakage; neither replaces the other.
- Safe isolation is a fixed sequence, and proving the indicator before and after the test is the step that catches a failed instrument.
- Arc flash severity depends on prospective fault current and clearing time, and stored energy remains after isolation.
- National wiring codes govern the detailed requirements, and the edition in force must be identified.

### Equations introduced in this chapter

| Relationship | Expression | When it applies |
|---|---|---|
| Fault current | $I = V / Z_{loop}$ | Earth fault loop |
| Loop impedance | Z = source + line + protective conductor | Complete fault path |
| Voltage on exposed part | $V = I_{fault} \times R_{protective}$ | High-resistance protective conductor |
| Residual detection | Operates when $I_{line} - I_{neutral}$ exceeds the rating | Residual current device |
| Stored capacitor energy | $E = \frac{1}{2} C V^2$ | Isolated but charged capacitor |
| Discharge time constant | $\tau = R C$ | Discharge resistor fitted |

### Vocabulary check

| Term | Meaning in this chapter |
|---|---|
| Fault loop impedance | The total impedance of the path a fault current takes back to source |
| Protective conductor | The conductor connecting exposed conductive parts for fault protection |
| Disconnection time | The time within which a protective device must clear a fault |
| Residual current | The difference between line and neutral current, indicating leakage |
| Prospective fault current | The current that would flow at a given point in a fault |
| Proving unit | A known source used to confirm a voltage indicator is working |
| Arc flash | The thermal and pressure hazard produced by an arcing fault |
| Discrimination | Coordination so that only the device nearest the fault operates |

LOOKING AHEAD

Safety and fault clearance depend on measurement, and measurement depends on both the instruments and the judgement behind them. The next chapter turns to control, sensing and signal conversion: how a circuit measures the world, compares the result against what was wanted, and corrects the difference. The chapter after that closes the book with the diagnostic method that draws all of this together.

CHAPTER 19

# Control, sensing and signal conversion

Closed-loop control, the sensors that report the real world, and the conversion between analogue and digital

WHY THIS CHAPTER MATTERS

A circuit that simply switches a load on is open-loop: it acts without checking the result. A control loop measures the result and adjusts. This chapter develops the sensing, comparison and correction chain that appears in every thermostat, motor drive and process controller.

## What you will be able to do

- Draw a closed control loop and name the function of each block.
- Distinguish open-loop from closed-loop control and state when each is appropriate.
- Describe on-off control with hysteresis and compute the switching band.
- Explain proportional control and the meaning of offset.
- Select a sensor for temperature, light or position and describe its characteristic.
- Compute the resolution and quantisation step of an analogue-to-digital converter.
- Scale a sensor signal to match a converter input range.

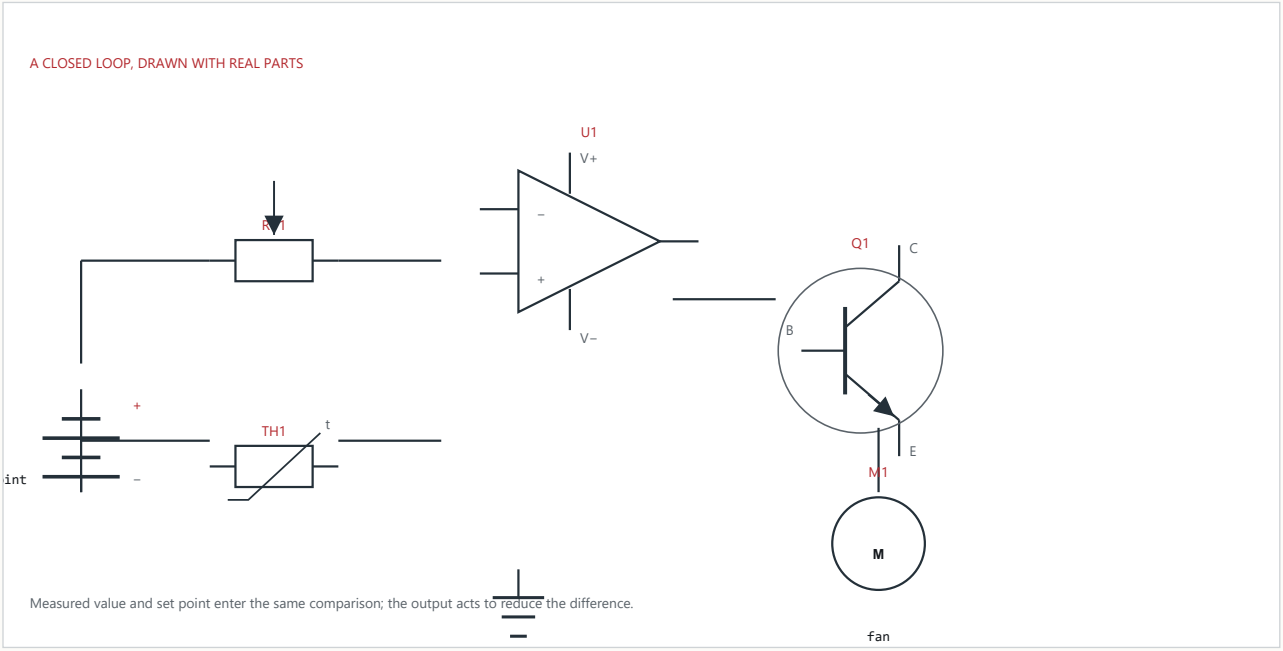



Fig. 19.1 A closed control loop. The measured value is compared with the set point and the difference drives the correcting element until the error is reduced.

## Open loop and closed loop

In open-loop control the output is set and no account is taken of the result. A lamp on a timer is open-loop: the timer does not know whether the room is bright. Open-loop control is simple, cheap and adequate whenever the relationship between input and output is predictable and disturbances are small.

Closed-loop control measures the controlled quantity, compares it with the desired value, and uses the difference to drive the correcting element. The measured value is called the process variable, the desired value is the set point, and their difference is the error. Closed-loop control tolerates disturbances and component drift because it responds to the result rather than to the command.

### The blocks of a control loop

- Set point: the value the system should reach, supplied by the operator or a schedule.
- Process variable: the measured value of the controlled quantity.
- Error: set point minus process variable, the signal that drives correction.
- Controller: the block that converts error into a correcting output.
- Correcting element: the valve, heater, contactor or drive that acts on the process.
- Feedback path: the sensor and conditioning circuit that reports the result.

```
error = set point - process variable

e = SP - PV
```

## On-off control and the need for hysteresis

The simplest controller switches the correcting element fully on when the error is positive and fully off when it is negative. A heater controlled this way would switch at exactly the set point, and because the measurement always carries a little noise the output would chatter rapidly between states, wearing the contacts and the load.

The cure is hysteresis: two thresholds instead of one. The output turns on below a lower threshold and off above an upper threshold, and between them it holds its last state. The separation is the switching band, or differential. A wide band gives slow, infrequent switching and a large swing in the controlled quantity; a narrow band holds the quantity closer to the set point but switches more often.

```
upper threshold = SP + band/2

lower threshold = SP - band/2

differential = upper - lower
```

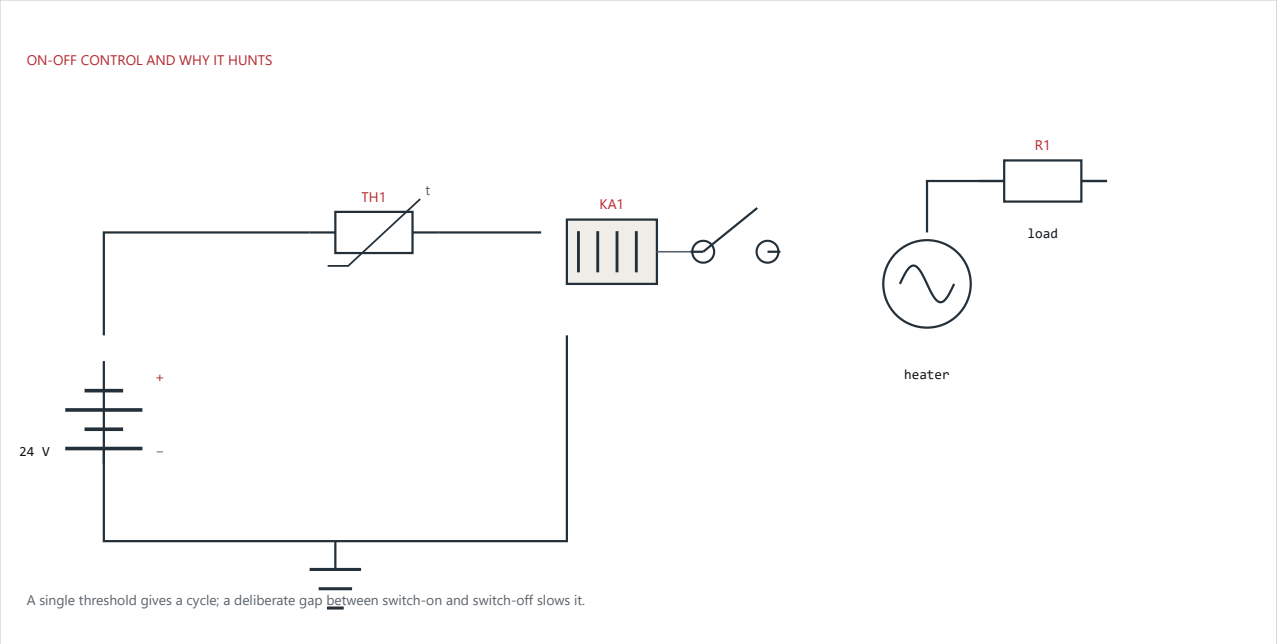



Fig. 19.2 On-off control with hysteresis. The two thresholds prevent chatter by requiring the measured value to travel the width of the band before the output changes.

## Proportional control and offset

Proportional control makes the correcting output a multiple of the error rather than a simple on or off. Small errors produce small corrections, so the system settles smoothly instead of overshooting. The multiplier is the proportional gain.

A purely proportional controller has a characteristic limitation. Because the output is proportional to the error, a non-zero output requires a non-zero error. A heater that must run at thirty per cent to hold temperature against heat loss therefore settles at a temperature slightly below the set point. That steady difference is the offset, or droop. Raising the gain reduces the offset but moves the loop towards instability, which is why practical controllers add integral action to remove the offset entirely.

```
output = K_p × e

offset ≈ required output / K_p
```

| Control action | What it responds to | What it fixes | What it costs |
|---|---|---|---|
| On-off with hysteresis | Error sign only | Simplicity, no tuning | Continuous cycling about the set point |
| Proportional | Present error | Smooth approach, no chatter | Leaves a steady offset |
| Proportional plus integral | Present and accumulated error | Removes the offset | Slower response, risk of overshoot |
| Proportional, integral, derivative | Error, its history and its rate | Fast settling with no offset | Three parameters to tune, noise sensitivity |

## Sensors: turning a physical quantity into a voltage

A control loop can only regulate what it can measure. Every sensor is a transducer with a characteristic relating the physical quantity to an electrical one, and that characteristic is rarely a straight line over the whole range.

A thermistor is a resistor whose resistance falls sharply with temperature. It is cheap and sensitive but markedly non-linear, so it is used either over a narrow span or with linearisation. A resistance thermometer uses the nearly linear rise of platinum resistance and is used where accuracy matters. A thermocouple produces a small voltage from the junction of two dissimilar metals and suits high temperatures, but it needs cold-junction compensation. A light-dependent resistor falls in resistance with illumination. A potentiometer reports shaft or slider position directly as a voltage ratio.

Most of these are resistive, and a resistance is not a voltage. The usual conditioning circuit is therefore a divider or a bridge, which converts the resistance change into a voltage change that the controller can read.

| Sensor | Quantity | Characteristic | Typical use |
|---|---|---|---|
| Thermistor | Temperature | Strongly non-linear, resistance falls as temperature rises | Appliance and board-level temperature limits |
| Resistance thermometer | Temperature | Nearly linear rise with temperature | Process measurement where accuracy matters |
| Thermocouple | Temperature | Small voltage, needs cold-junction compensation | Furnace and exhaust temperatures |
| Light-dependent resistor | Illumination | Resistance falls with light, slow response | Lighting control and dusk switching |
| Potentiometer | Position | Linear ratio of output to supply | Valve and actuator position feedback |
| Current transformer | Alternating current | Secondary current proportional to primary | Load monitoring without breaking the circuit |

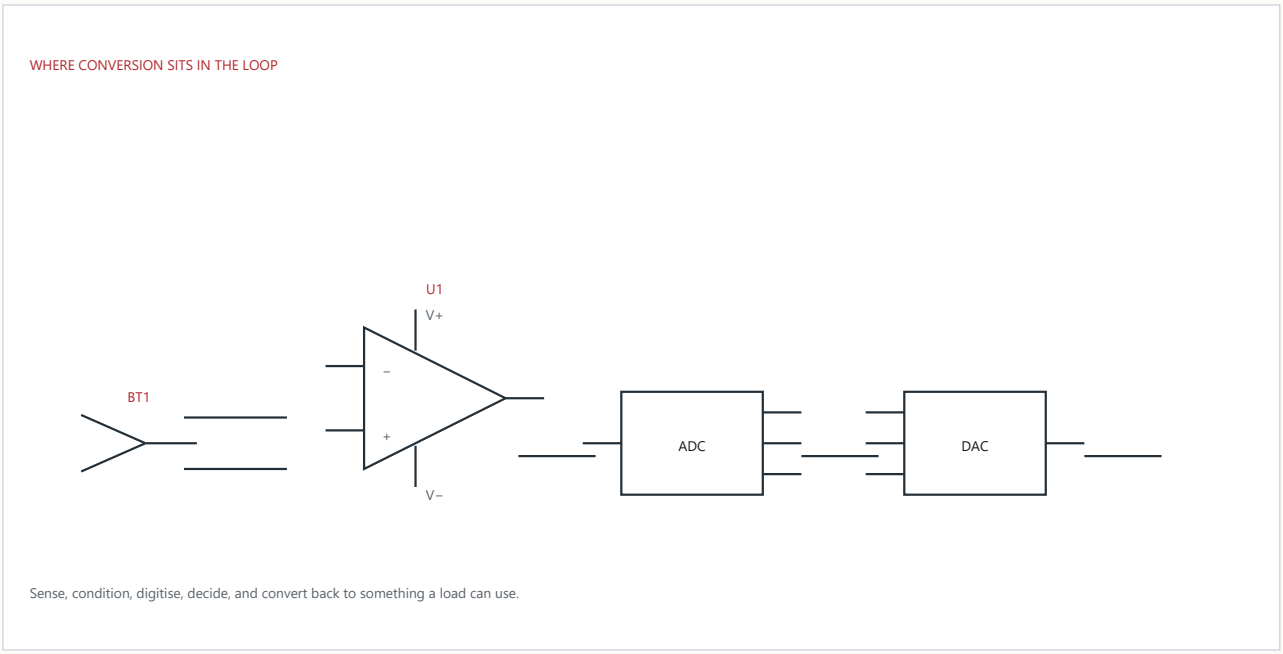



Fig. 19.3 Sensor conditioning. The resistive sensor forms one arm of a divider, and the resulting voltage is scaled to match the converter input range.

## Analogue to digital conversion

A digital controller works on numbers, so the conditioned sensor voltage must be converted. An analogue-to-digital converter divides its input range into a number of equal steps set by its resolution in bits, and reports which step the input falls in. An n-bit converter has two to the power n steps.

The size of one step is the quantisation step, and it sets the finest change the system can detect. No amount of averaging recovers detail smaller than one step, so the sensor signal should be scaled to use as much of the converter range as possible. A sensor that swings only over a tenth of the input range wastes more than three bits of resolution.

Conversion in the other direction uses a digital-to-analogue converter, which reconstructs a voltage from a number. The same resolution argument applies to its output steps.

```
steps = 2^n

step size = V_range / 2^n

resolution (10 bit, 5 V) = 5 / 1024 = 4.88 mV
```

SCALING BEFORE CONVERSION

Measure the full swing of the sensor output across the range you actually need.
Choose a divider or amplifier that maps that swing onto nearly the whole converter input range.
Leave a small margin at both ends so that tolerance and drift do not clip the signal.
Check that the source resistance seen by the converter is low enough for its sampling circuit.

## Worked examples

Work each example with a pencil before reading the solution. Sketch the circuit, mark the reference node, and write the governing law before substituting numbers.

EXAMPLE 19.1 Switching band of an on-off controller

**Given.** A thermostat has a set point of 20 °C and a differential of 1.5 °C.

**Find.** The two switching thresholds and the resulting swing in room temperature.

Half the differential

```
1.5/2 = 0.75 °C
```

Upper threshold

```
20 + 0.75 = 20.75 °C
```

Lower threshold

```
20 - 0.75 = 19.25 °C
```

Swing between switching points

```
20.75 - 19.25 = 1.5 °C
```

Effect of narrowing the band

```
Closer control, more frequent switching
```

**Check.** The room temperature cycles over the width of the differential, not about a single value. Choosing the band is a direct trade between control accuracy and switching frequency.

EXAMPLE 19.2 Offset in a proportional controller

**Given.** A proportional heater controller has a gain of 8 per cent output for each degree of error, and the heater must run at 24 per cent to hold the set point against heat loss.

**Find.** The steady-state offset below the set point.

Relationship

```
output = K_p × e
```

Rearranged for error

```
e = output / K_p
```

Substituting

```
e = 24 / 8 = 3 °C
```

Settled temperature

```
3 °C below the set point
```

Effect of doubling the gain

```
Offset halves to 1.5 °C, stability margin falls
```

**Check.** The offset is real and predictable, not a fault. It is the price of proportional-only control, and it is why integral action exists.

EXAMPLE 19.3 Resolution of an analogue-to-digital converter

**Given.** A 10-bit converter has a 0 to 5 V input range. A temperature sensor delivers 10 mV per °C over a 0 to 100 °C span.

**Find.** The voltage step, the temperature step, and the fraction of range used.

Number of steps

$2^{10} = 1024$

Voltage per step

5 / 1024 = 4.88 mV

Temperature per step

4.88 mV / 10 mV per °C = 0.49 °C

Sensor output at 100 °C

100 × 10 mV = 1.0 V

Fraction of the 5 V range used

1.0 / 5.0 = 20 per cent

**Check.** The converter can resolve about half a degree, but only a fifth of its range is in use. Amplifying the sensor output by five would improve the temperature resolution to about 0.1 °C without changing the converter.

# Practical activity: building and tuning an on-off temperature controller

AIM

Build a thermistor-sensed on-off controller, measure its switching thresholds, observe the effect of changing the differential, and record the resulting temperature cycling.

## Equipment

- Regulated direct-voltage supply, 12 V.
- Thermistor, nominal 10 kΩ at 25 °C, with its stated characteristic.
- Resistors to form the divider and the feedback path, 0.25 W.
- Comparator or operational amplifier, single supply type.
- Relay or transistor output stage with a small lamp as the simulated heater.
- Digital multimeter and a reference thermometer.

## Procedure

1. Record the thermistor resistance at room temperature and compare it with its stated value.
2. Build the divider with the thermistor and compute the expected output voltage.
3. Set up the comparator with a fixed reference to establish the set point.
4. Confirm the output switches as you warm the thermistor gently between your fingers.
5. Add a feedback resistor from output to the reference input to create hysteresis.
6. Measure the divider voltage at the instant the output turns on, and again when it turns off.
7. Compute the differential from those two voltages and convert it to a temperature span.
8. Change the feedback resistor to widen the band, then repeat the two threshold measurements.
9. Record the lamp on-time and off-time in each case and compare the switching frequency.
10. Return the circuit to the narrower band and confirm the thresholds repeat on a second run.

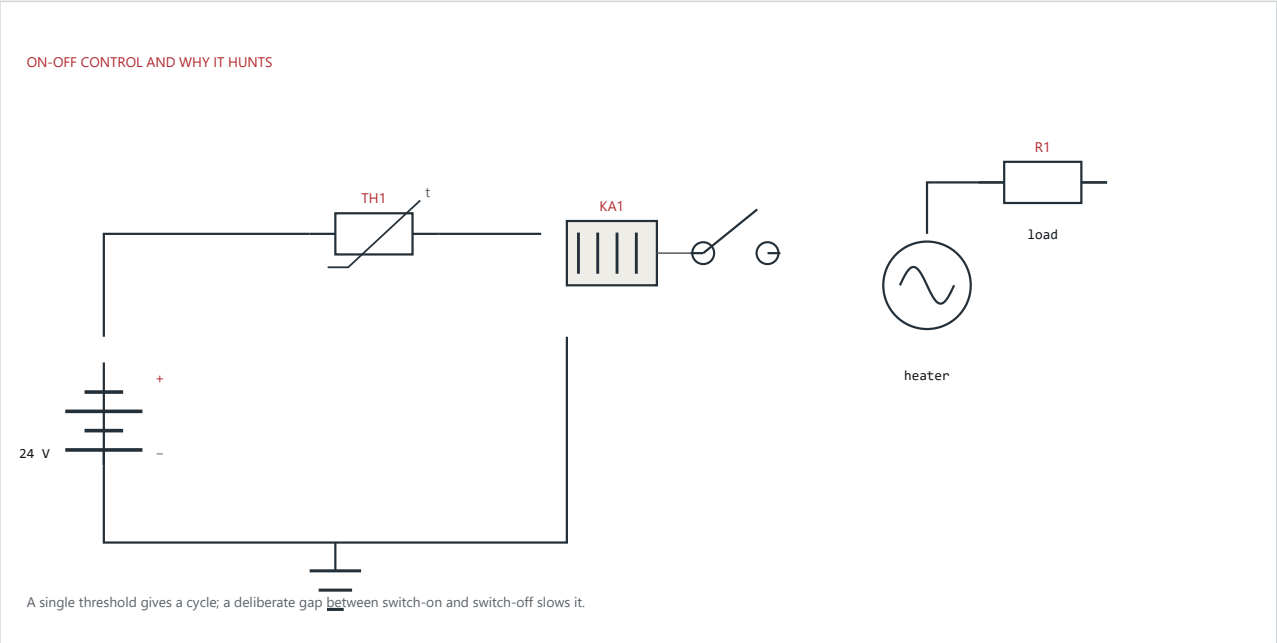



Fig. 19.L The on-off control test circuit. The feedback resistor sets the width of the switching band, which you measure directly as two divider voltages.

### Results table

| Feedback resistor | V at turn-on | V at turn-off | Differential (V) | On-time (s) |
|---|---|---|---|---|
| none | | | | |
| 1 MΩ | | | | |
| 470 kΩ | | | | |
| 220 kΩ | | | | |
| 100 kΩ | | | | |
| 47 kΩ | | | | |

### Discussion

1. Which feedback resistor produced the widest switching band, and why?
2. How did the switching frequency change as the band narrowed?
3. What happened to the output with no feedback resistor fitted, and why?
4. Convert one measured differential into a temperature span using the thermistor characteristic.
5. If this controlled a real heater, what band would you choose and what would you trade away?
6. Why does a narrow band not give arbitrarily good control in a system with slow response?

> SAFE-WORK REMINDERS
> Keep the supply voltage low; there is no reason to prove this principle at mains voltage.
> Warm the thermistor with your fingers or warm air, never with a flame.
> Fit a flyback diode across any relay coil before applying power.
> Check the output stage current rating against the lamp before connecting it.

## Troubleshooting guide

Faults are found by narrowing the circuit, not by replacing parts. Each row below pairs an observed symptom with the measurement that separates the likely causes.

| Symptom | Likely cause | Decisive test | Correct action |
|---|---|---|---|
| Output chatters rapidly at the set point | No hysteresis, or band narrower than the measurement noise | Observe the comparator input on a scope while it switches | Add or increase the feedback resistor to widen the band |
| Controlled quantity settles below the set point | Proportional offset, not a fault | Compare the steady error with required output divided by gain | Accept it, raise the gain slightly, or add integral action |
| Loop oscillates with a growing swing | Gain too high for the system's delay | Reduce the gain and observe whether the swing decays | Lower the gain, then add integral action to recover accuracy |
| Reading jumps by several converter steps | Sensor signal too small for the converter range | Measure the sensor swing and compare it with the full input range | Amplify or rescale the sensor signal before conversion |
| Temperature reading drifts over hours | Self-heating in the sensor, or reference drift | Reduce the sensor current and watch whether the drift falls | Lower the divider current, use a stable reference |
| Output never switches | Set point outside the sensor's achievable range | Compare the divider voltage with the reference voltage directly | Rescale the divider so the set point falls inside the swing |
| Correct response but very slow | Sensor thermally isolated from what it should measure | Compare sensor response time with the process response time | Improve thermal contact and reduce the sensor's own mass |

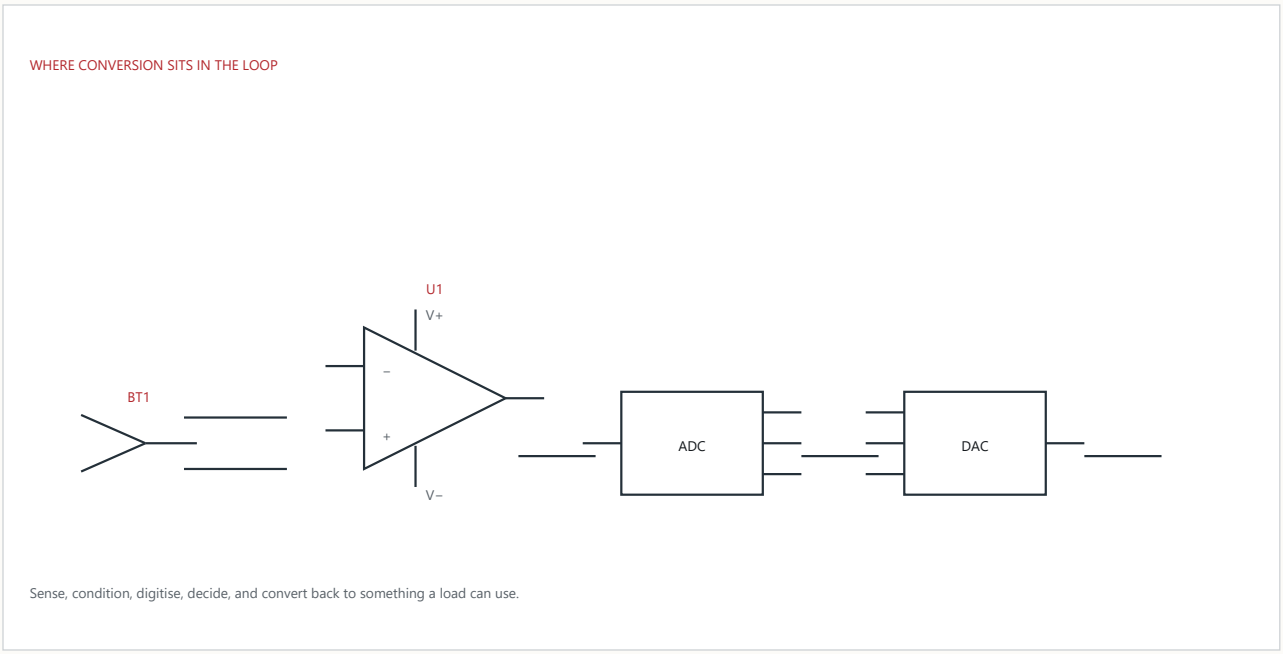



Fig. 19.T The sensing and conversion chain. Each stage must match the range of the next, or resolution is lost before the controller ever sees the signal.

FIELD HABITS THAT PREVENT CALLBACKS

State the set point, the measured quantity and the correcting element before drawing a loop.
Record a sensor's actual characteristic rather than trusting the nominal figure.
Scale a sensor signal to fill the converter range; resolution thrown away cannot be recovered.
Set the hysteresis band deliberately from the acceptable swing, not by trial and error.
Check that the sensor measures the quantity you care about, at the place that matters.
Change one tuning parameter at a time and record the result before changing another.

## Exercises

Answer from first principles. Selected answers follow; the reasoning matters more than the number.

1 Define open-loop and closed-loop control, and give a practical example of each.
2 Name the five blocks of a closed control loop and state the function of each.
3 A thermostat has a set point of 60 °C and a differential of 4 °C. Find both thresholds.
4 Explain why an on-off controller with no hysteresis is unsatisfactory in practice.
5 A proportional controller has a gain of 5 per cent per degree and needs 40 per cent output. Find the offset.
6 Explain in your own words why proportional-only control leaves a steady offset.
7 An 8-bit converter has a 0 to 10 V range. Find the step size.
8 A sensor swings over 0 to 2 V into a 0 to 5 V, 12-bit converter. How many bits are effectively wasted?
9 Compare a thermistor and a resistance thermometer for a process requiring accuracy over 0 to 200 °C.
10 Why does a thermocouple need cold-junction compensation?
11 Sketch a divider that converts a resistive sensor into a voltage, and state what sets the sensitivity.
12 A loop oscillates with increasing amplitude. State two possible causes and the test for each.
13 Explain why narrowing the hysteresis band does not improve control of a slow process indefinitely.

14 Describe one case where open-loop control is the correct engineering choice.

Selected answers

| # | Answer | Reason |
|---|---|---|
| 3 | 63 °C and 57 °C | Half the differential is 2 °C, applied either side of the set point. |
| 5 | 8 °C | Error equals required output divided by gain, 40 divided by 5. |
| 7 | 39.1 mV | Ten volts divided by 256 steps. |
| 8 | Between two and three bits | Only two fifths of the range is used, so the top bits never change. |
| 10 | The measured voltage depends on both junctions | The reference junction temperature must be known to recover the hot junction temperature. |

## Chapter summary

- Open-loop control acts without checking the result; closed-loop control measures the result and corrects.
- A control loop comprises a set point, a sensor reporting the process variable, a comparison producing error, a controller, and a correcting element.
- On-off control requires hysteresis to prevent chatter; the differential sets the swing in the controlled quantity.
- Proportional control is smooth but leaves a steady offset, because a non-zero output needs a non-zero error.
- Integral action removes the offset at the cost of response speed and overshoot margin.
- Every sensor has a characteristic that is usually non-linear, and conditioning converts it into a usable voltage.
- Converter resolution sets the finest detectable change; scaling the signal to fill the input range preserves it.
- Resolution lost to poor scaling cannot be recovered by filtering or averaging.

### Equations introduced in this chapter

| Relationship | Expression | When it applies |
|---|---|---|
| Control error | $e = SP - PV$ | The driving signal for every closed-loop controller |
| Proportional output | $output = K_p \times e$ | Proportional control, within the output limits |
| Proportional offset | $e \approx output / K_p$ | Steady state of a proportional-only loop |
| Switching thresholds | $SP \pm band/2$ | On-off control with a symmetrical differential |
| Converter steps | $2^n$ | Any converter of n bits resolution |
| Quantisation step | $V_{range} / 2^n$ | Uniform converters across their stated range |
| Divider output | $V_{out} = V_{in} R_2/(R_1+R_2)$ | Resistive sensor conditioning, light loading |

### Vocabulary check

| Term | Meaning in this chapter |
|---|---|
| Set point | The value the controlled quantity should reach |
| Process variable | The measured value of the controlled quantity |
| Error | Set point minus process variable |
| Hysteresis | Deliberate separation of the on and off thresholds |
| Differential | The width of the switching band in an on-off controller |
| Offset | The steady error left by proportional-only control |
| Transducer | A device converting one physical quantity into another |
| Quantisation step | The voltage change corresponding to one converter count |
| Resolution | The number of distinguishable steps in a converter output |

LOOKING AHEAD

The remaining chapter draws the whole book together. It takes the measurement habits, the circuit models, the machines, the electronics and the safety discipline, and sets them inside a single ordered method for diagnosing a fault and reviewing a design.

CHAPTER 20

# Diagnosis, design review and the habits that last

A single ordered method for finding a fault, and a checklist for reviewing work before it is energised

WHY THIS CHAPTER MATTERS

The previous nineteen chapters supplied models, equations and instruments. This one supplies method. A competent electrical worker is not distinguished by knowing more formulae but by following an order of operations that converges on the answer instead of wandering towards it.

## What you will be able to do

- Apply a structured diagnostic sequence rather than replacing parts speculatively.
- Divide a circuit at a measurable boundary to halve the search space at each step.
- Choose the measurement that distinguishes between competing hypotheses.
- Distinguish an open circuit, a short circuit, a high-resistance joint and a wrong value by measurement.
- Review a completed design against supply, loading, protection, and safety criteria.
- Record a fault and its resolution so that the next person benefits.
- Identify which code edition applies and verify a requirement at its source.

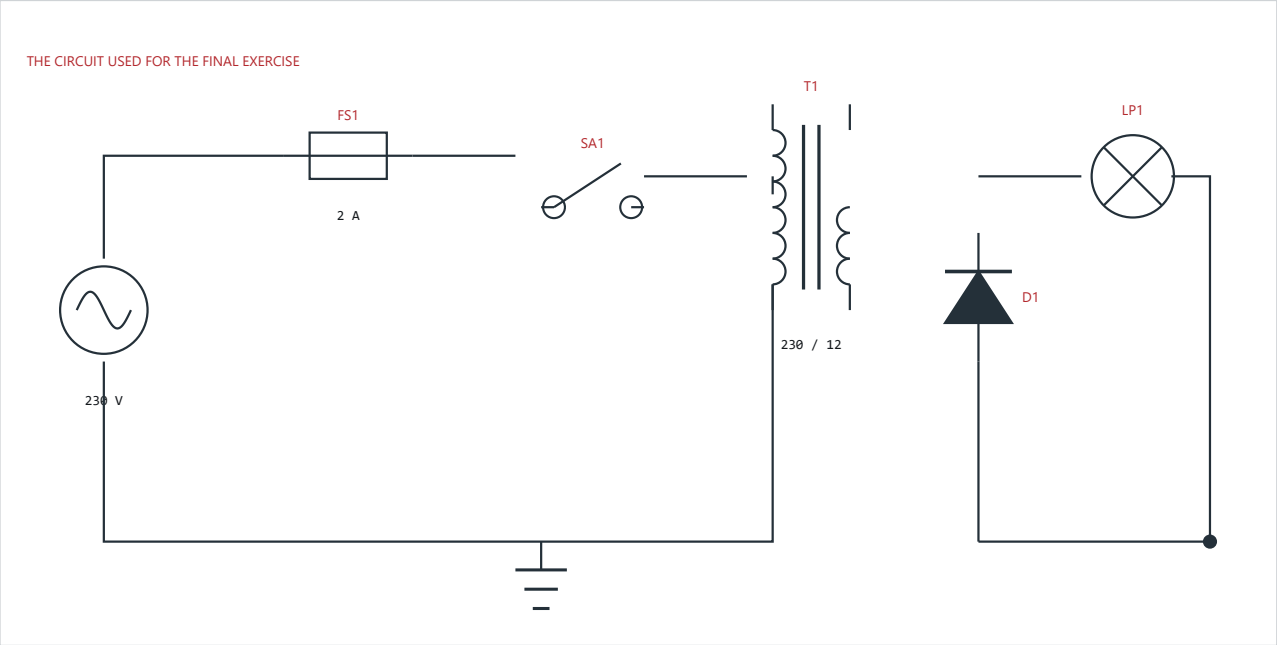



Fig. 20.1 The diagnostic sequence. Each measurement is chosen to divide the remaining possibilities roughly in half rather than to confirm a guess.

## Why speculative replacement fails

The commonest diagnostic error is to form a hypothesis from the symptom alone and then replace the suspected component. Sometimes this works, which is precisely why the habit survives. When it fails it costs the price of the part, the time to fit it, and the loss of information, because the original fault condition has now been disturbed.

A structured sequence costs a few minutes of measurement and returns a definite answer. It also leaves a record: the measurements themselves tell the next person what was ruled out. The sequence below is not specific to any one kind of circuit, which is what makes it worth learning once.

1. Establish the symptom precisely. What does the circuit do, and under what conditions? A fault that appears only when warm or only under load is a different problem from one that is always present.
2. Confirm the supply. Measure the voltage at the point of use, not at the source, and measure it under load. More apparent faults are supply faults than any other single cause.
3. Establish what the circuit should do. Write the expected voltage or current at two or three accessible points from the circuit model before measuring anything.
4. Divide the circuit at an accessible boundary and measure there. The result eliminates one half of the remaining circuit.
5. Repeat the division within the half that failed, until the fault is localised to a single component or connection.
6. Identify the mechanism, not just the component. A resistor that has failed open was probably overloaded by something else.
7. Correct the cause, then retest under the original conditions that produced the symptom.
8. Record the symptom, the decisive measurement, the cause and the correction.

## Dividing the circuit: the halving principle

A chain of ten elements with one fault requires ten tests if each is checked in turn, but only four if the chain is halved each time. The principle is the same one used in searching a sorted list, and it applies to any series arrangement: a supply chain, a control circuit, a signal path through several stages.

The requirement is an accessible boundary and a known expected value at that boundary. A terminal block, a connector, a test point or the input of an identifiable stage will all serve. Where the circuit is not a simple chain, divide it functionally instead: supply, input conditioning, processing, output stage, load.

### Choosing the quantity to measure

- Measure voltage with the circuit live and the load connected; it reveals what the circuit is actually doing.
- Measure resistance only with the circuit isolated and discharged; it reveals what the circuit is made of.
- Measure current when you need to know whether a path is carrying what it should.
- Compare against a written expected value, never against a vague impression of normality.
- Record each measurement as you take it; a diagnosis reconstructed from memory is unreliable.

```
tests in sequence ≈ n

tests by halving ≈ log₂ n

n = 64:  64 tests versus 6
```

## The four basic fault conditions

Almost every electrical fault reduces to one of four conditions, and each has a measurement that identifies it unambiguously. Learning these four and their signatures removes most of the guesswork from the final stage of a diagnosis.

An open circuit carries no current and shows the full supply voltage across the break. A short circuit carries excessive current and shows almost no voltage across the shorted part. A high-resistance joint is the most troublesome: it passes some current, develops a voltage drop proportional to that current, and therefore appears and disappears with load. A wrong value gives a circuit that works but not correctly, and is found by comparing measured values against calculated ones.

| Condition | Current | Voltage signature | Decisive test |
|---|---|---|---|
| Open circuit | Zero in that branch | Full supply across the break | Voltage across the suspected break, live |
| Short circuit | Excessive, device operates | Near zero across the shorted section | Resistance with the circuit isolated |
| High-resistance joint | Reduced, falls under load | Drop across the joint, rises with current | Voltage across the joint while on load |
| Wrong value fitted | Wrong but steady | Wrong but consistent with the fitted value | Compare measured value with calculated |
| Intermittent | Varies with time or temperature | Changes when disturbed | Monitor while flexing, warming or loading |

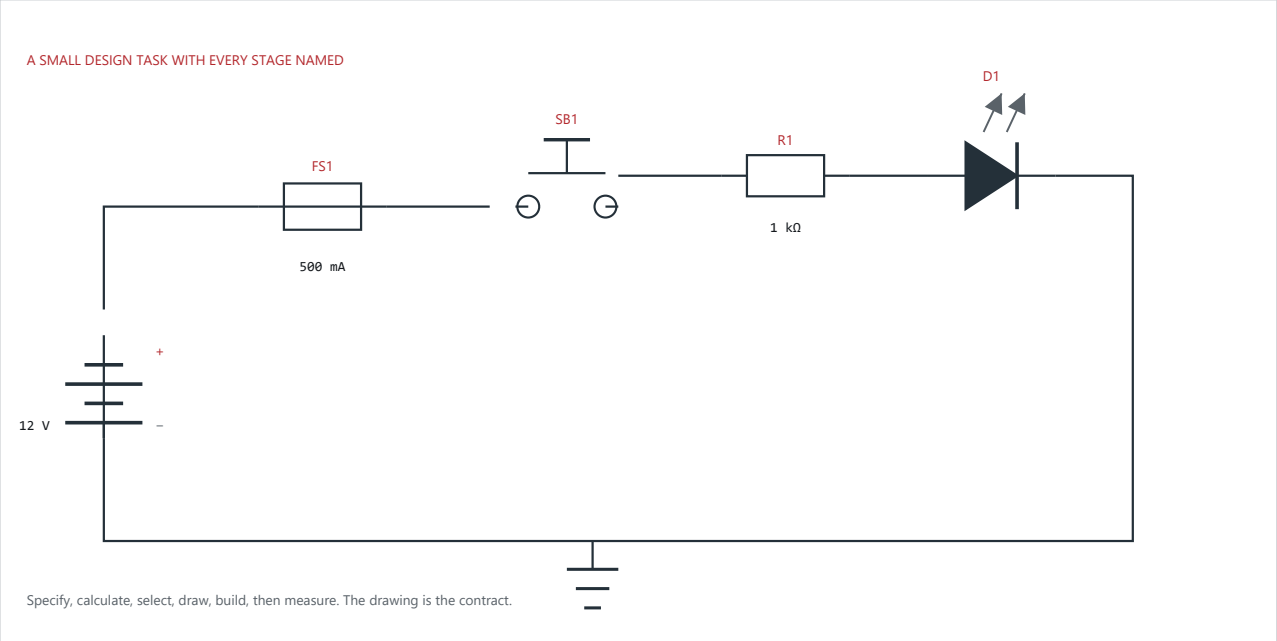



Fig. 20.2 The four fault conditions on one circuit. Each has a distinct voltage and current signature that identifies it without removing components.

## Reviewing a design before it is energised

The same discipline applies before the circuit is built. A design review is a short, ordered set of questions whose purpose is to find the error that would otherwise be found by the protective device. The questions are the ones a competent assessor would ask.

The order matters. Supply before load, load before conductor, conductor before protection, protection before enclosure, and safety considerations throughout rather than appended at the end. An error in the supply assumption invalidates everything downstream of it.

1. Supply: what voltage, what frequency, what phase arrangement, and what is the available fault current at the point of connection?
2. Load: what is the full-load current, what is the starting or inrush current, what is the duty cycle, and is the load inductive, capacitive or resistive?
3. Conductor: is the cross-sectional area adequate for the current after all derating, and is the voltage drop at full load within the permitted limit over the actual route length?
4. Protection: does the device carry the normal load without operating, and does it clear a fault at the furthest point within the required time?
5. Isolation: can the circuit be isolated, locked off and proved dead by one person at one place?
6. Earthing: is the protective conductor continuous, correctly sized and terminated throughout?
7. Environment: temperature, moisture, dust, vibration, and mechanical damage risk along the route.
8. Documentation: is the circuit recorded so that someone unfamiliar with it can work on it safely?

> QUESTIONS THAT CATCH THE MOST ERRORS
> Was the voltage drop computed over the real route length, not the straight-line distance?
> Does the protective device characteristic actually clear a fault at the far end of the run?
> Were derating factors applied for grouping, ambient temperature and insulation together?
> Does the starting current pass without nuisance operation of the protective device?
> Is every protective conductor sized against the fault current it may have to carry?

## Codes, editions and verifying at the source

Technical requirements are set by codes that are revised on a cycle, and a requirement remembered from an earlier edition may no longer be current. The habit worth forming is to identify which edition applies to the work in hand, then verify the specific requirement in that edition rather than relying on recollection.

The principles developed in this book do not change with an edition. Ohm's law, the conservation laws, the behaviour of reactive elements and the physiological effect of current are stable. What changes is the particular numerical limit, the required disconnection time, the permitted method. Keep the principles in your head and look the numbers up.

Where this book states a numerical limit, treat it as illustrative of the method rather than as the authority. The authority is the edition of the code in force in the jurisdiction where the work is carried out.

### Working with a code you do not have memorised

- Identify the code and edition that applies before quoting any numerical requirement.
- Verify the clause in that edition rather than reproducing it from memory or from a course note.
- Record which edition a design was assessed against, so later reviewers know the basis.
- Expect limits, disconnection times and permitted methods to change between editions.
- Expect the underlying physics and the diagnostic method not to change at all.

## Worked examples

Work each example with a pencil before reading the solution. Sketch the circuit, mark the reference node, and write the governing law before substituting numbers.

EXAMPLE 20.1 Locating a fault by halving a chain

**Given.** A lighting circuit has eight luminaires in a chain through seven accessible junction boxes. Nothing operates, and the supply measures 230 V at the distribution board.

**Find.** The number of measurements needed to find a single open circuit, and the sequence.

Measure at the middle box, box 4

```
230 V present: fault is downstream
```

Measure at box 6

```
0 V present: fault is between boxes 4 and 6
```

Measure at box 5

```
230 V present: fault is between boxes 5 and 6
```

Inspect that section

```
Open conductor or loose terminal located
```

Total measurements

```
3 rather than 7
```

**Check.** Three measurements replaced seven. The saving grows with the length of the chain, and the method needs no assumption about which component is most likely to fail.

EXAMPLE 20.2 Identifying a high-resistance joint

**Given.** A 12 V, 2 A lamp circuit gives a dim lamp. The supply measures 12.0 V at no load and 9.4 V at the lamp terminals with the lamp connected.

**Find.** The nature of the fault and the resistance of the offending joint.

Voltage lost in the circuit

```
12.0 - 9.4 = 2.6 V
```

Current actually flowing

```
Measured as 1.6 A, reduced by the added resistance
```

Resistance causing the loss

```
R = 2.6 / 1.6 = 1.63 Ω
```

Why resistance testing missed it

```
1.63 Ω is hard to see in series with the lamp cold resistance
```

Confirmation

```
Voltage measured directly across each joint while on load
```

**Check.** The decisive test is a voltage measurement across each joint while current flows. A resistance test with the circuit dead would not have found this reliably.

EXAMPLE 20.3 Design review of a motor supply

**Given.** A 400 V three-phase motor draws 18 A at full load with a starting current of six times full load. The run is 70 m. The permitted voltage drop is 5 per cent.

**Find.** The review findings in order.

Supply check

```
400 V three-phase confirmed at the point of connection
```

Full-load current

```
18 A continuous, 108 A for a few seconds at start
```

Permitted voltage drop

```
5 per cent of 400 V = 20 V
```

Conductor check

```
Area selected so that drop over 70 m at 18 A stays under 20 V
```

Protection check

```
Device must pass 108 A briefly yet clear a far-end fault in time
```

**Check.** The starting current governs the choice of protective device while the full-load current and route length govern the conductor. Reviewing them in order prevents the common error of sizing the conductor for the starting current or the device for the running current.

## Practical activity: a structured fault-finding exercise on a prepared board

> AIM
> Diagnose three faults introduced into a prepared low-voltage board using the halving method, recording the measurement sequence and identifying the fault condition in each case.

### Equipment

- Prepared fault board with switchable faults, 12 V direct supply.
- Regulated direct-voltage supply, 12 V, current limited.
- Digital multimeter with test leads in good condition.
- Circuit diagram of the board with expected voltages marked.
- Record sheet for the measurement sequence.

### Procedure

1. Study the diagram and write the expected voltage at every accessible test point before any measurement.
2. Energise the board with no fault set and verify your expected values. Resolve any disagreement now.
3. Have a partner set fault one without telling you which it is.
4. Record the symptom precisely before measuring anything.
5. Confirm the supply voltage at the board under load.
6. Choose a mid-circuit test point, predict the reading, then measure it and record both.
7. Continue halving, recording prediction and measurement at each step, until the fault is localised.
8. State the fault condition: open, short, high resistance or wrong value, and the evidence for it.
9. Repeat the whole sequence for faults two and three.
10. Count your measurements for each fault and compare with the number a sequential search would need.

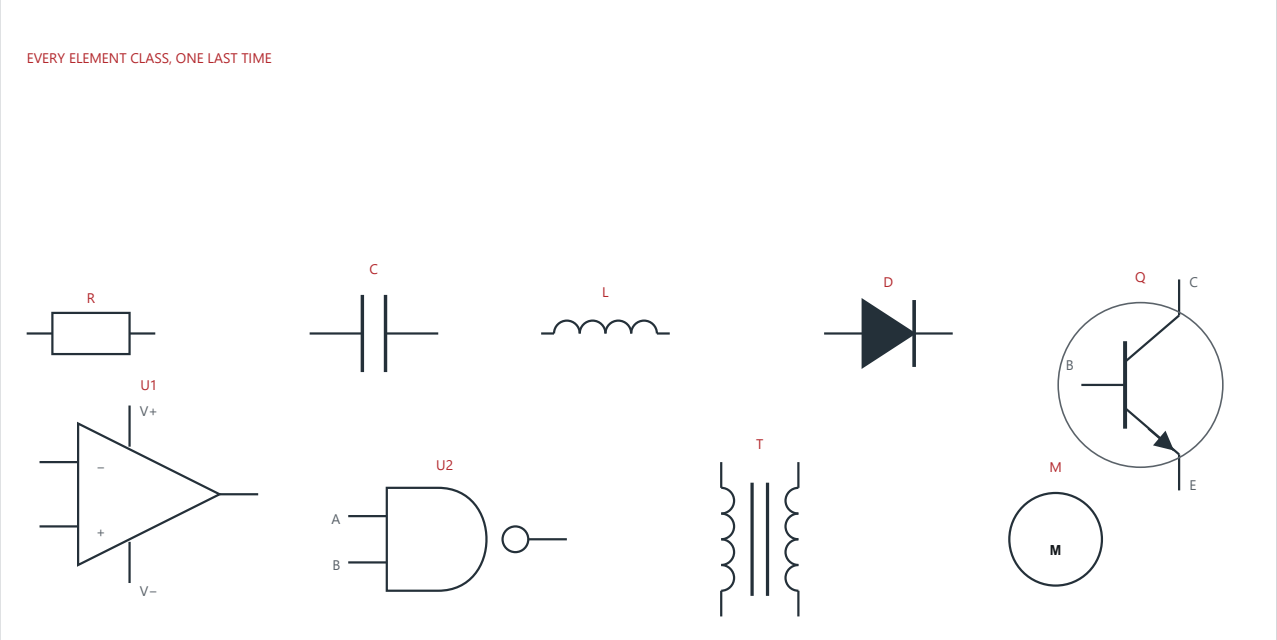



Fig. 20.L The prepared fault board. Test points divide the circuit into sections that can each be eliminated by a single measurement.

### Results table

| Fault | Symptom | Measurements used | Condition identified | Decisive reading |
|---|---|---|---|---|
| 1 | | | | |
| 2 | | | | |
| 3 | | | | |
| | | | | |
| | | | | |
| | | | | |

### Discussion

1. For which fault did the halving method save the most measurements, and why?
2. Which fault condition was hardest to identify by measurement, and what made it difficult?
3. Did any prediction disagree with a measurement on the healthy board? What did that reveal?
4. How would your sequence change if the board were a parallel rather than a series arrangement?
5. Which single measurement in each sequence was genuinely decisive?
6. Write the fault record for one fault as you would leave it for the next person.

> **SAFE-WORK REMINDERS**
> Keep the exercise at 12 V; the method is identical at any voltage and the risk is not.
> Current-limit the supply so that a set short circuit cannot overheat the board.
> Inspect test leads for damaged insulation before starting.
> Isolate and discharge before any resistance measurement.

## Troubleshooting guide

Faults are found by narrowing the circuit, not by replacing parts. Each row below pairs an observed symptom with the measurement that separates the likely causes.

| Symptom | Likely cause | Decisive test | Correct action |
|---|---|---|---|
| Nothing operates at all | Supply, main protective device, or a common open circuit | Measure voltage at the point of use under load | Restore supply, then find why the device operated |
| Works unloaded, fails under load | High-resistance joint or undersized conductor | Voltage across each joint and along the run while loaded | Remake the joint, or recalculate the conductor size |
| Protective device operates immediately on energising | Short circuit or wrong connection | Resistance between conductors with the circuit isolated | Find and clear the short before re-energising |
| Protective device operates after several minutes | Overload rather than a fault | Measure the actual load current and compare with the rating | Reduce the load or correct the device and conductor sizing |
| Correct voltage but wrong behaviour | Wrong component value fitted | Compare every measured value against the calculated value | Fit the correct value and recheck the whole stage |
| Fault appears only when warm | Thermal effect on a marginal joint or component | Monitor the suspect point while warming it gently | Remake the joint or replace the marginal component |
| Fault clears when the enclosure is disturbed | Mechanical: loose terminal or fractured conductor | Monitor voltage while gently flexing each section | Remake the termination, support the cable properly |
| Intermittent with no pattern | Vibration, moisture ingress, or a cracked joint | Monitor over time while varying one condition | Address the environmental cause, not just the symptom |

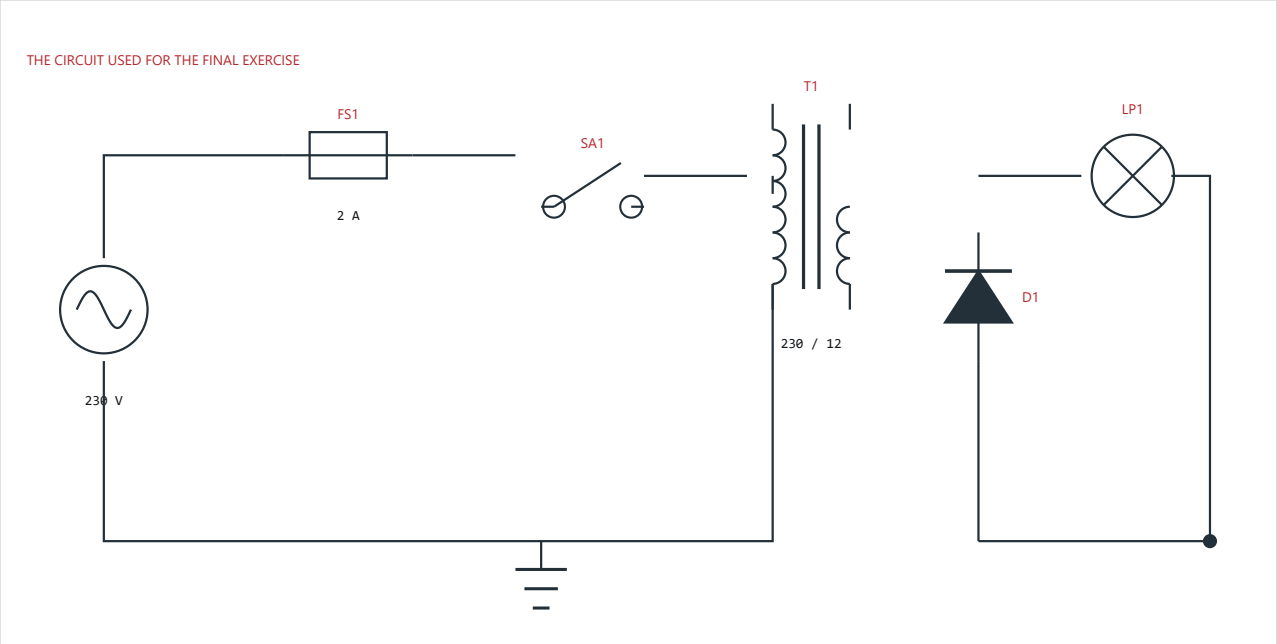



Fig. 20.T The complete method, from symptom to record. The final step is the one most often omitted and the one that saves the most time later.

FIELD HABITS THAT PREVENT CALLBACKS

Write the expected value before taking the measurement; otherwise any reading looks plausible.
Confirm the supply at the point of use and under load before suspecting anything downstream.
Divide the circuit to halve the search space; do not test components in the order you remember them.
Measure voltage live and under load, resistance only when isolated and discharged.
Find the mechanism, not only the failed component, or the same fault will return.
Retest under the conditions that produced the original symptom, not under easier ones.
Record the symptom, the decisive measurement, the cause and the correction.
Verify any numerical requirement in the edition of the code that applies to the work.

## Exercises

Answer from first principles. Selected answers follow; the reasoning matters more than the number.

1 List the eight steps of the diagnostic sequence in order, and justify the position of each.
2 Explain why speculative component replacement sometimes works and why it is still poor practice.
3 A chain has 32 accessible test points and one open circuit. How many measurements does halving need?
4 State the voltage and current signature of an open circuit, and the test that confirms it.
5 State the voltage and current signature of a high-resistance joint, and why load is needed to find it.
6 Why must resistance measurements be made only on an isolated and discharged circuit?
7 A 24 V, 3 A circuit loses 4.2 V somewhere in the supply path at 2.4 A. Find the offending resistance.
8 List the design review questions in order and explain why supply must precede load.
9 Explain why starting current governs the protective device while route length governs the conductor.
10 A circuit works but a measured voltage disagrees with calculation by thirty per cent. How do you proceed?
11 Describe how you would diagnose a fault that appears only after twenty minutes of operation.

12 Write a fault record for a loose neutral terminal found in a lighting circuit.

13 Explain why a requirement remembered from an earlier code edition should not be relied upon.

14 Name three things this book has taught that will not change with a code revision.

15 Choose any chapter and state how its content would appear in a diagnostic sequence.

## Selected answers

| # | Answer | Reason |
|---|---|---|
| 3 | Five measurements | Log to base two of 32 is 5. |
| 4 | No current, full supply voltage across the break | The break carries the whole source voltage because no current flows elsewhere. |
| 7 | 1.75 Ω | Resistance equals voltage lost divided by current, 4.2 divided by 2.4. |
| 9 | Starting current is brief but large, running current is continuous | The device must tolerate the brief peak; the conductor must carry the continuous current with acceptable drop. |
| 13 | Limits and methods change between editions | Only the edition in force in that jurisdiction is authoritative. |

## Chapter summary

- A structured diagnostic sequence converges on a fault; speculative replacement does not.
- Confirming the supply at the point of use and under load eliminates the commonest cause first.
- Halving the circuit at an accessible boundary reduces the number of tests from n to about log base two of n.
- Four conditions cover almost every fault: open, short, high resistance and wrong value.
- A high-resistance joint is found by measuring voltage across it while current flows, not by resistance testing.
- A design review asks supply, load, conductor, protection, isolation, earthing, environment and documentation, in that order.
- Codes change by edition; principles do not. Keep the method in your head and verify the numbers at source.
- Recording a fault and its resolution is part of the repair, not an optional addition.

### Equations introduced in this chapter

| Relationship | Expression | When it applies |
|---|---|---|
| Sequential search | tests ≈ n | Testing elements one at a time along a chain |
| Halving search | tests ≈ $\log_2 n$ | Dividing at an accessible boundary each time |
| Joint resistance from loss | R = V_lost / I | Voltage measured across a joint while on load |
| Permitted voltage drop | V_drop = percentage × V_supply | Checking conductor adequacy over the route |
| Power lost in a joint | $P = I^2 R$ | Assessing whether a joint will overheat |
| Starting current | I_start = k × I_FL | Sizing protection for motor and transformer loads |

### Vocabulary check

| Term | Meaning in this chapter |
|---|---|
| Symptom | The precisely stated observable behaviour of the faulty circuit |
| Halving | Dividing the search space in two with each measurement |
| Decisive test | The measurement that distinguishes between competing causes |
| Open circuit | A break carrying no current, with full voltage across it |
| Short circuit | An unintended low-resistance path carrying excessive current |
| High-resistance joint | A connection passing current but developing a voltage drop |
| Design review | An ordered set of checks applied before energising |
| Derating | Reducing a conductor's permitted current for its installed conditions |
| Fault record | The written symptom, measurement, cause and correction |

LOOKING AHEAD

That completes the twenty chapters. The reference section that follows collects the equations, units, symbols, checklists and glossary in one place, so that the method in this chapter can be applied without turning back through the book.

# Reference section

Everything in this section appears somewhere in the chapters. It is gathered here so that it can be used without searching. The equation list is ordered by subject, and every entry states the conditions under which it applies.

## Fundamental relationships

The relationships that follow directly from the definitions of charge, current, voltage and energy. These hold for any circuit at any instant.

| Quantity | Expression | Conditions |
|---|---|---|
| Current | $I = Q / t$ | Steady current; otherwise the instantaneous rate of charge flow |
| Voltage | $V = W / Q$ | Energy per unit charge between two points |
| Resistance | $R = V / I$ | Defines resistance; constant only for a linear element |
| Ohm's law | $V = I R$ | Linear resistive element at constant temperature |
| Power | $P = V I$ | Any element, instantaneous values |
| Power, resistive | $P = I^2 R = V^2 / R$ | Resistive element only |
| Energy | $E = P t$ | Constant power over the interval t |
| Resistivity | $R = \rho L / A$ | Uniform conductor of length L and area A |
| Temperature effect | $R_t = R_0(1 + \alpha \Delta T)$ | Over a modest temperature range |

## Network analysis

The conservation laws and the standard network reductions derived from them.

| Relationship | Expression | When it applies |
|---|---|---|
| Current law | $\Sigma I_{in} = \Sigma I_{out}$ | At any node, at every instant |
| Voltage law | $\Sigma V = 0$ | Around any closed loop |
| Series resistance | $R = R_1 + R_2 + \dots$ | Elements carrying the same current |
| Parallel resistance | $1/R = 1/R_1 + 1/R_2 + \dots$ | Elements across the same voltage |
| Two in parallel | $R = R_1 R_2 / (R_1 + R_2)$ | Convenient form for exactly two |
| Voltage divider | $V_2 = V R_2 / (R_1 + R_2)$ | Series pair, negligible load current |
| Current divider | $I_1 = I R_2 / (R_1 + R_2)$ | Two parallel branches |
| Star to delta | $R_d = 3 R_s$ | Balanced three-element conversion only |

## Capacitance, inductance and transients

Energy storage and the exponential response that follows from it. Time constants assume a single storage element and a single resistance.

| Relationship | Expression | When it applies |
|---|---|---|
| Charge stored | $Q = C\,V$ | Any capacitor at any instant |
| Capacitor energy | $E = \tfrac{1}{2}\,C\,V^2$ | Energy stored in the electric field |
| Capacitors in parallel | $C = C_1 + C_2 + \ldots$ | Same voltage across each |
| Capacitors in series | $1/C = 1/C_1 + 1/C_2 + \ldots$ | Same charge on each |
| Resistive-capacitive constant | $\tau = R\,C$ | Single resistance and capacitance |
| Charging | $v = V(1 - e^{(-t/\tau)})$ | Step applied to an initially discharged capacitor |
| Discharging | $v = V\,e^{(-t/\tau)}$ | No source, initial voltage V |
| Induced voltage | $e = -L\,(di/dt)$ | Self-inductance opposing current change |
| Inductor energy | $E = \tfrac{1}{2}\,L\,I^2$ | Energy stored in the magnetic field |
| Resistive-inductive constant | $\tau = L / R$ | Single inductance and resistance |
| Practical settling | $5\tau$ | Within one per cent of the final value |

## Alternating quantities, impedance and power

Sinusoidal steady state. Root-mean-square values are assumed unless stated otherwise, and the form factors given apply to sine waves only.

| Relationship | Expression | When it applies |
|---|---|---|
| Period and frequency | $T = 1 / f$ | Any periodic waveform |
| Angular frequency | $\omega = 2\pi f$ | Sinusoidal quantities |
| Root-mean-square value | $V_{rms} = V_{pk} / \sqrt{2}$ | Sine waves only |
| Average value | $V_{av} = 0.637\,V_{pk}$ | Rectified sine, half cycle |
| Inductive reactance | $X_L = 2\pi f L$ | Ideal inductor |
| Capacitive reactance | $X_C = 1/(2\pi f C)$ | Ideal capacitor |
| Series impedance | $Z = \sqrt{(R^2 + X^2)}$ | Series resistance and net reactance |
| Phase angle | $\varphi = \arctan(X / R)$ | Series circuit |
| Power factor | $pf = \cos\varphi = P / S$ | Sinusoidal conditions |
| True power | $P = V\,I\cos\varphi$ | Power converted to another form |
| Apparent power | $S = V\,I$ | Determines conductor and supply rating |
| Reactive power | $Q = V\,I\sin\varphi$ | Power exchanged with storage elements |

## Resonance and filters

The frequency at which reactances cancel, and the single-stage filter relationships.

| Relationship | Expression | When it applies |
|---|---|---|
| Resonant frequency | $f_0 = 1/(2\pi \sqrt{(L\,C)})$ | Series or parallel, lightly damped |
| Condition at resonance | X_L = X_C | Definition of resonance |
| Series resonance impedance | Z = R | Reactances cancel, current is maximum |
| Quality factor | $Q = 2\pi f_0 L / R$ | Series resonant circuit |
| Bandwidth | $BW = f_0 / Q$ | Between the half-power points |
| Voltage magnification | V_L = Q V_in | Across either reactive element at resonance |
| Filter cut-off | f_c = 1/(2π R C) | Single-stage resistive-capacitive filter |
| Half-power point | 0.707 of maximum | Defines the cut-off frequency |

## Magnetism, transformers and machines

Magnetic circuit quantities and the machine relationships that follow from them.

| Relationship | Expression | When it applies |
|---|---|---|
| Magnetomotive force | F = N I | Coil of N turns carrying current I |
| Flux density | B = Φ / A | Uniform flux through area A |
| Induced voltage | e = -N (dΦ/dt) | Faraday, flux linking N turns |
| Force on a conductor | F = B I L | Conductor perpendicular to the field |
| Turns ratio | $V_1/V_2 = N_1/N_2$ | Ideal transformer, no losses |
| Current ratio | $I_1/I_2 = N_2/N_1$ | Ideal transformer, power conserved |
| Impedance ratio | $Z_1/Z_2 = (N_1/N_2)^2$ | Impedance referred across a transformer |
| Synchronous speed | n_s = 120 f / p | Alternating machine, p poles, speed in rev/min |
| Slip | s = (n_s - n)/n_s | Induction machine running below synchronous speed |
| Efficiency | η = P_out / P_in | Any machine, consistent units |

## Electronics and conversion

Semiconductor, amplifier, logic and converter relationships.

| Relationship | Expression | When it applies |
|---|---|---|
| Diode forward drop | ≈ 0.7 V silicon, 0.3 V germanium | Conducting, order of magnitude |
| Half-wave ripple frequency | $f_r = f$ | One pulse per supply cycle |
| Full-wave ripple frequency | $f_r = 2\,f$ | Bridge or centre-tapped rectifier |
| Reservoir capacitor ripple | $V_r \approx I / (f_r\,C)$ | Light ripple approximation |
| Transistor current gain | $\beta = I_C / I_B$ | Active region, common emitter |
| Saturation condition | $I_B > I_C / \beta$ | Transistor used as a switch |
| Inverting amplifier gain | $A = -R_f / R_{in}$ | Ideal operational amplifier |
| Non-inverting gain | $A = 1 + R_f / R_g$ | Ideal operational amplifier |
| Converter steps | $2^n$ | Converter of n bits |
| Quantisation step | $V_{range} / 2^n$ | Uniform converter |
| Regulator dissipation | $P = (V_{in} - V_{out})\,I$ | Series linear regulator |

## Three-phase systems

Balanced three-phase relationships. All voltages and currents are root-mean-square values.

| Relationship | Expression | When it applies |
|---|---|---|
| Star voltage relation | $V_{line} = \sqrt{3}\,V_{phase}$ | Balanced star connection |
| Star current relation | $I_{line} = I_{phase}$ | Balanced star connection |
| Delta voltage relation | $V_{line} = V_{phase}$ | Balanced delta connection |
| Delta current relation | $I_{line} = \sqrt{3}\,I_{phase}$ | Balanced delta connection |
| Three-phase power | $P = \sqrt{3}\,V_L\,I_L \cos\varphi$ | Balanced load, either connection |
| Apparent power | $S = \sqrt{3}\,V_L\,I_L$ | Balanced load |
| Neutral current | zero | Balanced linear load in star |

## Safety and fault clearance

Relationships used in assessing protection and shock risk.

| Relationship | Expression | When it applies |
|---|---|---|
| Body current | $I = V_{contact} / Z_{body}$ | Order-of-magnitude shock assessment |
| Earth fault loop current | $I_f = V / Z_{loop}$ | Fault current available to operate the device |

| Relationship | Expression | When it applies |
|---|---|---|
| Joint resistance from loss | R = V_lost / I | Measured across a joint while on load |
| Power lost in a joint | P = $I^2$ R | Assessing whether a joint will overheat |
| Permitted voltage drop | V_drop = percentage × V_supply | Conductor adequacy over the route |
| Starting current | I_start = k × I_FL | Sizing protection for motors and transformers |

## SI units and prefixes

The units used throughout the book and the prefixes applied to them. Writing the unit beside every number is the cheapest error check available.

| Quantity | Symbol | Unit | Unit symbol |
|---|---|---|---|
| Charge | Q | coulomb | C |
| Current | I | ampere | A |
| Voltage | V | volt | V |
| Resistance | R | ohm | Ω |
| Conductance | G | siemens | S |
| Capacitance | C | farad | F |
| Inductance | L | henry | H |
| Reactance | X | ohm | Ω |
| Impedance | Z | ohm | Ω |
| Frequency | f | hertz | Hz |
| Power, true | P | watt | W |
| Power, apparent | S | volt-ampere | VA |
| Power, reactive | Q | volt-ampere reactive | var |
| Energy | E | joule | J |
| Magnetic flux | Φ | weber | Wb |
| Flux density | B | tesla | T |
| Temperature | T | kelvin or degree Celsius | K, °C |

## Prefixes

Prefixes in common electrical use, with the multiplier each represents.

| Prefix | Symbol | Multiplier | Typical electrical use |
|---|---|---|---|
| giga | G | $10^{9}$ | Gigahertz clock and radio frequencies |
| mega | M | $10^{6}$ | Megohm insulation resistance |
| kilo | k | $10^{3}$ | Kilovolt distribution, kilohm resistors |
| unity | — | 1 | Volts, amperes, ohms |
| milli | m | $10^{-3}$ | Milliampere shock thresholds, millihenry coils |
| micro | μ | $10^{-6}$ | Microfarad capacitors, microvolt signals |
| nano | n | $10^{-9}$ | Nanofarad film capacitors, nanosecond delays |

| Prefix | Symbol | Multiplier | Typical electrical use |
|---|---|---|---|
| pico | p | $10^{-12}$ | Picofarad stray capacitance |

## Reading a value from a number and a prefix

The commonest arithmetic errors in electrical work are prefix errors, not algebra errors. These conversions are worth being able to do without hesitation.

| Written | Means | In base units | Common context |
|---|---|---|---|
| 4k7 | 4.7 kΩ | 4700 Ω | Resistor marking without a decimal point |
| 100n | 100 nF | 0.0000001 F | Film capacitor marking |
| 470µ | 470 µF | 0.00047 F | Electrolytic reservoir capacitor |
| 2M2 | 2.2 MΩ | 2200000 Ω | High-value bias resistor |
| 15mH | 15 millihenry | 0.015 H | Small choke or filter inductor |
| 30mA | 30 milliampere | 0.03 A | Residual current device rating |
| 400V | 400 volt | 400 V | Three-phase line voltage |
| 1.5kW | 1.5 kilowatt | 1500 W | Small motor rating |

## Measurement checklist

Applied before every measurement. The sequence prevents the two commonest errors: measuring the wrong quantity, and disturbing the circuit by measuring it.

| Stage | Check | Why it matters |
|---|---|---|
| Before | Identify the quantity you actually need | Voltage, current and resistance answer different questions |
| Before | Write the expected value | Any reading looks plausible without a prediction |
| Before | Select the function and range | Wrong function can damage the instrument or the circuit |
| Before | Prove the instrument on a known source | An instrument that reads zero may be faulty, not the circuit |
| Before | Inspect leads and probes | Damaged insulation is the commonest instrument hazard |
| During | Identify the reference point | A voltage without a reference is meaningless |
| During | Consider the loading effect | A low-impedance meter changes a high-impedance circuit |
| During | Measure under the relevant load | Many faults appear only when current flows |
| After | Compare with the expected value | Disagreement is information, not an inconvenience |
| After | Prove the instrument again | Confirms it was working during the measurement |
| After | Record the reading | A measurement not recorded will be taken again |

## Safe isolation sequence

The sequence is not negotiable and the order is the point. Each step depends on the one before it being complete.

| Step | Action | Reason |
|---|---|---|
| 1 | Identify the circuit and its source | Isolating the wrong circuit is worse than not isolating |
| 2 | Obtain permission to isolate | Other people may depend on the circuit |
| 3 | Isolate at the correct point | Switching the load off is not isolation |
| 4 | Secure the isolation, lock off, retain the key | Prevents re-energising by another person |
| 5 | Post a warning notice | Tells others why the circuit is off |
| 6 | Prove the test instrument on a known source | Establishes the instrument works now |
| 7 | Test all conductor combinations | A single test can miss a live conductor |
| 8 | Re-prove the instrument | Confirms it still worked during the test |
| 9 | Address stored energy | Capacitors and standing charge remain after isolation |
| 10 | Begin work | Only now is the circuit safe to work on |

## Glossary A to I

Terms as used in this book. Where general usage is looser, the definition here is the one intended in these chapters.

| Term | Meaning |
|---|---|
| Admittance | The reciprocal of impedance. |
| Alternating current | Current that reverses direction periodically. |
| Ampere | The unit of current, one coulomb per second. |
| Apparent power | The product of root-mean-square voltage and current, in volt-amperes. |
| Bandwidth | The frequency span between the half-power points of a resonant circuit. |
| Capacitance | The charge stored per unit voltage applied. |
| Conductance | The reciprocal of resistance. |
| Coulomb | The unit of charge. |
| Cut-off frequency | The frequency at which a filter output falls to 0.707 of its maximum. |
| Derating | Reducing a conductor’s permitted current for its installed conditions. |
| Dielectric | The insulating material between capacitor plates. |
| Differential | The width of the switching band in an on-off controller. |
| Duty cycle | The fraction of a period for which a switched output is on. |
| Eddy current | Circulating current induced in a conducting core, causing loss. |
| Efficiency | Output power divided by input power. |
| Electromotive force | The voltage generated by a source, before internal loss. |
| Error | In control, the set point minus the process variable. |
| Farad | The unit of capacitance. |
| Flux density | Magnetic flux per unit area, in tesla. |
| Form factor | The ratio of root-mean-square to average value of a waveform. |
| Henry | The unit of inductance. |
| Hysteresis | Lagging of a magnetic or control response behind its cause. |
| Impedance | The total opposition to alternating current, combining resistance and reactance. |
| Inductance | The property producing a voltage in opposition to a change of current. |
| Inrush current | The brief high current drawn when a load is first energised. |
| Insulation resistance | The resistance of insulation between conductors or to earth. |

## Glossary J to R

Continued. Cross-references are to the chapter in which the term is developed.

| Term | Meaning |
|---|---|
| Joule | The unit of energy. |
| Kirchhoff’s current law | The sum of currents into a node equals the sum leaving it. |
| Kirchhoff’s voltage law | The sum of voltages around a closed loop is zero. |
| Load | The element or circuit that converts supplied electrical energy. |
| Magnetomotive force | The product of turns and current driving flux in a magnetic circuit. |
| Node | A junction at which two or more circuit elements meet. |
| Offset | In control, the steady error left by proportional-only action. |
| Ohm | The unit of resistance. |
| Open circuit | A break in a path, carrying no current. |
| Overcurrent device | A fuse or circuit breaker interrupting excessive current. |
| Peak value | The maximum instantaneous value of a waveform. |
| Permeability | The ease with which a material carries magnetic flux. |
| Phase angle | The angular difference between voltage and current waveforms. |
| Power factor | The cosine of the phase angle; true power divided by apparent power. |
| Process variable | The measured value of the quantity under control. |
| Protective conductor | A conductor provided to carry fault current safely to earth. |
| Quality factor | A measure of the sharpness of resonance. |
| Quantisation step | The input change corresponding to one converter count. |
| Reactance | The opposition to alternating current offered by inductance or capacitance. |
| Reactive power | Power exchanged with storage elements, in volt-amperes reactive. |
| Rectification | Conversion of alternating to unidirectional current. |
| Residual current device | A device operating on an imbalance between live and neutral current. |
| Resistivity | The resistance of unit length and unit area of a material. |
| Resonance | The condition in which inductive and capacitive reactance cancel. |
| Ripple | The residual alternating component on a rectified and smoothed supply. |
| Root-mean-square | The equivalent steady value producing the same heating effect. |

## Glossary S to Z

Continued.

| Term | Meaning |
|---|---|
| Saturation | The condition in which a core carries little additional flux, or a transistor is fully on. |
| Series | Elements connected so that the same current passes through each. |
| Set point | The value a controlled quantity is required to reach. |
| Short circuit | An unintended low-resistance path carrying excessive current. |
| Slip | The fractional difference between synchronous and actual rotor speed. |
| Star connection | A three-phase arrangement with one end of each winding at a common point. |
| Delta connection | A three-phase arrangement with the windings connected end to end in a loop. |
| Synchronous speed | The rotational speed of the rotating field in an alternating machine. |
| Tesla | The unit of magnetic flux density. |
| Thermistor | A resistor whose resistance varies strongly with temperature. |
| Thermocouple | A junction of dissimilar metals producing a temperature-dependent voltage. |
| Time constant | The product of resistance and capacitance, or inductance over resistance. |
| Transducer | A device converting one physical quantity into another. |
| Transformer | A device transferring energy between windings by mutual induction. |
| Transient | The temporary response of a circuit between two steady states. |
| True power | Power actually converted to another form, in watts. |
| Turns ratio | The ratio of primary to secondary turns in a transformer. |
| Volt | The unit of potential difference, one joule per coulomb. |
| Voltage drop | The voltage lost along a conductor carrying current. |
| Watt | The unit of power, one joule per second. |
| Weber | The unit of magnetic flux. |

## Where to go next

Suggested routes through the material for different purposes, using chapters already in this book.

| Purpose | Route | Focus |
|---|---|---|
| Grounding in fundamentals | Chapters 1 to 4 | Models, conservation laws, networks and transients |
| Alternating current competence | Chapters 6 to 8 | Waveforms, impedance, power factor, resonance |

| Purpose | Route | Focus |
|---|---|---|
| Machines and power | Chapters 9 to 11 and 17 | Magnetism, transformers, motors, three-phase |
| Electronics | Chapters 12 to 16 | Diodes, transistors, amplifiers, logic, supplies |
| Field practice | Chapters 18 to 20 | Safety, fault paths, control, diagnosis |
| Measurement discipline | The practical activity in every chapter | Prediction, measurement, explanation |

A CLOSING NOTE

The method in the final chapter is the most portable thing in this book. Equations can be looked up and code limits change between editions, but the habit of writing the expected value before taking the measurement, of dividing the circuit rather than guessing at it, and of recording what was found, will serve on every circuit you ever meet.